

ÁLVARO YÁNGÜEZ BACHILLER

COMPUTATIONAL QUANTUM INFORMATION

FROM FOUNDATIONS TO APPLICATIONS



SORBONNE UNIVERSITÉ

COMPUTATIONAL QUANTUM INFORMATION

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Wir müssen wissen, wir werden wissen.

David Hilbert

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Publications Underlying This Thesis

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- [1] Eleni Diamanti, Alex B. Grilo, Adriano Innocenzi, Pascal Lefebvre, Verena Yacoub, and Álvaro Yángüez. *A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions*. Quantum 10, 2124, 2026.
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- [4] Álvaro Yángüez, Noam Avidan, Jan Kochanowski, and Thomas A. Hahn. *Accessible Quantum Correlations Under Complexity Constraints*. arXiv:2604.15540, 2026.

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Chapter 1

Introduction

The most obvious, important realities are often the ones that are hardest to see and talk about.

David Foster Wallace, *This Is Water*

The frontier between mathematics and physics is often subtle. When working from a theoretical perspective, we, as scientists, are tempted to prioritize simplicity and elegance over reality. Nevertheless, what ultimately matters for a physical model is that it remains connected to reality. For quantum systems, this means looking beyond what is theoretically allowed and asking what can actually be implemented. This is the underlying motivation of this thesis. Its aim is to develop a mathematical framework for studying how *complexity*¹ reshapes which properties can be observed, which states can be distinguished, and which resources can be extracted from quantum systems.

In the twentieth century, two of the most relevant intellectual revolutions came from, a priori, very different theories: *quantum theory* and *information theory*. The first changed our perception of physical systems [Hei25, Sch26, vN27], while the second provided a mathematical language for processing and communicating information, eventually leading to a technological revolution [Sha48]. For several decades, these two fields evolved mostly in parallel. Nevertheless, if information is physical [Lan61], and physical processes are ultimately implemented by systems governed by quantum mechanics, then it is natural to ask how information processing changes when it is described by the laws of quantum theory. This question, emphasized by Feynman among others [Fey82], lies at the origin of *quantum information theory* (QIT) [NC10, Wat18].

The success of QIT comes, in part, from its ability to reduce the study of a wide range of physical processes and phenomena to a small number of fundamental quantities. Tasks such as hypothesis testing [ON00, ACMnT⁺07], resource manipulation [BG15, CG19], quantum thermodynamics [BHN⁺15, ECP10, CRF21], and quantum cryptography [Ren08, DW05, Wol21, VW23] can often be described through mathematical objects such as quantum divergences, entropies, and distinguishability measures [NC10, Wat18]. This abstraction

¹Here and throughout the thesis, complexity refers to computational complexity, understood through the resources needed to prepare a state or implement an operation in a circuit model. The connection between this notion and physical implementability is discussed below through the Church–Turing thesis perspective and the quantum circuit model.

is one of the strengths of QIT, but it also hides an idealization. Most of these operational quantities are defined by assuming the availability of all possible measurements or transformations, as if every operation allowed by the formalism were physically accessible.

This idealization becomes problematic when one asks which operations can actually be implemented. A transformation is not available merely because it is allowed by the theoretical model; it has to be physically realized in a concrete experimental setting. Quantum systems can be realized in many different ways, including integrated photonics, neutral-atom arrays, trapped ions, and superconducting circuits [WSLT20, BL20, BCMS19, KSB⁺20]. Although these realizations are very different, they share similar constraints: measurements and transformations have to be constructed from a limited set of physical operations. Moreover, these operations have locality constraints that affect the spatial and temporal complexity of the implementation. As the size of the studied system grows, these restrictions become increasingly relevant: the question is no longer only whether a measurement or transformation is mathematically allowed, but whether it can be constructed from the operations available in practice. The theoretical description should therefore incorporate these restrictions from the outset.

To incorporate this limitation, one needs an abstract description of what can actually be performed that does not depend on a specific experimental setting. The approach adopted in this thesis is to use *computational complexity* to provide such a description. Church and Turing made precise the idea that any function calculable by a completely mechanical procedure can be computed by a Turing machine². Physical formulations of the Church-Turing thesis extend the same question from abstract calculations to physical processes, asking which computations can be carried out by systems subject to natural finiteness and locality assumptions [Gan80, AD12]. The extended, or complexity-theoretic, Church-Turing thesis also considers efficiency: in the classical setting, every realistic model of computation is expected to be efficiently simulable by a probabilistic Turing machine, i.e., with at most polynomial overhead [BV97, AB09]. One may then ask how this picture changes when quantum processes are taken into account. Finite descriptions of quantum computations are still Turing-computable, but quantum computers are not expected to be efficiently simulable by classical probabilistic machines. Thus, when the underlying physical systems are quantum, a quantum Turing machine becomes the natural reference model for efficient implementation [Deu85, BV97]. Inspired by these ideas, this thesis adopts the following modelling principle: scalable implementability in quantum systems is represented by the existence of *polynomial-size* quantum circuit descriptions.

This modelling principle already separates our approach from standard QIT. In the unconstrained information-theoretic setting, mathematically admissible states are treated as preparable, and mathematically admissible measurements and channels as implementable. Nevertheless, a simple counting argument shows that almost all states, and likewise most measurements and transformations, require exponential-size circuits to be generated or

²Turing formulates the underlying intuition as follows: “A function is said to be effectively calculable if its values can be found by some purely mechanical process” [Tur39]; see also the original works of Church and Turing [Chu36, Tur37].

implemented [Kni95, Aar16]. More recent results go beyond this typicality statement by constructing explicit families of states with exponential circuit complexity [JW23]. Hence, complexity does not merely add a cost to an otherwise unchanged operational theory: it changes which states, measurements and transformations are valid. This distinction and its operational consequences are the central objects of study throughout this thesis.

This question has recently become the focus of several related approaches. A first line of work studies quantum states and transformations under computational restrictions. A central example is the quantum analogue of pseudorandomness, known as *pseudorandom quantum states* (PRS): efficiently generated quantum states that are computationally indistinguishable from Haar-random states [JLS18]. This same idea has since been applied to quantum resources, showing that highly nonclassical states can be indistinguishable from simpler ones for every polynomial-time observer [ABF⁺24, ABV23, GE24, LREJ25, HBK25, BMB⁺25, MKH⁺25]. In entanglement theory, this has led to one-shot notions of computational entanglement cost and distillable entanglement under efficient LOCC constraints [ABV23], as well as asymptotic results showing large separations between information-theoretic and computational notions of entanglement manipulation [LREJ25]. Together, these works show that computational restrictions can hide resources such as entanglement: a state may contain them in the information-theoretic sense, while efficient observers or efficient protocols fail to access them.

A second line of work focuses on the information measures themselves. Computational quantum entropies aim to retain the operational meaning of their information-theoretic counterparts while restricting the adversary or observer to efficient strategies [CCL⁺17, AA26, AHRA25]. Related complexity-constrained divergences have been connected to efficient data compression, decoupling, thermodynamics, and hypothesis testing [YHKH⁺22, MKH⁺25]. In this direction, the central question is how entropies, divergences, and distinguishability measures should be formulated when the relevant measurements and transformations are required to be efficiently implementable.

The results presented in this thesis contribute to this emerging programme by developing a unified framework for computational quantum information. We begin from the second line of work, by studying how computational restrictions redefine core operational quantities of quantum information theory. To do so, we focus on computationally constrained observers: if only efficient measurements and channels are accessible, then the norms, divergences, and entropies used to describe distinguishability and correlations have to be rebuilt with this restriction in mind. In this way, efficient-measurement divergences and computational entropies are placed within the same mathematical formalism [CCL⁺17, AA26, AHRA25, MRR⁺25, MRR⁺26].

This computational framework also allows us to study computationally hidden quantum resources. In our setting, this phenomenon appears through quantitative separations between information-theoretic resources and the part of them that can be accessed by efficient observers. In this sense, the results connect naturally with computational entanglement theory [ABV23, LREJ25] and with pseudoentanglement [ABF⁺24, GE24].

The same point of view also leads to *cryptographic* consequences. In classical cryptography, randomness and pseudorandomness play a central role; in the quantum setting, recent work suggests that quantum resources may play an analogous role through pseudoresources, i.e., families of states with different resource content that remain computationally indistinguishable among themselves [HBK25, BMB⁺25]. This thesis explores this connection by showing how computationally hidden resources can imply cryptographic primitives such as *quantum commitments*. This builds on the relation between EFI pairs and commitments³. Finally, we focus on a more concrete near-term direction by constructing a quantum protocol for *oblivious transfer*, a core cryptographic primitive from which powerful functionalities such as secure multiparty computation can be obtained [Kil88, IPS08]. By using quantum states, the protocol obtains oblivious transfer from one-way functions⁴ [BCKM21, GLSV21], an assumption that is not sufficient to obtain oblivious transfer in the classical setting by standard black-box constructions [IR90]. Thus, computational assumptions and quantum communication can be combined to develop protocols whose resource requirements are closer to current experimental capabilities.

Contributions and Organization

We now describe how these ideas are organized throughout the thesis. Before the main results, Chapter 2 introduces the notation and recalls the mathematical background, the basic concepts of quantum information theory, and the cryptographic notions used throughout the thesis.

The first part, *Foundations of Computational Quantum Information*, based on [YHK26, YAKH26], introduces the proposed framework for computational quantum information. Chapter 3 starts from complexity-constrained quantum channels, constructs the associated cones through efficiently implementable Choi operators, and then specializes the formalism to efficient POVMs, leading to a computational trace norm and trace distance. Then, Chapter 4 develops the corresponding computational divergences and entropies. We first consider single-system quantities defined through efficient binary measurements, such as computational max-divergence and measured Rényi divergences, and then move to bipartite systems, where the same viewpoint leads to the computational conditional min-entropy, and its operational interpretations in entanglement recovery and guessing with quantum side information.

³EFI pairs, introduced in [BCQ23], are pairs of efficiently generated quantum states that are statistically far but computationally indistinguishable. They are equivalent to canonical quantum commitments [Yan22, BCQ23] and serve as a candidate minimal primitive for much of computational quantum cryptography, since commitments, oblivious transfer, secure multiparty computation, and computational zero-knowledge proofs can be based on them or on the corresponding commitment notions [BCKM21, GLSV21, BCQ23]. This is part of a broader picture in which quantum cryptography may rely on assumptions weaker than classical one-way functions; for instance, pseudorandom quantum states can exist relative to an oracle with $P = NP$ [KQST23].

⁴Informally, one-way functions are functions that can be efficiently evaluated but cannot be efficiently inverted on a typical input. They are the standard minimal computational assumption underlying many central primitives in classical cryptography [Gol04].

The second part, *Computational Separations and Applications*, studies the operational consequences of this formalism. Chapter 5, based on [YHK26, YAKH26], proves explicit separations between information-theoretic and computational measures, first for single systems and then for bipartite correlations in both unconstrained and efficiently generated settings. Chapter 6, based on [YHK26], applies the efficient-measurement framework to computational hypothesis testing, computational resource theories, asymptotic continuity bounds, and computational entanglement measures. Finally, Chapter 7, based on [GY26], studies quantum pseudoresources as resources that are present information-theoretically but hidden from efficient observers, and shows how such hidden resources imply cryptographic primitives such as EFI pairs and quantum commitments.

The third part, *Near-Term Applications: Quantum Cryptography Beyond QKD*, turns to a concrete cryptographic protocol. Chapter 8, based on [DGI⁺26], presents a practical quantum oblivious-transfer protocol from one-way functions. It builds on the quantum approach of [GLSV21, BCKM21], where BB84 states [BB84] and equivocal and extractable commitments allow the construction of oblivious transfer from one-way functions, but addresses the main obstacles to an implementation of these protocols. This leads to a noise-tolerant protocol whose resource requirements are reduced from the order of 10^{13} BB84 states to around 10^7 , bringing the construction closer to current quantum communication technology. Thus, the final part connects the computational viewpoint developed in the thesis with a concrete protocol whose resource requirements are compatible with near-term experimental implementations.

Finally, *Discussion and Open Problems* summarizes the main results of the thesis and collects the open questions that arise from them.

Chapter 2

Mathematical Preliminaries

This chapter fixes the mathematical notation used throughout the thesis and recalls some basic concepts in quantum information theory. In Section 2.1, we settle the basic notation and conventions. Then, Sections 2.2 to 2.5 introduce the main notions of quantum information theory that will appear in the rest of the thesis. Section 2.6 provides the background on cone theory that will be crucial for the framework constructed in Chapter 3. Lastly, Section 2.7 introduces the basic concepts in representation theory that will be relevant for Chapter 5.

2.1. Notation and Basic Conventions

Throughout this thesis, we work with finite-dimensional *Hilbert spaces*, denoted by $\mathcal{H}$. More specifically, $\mathcal{H}$ is a complex space $\mathbb{C}^d$ equipped with the inner product

$$\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C},$$

and which is complete with respect to the norm induced by this inner product. Two different Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$ are isomorphic if there is a bijection between them that preserves the inner product. In finite dimension, $\mathcal{H} \simeq \mathbb{C}^d$, and the two notations will be used interchangeably when no ambiguity can arise. When considering n qudits of local dimension d , we write $\mathcal{H}_n \simeq (\mathbb{C}^d)^{\otimes n}$.

A *continuous linear functional* is defined as a linear map from a space X into the underlying field $\mathbb{F}$ of $\mathcal{H}$ which is continuous. The space of all continuous linear functionals on $\mathcal{H}$ is known as the dual space $\mathcal{H}'$. In other words, $\mathcal{H}'$ is the space of all linear maps from $\mathcal{H}$ to $\mathbb{F}$ which are continuous. The Hilbert space $\mathcal{H}$ is related to its dual space $\mathcal{H}'$ by the *Riesz representation theorem*. This theorem states that every element $f \in \mathcal{H}'$ can be represented as an inner product with an element $v \in \mathcal{H}$, i.e., $f(x) = \langle v, x \rangle$ for all $x \in \mathcal{H}$, where the inner product is linear in the second argument. It is worth bearing in mind that the dual space $\mathcal{H}'$ is also a Hilbert space, equipped with its own inner product and norm induced by the Riesz representation theorem.

Dirac's notation is especially useful when distinguishing at a glance between elements of $\mathcal{H}$ and $\mathcal{H}'$. The elements of $\mathcal{H}$ are represented by kets $|\cdot\rangle$ and those of $\mathcal{H}'$ are represented by bras $\langle \cdot|$. In Euclidean spaces, kets and bras correspond to column and row vectors, respectively.

The *tensor product* of two Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$ is defined as the space of all linear combinations of the form $\sum_{i=1}^n \psi_i \otimes \phi_i$, where $\psi_i \in \mathcal{H}_A$, $\phi_i \in \mathcal{H}_B$, and $n \in \mathbb{N}$. The product Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$ is denoted by the space of all finite linear combinations of tensor products with respect to the inner product $\langle \psi_1 \otimes \phi_1, \psi_2 \otimes \phi_2 \rangle = \langle \psi_1, \psi_2 \rangle \langle \phi_1, \phi_2 \rangle$. In other words, $\mathcal{H}_A \otimes \mathcal{H}_B$ contains not only the finite linear combinations of tensor products, but also the limits of Cauchy sequences of such combinations that converge in the norm induced by the inner product.

In mathematics, a *Banach space* is a normed linear space that is complete, meaning that every Cauchy sequence in it converges to a limit that also belongs to the space. An important class of linear maps between Banach spaces is the class of *operators*, which are functions that preserve addition and scalar multiplication. The set of all operators from one Banach space $\mathcal{X}$ to another Banach space $\mathcal{Y}$ is denoted by $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ and is itself a Banach space. When $\mathcal{X} = \mathcal{Y}$, we will simply write $\mathcal{B}(\mathcal{X})$ in order to simplify the notation if it is clear that we are working with the operator class. An operator $A \in \mathcal{B}(\mathcal{X}, \mathcal{Y})$ is continuous if, and only if, it is bounded, i.e., there exists a number $a \geq 0$ such that $\|A\psi\| \leq a\|\psi\|$ for all $\psi \in \mathcal{X}$. All the Banach spaces studied in this thesis are finite dimensional.

Given an operator $A \in \mathcal{B}(\mathcal{H}_A, \mathcal{H}_B)$, where $\mathcal{H}_A$ and $\mathcal{H}_B$ are Hilbert spaces, we can define its *adjoint* ($A^* \in \mathcal{B}(\mathcal{H}_B, \mathcal{H}_A)$). The adjoint of an operator A is an involution that satisfies the equation $\langle \phi, A\psi \rangle = \langle A^*\phi, \psi \rangle$ for all vectors ϕ and ψ in their respective Hilbert spaces. Therefore, the Banach algebra defined by the bounded operators and equipped with the adjoint involution is a *C^* -algebra*.

If $A = A^*$, we say that the operator is *Hermitian* (or *self-adjoint*). This type of operators will be of special interest in our study.

Another important definition is the concept of positivity for operators. An operator $A \in \mathcal{B}(\mathcal{H})$ is said to be *positive semi-definite* if $\langle \psi, A\psi \rangle \geq 0$ for all vectors ψ in $\mathcal{H}$, and *positive definite* if $\langle \psi, A\psi \rangle > 0$ for all non-zero vectors ψ in $\mathcal{H}$. We denote the set of positive semi-definite operators by

$$\text{Pos}(\mathcal{H}) := \{A \in \mathcal{B}(\mathcal{H}) : A \geq 0\}.$$

This allows us to introduce an order in the operators, such that $A > B$ if $A - B > 0$.

We denote the identity operator on a Hilbert space $\mathcal{H}$ by $\mathbb{1}_{\mathcal{H}}$, or simply by $\mathbb{1}$ when the Hilbert space is clear from context.

We can also define the *Hilbert-Schmidt inner product* between operators A and B in $\mathcal{B}(\mathcal{H})$ as $\langle A, B \rangle = \text{Tr}[A^*B]$, where ‘Tr’ denotes the trace of a matrix. The partial trace is the unique map $\text{tr}_B : \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B) \rightarrow \mathcal{B}(\mathcal{H}_A)$ such that

$$\text{Tr}[\text{tr}_B[X]Y] = \text{Tr}[X(Y \otimes \mathbb{1}_B)]$$

for all $X \in \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B)$ and $Y \in \mathcal{B}(\mathcal{H}_A)$.

There are different ways to characterize spaces of operators, depending on the norm with which they are equipped. Since throughout we work in finite dimension, all Schatten classes

coincide as vector spaces with $\mathcal{B}(\mathcal{H})$. Nevertheless, we keep the notation $\mathcal{B}_1(\mathcal{H})$ and $\mathcal{B}_2(\mathcal{H})$ to indicate that this same vector space is equipped with the trace norm and the Hilbert–Schmidt norm, respectively. The trace norm is defined as $\|A\|_1 := \text{Tr}[\sqrt{AA^*}]$, the Hilbert–Schmidt norm as $\|A\|_2 := \sqrt{\text{Tr}[AA^*]}$, and the operator norm as $\|A\|_\infty := \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}$. These three norms belong to the same family: for $p \geq 1$, the *Schatten p -norm* of an operator A is defined as $\|A\|_p := (\text{Tr}[|A|^p])^{1/p}$, with $\|A\|_\infty$ obtained as the limit $p \rightarrow \infty$. Hence, the trace norm, the Hilbert–Schmidt norm, and the operator norm correspond respectively to $p = 1$, $p = 2$, and $p = \infty$. For $0 < p < 1$, the same expression defines a Schatten p -quasi-norm.

A deeper insight into the mathematical description of Hilbert and operator spaces can be found in [HZ11].

Throughout the cryptographic chapters, we represent classical random variables with capital letters (e.g., X) while the value they take is represented by lower case letters (e.g., x). Bold characters (e.g., $\mathbf{x}$) are strings of values. The security parameter is denoted by λ , and when using the symbol $\perp$, it means to abort. A function $h : \mathcal{X} \times \mathcal{S}_{\text{seed}} \rightarrow \mathcal{Y}$ is called a *2-universal hash* function if, for every two strings $x \neq x' \in \mathcal{X}$, we have that $\Pr[h(x, s) = h(x', s)] \leq 1/|\mathcal{Y}|$ with s chosen uniformly at random. In the same way, a function $PRG : \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ with seed length $\ell < n$ is a *pseudorandom generator* if, given a non-uniform quantum polynomial-time (QPT) distinguisher $\mathcal{D}^* = \{\mathcal{D}_\lambda^*, \sigma_\lambda\}$, there exists a negligible function $\nu(\cdot)$ such that $\left| \Pr_{s \in \{0, 1\}^\ell}[\mathcal{D}^*(1^\lambda, PRG(s)) = 1] - \Pr_{r \in \{0, 1\}^n}[\mathcal{D}^*(1^\lambda, r) = 1] \right| \leq \nu(\lambda)$ with s and r chosen uniformly at random. A *non-uniform quantum polynomial-time distinguisher* $\mathcal{D}^* = \{\mathcal{D}_\lambda^*, \sigma_\lambda\}$ is composed of a polynomial-size quantum circuit $\mathcal{D}_\lambda^*$ and a polynomial-size quantum advice $\sigma_\lambda \in \mathcal{B}_1(\mathcal{H}_\mathcal{D})$.

2.2. Quantum States

In order to study a quantum system, we can divide into in the simplest and most intuitive way possible. To perform an experiment on the system, it must first be established or prepared in a specific state. Once such state is achieved, we can extract information from it to study its behavior and properties. Thus, any realization of an experiment in quantum mechanics can be divided into two parts, preparation and measurement. The reader may sometimes find this separation somewhat ambiguous, as it will be seen in later examples.

Let us first focus on the preparation of a quantum system. It is as of this step when the probabilistic nature of quantum mechanics arises. Throughout preparation, the probability distribution of the measurement outcomes is determined. We refer to these statistical ensembles as *states*. Hence, a quantum state is not a representation of a real attribute of the system itself, but a statistical ensemble that leads to the probability distribution of the possible outcomes of a certain measurement. It is also important to remark that different preparation procedures can result in the same probability distribution. Therefore,

if different preparation methods yield indistinguishable outcomes once measured, quantum states can be seen as equivalence classes of them.

Definition 2.1 (Density operator). An operator $\rho \in \mathcal{B}_1(\mathcal{H})$ is a *density operator* if it is positive semi-definite (thus it is Hermitian) and has $\text{Tr}[\rho] = 1$. If there is a unit vector $\psi \in \mathcal{H}$ such that $\rho = |\psi\rangle\langle\psi|$ (rank-one operator), the state is called a pure state. If this is not the case (higher rank operator), it is called a mixed state.

We denote the set of density operators on $\mathcal{H}$ by $\mathcal{S}(\mathcal{H})$.

Both vectors and density matrices are valid quantum state representations. Any state vector $|\psi\rangle$ can be expressed as a linear combination of two different state vectors $|\psi\rangle = c_1 |\phi\rangle_1 + c_2 |\phi\rangle_2$, with $c_1, c_2 \in \mathbb{C}$ chosen so that $\|\psi\| = 1$; when the states are orthonormal, this corresponds to $\sum_i |c_i|^2 = 1$. This linear combination is referred to as *superposition* in quantum information theory.

Indeed, the motivation behind using density operator notation over vector notation comes from the fact that statistical mixtures of pure states are allowed in quantum information theory, which will be very convenient as this is the most common —and general— case. Owing to the fact that the set of density matrices is the convex hull of pure states, we can express any density matrix as a convex combination of rank-1 projectors,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad (2.2.1)$$

with $p_i \in \mathbb{R}^+$ and normalization to one.

We call this decomposition a *convex decomposition* of a density operator, and it is a direct consequence of Carathéodory's theorem¹. Therefore, the aforementioned condition of the trace of density operators is a simple consequence of the normalization of the probability distribution. States $|\psi_i\rangle \in \mathcal{H}$ do not necessarily have to be orthogonal to each other, subsequently the ensemble representation of pure states is not necessarily unique. The completely mixed state on a d -dimensional Hilbert space is $\mathbb{1}/d$.

For a bipartite system AB with Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, a joint state is an operator $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$. The reduced state on A is obtained by the partial trace, i.e., $\rho_A = \text{tr}_B[\rho_{AB}]$. For two positive semidefinite operators $\rho, \sigma \in \text{Pos}(\mathcal{H})$, we write $\rho \ll \sigma$ if $\ker \sigma \subseteq \ker \rho$. We write $\rho \perp \sigma$ if $\text{Tr}[\rho\sigma] = 0$.

¹The Carathéodory's theorem states that any point x in a convex hull $S \subset \mathbb{R}^n$, can be expressed as a convex combination of at most $n+1$ points from S [Car11].

2.3. Measurement Formalism

Let us now focus on the measurement process. What will be referred to as the measurement process, from this moment on, is a statistical measurement, which yields an ideal probability distribution. In order to formalize measurements in quantum theory, we have to define a measurable space $(X, \mathbb{B}(X))$ in which X denotes the set of all possible measurement outcomes and $\mathbb{B}(X)$, a σ -algebra² over X . Then, we can define the most general object for describing measurements, *generalized measurements* or *POVMs*.

Definition 2.2 (POVM). A *positive operator-valued measure* (POVM) is a map M acting on a measurable space $(X, \mathbb{B}(X))$ such that $M : \mathbb{B}(X) \rightarrow \mathcal{B}(\mathcal{H})$ with $M(J) \geq 0 \forall J \in \mathbb{B}(X)$ and $\sum_k M(Y_k) = \mathbb{1}$ for all partitions such that $X = \cup_k Y_k$. If all elements of the tuple $M = (M(Y_1), \dots, M(Y_m))$ are orthogonal projections, it is called PVM, *projection-valued measure* or *von Neumann measurements*.

As a means to simplify the notation, let us refer to the elements of the POVM $M(Y)$ with the symbols $(E_i)_{i \in X}$, where the operators E_i are called *effect operators* or *POVM elements*. It is known that any POVM on $\mathcal{H}$ is implemented by appending ancillas to the quantum system, applying a quantum circuit on the entire system, applying a projective measurement on the enlarged space, and tracing out the remaining subsystems. The formal way to express one specific implementation of M is typically via Naimark's dilation theorem, see e.g. [Hol11, Theorem 2.5.1].

Theorem 2.3 (Naimark's Dilation Theorem in Finite Dimensions). *Any POVM $M = (E_i)_{i=1}^K$ on $\mathcal{H}$ can be implemented via an isometry and a projective measurement on ancilla systems. In particular, there exists an isometry $V : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}_K$, where $\mathcal{H}_K$ denotes a Hilbert space of dimension K , such that*

$$E_i = V^* \left(\mathbb{1}_{\mathcal{H}} \otimes |i\rangle\langle i|_{\mathcal{H}_K} \right) V \quad \forall i \in [K]. \quad (2.3.1)$$

Remark 2.4. One can alternatively extend the isometry $|\psi\rangle \otimes |0\rangle_{\mathcal{H}_K} \mapsto V |\psi\rangle$ to a unitary U on the Hilbert space $\mathcal{H} \otimes \mathcal{H}_K$, initialized with states of the form $\rho \otimes |0\rangle\langle 0|_{\mathcal{H}_K}$. A simple consequence of Theorem 2.3 is that for any state ρ ,

$$\text{Tr}[E_i \rho] = \text{Tr} \left[\left(\mathbb{1}_{\mathcal{H}} \otimes |i\rangle\langle i|_{\mathcal{H}_K} \right) \left(U \left(\rho \otimes |0\rangle\langle 0|_{\mathcal{H}_K} \right) U^* \right) \right]. \quad (2.3.2)$$

Since only the ancillary system is being measured, one can trace out $\mathcal{H}$, thus retrieving the aforementioned interpretation of Naimark's dilation.

Once POVMs have been introduced, we can describe the measurement procedure. POVMs and states determine the probability distribution of measurement outcomes. This relation between POVMs, states and probabilities is modelled by the following postulate:

²A σ -algebra over a set X is a collection of subsets of X that includes the empty set, is closed under the complement, and is closed under countable unions.

Postulate 2.5 (Born's rule). *Given a density operator $\rho \in \mathcal{B}_1(\mathcal{H})$ and a POVM $M : \mathbb{B}(X) \rightarrow \mathcal{B}(\mathcal{H})$, the probability of measuring an outcome in $Y \in \mathbb{B}(X)$ is given by*

$$p(Y|\rho, M) = \text{Tr}[\rho M(Y)] . \quad (2.3.3)$$

Equivalently one can associate to any m -outcome POVM $M = (E_1, \dots, E_m)$ a channel that maps quantum states to diagonal classical registers over $\{0, \dots, m-1\}$, via

$$\mathcal{M}_M : \rho \mapsto \sum_{i=0}^{m-1} \text{Tr}[\rho E_{i+1}] |i\rangle\langle i| . \quad (2.3.4)$$

We will sometimes drop the subscript M when the POVM to which this measurement map corresponds is clear from context.

Definition 2.6 (Observable). An *observable* is represented by a Hermitian operator $\hat{M} = \hat{M}^* \in \mathcal{B}(\mathcal{H})$.

Thus, Hermitian operators provide the physically meaningful representation of observables. Any measurement process usually has a numerical value assigned to the outcome $x \in X$. This outcome is characterized by a Borel measurable function $m(x) \in \mathbb{R}$ such that $\hat{M} = \sum_x m(x) E_x$ for the discrete variable case. Moreover, Born's rule can also be applied to observables, obtaining what is commonly known as *expectation value* $\langle m \rangle := \text{Tr}[\rho \hat{M}]$.

When using a Hermitian operator, we are usually interested in the expectation value of the observable being measured, which is a single number, rather than the full statistical distribution of measurement outcomes obtained through POVMs. The relation between a Hermitian operator and a sharp POVM (PVM) is thus given by the spectral decomposition of $\hat{M}$. As a consequence, a Hermitian operator unequivocally determines the tuple of elements of the PVM, $(E_x)_{x=1}^m$, as well as the numerical values $m(x)$ assigned to each of them.

2.4. Evolution of a Quantum System

Once we have introduced a mathematical description of preparation and measurement of quantum states, we ought to model any phenomenon that might happen in between. As previously mentioned, any quantum process is divided into either of two parts: preparation and measurement. Owing to that, any further evolution or interaction has to be included in one of these two processes. This leads to two compatible descriptions: the *Schrödinger picture* and the *Heisenberg picture*. The Schrödinger picture incorporates evolution of states in the preparation procedure, that is, the components of the system which evolve are quantum states. By contrast, in the Heisenberg picture, measurement operators undergo an evolution. A mathematical object describing these evolutions is a linear map:

Definition 2.7. Let V and W be two vector spaces over the ground field $\mathbb{F}$. A function $T : V \rightarrow W$ is a linear map if it satisfies for all $u, v \in V$ and $a \in \mathbb{F}$:

- $T(u + v) = T(u) + T(v)$
- $T(au) = aT(u)$

Definition 2.8 (Positive, completely positive, trace-preserving, and unital maps). A linear map $T : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$ is called

- *positive* if $T(A) \geq 0 \forall A \geq 0 \in \mathcal{B}(\mathcal{H}_A)$.
- *completely positive* if $T \otimes \text{Id}_n$ is positive for all $n \in \mathbb{N}$, with Id_n denoting the identity channel, or identity map, on $\mathcal{B}(\mathbb{C}^n)$.
- *trace-preserving* if $\text{Tr}[T(A)] = \text{Tr}[A] \forall A \in \mathcal{B}(\mathcal{H}_A)$.
- *unital* if $T(\mathbb{1}_A) = \mathbb{1}_B$.

We refer to a completely positive unital map as a *CPU map*.

Definition 2.9 (Quantum channel). A completely positive trace-preserving (CPTP) linear map $T : \mathcal{B}_1(\mathcal{H}_A) \rightarrow \mathcal{B}_1(\mathcal{H}_B)$ is called a *quantum channel*.

Quantum channels are the most general representation of evolution of a quantum system. Since all vector spaces in this thesis are finite dimensional, all linear maps considered below are bounded. This enables us to define the dual map of a quantum channel, that is, the mathematical description of the Heisenberg picture.

Theorem 2.10 (Equivalence between Schrödinger and Heisenberg pictures). *Given a bounded linear map $T : \mathcal{B}_1(\mathcal{H}_A) \rightarrow \mathcal{B}_1(\mathcal{H}_B)$, there is a unique linear map $T^* : \mathcal{B}(\mathcal{H}_B) \rightarrow \mathcal{B}(\mathcal{H}_A)$ defined as the dual map, such that $\forall X \in \mathcal{B}(\mathcal{H}_B), \forall \rho \in \mathcal{B}_1(\mathcal{H}_A)$:*

$$\text{Tr}[T(\rho)X] = \text{Tr}[\rho T^*(X)]. \quad (2.4.1)$$

Corollary 2.11. *Any linear map T and its dual T^* present the following relations:*

- *T is positive if and only if T^* is positive.*
- *T is completely positive if and only if T^* is completely positive.*
- *T is trace-preserving if and only if T^* is unital.*

The use of both equivalent pictures will be used throughout the following chapters, playing a crucial role in the definition of the complexity constrained formalism.

2.4.1. Choi-Jamiołkowski Representation

Quantum channels, and equivalently their Heisenberg-picture dual CPU maps, admit several mathematical representations. In this thesis, we focus on the *Choi–Jamiołkowski representation*, in which a linear map is associated with an operator, commonly called its *Choi operator* or *Choi matrix*.

Definition 2.12 (Choi-Jamiołkowski representation [Cho75, Jam72]). Let $\mathcal{H}_A, \mathcal{H}_{A'}$ be finite-dimensional Hilbert spaces with $\dim \mathcal{H}_A = d_A$, and let $\mathcal{H}_A \simeq \mathcal{H}_{A'}$. Let $\Phi : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$ be a linear map. For such a Φ , its *Choi-Jamiołkowski operator representation* on $\mathcal{H}_{A'} \otimes \mathcal{H}_B$ is given by

$$J(\Phi) := d_A(\text{Id}_{A'} \otimes \Phi)(|\Omega_{A'A}\rangle\langle\Omega_{A'A}|),$$

where $\text{Id}_{A'}$ denotes the identity channel, or identity map, on $\mathcal{B}(\mathcal{H}_{A'})$. Moreover, $|\Omega_{A'A}\rangle = \frac{1}{\sqrt{d_A}} \sum_{x \in [d_A]} |x\rangle_{A'} \otimes |x\rangle_A \in \mathcal{H}_{A'} \otimes \mathcal{H}_A$.

We will use this convention both for CPTP maps and for their dual CPU maps. In later sections, the reference system will be denoted by A and the input copy of the CPU map by A' . Thus, for a CPU map $T^* : \mathcal{B}(\mathcal{H}_{A'}) \rightarrow \mathcal{B}(\mathcal{H}_B)$, we write $J(T^*) = d_A(\text{Id}_A \otimes T^*)(|\Omega_{AA'}\rangle\langle\Omega_{AA'}|)$ on $\mathcal{H}_A \otimes \mathcal{H}_B$. Equivalently, the corresponding CPTP map acts from B to A' .

The Choi representation will be of special relevance in the first section of this dissertation. Let us recall some main properties of this representation.

Lemma 2.13. *Let $J \in \mathcal{B}_1(\mathcal{H}_A \otimes \mathcal{H}_B)$ be the Choi matrix associated with the linear map $\Phi : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$. Then the following relations hold:*

1. *As the map $\Phi \mapsto J$ is bijective, the following relation holds,*

$$\text{Tr}[\Phi(X)Y] = \text{Tr}[J(X^T \otimes Y)] \quad \forall X \in \mathcal{B}(\mathcal{H}_A), \quad \forall Y \in \mathcal{B}(\mathcal{H}_B), \quad (2.4.2)$$

where X^T is the transpose matrix.

2. *$J = J^*$ if and only if the map is hermiticity preserving, i.e., $\Phi(A)^* = \Phi(A^*)$.*
3. *$J \geq 0$ if and only if Φ is completely positive.*
4. *$\text{tr}_A J = \mathbb{1}_B$ if and only if Φ is unital.*
5. *$\text{tr}_B J = \mathbb{1}_A$ if and only if Φ is trace-preserving.*

The proof of these properties can be found in [Cho75] [Jam72]. Therefore, due to the first property of Lemma 2.13, the action of the map Φ can be recovered from its Choi operator $J(\Phi)$ in the following way,

$$\Phi(X) = \text{tr}_A[J(\Phi)(X^T \otimes \mathbb{1}_B)], \quad (2.4.3)$$

where the transpose is taken with respect to the computational basis used to define the state $|\Omega_{AA'}\rangle$.

2.4.2. Stinespring Dilation

We must take into account that the *environment* generally influences the evolution of the system. In order to account for these effects, we introduce a widely used representation. In the Schrödinger picture, the Stinespring dilation represents a quantum channel as an isometric evolution on a larger space followed by tracing out the environment. Equivalently, in the Heisenberg picture, its dual CPU map is represented by extending each observable X with the identity on the environment and mapping the resulting operator back to the input space through the isometry.

Theorem 2.14 (Stinespring representation[Sti55]). *Let $T : \mathcal{B}_1(\mathcal{H}_A) \rightarrow \mathcal{B}_1(\mathcal{H}_B)$ be a completely positive trace-preserving (CPTP) linear map, and let $T^* : \mathcal{B}(\mathcal{H}_B) \rightarrow \mathcal{B}(\mathcal{H}_A)$ denote its dual map. Then there exists an integer $r \leq d_A d_B$ and an isometry $V : \mathcal{H}_A \rightarrow \mathcal{H}_B \otimes \mathbb{C}^r$ such that*

$$\begin{aligned} T^*(X) &= V^*(X \otimes \mathbb{1}_r)V \\ T(\rho) &= \text{tr}_r[V\rho V^*] \end{aligned}$$

for all $X \in \mathcal{B}(\mathcal{H}_B)$ and $\rho \in \mathcal{B}_1(\mathcal{H}_A)$.

In later sections, when the channel acts on the conditioning system, we will use the same statement with input space $\mathcal{H}_B$ and output space $\mathcal{H}_{A'}$. In that notation, the dual CPU map has the form $T^* : \mathcal{B}(\mathcal{H}_{A'}) \rightarrow \mathcal{B}(\mathcal{H}_B)$.

2.5. Information Theoretic Concepts

Once the basic objects of quantum information theory have been introduced, we can define the quantities that will be used to quantify distinguishability between quantum states, resources and other operationally meaningful quantities. We start with divergences between quantum states, then recall their measured and regularized versions, and finally introduce the entropic quantities that follow from them.

Definition 2.15 (Sandwiched Rényi Divergence [MLDS⁺13, WWY14]). Given any two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H})$ with $\text{Tr}[\rho] > 0$, and $\alpha \in (0, 1) \cup (1, \infty)$, the *sandwiched Rényi divergence* between ρ, σ is given by:

$$D_\alpha(\rho||\sigma) := \begin{cases} \frac{1}{\alpha-1} \log \frac{\left\| \sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}} \right\|_\alpha^\alpha}{\text{Tr}[\rho]} & \text{if } \alpha < 1 \text{ and } \rho \not\ll \sigma, \\ & \text{or if } \alpha > 1 \text{ and } \rho \ll \sigma \\ +\infty & \text{else} \end{cases} \quad (2.5.1)$$

Here $\|\cdot\|_\alpha$ denotes the Schatten α -(quasi-)norm. In this work, the divergence of interest is the sandwiched Rényi divergence. For $\alpha \geq \frac{1}{2}$, it satisfies the data-processing inequality

(DPI), and it is monotone increasing in the order, i.e., $D_\alpha(\rho\|\sigma) \leq D_\beta(\rho\|\sigma)$ whenever $0 < \alpha \leq \beta$ and both sides are defined. Moreover, we use Q_α to denote

$$Q_\alpha(\rho\|\sigma) := \left\| \sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}} \right\|_\alpha^\alpha. \quad (2.5.2)$$

Definition 2.16 (Relative Entropy and Max-divergence [Dat09b]). Given any two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H})$ with $\text{Tr}[\rho] > 0$, the (Umegaki) *relative entropy* and *max-divergence* between ρ, σ are, respectively, given by:

$$D(\rho\|\sigma) := \begin{cases} \frac{\text{Tr}[\rho(\log \rho - \log \sigma)]}{\text{Tr}[\rho]} & \text{if } \rho \ll \sigma, \\ +\infty & \text{otherwise,} \end{cases} \quad (2.5.3)$$

$$D_{\max}(\rho\|\sigma) := \log \inf\{\lambda \in \mathbb{R} \mid \rho \leq \lambda \sigma\}. \quad (2.5.4)$$

Definition 2.17 (Fidelity). Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, the *fidelity* between ρ, σ is given by:

$$F(\rho, \sigma) := \|\sqrt{\rho}\sqrt{\sigma}\|_1^2. \quad (2.5.5)$$

We will also use the *trace distance* between two quantum states, denoted by

$$\Delta(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1.$$

When a negligible function $\mu(\lambda)$ exists such that $\Delta(\rho^1, \rho^2) \leq \mu(\lambda)$, ρ^1 and ρ^2 are said to be statistically close.

Theorem 2.18 (Holevo-Helstrom theorem [Hol73, Hel69]). *Given two equiprobable quantum states ρ and σ , the optimal probability of distinguishing them is*

$$p_{\text{guess}}(\rho, \sigma) = \frac{1}{2} (1 + \Delta(\rho, \sigma)).$$

Moreover, if $\Delta(\rho, \sigma) \geq 1/\text{poly}(\lambda)$, then there exists a polynomial $m(\lambda)$ such that $\Delta(\rho^{\otimes m(\lambda)}, \sigma^{\otimes m(\lambda)}) \geq 1 - \text{negl}(\lambda)$.

Theorem 2.19 (Uhlmann's theorem [Uhl76]). *Let $\rho, \sigma \in \mathcal{S}(\mathcal{H}_A)$ and let $|\psi\rangle_{AB}$ be a purification of ρ . Then*

$$F(\rho, \sigma) = \max_{|\eta\rangle_{AB}} |\langle \psi | \eta \rangle|^2,$$

where the maximum is taken over all purifications $|\eta\rangle_{AB}$ of σ on a sufficiently large purifying system B .

We are particularly interested in the standard extensions of the sandwiched Rényi divergence at $\alpha \in \{\frac{1}{2}, 1, \infty\}$. These three cases are of particular use and satisfy

$$\begin{aligned}\lim_{\alpha \rightarrow 1} D_\alpha(\rho\|\sigma) &= D(\rho\|\sigma), \\ \lim_{\alpha \rightarrow \infty} D_\alpha(\rho\|\sigma) &= D_{\max}(\rho\|\sigma), \\ D_{\frac{1}{2}}(\rho\|\sigma) &= -\log F(\rho, \sigma).\end{aligned}$$

A closely related divergence is the measured Rényi divergence.

Definition 2.20 (Measured Divergence). Let $\mathbf{M}$ be a set of POVMs. Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, for any divergence $\mathbb{D}$ defined on classical probability distributions, the corresponding *measured divergence* between ρ, σ is given by:

$$\mathbb{D}^{\mathbf{M}}(\rho\|\sigma) = \sup_{M \in \mathbf{M}} \mathbb{D}(\mathcal{M}_M(\rho)\|\mathcal{M}_M(\sigma)). \quad (2.5.6)$$

We note that the RHS of Eq. (2.5.6) is evaluated on classical (commuting) states and hence it suffices to define the divergence on probability distributions. In particular, for $\alpha \in (0, 1) \cup (1, \infty)$, if

$$\mathbb{D}(p\|q) = D_\alpha(p\|q) := \frac{1}{\alpha - 1} \log \sum_i p_i^\alpha q_i^{1-\alpha} \quad (2.5.7)$$

is the classical Rényi divergence, with terms such that $p_i = 0$ taken to be zero and value $+\infty$ for $\alpha > 1$ whenever $\text{supp}(p) \not\subseteq \text{supp}(q)$, then we call $D_\alpha^{\mathbf{M}}(\rho\|\sigma)$ a *measured Rényi divergence*. If the POVMs in $\mathbf{M}$ have two outputs, these divergences are sometimes called *test-measured divergences*.

Note, though, that if we regularize this quantity, the divergence which emerges is the sandwiched Rényi divergence, i.e., a particular quantum extension of the classical Rényi divergence. Indeed, for any $\alpha \geq \frac{1}{2}$ and $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, if $\mathbf{M}_n = \mathbf{M}_{n, \text{ALL}}$, where $\mathbf{M}_{n, \text{ALL}}$ denotes the set of all POVMs that can act on $\rho^{\otimes n}$ (or $\sigma^{\otimes n}$), then (see, e.g., [MO14])

$$D_\alpha(\rho\|\sigma) = \lim_{n \rightarrow \infty} \frac{1}{n} D_\alpha^{\mathbf{M}_n}(\rho^{\otimes n}\|\sigma^{\otimes n}).$$

This naturally leads to the question of how one generally defines the regularization of measured divergences.

Definition 2.21 (Regularized Measured Divergence [RSB25, MH23]). Let $\widetilde{\mathbf{M}} = (\mathbf{M}_1, \mathbf{M}_2, \dots)$ be a family of POVM sets. Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, for any divergence $\mathbb{D}$ defined on classical probability distributions, the corresponding *regularized measured divergence* between ρ, σ is given by:

$$\mathbb{D}^{\widetilde{\mathbf{M}}, \infty}(\rho\|\sigma) := \limsup_{n \rightarrow \infty} \frac{1}{n} \mathbb{D}^{\mathbf{M}_n}(\rho^{\otimes n}\|\sigma^{\otimes n}). \quad (2.5.8)$$

If the corresponding measured divergence satisfies super-additivity, i.e.,

$$\mathbb{D}^{\mathbf{M}_{k+l}}(\rho^{\otimes(k+l)} \parallel \sigma^{\otimes(k+l)}) \geq \mathbb{D}^{\mathbf{M}_k}(\rho^{\otimes k} \parallel \sigma^{\otimes k}) + \mathbb{D}^{\mathbf{M}_l}(\rho^{\otimes l} \parallel \sigma^{\otimes l})$$

for any $k, l \in \mathbb{N}$, then it follows from, e.g., [RSB25, Lemma 4], that

$$\begin{aligned} \mathbb{D}^{\tilde{\mathbf{M}}, \infty}(\rho \parallel \sigma) &= \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{D}^{\mathbf{M}_n}(\rho^{\otimes n} \parallel \sigma^{\otimes n}) \\ &= \sup_{n \in \mathbb{N}} \frac{1}{n} \mathbb{D}^{\mathbf{M}_n}(\rho^{\otimes n} \parallel \sigma^{\otimes n}). \end{aligned}$$

Moreover, divergences are also relevant because several entropic quantities can be defined from them.

Definition 2.22 (Conditional min-entropy [KRS09]). Let $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ be a bipartite quantum state. The *min-entropy* of A conditioned on B is

$$H_{\min}(A|B)_\rho := - \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B] = 1}} D_{\max}(\rho_{AB} \parallel \mathbb{1}_A \otimes \sigma_B).$$

Moreover, as shown in [KRS09], the conditional min-entropy admits the following operational interpretation:

$$H_{\min}(A|B)_\rho = -\log \left(d_A \max_T F((\mathbb{1}_A \otimes T)(\rho_{AB}), |\Omega_{AA'}\rangle\langle\Omega_{AA'}|) \right),$$

where the maximization is over channels $T : B \rightarrow A'$, and $|\Omega_{AA'}\rangle = \frac{1}{\sqrt{d_A}} \sum_{x \in [d_A]} |x\rangle_A \otimes |x\rangle_{A'}$. Thus, the conditional min-entropy is a one-shot measure that quantifies how well one can recover a maximally entangled state on AA' from ρ_{AB} by acting locally on B . For states with quantum correlations, $H_{\min}(A|B)_\rho$ can be negative, reaching $-\log d_A$ for a maximally entangled state.

We define a *classical-quantum hybrid state* as a bipartite state of the subsystems XB such that $\rho_{XB} = \sum_{x \in \mathcal{X}} p_x |x\rangle\langle x| \otimes \rho_B^x$, where subsystem X is classical. In the case of the input state being a classical-quantum state, the operational meaning of the conditional min-entropy is the following:

$$H_{\min}(X|B)_\rho = -\log \max_{\{E_B^x\}_x} \sum_x p_x \text{Tr}[E_B^x \rho_B^x],$$

where $\{E_B^x\}_x$ is a POVM on B . Then, in this case, the conditional min-entropy quantifies the predictability of the classical register X given access to B .

Besides the conditional min-entropy, another entropy measure that will be used later is the von Neumann entropy $S(\rho) := -\text{Tr}[\rho \log \rho]$. Its dependence on the state is controlled by the trace distance through the following asymptotic continuity bound.

Lemma 2.23 (Fannes inequality [Fan73]). *Let $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, with $\dim \mathcal{H} = d$, and let $\varepsilon = \Delta(\rho, \sigma)$. Then*

$$|S(\rho) - S(\sigma)| \leq 2\varepsilon \log d + c(\varepsilon),$$

where $c(x) := \min\{-x \log x, 1/(2e)\}$.

One can also define relative entropies from operational tasks. In this work, we will use the hypothesis-testing relative entropy.

Definition 2.24 (ε -hypothesis testing relative entropy [KW24]). Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H})$ and $\varepsilon \in [0, 1]$, the corresponding ε -hypothesis-testing relative entropy is given by

$$D_H^\varepsilon(\rho \| \sigma) := -\log \inf_E \{\text{Tr}[E\sigma] : 0 \leq E \leq \mathbb{1}, \text{Tr}[(1 - E)\rho] \leq \varepsilon\}. \quad (2.5.9)$$

Equivalently, replacing ε by $1 - \varepsilon$, we have

$$D_H^{1-\varepsilon}(\rho \| \sigma) := -\log \inf_E \{\text{Tr}[E\sigma] : 0 \leq E \leq \mathbb{1}, \text{Tr}[E\rho] \geq \varepsilon\}. \quad (2.5.10)$$

2.5.1. Entanglement and Resource-Theoretic Quantities

The pseudoresource constructions used later require some basic terminology from entanglement theory and quantum resource theories. We first recall the standard entanglement quantities and then place them in the more general language of resource theories. A bipartite state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ is *separable* if it can be written as

$$\rho_{AB} = \sum_i p_i \rho_A^i \otimes \rho_B^i,$$

for a probability distribution $(p_i)_i$ and local states $\rho_A^i \in \mathcal{S}(\mathcal{H}_A)$, $\rho_B^i \in \mathcal{S}(\mathcal{H}_B)$. If this is not the case, the state is entangled. We denote the set of separable states by $\text{SEP}(A : B)$. The maximally entangled state of Schmidt rank 2^m is denoted by

$$|\Phi_m\rangle := \frac{1}{\sqrt{2^m}} \sum_{i=0}^{2^m-1} |i\rangle |i\rangle,$$

and we write $\Phi_m := |\Phi_m\rangle\langle\Phi_m|$. In particular, $\Phi := \Phi_1$ denotes one EPR pair, and $\Phi^{\otimes m} = \Phi_m$.

Definition 2.25 (LOCC map [ABV23]). A quantum channel $\Gamma : \mathcal{B}_1(\mathcal{H}_A \otimes \mathcal{H}_B) \rightarrow \mathcal{B}_1(\mathcal{H}_{\bar{A}} \otimes \mathcal{H}_{\bar{B}})$ is an *LOCC map* if it can be implemented by a finite-round protocol in which the two parties perform local quantum operations and exchange classical messages. An LOCC map family is called *efficient* if it admits such a protocol with a circuit description of polynomial size.

LOCC maps form the canonical class of free operations in entanglement theory: they cannot create entanglement from separable states and they directly model a two-party setting with local laboratories and classical communication.

There exists different measures of entanglement. Let us first introduce the most basic one.

Definition 2.26 (Entanglement entropy). Given a bipartite pure state $\rho_{AB} = |\psi\rangle\langle\psi|_{AB}$, its *entanglement entropy* is defined as

$$E_{A/B}(\rho_{AB}) := S(\rho_A) = S(\rho_B),$$

where $\rho_A = \text{tr}_B[\rho_{AB}]$ and $\rho_B = \text{tr}_A[\rho_{AB}]$.

Nevertheless, in the case of mixed states, the entanglement entropy is not operationally meaningful in general. This leads to several measures of entanglement [HHHH09]. Two standard one-shot quantities are the entanglement cost and the distillable entanglement.

Definition 2.27 (One-shot entanglement cost and distillable entanglement [ABV23]). Let $\varepsilon \in [0, 1]$ and let $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$. Its one-shot entanglement cost and distillable entanglement under LOCC are respectively given by

$$\begin{aligned} E_C^\varepsilon(\rho_{AB}) &:= \inf_{m, \Gamma} \{m : 1 - F(\rho_{AB}, \Gamma(\Phi_m)) \leq \varepsilon\}, \\ E_D^\varepsilon(\rho_{AB}) &:= \sup_{m, \Gamma} \{m : 1 - F(\Gamma(\rho_{AB}), \Phi_m) \leq \varepsilon\}, \end{aligned}$$

where the optimizations are over LOCC maps with the corresponding input and output spaces.

The asymptotic IID versions describe the rate at which entanglement can be diluted or extracted when many identical copies of the same state are available.

Definition 2.28 (Asymptotic IID entanglement cost and distillable entanglement). Given a bipartite state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$, its asymptotic IID entanglement cost and distillable entanglement are defined as

$$\begin{aligned} E_C^\infty(\rho_{AB}) &:= \lim_{\varepsilon \rightarrow 0} \limsup_{t \rightarrow \infty} \frac{1}{t} E_C^\varepsilon(\rho_{AB}^{\otimes t}), \\ E_D^\infty(\rho_{AB}) &:= \lim_{\varepsilon \rightarrow 0} \liminf_{t \rightarrow \infty} \frac{1}{t} E_D^\varepsilon(\rho_{AB}^{\otimes t}). \end{aligned}$$

These measures play an extremal role in the asymptotic manipulation of entanglement: distillable entanglement gives the extractable rate, while entanglement cost gives the dilution rate [HHHH09]. One entropic measure that will be used later is the regularised relative entropy of entanglement.

Definition 2.29 (Regularised relative entropy of entanglement). Given a bipartite state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$, its *relative entropy of entanglement* is

$$E_R(\rho_{AB}) := \inf_{\sigma_{AB} \in \text{SEP}(A:B)} D(\rho_{AB} \| \sigma_{AB}).$$

The corresponding regularised relative entropy of entanglement is

$$E_R^\infty(\rho_{AB}) := \lim_{n \rightarrow \infty} \frac{1}{n} E_R(\rho_{AB}^{\otimes n}),$$

whenever the limit exists.

Lemma 2.30 (Fannes type inequality for the regularised relative entropy of entanglement [Win16, DH99, Chr06]). Let $\rho_{AB}, \sigma_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ and assume that $\Delta(\rho_{AB}, \sigma_{AB}) \leq \varepsilon$. Then

$$|E_R^\infty(\rho_{AB}) - E_R^\infty(\sigma_{AB})| \leq \varepsilon \log d + (1 + \varepsilon) h_2\left(\frac{\varepsilon}{1 + \varepsilon}\right),$$

where $d = \dim(\mathcal{H}_A \otimes \mathcal{H}_B)$ and $h_2(x) := -x \log x - (1 - x) \log(1 - x)$ is the binary entropy.

Definition 2.31 (Quantum resource theory). A quantum resource theory $\mathcal{R} = (\mathcal{F}, \mathcal{O})$ is specified by a set of free states $\mathcal{F}(\mathcal{H}) \subseteq \mathcal{S}(\mathcal{H})$ and a set of free operations $\mathcal{O}$. A CPTP map $\Lambda : \mathcal{B}_1(\mathcal{H}_1) \rightarrow \mathcal{B}_1(\mathcal{H}_2)$ belongs to $\mathcal{O}$ if $\Lambda(\sigma) \in \mathcal{F}(\mathcal{H}_2)$ for all $\sigma \in \mathcal{F}(\mathcal{H}_1)$.

For the resource-theoretic constructions considered later, we will assume that the set of free states is convex and closed, and that it contains a full-rank state. Since all systems are finite dimensional, closedness also implies boundedness. These assumptions are a mild relaxation of the Brandao–Plenio axioms [BP10], and they are satisfied by many resource theories, including entanglement, coherence, magic, and athermality [CG19]. In the resource theory of entanglement, the free states are the separable states and the free operations can be chosen to be LOCC maps.

There are several ways to quantify resources, ranging from geometric to witness-based approaches. In this work, we focus on entropic measures, and more specifically on the relative entropy of resource.

Definition 2.32 (Relative entropy of resource). Given a state $\rho \in \mathcal{S}(\mathcal{H})$ and a set of free states $\mathcal{F}(\mathcal{H}) \subseteq \mathcal{S}(\mathcal{H})$, its *relative entropy of resource* is defined as

$$R_{\text{rel}}(\rho) := \inf_{\sigma \in \mathcal{F}(\mathcal{H})} D(\rho \| \sigma).$$

Lemma 2.33 (Fannes type inequality for the relative entropy of resource [Win16]). Let $\mathcal{F}$ be a closed, convex, and bounded set of positive semidefinite operators containing at least one full-rank operator, and let $D_{\mathcal{F}}(\rho) := \inf_{\gamma \in \mathcal{F}} D(\rho \| \gamma)$. If $\Delta(\rho, \sigma) \leq \varepsilon$, then

$$|D_{\mathcal{F}}(\rho) - D_{\mathcal{F}}(\sigma)| \leq \varepsilon \kappa + (1 + \varepsilon) h_2\left(\frac{\varepsilon}{1 + \varepsilon}\right), \quad (2.5.11)$$

where $\kappa := \sup_{\tau, \tau'} |D_{\mathcal{F}}(\tau) - D_{\mathcal{F}}(\tau')|$.

2.6. Cone Theory

In this section we summarize some basic notions of cone theory. For a deeper understanding, the reader may refer to [Bus73a, Bus73b, Eve95, RKW11, GC24].

Definition 2.34 (Convex Cone). Let $\mathcal{V}$ be a finite-dimensional real vector space with inner product $\langle \cdot, \cdot \rangle$. A *convex cone* $\mathcal{C} \subset \mathcal{V}$ is a subset that satisfies

$$a, b \in \mathcal{C} \implies \lambda a + \mu b \in \mathcal{C} \quad \forall \lambda, \mu \geq 0.$$

Moreover, for any $K \subset \mathcal{V}$, we denote the smallest (convex) cone that contains K by

$$\text{cone}(K) := \text{conv}\left(\bigcup_{\lambda \geq 0} \lambda K\right),$$

where $\text{conv}(S)$ denotes the convex hull of a set $S \in \mathcal{V}$.

A cone is called *closed* if it is closed with respect to the norm topology induced by the inner product of $\mathcal{V}$, and, as a consequence, includes its boundary. Following the notation of [RKW11], we further call a cone *pointed* if $\mathcal{C} \cap (-\mathcal{C}) = \{0\}$, where the elements of $-\mathcal{C}$ are given by

$$a \in -\mathcal{C} \iff -a \in \mathcal{C}.$$

Similarly, a cone is *solid* if $\text{span}(\mathcal{C}) = \mathcal{V}$. In finite-dimensional $\mathcal{V}$ this is equivalent to $\mathcal{C}$ having a non-empty interior $\text{int}(\mathcal{C}) \neq \emptyset$.

Definition 2.35 (Proper Cone). A convex, closed, pointed, and solid cone is called a *proper cone*.

Definition 2.36 (Cone Base). Given some convex cone $\mathcal{C}$, a convex set B is called a *base of the cone* $\mathcal{C}$ if $\mathcal{C} = \bigcup_{\lambda \geq 0} \lambda B$ and if every $c \in \mathcal{C} \setminus \{0\}$ has a unique representation $c = \lambda b$ for some $b \in B, \lambda > 0$. If $\mathcal{C}$ is in addition closed and with non-empty interior, then B is the *base of a proper cone*.

One can also assign to each cone a dual cone.

Definition 2.37 (Dual Cone). Given a cone $\mathcal{C} \subset \mathcal{V}$, its *dual cone* $\mathcal{C}^*$ is given by

$$\mathcal{C}^* := \{x \in \mathcal{V}^* \mid \langle x, c \rangle \geq 0, \forall c \in \mathcal{C}\}.$$

One natural consequence of this definition is that if $\mathcal{C}_1 \subset \mathcal{C}_2$, then $\mathcal{C}_2^* \subset \mathcal{C}_1^*$. We note that the dual cone of any (possibly not convex, nor closed cone) in finite dimensions is always closed and convex.³ Moreover, in finite-dimensional vector spaces, closed convex cones coincide with their bi-dual cones, meaning that the corresponding dual cone is also its pre-dual cone, i.e., $(\mathcal{C}^*)^* = \mathcal{C}$. Additionally it holds that if $\mathcal{C}$ is a proper cone, then so is $\mathcal{C}^*$. Given a real vector space $\mathcal{V}$ and a proper cone $\mathcal{C} \subset \mathcal{V}$, one can define a natural partial order on $\mathcal{V}$.

Definition 2.38 (Partial Order). Let $\mathcal{V}$ be a finite-dimensional real vector space and $\mathcal{C} \subset \mathcal{V}$ be a proper cone. We say that $a \succeq_{\mathcal{C}} b$ if and only if $a - b \in \mathcal{C}$.

In the following, we will take $\mathcal{V}$ to be the real vector space of Hermitian matrices $\mathcal{B}^\dagger(\mathcal{H}) := \{A \in \mathcal{B}(\mathcal{H}) : A = A^*\}$, over some Hilbert space $\mathcal{H}$. This vector space is isomorphic to $\mathbb{R}^{d^2}$, where $d = \dim(\mathcal{H})$. Over the space of Hermitian matrices, one of the most common examples of (proper) cones in quantum information theory is the cone $\text{Pos}(\mathcal{H})$ introduced in Section 2.1. Notably, this cone is self-dual in $\mathcal{V}$ and we denote its induced (positive operator) partial order simply by $\geq$. This particular partial order is relevant for the definition of the max-divergence [Dat09b, RKW11], which satisfies

$$\begin{aligned} D_{\max}(a||b) &= \log \inf\{\lambda \in \mathbb{R} | a \leq \lambda b\} \\ &= \log \sup_{v \in \text{Pos}(\mathcal{H})} \frac{\langle v, a \rangle}{\langle v, b \rangle}. \end{aligned}$$

Here and below, the quotient is understood as $+\infty$ if $\langle v, b \rangle = 0$ and $\langle v, a \rangle > 0$, and as 0 if both quantities vanish.

Remark 2.39. Note that in the above optimization one should interpret the v as being an effect operator living in the (pre-) dual of the cone of $\text{Pos}(\mathcal{H})$, in which a, b live, rather than that cone itself. However, since $\text{Pos}(\mathcal{H})$ is self-dual with respect to the Hilbert-Schmidt inner product these cones are the same. This differentiation will become important for arbitrary cones in Definition 2.40.

If one considers other proper cones inside of $\mathcal{B}^\dagger(\mathcal{H})$, then one can use this relation between the max-divergence and the conic structure of $\text{Pos}(\mathcal{H})$ to derive a more general notion of max-divergences.

Definition 2.40 ($\mathcal{C}$ -Max-divergence [RKW11, GC24]). Let $\mathcal{C}$ be a proper cone. For any $a, b \in \mathcal{C} \setminus \{0\}$, the $\mathcal{C}$ -max-divergence between a, b is given by

$$D_{\max}^{\mathcal{C}}(a||b) := \log \inf\{\lambda \in \mathbb{R} | a \leq_{\mathcal{C}} \lambda b\}. \quad (2.6.1)$$

If no $\lambda \in \mathbb{R}$ exists such that $\lambda b - a \in \mathcal{C}$, i.e., the infimum is over an empty set, then we set $D_{\max}^{\mathcal{C}}(a||b) = \infty$.

³Convexity follows directly from definition, since if $x, y \in \mathcal{C}^*$, then for any $\lambda \in [0, 1]$ it follows that $\langle \lambda x + (1 - \lambda)y, c \rangle = \lambda \langle x, c \rangle + (1 - \lambda) \langle y, c \rangle \geq 0$ for any $c \in \mathcal{C}$. Closedness effectively follows from continuity of the inner product in norm topology, as all $c \in \mathcal{C}$ are bounded.

Moreover, there exists a dual formulation of the $\mathcal{C}$ -max-divergence.

Lemma 2.41 (Dual Expression [RKW11, GC24]). *Let $\mathcal{C}$ be a proper cone. Then*

$$D_{\max}^{\mathcal{C}}(a||b) = \log \inf\{\lambda \in \mathbb{R} | a \leq_{\mathcal{C}} \lambda b\} = \log \sup_{v \in \mathcal{C}^*} \frac{\langle v, a \rangle}{\langle v, b \rangle}. \quad (2.6.2)$$

With these quantities, one can also additionally define a *Hilbert's projective metric* [Bus73a, Eve95] on a proper cone $\mathcal{C}$ as

$$d_{\mathbb{H}}^{\mathcal{C}}(a, b) := D_{\max}^{\mathcal{C}}(a||b) + D_{\max}^{\mathcal{C}}(b||a).$$

For the applications we are considering in this work, it will be natural to consider the following setup. Let $\mathbb{M} \subset \mathcal{V}^*$ be a closed, convex, and pointed set with a non-empty interior in the dual space, which contains a distinguished element $e \in \mathbb{M}$ such that

$$E \in \mathbb{M} \implies e - E \in \mathbb{M} \quad \forall E \in \mathbb{M}. \quad (2.6.3)$$

We refer to $e \in \mathbb{M}$ as the unit effect, and it upper bounds all elements in the set $\mathbb{M}$ in the partial order induced by the cone $\mathcal{C}_{\mathbb{M}}^*$. For the sets $\mathbb{M}$ we consider in this work, the unit effect can be thought of as being the identity operator $\mathbb{1}$.

In particular, this ensures that $\mathbb{M}$ generates a proper cone $\mathcal{C}_{\mathbb{M}}^*$.

Definition 2.42 (Distinguishability Norm). Let $\mathbb{M} \subset \mathcal{V}^*$ be a closed, convex, and pointed set with a non-empty interior. Moreover let $\mathbb{M}$ contain an element $e \in \mathbb{M}$ for which Eq. (2.6.3) holds. For any element $v \in \mathcal{V}$, we denote the *distinguishability norm* of v by

$$\|v\|_{(\mathbb{M})} := \sup_{E \in \mathbb{M}} \langle 2E - e, v \rangle. \quad (2.6.4)$$

In particular, it is known that the distinguishability norm defines a norm on $\mathcal{V}$ [RKW11].

Remark 2.43. In quantum information theory, this norm is simply the trace norm. Here, one chooses $\mathbb{M}$ to be the set of all effect operators, i.e., $E \in \mathbb{M}$ if and only if $0 \leq E \leq \mathbb{1}$. The operational interpretation of the trace norm is given by the task of distinguishing two equiprobable quantum states ρ and σ . In this case, one chooses $v = \rho - \sigma$ and $e = \mathbb{1}$, retrieving

$$\begin{aligned} \|\rho - \sigma\|_{(\mathbb{M})} &= 2 \sup_{E \in \mathbb{M}} \text{Tr}[E(\rho - \sigma)] - \text{Tr}[\rho - \sigma] \\ &= 2 \sup_{E \in \mathbb{M}} \text{Tr}[E(\rho - \sigma)] \\ &= \|\rho - \sigma\|_1. \end{aligned}$$

If one additionally chooses $\mathbb{M}$ to be a subset of all effect operators, then $\mathbb{M}$ can be related to a set of binary (POVM) measurements. The corresponding *minimal error probability* for discriminating between two equiprobable states ρ and σ is given by

$$\mathcal{P}_{err}^{\mathbb{M}}(\rho, \sigma) = \frac{1}{2} - \sup_{E \in \mathbb{M}} \frac{1}{2} |\text{Tr}[(\rho - \sigma)E]| = \frac{1}{2} - \frac{1}{4} \|\rho - \sigma\|_{\mathbb{M}}. \quad (2.6.5)$$

In order to exploit the duality between cones, it is convenient to use tools from *conic* (and in particular *semidefinite*) programming [BV04]. Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be finite-dimensional Euclidean spaces equipped with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$. In the same sense as the dual map in Theorem 2.10, given a linear map $\Phi : \mathcal{V}_1 \rightarrow \mathcal{V}_2$, its (Hilbert-space) adjoint $\Phi^* : \mathcal{V}_2 \rightarrow \mathcal{V}_1$ is the unique linear map satisfying

$$\langle \Phi(v_1), v_2 \rangle_2 = \langle v_1, \Phi^*(v_2) \rangle_1 \quad \forall v_1 \in \mathcal{V}_1, \forall v_2 \in \mathcal{V}_2.$$

Let $\mathcal{C}_1 \subseteq \mathcal{V}_1$ and $\mathcal{C}_2 \subseteq \mathcal{V}_2$ be proper cones, and let $c \in \mathcal{V}_1$ and $b \in \mathcal{V}_2$. A standard conic program (the *primal* problem) and its associated *dual* problem are given by

$$\gamma^{\text{primal}} := \inf_{x \in \mathcal{V}_1} \langle c, x \rangle_1 \quad \text{such that} \quad \Phi(x) - b \in \mathcal{C}_2, \quad x \in \mathcal{C}_1, \quad (2.6.6)$$

and

$$\gamma^{\text{dual}} := \sup_{y \in \mathcal{V}_2} \langle b, y \rangle_2 \quad \text{such that} \quad c - \Phi^*(y) \in \mathcal{C}_1^*, \quad y \in \mathcal{C}_2^*, \quad (2.6.7)$$

where $\mathcal{C}_1^*$ and $\mathcal{C}_2^*$ denote the dual cones with respect to $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, respectively. When $\mathcal{C}_1$ and $\mathcal{C}_2$ are positive semidefinite cones, these become *semidefinite programs*.

If the primal and dual problems are feasible, then *weak duality* holds: for every primal-feasible $x \in \mathcal{V}_1$ and dual-feasible $y \in \mathcal{V}_2$ one has

$$\langle c, x \rangle_1 \geq \langle b, y \rangle_2,$$

and in particular $\gamma^{\text{primal}} \geq \gamma^{\text{dual}}$.

Moreover, under a standard constraint qualification (Slater's interiority condition) there is no duality gap. One convenient conic formulation is the following.

Theorem 2.44 (Slater's interiority condition). *Assume that (2.6.6) is feasible and that there exists a strictly feasible point $\bar{x} \in \text{ri}(\mathcal{C}_1)$ such that $\Phi(\bar{x}) - b \in \text{ri}(\mathcal{C}_2)$. If γ^{primal} is finite, then strong duality holds, i.e.,*

$$\gamma^{\text{primal}} = \gamma^{\text{dual}},$$

and the dual optimum is attained, where ri denotes relative interior (see Equation (2.8.2)).

2.7. Representation Theory

Representation theory provides a natural framework for exploiting symmetries in composite quantum systems, in particular when dealing with many identical copies. A key instance is Schur-Weyl duality, which describes the commuting actions of the unitary group and the symmetric group on tensor powers $\mathcal{H}^{\otimes k}$, yielding a canonical decomposition into irreducible sectors and the associated Schur basis. These tools underpin several standard primitives in quantum information theory—such as spectrum estimation, permutation-invariant tomography, and entanglement concentration—and they will be used throughout this work. Let us briefly recall the notions and notation we need. For a more detailed treatment, we refer the reader to [Wey39, FH91].

Given a group G , a (unitary) representation of G on a Hilbert space $\mathcal{H}$ is a homomorphism $R : G \rightarrow \mathcal{U}(\mathcal{H})$, i.e., a family of unitaries $\{R_g\}_{g \in G} \subset \mathcal{B}(\mathcal{H})$ satisfying $R_e = \mathbf{1}$ and $R_{gh} = R_g R_h$ for all $g, h \in G$. A subspace $K \subseteq \mathcal{H}$ is called G -invariant (or simply *invariant*) if $R_g K \subseteq K$ for all $g \in G$. In this case, the restricted maps $\{R_g|_K\}_{g \in G}$ form a *subrepresentation*. The representation is *irreducible* if its only invariant subspaces are $\{0\}$ and $\mathcal{H}$. These irreducible representations of a group G play a fundamental role in representation theory.

In particular, every finite-dimensional unitary representation of a compact group is *completely reducible*, i.e., it decomposes as a direct sum of irreducible representations (or *irreps*), possibly with multiplicities. To relate two representations $R : G \rightarrow \mathcal{U}(\mathcal{H}_A)$ and $S : G \rightarrow \mathcal{U}(\mathcal{H}_B)$, we introduce *intertwiners*: a linear map $T : \mathcal{H}_A \rightarrow \mathcal{H}_B$ is an intertwiner if $TR_g = S_g T$ for all $g \in G$. Schur's lemma characterizes intertwiners between irreducible representations.

Lemma 2.45 (Schur's Lemma [FH91]). *Let $R : G \rightarrow \mathcal{U}(\mathcal{H}_A)$ and $S : G \rightarrow \mathcal{U}(\mathcal{H}_B)$ be irreducible representations, and let $T : \mathcal{H}_A \rightarrow \mathcal{H}_B$ be an intertwiner, i.e. $TR_g = S_g T$ for all $g \in G$. Then:*

- *If R and S are not isomorphic, then $T = 0$.*
- *If R and S are isomorphic, then T is an isomorphism; in particular, if $\mathcal{H}_A = \mathcal{H}_B$ and $R = S$, then $T = \alpha \mathbf{1}_{\mathcal{H}_A}$ for some $\alpha \in \mathbb{C}$.*

Let us now focus on the two groups that we will use in this work: the unitary group $\mathcal{U}(d)$ acting *collectively* on all copies, and the symmetric group S_k permuting the k copies [Wey39, FH91]. Schur-Weyl duality describes the joint representation of these commuting actions on the tensor-power space $(\mathbb{C}^d)^{\otimes k}$. Let $d = \dim(\mathbb{C}^d)$.

The *unitary group* on $\mathbb{C}^d$ is given by

$$\mathcal{U}(d) := \{U \in \mathcal{B}(\mathbb{C}^d) : U^* U = \mathbf{1}\},$$

and it acts on k copies via the tensor-power representation $U \mapsto U^{\otimes k}$ on $(\mathbb{C}^d)^{\otimes k}$.

The *symmetric group* S_k is the group of permutations of $[k] := \{1, \dots, k\}$ under composition. It acts on $(\mathbb{C}^d)^{\otimes k}$ through the permutation unitaries $\{R_\pi\}_{\pi \in S_k}$ defined by

$$R_\pi(|\psi_1\rangle \otimes \dots \otimes |\psi_k\rangle) = |\psi_{\pi^{-1}(1)}\rangle \otimes \dots \otimes |\psi_{\pi^{-1}(k)}\rangle.$$

These two actions commute, i.e. $(U^{\otimes k})R_\pi = R_\pi(U^{\otimes k})$ for all $U \in \mathcal{U}(d)$ and $\pi \in S_k$. Since $\mathcal{U}(d)$ is compact and S_k is finite, their representations are completely reducible. Consequently, Schur-Weyl duality yields the decomposition

$$(\mathbb{C}^d)^{\otimes k} \cong \bigoplus_{\lambda \in \mathcal{I}_{(k,d)}} \mathcal{U}_\lambda^{(d)} \otimes \mathcal{V}_\lambda,$$

where $\mathcal{U}_\lambda^{(d)}$ is an irreducible subspace for the action of $\mathcal{U}(d)$ and $\mathcal{V}_\lambda$ is the corresponding irreducible subspace for the action of S_k . In this decomposition, $\mathcal{U}(d)$ acts non-trivially only on $\mathcal{U}_\lambda^{(d)}$ and S_k acts non-trivially only on $\mathcal{V}_\lambda$. The index set $\mathcal{I}_{(k,d)}$ consists of partitions, commonly known as Young coefficients, $\lambda \vdash k$ with at most d parts, i.e.,

$$\mathcal{I}_{(k,d)} := \left\{ \lambda = (\lambda_1, \dots, \lambda_d) \in \mathbb{N}^d : \lambda_1 \geq \dots \geq \lambda_d \geq 0, \sum_{i=1}^d \lambda_i = k \right\}.$$

We identify such a partition $\lambda \vdash k$ with its associated Young diagram, consisting of k boxes arranged in left-justified rows of lengths $\lambda_1, \lambda_2, \dots$. The constraint “at most d parts” is equivalently $\ell(\lambda) \leq d$, i.e. the diagram has at most d rows. We refer to λ as the *Young coefficient* indexing the corresponding Schur-Weyl block. For further background on Young diagrams and their properties, see [Ful97].

Therefore, by Schur-Weyl duality there exists an orthonormal basis of $(\mathbb{C}^d)^{\otimes k}$, the *Schur basis*, which we label as $|\lambda, u, v\rangle$ with $\lambda \in \mathcal{I}_{(k,d)}$, $u \in [\dim \mathcal{U}_\lambda^{(d)}]$, and $v \in [\dim \mathcal{V}_\lambda]$. With respect to this basis, the commuting actions of S_k and $\mathcal{U}(d)$ are block-diagonal:

$$\begin{aligned} R_\pi &\cong \bigoplus_{\lambda \in \mathcal{I}_{(k,d)}} \mathbb{1}_{\mathcal{U}_\lambda^{(d)}} \otimes R_\pi^\lambda, \\ R_U &\cong \bigoplus_{\lambda \in \mathcal{I}_{(k,d)}} R_U^\lambda \otimes \mathbb{1}_{\mathcal{V}_\lambda}, \end{aligned}$$

where R_π^λ denotes the irreducible representation of S_k on $\mathcal{V}_\lambda$ and R_U^λ denotes the irreducible representation of $\mathcal{U}(d)$ on $\mathcal{U}_\lambda^{(d)}$. The unitary transformation that maps the computational basis to the Schur basis is known as *Schur transform*:

$$U_{Sch} : \{|k\rangle \otimes |l\rangle\} \mapsto \{|\lambda, u_i\rangle \otimes |\lambda, v_j\rangle\}.$$

For each $\lambda \in \mathcal{I}_{(k,d)}$, let Π_λ denote the orthogonal projector onto the Schur-Weyl block $\mathcal{U}_\lambda^{(d)} \otimes \mathcal{V}_\lambda$. This projector can be written as

$$\Pi_\lambda = \frac{\dim \mathcal{V}_\lambda}{k!} \sum_{\pi \in S_k} \chi^\lambda(\pi) R_\pi,$$

where $\chi^\lambda(\pi) := \text{Tr}(R_\pi^{(\lambda)})$ is the character of the irrep λ of S_k . The family $\{\Pi_\lambda\}_{\lambda \in \mathcal{I}_{(k,d)}}$ forms a projective measurement, i.e. $\Pi_\lambda \Pi_\mu = \delta_{\lambda\mu} \Pi_\lambda$ and $\sum_\lambda \Pi_\lambda = \mathbb{1}_{(\mathbb{C}^d)^{\otimes k}}$.

Moreover, the Schur transform (and hence the implementation of this measurement) can be approximated on a quantum computer to precision δ in polynomial time.

Theorem 2.46 ([CHW07, Kro19]). *Both the Schur transform and its corresponding measurement, commonly known as weak Schur sampling, can be implemented in a quantum computer up to precision δ with running time $O(\text{poly}(k, n, \log(1/\delta)))$.*

Then, it allows us to decompose tensor powers of pure states as follows.

Lemma 2.47 ([MH07, LREJ25]). *Given a bipartite pure state $|\psi_{AB}\rangle$, its k -fold tensor product admits the decomposition*

$$|\psi_{AB}\rangle^{\otimes k} = \sum_{\lambda \in \mathcal{I}_{(k,d)}} \sqrt{\text{Pr}(\lambda)} |\Phi_{AB}^\lambda\rangle \otimes (U_{\mathcal{V}_\lambda}^A \otimes U_{\mathcal{V}_\lambda}^B) |\phi_{AB}^+\rangle,$$

where $\text{Pr}(\lambda) = \text{Tr}[\Pi_\lambda^A \rho_A^{\otimes k}]$, $|\Phi_{AB}^\lambda\rangle \in \mathcal{U}_\lambda^A \otimes \mathcal{U}_\lambda^B$, and $|\phi_{AB}^+\rangle \in \mathcal{V}_\lambda^A \otimes \mathcal{V}_\lambda^B$ is a maximally entangled state of Schmidt rank $\dim \mathcal{V}_\lambda$. Here $U_{\mathcal{V}_\lambda}^A$ denotes the Schur transform restricted to the subspace $\mathcal{V}_\lambda$ on system A (and analogously for B).

The dimensions of the irreps of S_k and $\mathcal{U}(d)$ can be expressed as [Wey39, LREJ25]

$$\begin{aligned} \dim \mathcal{U}_\lambda^{(d)} &= \prod_{1 \leq i < j \leq d} \frac{\lambda_i - \lambda_j + j - i}{j - i}, \\ \dim \mathcal{V}_\lambda &= \frac{k! \prod_{1 \leq i < j \leq d} (\lambda_i - \lambda_j + j - i)}{\prod_{i=1}^d (\lambda_i + d - i)!}. \end{aligned}$$

We lastly focus on the *symmetric subspace* of $(\mathbb{C}^d)^{\otimes k}$, i.e. the subspace invariant under all permutations of the k tensor factors. In the Schur–Weyl decomposition, it corresponds to the one-row partition $\lambda = (k, 0, 0, \dots)$. This subspace is particularly important because Haar averages of k -fold tensor powers are supported entirely on it. Then, the *symmetric subspace* of $(\mathbb{C}^d)^{\otimes k}$ is defined as

$$\text{Sym}^k(\mathbb{C}^d) := \{|\varphi\rangle \in (\mathbb{C}^d)^{\otimes k} : R_\pi |\varphi\rangle = |\varphi\rangle \quad \forall \pi \in S_k\}.$$

Its orthogonal projector is

$$\Pi_{\text{sym}}^{(d,k)} := \frac{1}{k!} \sum_{\pi \in S_k} R_\pi,$$

and it satisfies $\text{Tr}[\Pi_{\text{sym}}^{(d,k)}] = \dim \text{Sym}^k(\mathbb{C}^d) = \binom{d+k-1}{k}$.

2.8. Asymptotic and Convex Notation

With $\log$ we denote the logarithm to base 2, and with $\ln$ the logarithm to base e . For $n \in \mathbb{N}$, we write $[n] = \{1, \dots, n\}$. For a set S , its affine hull is

$$\text{aff}(S) := \left\{ \sum_{i=1}^m \lambda_i x_i : x_i \in S, \lambda_i \in \mathbb{R}, \sum_{i=1}^m \lambda_i = 1 \right\}, \quad (2.8.1)$$

and its relative interior is

$$\text{ri}(S) := \{x \in S : \exists \varepsilon > 0 \text{ such that } B(x, \varepsilon) \cap \text{aff}(S) \subseteq S\}. \quad (2.8.2)$$

Here $B(x, \varepsilon)$ denotes the open ball of radius ε around x .

In Landau notation, given two functions $f(n)$ and $g(n)$, we write $f(n) = o(g(n))$ if $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$. In the same way, $f(n) = \omega(g(n))$ if $\lim_{n \rightarrow \infty} f(n)/g(n) = \infty$. $f(n) = O(g(n))$ if there exist two constant $C, N > 0$ such that $|f(n)| \leq C|g(n)|$ for all $n \geq N$. Similarly, $f(n) = \Omega(g(n))$ if there exist two constant $C, N > 0$ such that $|f(n)| \geq C|g(n)|$ for all $n \geq N$. Lastly, $f(n) = \Theta(g(n))$ if both $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$. A function $f(n)$ is **negl**(n), i.e., negligible, if, for every fixed c , $f(n) = o(1/n^c)$.

PART I

**FOUNDATIONS OF
COMPUTATIONAL QUANTUM
INFORMATION**

Chapter 3

Complexity-Constrained Quantum Information

Quantum information processing implementations are inherently limited to operations that admit efficient circuit descriptions. This has recently motivated complexity-aware variants of standard information measures, defined by restricting the class of physically realizable procedures [ABV23, CCL⁺17, AA26, AHRA25, MRR⁺26, MKH⁺25, YHKH⁺22, MRR⁺25]. This chapter introduces the complexity-constrained framework that will be used throughout the thesis, following [YHK26, YAKH26]. We begin with *complexity-constrained quantum channels*: in Section 3.1, we formalize the restriction and define the sets of polynomially implementable channels. We then specialize to binary-outcome measurements in Section 3.2, which are especially relevant for state discrimination. In Section 3.3, we show how efficiently generated quantum channels induce proper cones; the analogous construction for binary POVMs is treated in Section 3.4. Lastly, this binary setting leads to the computational counterparts of the trace norm and trace distance in Section 3.5.

3.1. Computationally Bounded Quantum Channels

In this section, we formalize the notion of complexity-constrained quantum channels. We measure complexity in terms of *gate complexity*: informally, given a fixed gate set $\mathcal{G}$, the cost of a transformation is the number of gates from $\mathcal{G}$ required to implement it. Since complexity-theoretic notions are asymptotic, we work with a scaling parameter n , denoting the number of registers in the circuit. We consider n -qudit systems, denoting the Hilbert space with $\mathcal{H}_n \simeq (\mathbb{C}^d)^{\otimes n}$ and $\dim(\mathcal{H}_n) = d^n$, where d is the local dimension.

Our complexity measure counts only gates. Accordingly, one has unrestricted access to ancillary registers and measurements in the computational basis. Moreover, for simplicity, polynomial-time classical (post-)processing is treated as free. When we refer to efficient channels or measurements, we mean those implementable with at most $p(n)$ gates for some polynomial p .

Let us first introduce the complexity-constrained framework for quantum channels. For this, we use the Choi–Jamiołkowski representation, but in a form tailored to our perspective: the relevant Choi operators are those of Heisenberg-picture maps, i.e., of completely positive unital (CPU) maps, rather than of Schrödinger-picture CPTP maps as in the standard convention (see Definition 2.12). Since our notion of efficient implementability is defined from the viewpoint of a bounded observer, the Heisenberg picture is the natural one to use.

Definition 3.1 (Choi-Jamiołkowski representation of CPU maps). Let $\mathcal{H}_n^A, \mathcal{H}_n^{A'}$ be finite-dimensional Hilbert spaces with $\dim \mathcal{H}_n^A = d_A$, and let $\mathcal{H}_n^A \simeq \mathcal{H}_n^{A'}$. Let $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ be a completely positive unital (CPU) map. For such a T^* , its Choi-Jamiołkowski operator representation on AB is given by

$$J(T^*) := d_A (\text{Id}_A \otimes T^*) (|\Omega_{AA'}\rangle\langle\Omega_{AA'}|),$$

where $|\Omega_{AA'}\rangle = \frac{1}{\sqrt{d_A}} \sum_{x \in [d_A]} |x\rangle_A \otimes |x\rangle_{A'}$.

Since the Choi-Jamiołkowski representation is a linear bijection, the action of the CPU map T^* can be recovered from its Choi-operator $J(T^*)$ in the following way,

$$T^*(X) = \text{tr}_A [J(T^*)(X^T \otimes \mathbb{1}_B)], \quad (3.1.1)$$

where the transpose is taken with respect to the computational basis used to define the state $|\Omega_{AA'}\rangle$.

The Stinespring representation introduced in Theorem 2.14 allows us to model the circuit complexity of the CPTP and CPU maps.

Definition 3.2 (Gate complexity of quantum maps). Let $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ be a completely positive and unital (CPU) linear map. Given a finite gate set $\mathcal{G}$, the *gate complexity* $C(T^*)$ is the minimum number of $\mathcal{G}$ -gates such that there exist ancillary spaces $\mathcal{H}_E, \mathcal{H}_K$ such that $\dim(\mathcal{H}_n^B) \dim(\mathcal{H}_E) = \dim(\mathcal{H}_n^{A'}) \dim(\mathcal{H}_K)$ and a unitary circuit $U : \mathcal{H}_n^B \otimes \mathcal{H}_E \rightarrow \mathcal{H}_n^{A'} \otimes \mathcal{H}_K$ generated by $C(T^*)$ gates from $\mathcal{G}$, such that

$$T^*(X) = \langle 0|_E U^* (X \otimes \mathbb{1}_K) U |0\rangle_E \quad \forall X \in \mathcal{B}(\mathcal{H}_n^{A'}). \quad (3.1.2)$$

Completely analogously we say that the *gate complexity* $C(T)$ of a completely positive trace-preserving (CPTP) map $T : \mathcal{B}_1(\mathcal{H}_n^B) \rightarrow \mathcal{B}_1(\mathcal{H}_n^{A'})$ is the minimal number of $\mathcal{G}$ -gates required to implement a Stinespring dilation of T , i.e., such that there exist ancillary spaces $\mathcal{H}_E, \mathcal{H}_K$ such that $\dim(\mathcal{H}_n^B) \dim(\mathcal{H}_E) = \dim(\mathcal{H}_n^{A'}) \dim(\mathcal{H}_K)$ and a unitary circuit $U : \mathcal{H}_n^B \otimes \mathcal{H}_E \rightarrow \mathcal{H}_n^{A'} \otimes \mathcal{H}_K$ generated by $C(T)$ gates from $\mathcal{G}$, such that

$$T(\rho) = \text{tr}_K [U(\rho \otimes |0\rangle\langle 0|_E)U^*] \quad \forall \rho \in \mathcal{B}(\mathcal{H}_n^B). \quad (3.1.3)$$

Remark 3.3. This notion of complexity of linear maps, allowing for arbitrary ancillary systems initiated in $|0\rangle\langle 0|$ which in the end are traced out and only counting the gate complexity of the unitary, is the channel analogue of the measurement complexity defined in Definition 3.7 and is also used in [MRR⁺25]. It is closely related to that of [AHRA25, Definition 18]; for the latter see also [AA26, CCL⁺17]. There, the authors consider circuit complexity of adaptive quantum circuits allowing for arbitrary ancillas, polynomial unitary gates, mid-circuit computational measurements, and arbitrary final computational measurements. Up to at most a polynomial overhead, the notion of complexity we use is equivalent to [AHRA25, Definition 18]. This is because the additional operations of mid-circuit measurement can be emulated using, e.g., extra ancillas and polynomially many CNOT gates.

In particular, in this model, the gate complexity of a CPU map T^* coincides with that of its dual CPTP map T .

Lemma 3.4. *Let $T : \mathcal{B}_1(\mathcal{H}_n^B) \rightarrow \mathcal{B}_1(\mathcal{H}_n^{A'})$ be a CPTP map then $C(T) = C(T^*)$.*

Proof. Let the CPTP map $T : \mathcal{B}(\mathcal{H}_n^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'})$ be efficiently implementable with polynomial complexity $p(n)$. I.e. there exist ancillary spaces $\mathcal{H}_E, \mathcal{H}_K$, an ancilla state $|0\rangle_E \in \mathcal{H}_E$, and a $p(n)$ -depth unitary

$$U : \mathcal{H}_n^B \otimes \mathcal{H}_E \longrightarrow \mathcal{H}_n^{A'} \otimes \mathcal{H}_K$$

such that

$$T(\rho) = \text{tr}_K[U(\rho \otimes |0\rangle\langle 0|_E)U^*] = \text{tr}_K[V\rho V^*] \quad (3.1.4)$$

where the Stinespring isometry $V : \mathcal{H}_n^B \rightarrow \mathcal{H}_n^{A'} \otimes \mathcal{H}_K$ is given by

$$V|\psi\rangle_B := U(|\psi\rangle_B \otimes |0\rangle_E) .$$

Therefore the dual map $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ is given by

$$T^*(X) = V^*(X \otimes \mathbb{1}_K)V = \langle 0|_E U^*(X \otimes \mathbb{1}_K)U|0\rangle_E .$$

Moreover, T^* is specified by the same isometry V (equivalently, by the same unitary U and ancilla initialization) and thus satisfies $C(T^*) \leq p(n)$. The converse follows by minimality of $C(T)$, in the sense that if there was some Stinespring dilation of T^* with a complexity $C < p(n)$, then this would, by the above argument in reverse, yield a Stinespring dilation of T with the a complexity $C < p(n) = C(T)$, which is not possible. $\square$

Moreover, after a suitable rescaling, Choi–Jamiołkowski operators of CPU maps can be viewed as effect operators on the space $\mathcal{B}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$.

Lemma 3.5 (Choi–Jamiołkowski representation of CPU maps and effect operators). *Let $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ be a completely positive and unital (CPU) map, and let $J(T^*)$ denote its Choi operator. Define the normalized Choi operator*

$$E_{AB} := \frac{J(T^*)}{d_A} \in \mathcal{B}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B) .$$

Then E_{AB} is an effect operator which satisfies

$$i) \ 0 \leq E_{AB} \leq \mathbb{1}_{AB} , \quad (3.1.5)$$

$$ii) \ \text{tr}_A[E_{AB}] = \frac{1}{d_A} \mathbb{1}_B . \quad (3.1.6)$$

Proof. Let $|\Omega_{AA'}\rangle := \frac{1}{\sqrt{d_A}} \sum_{i=1}^{d_A} |i\rangle_A \otimes |i\rangle_{A'}$, so that

$$J(T^*) := d_A(\mathbb{1}_A \otimes T^*)(|\Omega_{AA'}\rangle\langle\Omega_{AA'}|).$$

Since $|\Omega_{AA'}\rangle\langle\Omega_{AA'}| \leq \mathbb{1}_{AA'}$, complete positivity implies

$$0 \leq J(T^*) \leq d_A(\mathbb{1}_A \otimes T^*)(\mathbb{1}_{AA'}) = d_A \mathbb{1}_A \otimes T^*(\mathbb{1}_{A'}) = d_A \mathbb{1}_{AB},$$

where we use the fact that T^* is unital in the last step. Dividing by d_A yields $0 \leq E_{AB} \leq \mathbb{1}_{AB}$, proving (i).

For (ii), it follows from Eq. (3.1.1) that

$$\text{tr}_A[J(T^*)] = T^*(\mathbb{1}_{A'}) = \mathbb{1}_B.$$

Therefore, we have that $\text{tr}_A[E_{AB}] = \mathbb{1}_B/d_A$. $\square$

Next, we relate the gate complexity of a CPU map to that of the associated binary test on AB obtained from its (normalized) Choi operator.

Lemma 3.6 (Circuit complexity of Choi effect operators). *Let $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ be a CPU map with gate complexity $C(T^*)$ as in Definition 3.2, and define its normalized Choi effect operator*

$$E_{AB} := \frac{1}{d_A} J(T^*) = (\text{Id}_A \otimes T^*)(|\Omega_{AA'}\rangle\langle\Omega_{AA'}|) \in \mathcal{B}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B),$$

where $|\Omega_{AA'}\rangle = \frac{1}{\sqrt{d_A}} \sum_{x \in [d_A]} |x\rangle_A \otimes |x\rangle_{A'}$. Then $M_{AB} := \{E_{AB}, \mathbb{1}_{AB} - E_{AB}\}$ is a valid binary POVM and is implementable with at most

$$C(M_{AB}) \leq C(T^*) + C(U_\Omega)$$

$\mathcal{G}$ -gates, where U_Ω denotes a circuit implementing the Bell test $\{|\Omega_{AA'}\rangle, \mathbb{1}_{AA'} - |\Omega_{AA'}\rangle\langle\Omega_{AA'}|\}$ (equivalently, preparing $|\Omega_{AA'}\rangle$ and measuring in the computational basis). Moreover, for every state ρ_{AB} , the acceptance probability of this implementation equals $\text{Tr}[\rho_{AB} E_{AB}]$.

Proof. By Definition 3.2 there exist ancillas $\mathcal{H}_E, \mathcal{H}_K$, a fixed $|0\rangle_E \in \mathcal{H}_E$, and a unitary

$$U : \mathcal{H}_n^B \otimes \mathcal{H}_E \longrightarrow \mathcal{H}_n^{A'} \otimes \mathcal{H}_K$$

implemented with $C(T^*)$ gates. Define the isometry $V : \mathcal{H}_n^B \rightarrow \mathcal{H}_n^{A'} \otimes \mathcal{H}_K$ by

$$V |\psi\rangle_B := U(|\psi\rangle_B \otimes |0\rangle_E) \quad \forall |\psi\rangle_B \in \mathcal{H}_n^B,$$

so that

$$T^*(X) = V^*(X \otimes \mathbb{1}_K)V \quad \forall X \in \mathcal{B}(\mathcal{H}_n^{A'}).$$

Consider the following measurement procedure on input ρ_{AB} :

1. Apply V to register B , producing registers $A'K$.
2. Perform the two-outcome projective measurement $\{\Omega_{AA'}, \mathbb{1}_{AA'} - \Omega_{AA'}\}$ on AA' (discard K).

Step (1) can be implemented with $C(T^*)$ gates and step (2), with $C(U_\Omega)$ gates. The acceptance probability is

$$\begin{aligned} \text{Tr}[(\Omega_{AA'} \otimes \mathbb{1}_K)(\mathbb{1}_A \otimes V)\rho_{AB}(\mathbb{1}_A \otimes V^*)] &= \text{Tr}[\rho_{AB}(\mathbb{1}_A \otimes V^*)(\Omega_{AA'} \otimes \mathbb{1}_K)(\mathbb{1}_A \otimes V)] \\ &= \text{Tr}[\rho_{AB}(\text{Id}_A \otimes T^*)(\Omega_{AA'})] = \text{Tr}[\rho_{AB}E_{AB}], \end{aligned}$$

which proves the claim. $\square$

Therefore, the complexity of implementing the CPU map T^* upper bounds—up to the additive overhead $C(U_\Omega)$ for implementing the Bell measurement—the complexity of implementing the binary measurement $\{E_{AB}, \mathbb{1}_{AB} - E_{AB}\}$ on $\mathcal{B}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, where $\text{tr}_A[E_{AB}] = \mathbb{1}_B/d_A$.

When the admissible transformations are restricted to measurements, the same complexity constraint can be formulated directly in terms of POVM effects. This gives a simpler single-system formalism, which will later be recovered as a special case of the channel-based construction.

3.2. Complexity-Constrained POVMs

When considering generic two-output measurements, we write the corresponding POVMs as $M = (E_1, E_2)$. By Naimark's dilation theorem, introduced in Theorem 2.3, any such POVM can be implemented by appending an ancillary qubit to the quantum system, applying a unitary operator on the joint system, and performing a projective measurement of the ancilla in the computational basis. More generally, the theorem applies to arbitrary K -outcome POVMs by using an ancillary system of dimension K .

Similarly to Definition 3.2 for the channel-based case, the Naimark dilation introduced in Theorem 2.3 allows us to quantify the complexity of implementing a POVM in the following way.

Definition 3.7 (Gate complexity of a POVM). Given a finite gate set $\mathcal{G}$, the *gate complexity* $C(M)$ of a POVM $M := (E_i)_{i=1}^K$ is the minimal number of gates necessary to implement the unitary operator $U \in \mathcal{B}(\mathcal{H} \otimes \mathcal{H}_K)$ such that

$$\text{Tr}[E_i \rho] = \text{Tr} \left[\left(\mathbb{1}_{\mathcal{H}} \otimes |i\rangle\langle i|_{\mathcal{H}_K} \right) \left(U \left(\rho \otimes |0\rangle\langle 0|_{\mathcal{H}_K} \right) U^* \right) \right] \quad \forall i \in [K], \quad (3.2.1)$$

with $\sum_{i=1}^K E_i = \mathbb{1}$.

Therefore, one can define the set of quantum measurements restricted to some G gates and with K outcomes.

Definition 3.8 (Set of complexity and output constrained POVMs). Given a finite gate set $\mathcal{G}$, for $G \in \mathbb{N}$ and $K \geq 2$, the set of G -complexity K -outcome POVMs is defined as,

$$\mathbf{M}_{n,K}^G := \left\{ M = (E_i)_{i=1}^K \mid C(M) \leq G, \dim(\mathcal{H}_K) = K, E_i \in \text{Pos}(\mathcal{H}_n) \right\}, \quad (3.2.2)$$

where $C(M)$ is the complexity of the POVM M . When G is taken as a polynomial $p(n) : \mathbb{N} \rightarrow \mathbb{N}$, then we write $\mathbf{M}_{n,K}^{p(n)}$.

Remark 3.9 (Gate complexity of the POVM). When we refer to the complexity of a POVM, we are referring to the complexity of the (unitary) quantum circuit U that was necessary to implement it. We adopt the approach from, e.g., [AA26, AHRA25], where, in other words, one considers the process of adding ancillas, tracing out subsystems, and measuring in the computational basis, to be ‘free’ operations. To characterize the complexity of a quantum circuit, U , one first chooses an approximately universal gate set $\mathcal{G}$ consisting of $|\mathcal{G}|$ unique gates. The complexity of U is equal to the minimal number of gates that are necessary to generate U .

We fix a gate set $\mathcal{G}$ and a polynomial gate budget $p(n)$. Let us now focus on binary measurements.

Definition 3.10 (Polynomially Generated Sets of Binary POVMs). For each dimension n , let $\mathbf{M}_n^{\text{eff}}$ denote the set of *binary polynomially generated POVMs* implementable by a quantum circuit of size at most $p(n)$ over $\mathcal{G}$. We suppress the dependence on $\mathcal{G}$ and $p(n)$ in the notation. Then, the family $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ is the associated family of polynomially generated set of binary POVMs.

For binary measurements, there exists a bijection between POVMs $M = (E, \mathbb{1} - E)$ and effect operators E . We take advantage of this bijection for the remainder of this section and will in abuse of notation sometimes refer to both effect operators and POVMs as measurements, depending on context. In particular, whenever we consider the complexity of an effect operator E , we are referring to the complexity of the associated POVM M .

Definition 3.11 (The set of efficient binary effects). For each dimension n , let $\mathbf{E}_n^{\text{eff}} \subseteq \mathcal{B}(\mathcal{H}_n)$ denote the set of POVM effects E such that $0 \leq E \leq \mathbb{1}$ and the binary POVM $\{E, \mathbb{1} - E\}$ is implementable by a quantum circuit of size at most $p(n)$ over $\mathcal{G}$. We suppress the dependence on $\mathcal{G}$ and $p(n)$ in the notation. If each $\mathbf{E}_n^{\text{eff}}$ is informationally complete, i.e., it holds that $\text{span}(\mathbf{E}_n^{\text{eff}}) = \mathcal{B}^\dagger(\mathcal{H}_n)$, then we further call this family an *informationally complete polynomially generated set of effect operators*.

One should think of polynomially generated effect operators informally as ‘efficiently’ implementable measurement operators. To better understand this definition, a couple of observations and remarks are in order. Observe that we (almost) trivially have that $0, \mathbb{1} \in \mathbf{E}_n^{\text{eff}}$, since these measurements correspond to trivially outputting 0, 1, respectively, regardless of input. A way to implement this is to generate an ancilla qubit in the

state $|0\rangle$ (for 1 one additionally applies an X gate) and subsequently measuring it in the computational basis, which we assume to be free operations. We will usually assume that the polynomial p in the definition is strictly super-additive, satisfying $p(n_1 + n_2) \geq p(n_1) + p(n_2) + 1$, unless otherwise mentioned. Therefore, given $E_1 \in \mathbf{E}_{n_1}^{\text{eff}}$ and $E_2 \in \mathbf{E}_{n_2}^{\text{eff}}$, $E_1 \otimes E_2 \in \mathbf{E}_{n_1+n_2}^{\text{eff}}$. The same assumptions hold for the set of efficiently generated binary POVMs: if $M = (E, \mathbb{1} - E) \in \mathbf{M}_n^{\text{eff}}$, then so is $M' = (\mathbb{1} - E, E)$ and if $M_i = (E_i, \mathbb{1} - E_i) \in \mathbf{M}_{n_i}^{\text{eff}}$ for $i = 1, 2$, then $(E_1 \otimes E_2, \mathbb{1} - E_1 \otimes E_2) \in \mathbf{M}_{n_1+n_2}^{\text{eff}}$.

Remark 3.12 (Efficient effect operators, classical post-processing, and information completeness). Implicit in this definition is that we assume that polynomial classical post-processing with NOT and OR operations is also efficient. To motivate that $E \in \mathbf{E}_n^{\text{eff}} \implies \mathbb{1} - E \in \mathbf{E}_n^{\text{eff}}$, we highlight that the POVMs, which are the physical implementation of the effect operators, $M_1 = (E, \mathbb{1} - E)$ and $M_2 = (\mathbb{1} - E, E)$ differ only by a single classical NOT post-processing step. As such, if M_1 is ‘efficient’, then M_2 should also be considered efficient, which we enforce with this condition.

Likewise, the stability under tensor product condition (assuming the polynomial p grows fast enough, which is the case if it is super-additive) is motivated by our implicit assumption that polynomially many classical OR post-processing operations are also assumed to be efficient. For example, for two efficient effect operators E_1, E_2 the corresponding binary measurements are $M_1 = (E_1, \mathbb{1} - E_1)$ and $M_2 = (E_2, \mathbb{1} - E_2)$, and $M_3 = (E_1 \otimes E_2, \mathbb{1} - E_1 \otimes E_2)$ corresponds to the tensor product measurement $E_1 \otimes E_2$. It is implementable by implementing M_1, M_2 in parallel using at most $p(n_1) + p(n_2)$ gates and mapping the output ‘00’ to 0 and the other three outcomes to 1. This corresponds to a logical OR gate which is implementable via a single classical post-processing operation.

We point out that one way to ensure that the set of effect operators is informationally complete is if for every n ,

$$U |k\rangle\langle k| U^* \in \mathbf{E}_n^{\text{eff}} \quad \forall k \in [\dim(\mathcal{H}_n)] \quad \forall U \in \mathcal{U},$$

where $\mathcal{U}$ is the set of *Weyl operators* on $\mathcal{H}_n$ [Wat18, Section 4.1.2] and $|k\rangle\langle k|$ is the computational basis projector. In the case where $\dim(\mathcal{H}) = 2$, this can be reduced to the Pauli group. Therefore, it corresponds to first applying (for all $i \in \mathbb{N}$) a unitary $U_i \in \{X, Y, Z\}$ on the i -th qubit. Then one has to apply a measurement from some set, where the corresponding effect operators are of the form

$$|i_1 \cdots i_n\rangle\langle i_1 \cdots i_n|.$$

One way to generate the corresponding measurement would be to first apply the gate $X^{i_1} \otimes \cdots \otimes X^{i_n}$. Subsequently, one applies a computational measurement on all n qubits. Since we are only considering binary measurements, one has to then map the outcome $0 \cdots 0$ to 0 and all other outcomes to 1. This ‘ 2^n -output measurement + post-processing step’ is generally efficient, and can be implemented by an efficient quantum circuit and a computational measurement on a single qubit. So, in effect, the informationally complete condition means that the polynomial $p(n)$ cannot be too small.

Remark 3.13. Our definition of efficiently generated measurements encapsulates non-uniform quantum polynomial-time generated POVMs, which have found applications in cryptography due to their ability to model an adversary with quantum advice [AD14]. As such, any divergences we define in Section 4.1 via $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ may similarly be relevant for cryptographic applications.

Let us now connect the formalism for complexity-constrained binary measurements with the general channel-based framework by turning to the Choi operators associated with complexity-constrained channels.

3.3. The Cone of Efficiently Implementable Choi Operators

We return to the channel setting introduced in Section 3.1. Through the Choi–Jamiołkowski representation, a complexity bound on CPU maps determines a set of positive semidefinite operators. Taking its conic hull then provides a geometric representation of the corresponding complexity-constrained transformations. We first define this set, establish its relation with the efficiently generated effect operators introduced above, and then construct the associated cone.

Definition 3.14 (Set of complexity-constrained Choi operators). Given a finite, discrete gate set $\mathcal{G}$, the set of G -complexity Choi operators $\mathbf{J}_{n,A|B}^G$ is defined as

$$\mathbf{J}_{n,A|B}^G := \left\{ J(T^*) \mid J(T^*) := d_A(\text{Id}_A \otimes T^*) (|\Omega_{AA'}\rangle\langle\Omega_{AA'}|), \ C(T^*) \leq G \right\}, \quad (3.3.1)$$

where $C(T^*)$ denotes the circuit complexity of implementing the CPU map $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ given the gate-set $\mathcal{G}$. When G is a polynomial $p(n) : \mathbb{N} \rightarrow \mathbb{N}$, we write $\mathbf{J}_{n,A|B}^{p(n)}$.

Lemma 3.15 (Counting exact bounded-size channel circuits). *Fix a finite, discrete gate set $\mathcal{G}$. Since $\mathcal{G}$ is finite, there exists $\ell \in \mathbb{N}$ such that every gate in $\mathcal{G}$ acts nontrivially on at most ℓ registers. Then, for every $n, G \in \mathbb{N}$, the set $\mathbf{J}_{n,A|B}^G$ is finite. More precisely,*

$$\left| \mathbf{J}_{n,A|B}^G \right| \leq \sum_{m=0}^G \left(|\mathcal{G}|(n + \ell G)^\ell \right)^m \leq 2^{cG \log(n+G)} \quad (3.3.2)$$

for some constant $c > 0$ depending only on $\mathcal{G}$ and ℓ . In particular, if $G = p(n)$ is polynomial, then there exists a polynomial $r(n)$ such that

$$\left| \mathbf{J}_{n,A|B}^{p(n)} \right| \leq 2^{r(n)}. \quad (3.3.3)$$

Proof. By Definition 3.2, every element of $\mathbf{J}_{n,A|B}^G$ is the Choi operator of a CPU map $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ implemented by a unitary circuit U of size at most G , possibly using ancillary registers.

Although ancillary registers are unrestricted in principle, only those on which at least one gate acts are relevant. Hence every such circuit is equivalent to one in which all ancillas on which no gate acts have been removed. Moreover, the remaining ancillas may be relabelled according to the first gate in which they appear. Since each gate acts on at most ℓ registers, a circuit with at most G gates uses at most ℓG active ancilla registers. Therefore every such circuit admits a canonical description on at most

$$N := n + \ell G \quad (3.3.4)$$

registers.

For each of the $m \leq G$ gate positions, one chooses:

1. the gate in $\mathcal{G}$, giving at most $|\mathcal{G}|$ possibilities;
2. the ordered tuple of registers on which it acts, giving at most N^ℓ possibilities.

Hence the number of exact circuit descriptions of length m is at most

$$\left(|\mathcal{G}| N^\ell\right)^m = \left(|\mathcal{G}|(n + \ell G)^\ell\right)^m. \quad (3.3.5)$$

Summing over all $m \leq G$ gives

$$\left|\mathbf{J}_{n,A|B}^G\right| \leq \sum_{m=0}^G \left(|\mathcal{G}|(n + \ell G)^\ell\right)^m. \quad (3.3.6)$$

Since distinct Choi operators are fewer than distinct circuit descriptions, this proves finiteness.

Finally, because ℓ and $|\mathcal{G}|$ are constants,

$$\sum_{m=0}^G \left(|\mathcal{G}|(n + \ell G)^\ell\right)^m \leq (G + 1) \left(|\mathcal{G}|(n + \ell G)^\ell\right)^G \leq 2^{c G \log(n+G)} \quad (3.3.7)$$

for some constant $c > 0$. Taking $G = p(n)$ yields the claim. $\square$

Therefore, for every polynomial $p(n)$, the family $\{\mathbf{J}_{n,A|B}^{p(n)}\}_{n \in \mathbb{N}}$ consists of at most $2^{r(n)}$ Choi operators for some polynomial $r(n)$. In particular, for each fixed n , the set $\mathbf{J}_{n,A|B}^{p(n)}$ is finite. There is a direct connection between this family and the family of efficiently generated effect operators $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ define in Definition 3.11.

Lemma 3.16. *Given a gate set $\mathcal{G}$, let $\mathbf{E}_n^{\text{eff}} \subseteq \mathcal{B}(\mathcal{H}_n^B)$ denote the set of efficiently generated binary effect operators as in Definition 3.11 i.e. those E_B for which the binary POVM $\{E_B, \mathbb{1}_B - E_B\}$ is implementable by a circuit of size $q(n)$. Let $\mathcal{H}_n^{A'} \simeq \mathbb{C}^2$ with computational basis $\{|0\rangle, |1\rangle\}$. Then, for every $E_B \in \mathbf{E}_n^{\text{eff}}$ there exists a CPU map $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ with $C(T^*) \leq q(n)$ and Choi operator $J_{AB} \in \mathbf{J}_{n,A|B}^{q(n)}$ such that*

$$E_B = T^*(|0\rangle\langle 0|_{A'}) = \text{tr}_A[J_{AB}(|0\rangle\langle 0|_A \otimes \mathbb{1}_B)] = \langle 0|_A J_{AB} |0\rangle_A.$$

Similarly, $\mathbb{1}_B - E_B = T^*(|1\rangle\langle 1|_{A'})$. In particular,

$$\mathbf{E}_n^{\text{eff}} \subseteq \left\{ \langle i|_A J_{AB} |i\rangle_A \mid J_{AB} \in \mathbf{J}_{n,A|B}^{q(n)}, i \in \{0, 1\} \right\}.$$

The same inclusion holds more generally for any choice of $\dim(\mathcal{H}_n^{A'}) \geq 2$.

Proof. Fix $E_B \in \mathbf{E}_n^{\text{eff}}$. By definition, there exists a circuit of size $q(n)$ implementing the binary POVM $\{E_B, \mathbb{1}_B - E_B\}$. By Theorem 2.3, we may assume this circuit can be described in the following way

$$E_B = \langle 0|_{A'} U^*(\mathbb{1}_B \otimes |0\rangle\langle 0|_{A'}) U |0\rangle_{A'}.$$

Define $T^* : \mathcal{B}(\mathcal{H}_n^{A'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ by

$$T^*(X) := \langle 0|_{A'} U^*(\mathbb{1}_B \otimes X) U |0\rangle_{A'} \quad \forall X \in \mathcal{B}(\mathcal{H}_n^{A'}).$$

Then T^* is completely positive and unital, and it satisfies $T^*(|0\rangle\langle 0|_{A'}) = E_B$ and $T^*(|1\rangle\langle 1|_{A'}) = \mathbb{1}_B - E_B$. Moreover, $C(T^*) \leq q(n)$ since it is implemented by the same circuit. The rest of the proof follows from Equation (3.1.1). $\square$

Then, for every $n \in \mathbb{N}$, the set of efficient measurements $\mathbf{E}_n^{\text{eff}}$ implementable with $p(n)$ gates is included (as a representation of its corresponding quantum channel) in the set of complexity constrained Choi operators $\mathbf{J}_{n,A|B}^{p(n)}$.

Moreover, one can define the family of cones generated by the family of sets $\{\mathbf{J}_{n,A|B}^{p(n)}\}_{n \in \mathbb{N}}$ which capture the operator representations of efficiently implementable quantum channels $T : \mathcal{B}(\mathcal{H}_n^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'})$.

Definition 3.17 (Family of the cones of efficiently generated Choi operators). Given a Hilbert space $\mathcal{H}_n^A \otimes \mathcal{H}_n^B$, the cone of efficiently generated Choi operators is given by

$$\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} = \text{cone}(\mathbf{J}_{n,A|B}^{p(n)}) = \bigcup_{\lambda \geq 0} \lambda \cdot \text{conv}(\mathbf{J}_{n,A|B}^{p(n)}). \quad (3.3.8)$$

where $\text{conv}(\mathbf{J}_{n,A|B}^{p(n)})$ is the convex hull of the set $\mathbf{J}_{n,A|B}^{p(n)}$.

Lemma 3.18 (Normalization of Choi operators). *For any CPU map T^* , the Choi operator satisfies $J(T^*) \geq 0$ and $\text{tr}_A[J(T^*)] = \mathbb{1}_B$. Consequently every $F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ obeys $\text{tr}_A[F_{AB}] = \lambda \mathbb{1}_B$ for some $\lambda \geq 0$.*

Proof. Complete positivity gives $J(T^*) \geq 0$, and $\text{tr}_A[J(T^*)] = d_A T^*(\text{tr}_A[|\Omega_{AA'}\rangle\langle\Omega_{AA'}|]) = T^*(\mathbb{1}_{A'}) = \mathbb{1}_B$. The trace condition extends by linearity and the conic closure. $\square$

Throughout this thesis we will work with two normalized bases of the same cone $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$. The first one is the *channel-normalized slice*,

$$\text{conv}(\mathbf{J}_{n,A|B}^{p(n)}) := \left\{ F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} : \text{tr}_A[F_{AB}] = \mathbb{1}_B \right\},$$

which corresponds to the convex hull of efficiently generated Choi operators. The second one is the *effect-normalized slice*,

$$\mathbf{E}_{n,A|B}^{\text{eff}} := \left\{ F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} : \text{tr}_A[F_{AB}] = \mathbb{1}_B/d_A \right\},$$

which arises from the normalized Choi effect operators. Nevertheless, as discussed in Lemma 3.18, these two normalizations differ only by a positive scalar rescaling, and hence generate the same proper cone. Which normalization is preferable depends on the application.

Lemma 3.19. *Let*

$$\mathcal{V}_{n,A|B} := \left\{ X \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B) : \text{tr}_A[X] \in \mathbb{R} \cdot \mathbb{1}_B \right\}$$

be the subspace spanned by Hermitian operators whose partial trace is proportional to the identity. If

$$\text{span}_{\mathbb{R}}(\mathbf{J}_{n,A|B}^{p(n)}) = \mathcal{V}_{n,A|B}$$

for every $n \in \mathbb{N}$, then the family of cones $\{\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$ is a family of proper cones with respect to the subspaces $\mathcal{V}_{n,A|B}$, i.e., each $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ is closed, convex, pointed, and solid in $\mathcal{V}_{n,A|B}$.

Proof. Convexity is clear from the definition. By Lemma 3.15, the set $\mathbf{J}_{n,A|B}^{p(n)}$ is finite for every n . Hence $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} = \text{cone}(\mathbf{J}_{n,A|B}^{p(n)})$ is a finitely generated cone, and therefore it is closed. Pointedness follows since every generator $J(T^*)$ is positive semidefinite, since T^* are all completely positive, and $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} \subseteq \text{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, hence $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} \cap (-\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}) = \{0\}$. Moreover, for CPU maps one has $\text{tr}_{A_n} J(T^*) = \mathbb{1}_{B_n}$, so $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} \subseteq \mathcal{V}_{n,A|B}$.

Lastly, solidness holds in $\mathcal{V}_{n,A|B}$ as the assumption implies

$$\text{span}_{\mathbb{R}}(\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}) = \text{span}_{\mathbb{R}}(\mathbf{J}_{n,A|B}^{p(n)}) = \mathcal{V}_{n,A|B},$$

(using $\text{span}(\text{cone}(\text{conv}(S))) = \text{span}(S)$ for nonempty S). Hence, by the standard full-dimensionality argument (cf. [BV04, Section 2.1.3]), $\text{ri}_{\mathcal{V}_{n,A|B}}(\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}) \neq \emptyset$, where ri denotes the relative interior. $\square$

The solidness condition

$$\text{span}_{\mathbb{R}}(\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}) = \text{span}_{\mathbb{R}}(\mathbf{J}_{n,A|B}^{p(n)}) = \mathcal{V}_{n,A|B},$$

can be seen as an informational completeness condition at the level of Choi representations: every Choi operator associated with a CPU map (efficient or not) lies in $\mathcal{V}_{n,A|B}$ and can be spanned by Choi operators of efficiently implementable CPU maps, i.e.,

$$\forall X \in \mathcal{V}_{n,A|B} \exists c_i \in \mathbb{R}, J_{AB}^i \in \mathbf{J}_{n,A|B}^{p(n)} \text{ s.t. } X = \sum_i c_i J_{AB}^i.$$

Example 3.20 (An explicit example of a set of informationally complete quantum channels). In order to show the feasibility of the informational completeness assumption under complexity constraints, let us provide an example. Fix $n \in \mathbb{N}$ and set of symbols $\mathcal{S}_n := \{0, 1, +, +i\}^n$, where its corresponding quantum states are defined as $|+\rangle := 1/\sqrt{2}(|0\rangle + |1\rangle)$ and $|+i\rangle := 1/\sqrt{2}(|0\rangle + i|1\rangle)$. Then, let us define the product states on $\mathcal{H}_n^{A'} \simeq (\mathbb{C}^2)^{\otimes n}$ by

$$\sigma_s := \bigotimes_{k=1}^n |s_k\rangle\langle s_k|,$$

and set $\sigma_{0^n} := |0\rangle\langle 0|^{\otimes n}$. For each Pauli string $Q \in \{\pm 1, \pm X, \pm Y, \pm Z\}^{\otimes n}$, let us now define the binary POVM $\{E_Q, \mathbb{1} - E_Q\}$ with

$$E_Q := \frac{\mathbb{1} + Q}{2}.$$

Assuming that the POVMs $\{\{E_Q, \mathbb{1} - E_Q\}\}_Q$ and the states $\{\sigma_s\}_{s \in \mathcal{S}_n}$ are efficiently implementable, then the measurement map $T_{s,Q} : \mathcal{B}(\mathcal{H}_n^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'})$ defined by

$$T_{s,Q}(\rho) := \text{Tr}[E_Q \rho] \sigma_s + \text{Tr}[(\mathbb{1} - E_Q) \rho] \sigma_{0^n}$$

is also efficiently implementable for every $s \in \mathcal{S}_n$ and Q .

Note that this linear map is nothing but a measure-and-prepare channel, where the set of measurements is the tomographically complete set of Pauli measurements, and the prepared states are products of Pauli eigenstates. One can easily verify that it is a CPTP map and, thus, its dual, a CPU map. Its corresponding Choi operators can be written as

$$J(T_{s,Q}^*) = \sigma_s^\top \otimes E_Q + \sigma_{0^n}^\top \otimes (\mathbb{1} - E_Q) = \sigma_{0^n} \otimes \mathbb{1} + (\sigma_s - \sigma_{0^n})^\top \otimes E_Q,$$

where $\top$ denotes transpose in the computational basis. Therefore,

$$\begin{aligned} \text{span}_{\mathbb{R}}\{J(T_{s,Q}^*) : s \in \mathcal{S}_n, Q \in \{\pm 1, \pm X, \pm Y, \pm Z\}^{\otimes n}\} \\ = \mathbb{R} \cdot (\mathbb{1}_A \otimes \mathbb{1}_B) \oplus \text{Herm}_0(\mathcal{H}_n^A) \otimes \text{Herm}(\mathcal{H}_n^B) = \mathcal{V}_{n,A|B}. \end{aligned}$$

Let us now define the dual subspace of the channel subspace $\mathcal{V}_{n,A|B}$. Following [Jen13, Jen14], the algebraic dual $\mathcal{V}_{n,A|B}^*$ can be identified with a quotient of $\text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ via the Hilbert-Schmidt product. Let us define the orthogonal complement of $\mathcal{V}_{n,A|B}$:

$$\mathcal{V}_{n,A|B}^\perp := \left\{ Y \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B) \mid \text{Tr}[YX] = 0 \forall X \in \mathcal{V}_{n,A|B} \right\}.$$

Let $\pi : \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B) \rightarrow \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B) / \mathcal{V}_{n,A|B}^\perp$, be the quotient map $\pi(Y) = Y + \mathcal{V}_{n,A|B}^\perp$. Then π induces an isomorphism $\mathcal{V}_{n,A|B}^* \cong \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B) / \mathcal{V}_{n,A|B}^\perp$, where the duality between $\mathcal{V}_{n,A|B}$ and $\mathcal{V}_{n,A|B}^*$ is given by

$$\langle \pi(Y), X \rangle = \text{Tr}[YX], \quad Y \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B), X \in \mathcal{V}_{n,A|B}.$$

In particular, two operators X and $X' \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ define the same functional on the effect subspace $\mathcal{V}_{n,A|B}$ if

$$X - X' = \mathbb{1}_A \otimes Z, \quad \text{Tr}[Z] = 0.$$

Definition 3.21 (Dual cone of quantum states). Let $\mathcal{C}_{n,A|B}^{\text{Eff}} \subseteq \mathcal{V}_{n,A|B}$ be the cone of efficiently generated Choi operators. Its dual cone is defined as

$$\mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}} := \mathcal{C}_{n,A|B}^{\text{Eff}*} = \left\{ \pi(Y) \mid Y \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B), \text{Tr}[YX] \geq 0 \forall X \in \mathcal{C}_{n,A|B}^{\text{Eff}} \right\}, \quad (3.3.9)$$

where $\pi(Y) = Y + \mathcal{V}_{n,A|B}^\perp$ is the quotient representation of the operator $Y \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$.

Then, the dual cone contains (the image of) every quantum state: if $\mathcal{C}_{n,A|B}^{\text{Eff}} \subseteq \text{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, then $\pi(\rho_{AB}) \in \mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}}$ for every $\rho_{AB} \geq 0$, since $\text{Tr}[\rho_{AB} F_{AB}] \geq 0$ for all $\rho_{AB} \geq 0$ and all $F_{AB} \geq 0$. Thus, whenever $\mathcal{C}_{n,A|B}^{\text{Eff}}$ is a proper cone with respect to $\mathcal{V}_{n,A|B}$, then $\mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}}$ is also proper with respect to $\mathcal{V}_{n,A|B}^*$.

Under the informational-completeness condition, $\mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}}$ is a proper cone, and hence we can define a partial order on $\mathcal{V}_{n,A|B}^*$, via

$$\pi(\rho_{AB}) \leq_{\mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}}} \pi(\sigma_{AB}) \iff \text{Tr}[X\rho_{AB}] \leq \text{Tr}[X\sigma_{AB}] \forall X \in \mathcal{C}_{n,A|B}^{\text{Eff}}. \quad (3.3.10)$$

In a slight abuse of notation, we will often omit the quotient map $\pi(\cdot)$ and write $\rho_{AB} \leq_{\mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}}} \sigma_{AB}$ to mean $\pi(\rho_{AB}) \leq_{\mathcal{C}_{n,A|B}^{\mathcal{S}_{\text{eff}}}} \pi(\sigma_{AB})$.

3.4. The Proper Cone of Efficient Measurements

Given any family $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ as defined above, for each value $n \in \mathbb{N}$ one can naturally define the smallest convex set that contains all corresponding effect operators. The same also holds true for the corresponding family $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ of binary POVMs.

Definition 3.22 (Polynomially Generated Convex Sets of Effect Operators and Binary POVMs). Let $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ denote a (informationally complete) polynomially generated set

of effect operators. The corresponding (*informationally complete*) *polynomially generated convex set of effect operators* is given by the family $\{\bar{\mathbf{E}}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, where

$$\bar{\mathbf{E}}_n^{\text{eff}} := \text{conv}(\mathbf{E}_n^{\text{eff}}) = \left\{ \sum_{i=1}^k \lambda_i E_n^i \mid k \in \mathbb{N}, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1, E_n^i \in \mathbf{E}_n^{\text{eff}} \right\}$$

Similarly, for the corresponding polynomially generated set of binary POVMs $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, we define the (*informationally complete*) *polynomially generated convex set of binary POVMs* as $\{\bar{\mathbf{M}}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, where

$$\bar{\mathbf{M}}_n^{\text{eff}} := \text{conv}(\mathbf{M}_n^{\text{eff}}) = \left\{ (E, \mathbb{1} - E) \mid E \in \bar{\mathbf{E}}_n^{\text{eff}} \right\}.$$

Remark 3.23. Note that the convex sets $\bar{\mathbf{E}}_n^{\text{eff}}$ and $\bar{\mathbf{M}}_n^{\text{eff}}$ are closed $\forall n \in \mathbb{N}$. This is due to the fact that both $\mathbf{E}_n^{\text{eff}}$ and $\mathbf{M}_n^{\text{eff}}$ are finite sets in a finite-dimensional vector space. As such, their convex hulls are polytopes and therefore closed.

One key property of these sets is that every measurement in $\bar{\mathbf{M}}_n^{\text{eff}}$ can be efficiently approximated. This ensures that one can also view $\bar{\mathbf{M}}_n^{\text{eff}}$ as a set of efficiently approximable measurements. To prove this property, we require the following proposition. The proof is partly based on the techniques used in [Bar18].

Proposition 3.24. *For every effect operator $E \in \bar{\mathbf{E}}_n^{\text{eff}}$, there exists a subset of $k = \frac{5(n \ln \dim(\mathcal{H}) + 1)}{\varepsilon^2}$ effect operators $\{E_{j_1}, \dots, E_{j_k}\} \subset \mathbf{E}_n^{\text{eff}}$ such that*

$$\left\| E - \frac{1}{k} \sum_{i=1}^k E_{j_i} \right\|_{\infty} \leq \varepsilon. \quad (3.4.1)$$

Proof. Let $m = |\mathbf{E}_n^{\text{eff}}|$ denote the cardinality of $|\mathbf{E}_n^{\text{eff}}|$ and let $E_1, \dots, E_m$ denote the elements of $\mathbf{E}_n^{\text{eff}}$. By construction, there exist weights $\alpha_1, \dots, \alpha_m \geq 0$ such that

$$\sum_{i=1}^m \alpha_i E_i = E \quad \text{and} \quad \sum_{i=1}^m \alpha_i = 1.$$

Let X be a self-adjoint random matrix valued variable which with probability α_i satisfies

$$X = E - E_i.$$

Notably, this random variable satisfies

$$\mathbb{E}[X] = 0 \quad \text{and} \quad \|\mathbb{E}[X^2]\|_{\infty} \leq 1.$$

Consequently, for $X_1, X_2, \dots, X_k$, k IID copies of X , the variance satisfies

$$\sigma^2 := \left\| \sum_{i=1}^k \mathbb{E}[X_i^2] \right\|_{\infty} \leq k.$$

Applying the matrix Bernstein inequality [MJC⁺14, Cor. 5.2] for any $0 \leq \varepsilon \leq 1$, it holds that

$$\begin{aligned} \Pr \left[\lambda_{\max} \left(\frac{1}{k} \sum_{j=1}^k X_j \right) \geq \varepsilon \right] &= \Pr \left[\lambda_{\max} \left(\sum_{j=1}^k X_j \right) \geq k\varepsilon \right] \leq \dim(\mathcal{H})^n \exp \left(-\frac{k^2 \varepsilon^2}{3k + 2k\varepsilon} \right) \\ &\leq \dim(\mathcal{H})^n \exp \left(-\frac{k^2 \varepsilon^2}{5k} \right) \\ &= \dim(\mathcal{H})^n \exp \left(-\frac{k\varepsilon^2}{5} \right) \\ &= \exp \left(n \ln \dim(\mathcal{H}) - \frac{k\varepsilon^2}{5} \right), \end{aligned}$$

where $\lambda_{\max} \left(\frac{1}{k} \sum_{j=1}^k X_j \right)$ denotes the largest eigenvalue of $\frac{1}{k} \sum_{j=1}^k X_j$. Setting $k = \frac{5(n \ln \dim(\mathcal{H}) + 1)}{\varepsilon^2}$ hence ensures that

$$\Pr \left[\lambda_{\max} \left(\frac{1}{k} \sum_{j=1}^k X_j \right) \geq \varepsilon \right] \leq e^{-1} < \frac{1}{2}.$$

By the same argument for $-X_i$ it holds that, for the same value k , we have

$$\Pr \left[\lambda_{\min} \left(\frac{1}{k} \sum_{j=1}^k X_j \right) \leq -\varepsilon \right] \leq e^{-1} < \frac{1}{2},$$

where $\lambda_{\min} \left(\frac{1}{k} \sum_{j=1}^k X_j \right)$ denotes the smallest eigenvalue of $\frac{1}{k} \sum_{j=1}^k X_j$. Using a union bound argument, it follows from these two inequalities that

$$\Pr \left[\lambda_{\min} \left(\frac{1}{k} \sum_{j=1}^k X_j \right) \geq -\varepsilon \wedge \lambda_{\max} \left(\frac{1}{k} \sum_{j=1}^k X_j \right) \leq \varepsilon \right] \geq 1 - \frac{2}{e} > 0.$$

As such, there must exist values for $X_1, \dots, X_k$ such that the maximal eigenvalue of $|\frac{1}{k} \sum_{j=1}^k X_j|$ is less than or equal to ε . For these specific values of X_i , we know that there exists $\{E_{j_i}\}_{i=1}^k \subset \mathbf{E}_n^{\text{eff}}$ such that

$$X_i = E - E_{j_i}.$$

By construction, it must hold for these effect operators that

$$\left\| E - \frac{1}{k} \sum_{i=1}^k E_{j_i} \right\|_{\infty} \leq \varepsilon,$$

thus concluding the proof. $\square$

We now discuss two consequences that follow from Proposition 3.24. First, even if one chooses $\varepsilon = \frac{1}{\text{poly}(n)}$, i.e. the allowed error to scale inverse polynomially in n , one can

approximate any $E \in \overline{\mathbf{E}}_n^{\text{eff}}$ using an (in general non-unique and non-degenerate) polynomial subset of effect operators $\{E_{j_i}\}_{i=1}^k \subset \mathbf{E}_n^{\text{eff}}$. Secondly, our bound on the size of this set holds true for every effect operator $E \in \overline{\mathbf{E}}_n^{\text{eff}}$. This allows one to efficiently approximate every binary POVM $M \in \overline{\mathbf{M}}_n^{\text{eff}}$ to inverse-polynomial accuracy. Note that this in particular applies to αE for some $E \in \mathbf{E}_n^{\text{eff}}$ and $0 \leq \alpha \leq 1$, as $0 \in \mathbf{E}_n^{\text{eff}}$. This consequence is formally stated in Corollary 3.25.

Corollary 3.25. *Every $M \in \overline{\mathbf{M}}_n^{\text{eff}}$ can be implemented in poly-time up to inverse-polynomial accuracy.*

Proof. It suffices to show that any $E \in \overline{\mathbf{E}}_n^{\text{eff}}$ is polynomially preparable up to inverse polynomial error. For any $\tilde{E} = \frac{1}{k} \sum_{i=1}^k E_{j_i}$, with $E_{j_i} \in \mathbf{E}_n^{\text{eff}}$ and $k = \text{poly}(n)$, we can define a circuit D_n :

1. Given a control register, prepare a control state $|\lambda'\rangle_C = \sum_{i=1}^k \frac{1}{\sqrt{k}} |i\rangle_C$, which requires $\mathcal{O}(\log k)$ gates.
2. Apply the multiplexed unitary $\sum_{i=1}^k |i\rangle\langle i|_C \otimes U_i$, where each U_i corresponds to the unitary operator that implements E_{j_i} . This circuit can be implemented by $\mathcal{O}(\text{poly}(n) \cdot k \cdot \log(k))$ gates and $\mathcal{O}(\log(k))$ ancillas [SBM06, dCBdV⁺25].
3. Trace out the control registers, along with all other registers that will not be measured.
4. Measure the remaining qubit register.

Since the unitary operators corresponding to the effects are generated in poly-time by assumption and $k = \text{poly}(n)$ by Proposition 3.24, any $E \in \overline{\mathbf{E}}_n^{\text{eff}}$ can be approximated up to inverse polynomial precision in polynomial time. $\square$

While Proposition 3.24 provides an upper bound on the Schatten ∞ -norm, also known as the operator norm, this can straightforwardly be translated into an upper bound on the difference between the corresponding measurement channels in the diamond norm, see e.g. [KW24, Definition 6.18]. As discussed in Section 2.5, one should view any binary POVM $M = (E_1, E_2)$ as a channel

$$\mathcal{M}_M : \mathcal{B}(\mathcal{H}_n) \rightarrow \mathcal{B}(\mathbb{C}^2), \quad \rho \mapsto \sum_{i=0}^1 \text{Tr}[E_{i+1}\rho] |i\rangle\langle i|, \quad (3.4.2)$$

where $\mathbb{C}^2$ is a two-dimensional Hilbert space.

Corollary 3.26. *For every binary POVM $M \in \overline{\mathbf{M}}_n^{\text{eff}}$, there exists a subset of $k = \frac{5(n \ln \dim(\mathcal{H}) + 1)}{\varepsilon^2}$ effect operators $\{E_{j_i}\}_{i=1}^k \subset \mathbf{E}_n^{\text{eff}}$ such that the measurement maps $\mathcal{M}$ corresponding to M and $\mathcal{M}'$ to $M' = (\frac{1}{k} \sum_{i=1}^k E_{j_i}, \mathbb{1} - \frac{1}{k} \sum_{i=1}^k E_{j_i})$ respectively, satisfy*

$$\|\mathcal{M} - \mathcal{M}'\|_{\diamond} \leq 2\varepsilon. \quad (3.4.3)$$

The proof of this is standard and can be found in Section A.1.1. With these tools, we are now in a position to define the cone related to our sets of efficient effect operators.

Definition 3.27 (Cones of Polynomially Generated Effect Operators). Given an (informationally complete) polynomially generated convex set of effect operators $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, the family of cones of (informationally complete) polynomially generated effect operators is given by

$$\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}, \quad \text{where} \quad \mathcal{C}_n^{\mathbf{E}^{\text{eff}}} = \bigcup_{\lambda \geq 0} \lambda \overline{\mathbf{E}}_n^{\text{eff}} = \text{cone}(\mathbf{E}_n^{\text{eff}}). \quad (3.4.4)$$

Remark 3.28. We note that this cone is the computational analog to the cones used in, e.g., [RSB25, RKW11, GC24]. Apart from constraining the measurements to be efficiently generated, the main difference is that we consider binary measurements. Their work considers general m -output POVMs. In principle, we could define computational analogs to their sets of POVMs provided in [RSB25, Eq. (4.1)] or [RKW11, Eq. (76)]. In their setting, $\{E_1, \dots, E_m\} \subseteq \mathbf{E}_n^{\text{eff}}$ if there exists an efficiently implementable measurement given by $M = (E_1, \dots, E_m)$. Elements in $\overline{\mathbf{E}}_n^{\text{eff}}$ would still correspond to convex mixtures of elements in $\mathbf{E}_n^{\text{eff}}$. However, as there is no longer a one-to-one correspondence between the set of efficient effect operators $\mathbf{E}_n^{\text{eff}}$ and the set of efficient measurements $\mathbf{M}_n^{\text{eff}}$, Proposition 3.24 would no longer suffice to ensure that every measurement in $\overline{\mathbf{M}}_n^{\text{eff}}$ is efficiently preparable. This issue becomes particularly exacerbated when m is exponential in n , which is, e.g., the case when measuring n qubits in the computational basis. Finding techniques to generalize our theory to the case of m -output measurements while adequately capturing the computational constraints thus remains an interesting open question.

Lemma 3.29. *The family of cones of informationally complete polynomially generated effect operators $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$ is a family of proper cones.*

Note that the informational completeness condition is sufficient and necessary for the family to be proper, in particular to have a non-empty interior.

Proof. The cones are convex and closed because their generating sets $\overline{\mathbf{E}}_n^{\text{eff}}$ are convex and closed. To see that the cones must be pointed, first note that every element $E \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ is positive semi-definite. As such, if $F \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ and $-F \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$, it follows that $F = 0$. Thus, every cone $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ is pointed. Lastly, solidness, i.e. having a non-empty interior, follows from the informational completeness condition imposed on $\overline{\mathbf{E}}_n^{\text{eff}}$. To see this, one can show, using a standard counter argument, e.g., [BV04, Section 2.1.3], that closed convex sets with empty interior cannot span the whole space. Since this is an if and only if statement, the informational completeness is equivalent to the solidness of the corresponding cone. $\square$

To each $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$, one can generate its dual cone. This corresponding family of dual cones is defined below.

Definition 3.30 (Family of Dual Cones). The family of *dual cones* of $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$ is denoted by $\{\mathcal{C}_n^{\mathcal{S}^{\text{eff}}}\}_{n \in \mathbb{N}}$, where

$$\mathcal{C}_n^{\mathcal{S}^{\text{eff}}} := \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}^* = \left\{ Y \in \mathcal{B}^\dagger(\mathcal{H}_n) \mid \text{Tr}(YX) \geq 0 \ \forall X \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}} \right\}.$$

Remark 3.31. Since we work in finite dimensional vector spaces the dual cone is also the pre-dual cone, so its elements should be thought of as all ‘states’ which are positive w.r.t. the partial order $\leq_{\mathcal{C}_n^{\mathcal{S}^{\text{eff}}}}$ induced by measurements in $\mathbf{E}_n^{\text{eff}}$. It is easy to see that the family of dual cones $\{\mathcal{C}_n^{\mathcal{S}^{\text{eff}}}\}_{n \in \mathbb{N}}$ contains the family of cones of all positive semi-definite operators, i.e., $\text{Pos}(\mathcal{H}_n) \subseteq \mathcal{C}_n^{\mathcal{S}^{\text{eff}}}$, so in particular contains all quantum states.

One interesting question that arises is whether all valid (binary) POVMs with effect operators in $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ can be efficiently approximated. Since we can efficiently approximate convex mixtures of efficiently implementable effect operators, including $0, \mathbf{1}$, it follows that for any $0 \leq \alpha \leq 1$ one can approximate αE for any $E \in \mathbf{E}_n^{\text{eff}}$, as mentioned above. The question now is whether this also holds for αE when $\alpha > 1$, if αE is part of some POVM, i.e. $E \in \overline{\mathbf{E}}_n^{\text{eff}}$ and $\alpha E \leq \mathbf{1}$. We will now argue that for certain cases one should not expect this to be true.

As is discussed in, e.g., [Gol08], one cannot efficiently implement all functions of the form

$$f : \{0, 1\}^n \rightarrow \{0, 1\}.$$

As such, one binary measurement that would generally be inefficient is, given a (uniformly) random function f , to first measure n qubits in the computational basis and then map the output $x \in \{0, 1\}^n$ to $f(x) \in \{0, 1\}$. However, as we discussed in Remark 3.12, there exists a setting, i.e., a polynomial and an approximately universal gate-set, in which the effect operators

$$|x\rangle\langle x| := |x_1 \dots x_n\rangle\langle x_1 \dots x_n| \in \mathbf{E}_n^{\text{eff}}$$

are efficiently implementable. As such, the effect operator doing just that is

$$E_f = \sum_{x: f(x)=0} |x\rangle\langle x| \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$$

and lies in the cone $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$, but would correspond to a binary measurement which is not efficient. Let us consider such a computationally inefficient function f with $|\{x : f(x) = 0\}| = \exp(O(n))$. We highlight that the corresponding effect operators $E_f \notin \overline{\mathbf{E}}_n^{\text{eff}}$ as they are not convex mixtures of efficient effect operators. However, there exists an $E' \in \overline{\mathbf{E}}_n^{\text{eff}}$ such that $E_f = \alpha E'$ and $\alpha = |\{x : f(x) = 0\}|$, by construction of $\overline{\mathbf{E}}_n^{\text{eff}}$. In this case, since the average size of $|\{x : f(x) = 0\}|$ scales exponentially in n , the eigenvalues of E' scale inverse exponentially and it is due to this scaling that E' becomes efficiently implementable.

Remark 3.32 (Inefficient measurements vs. $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$). Our efficient cone of effect operators is non-uniform, in the sense that its elements are prepared by families of polynomial-size circuits without a uniform generation requirement. Many natural effects and measurements nevertheless lie outside this cone due to computational hardness. Next we give explicit non-membership examples.

Since $\mathbf{E}_n^{\text{eff}}$ is finite for each n , the cone $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ is generated by finitely many rays. A rank-one projector $P_\psi := |\psi\rangle\langle\psi|$ can belong to $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ only if one of these generators is proportional to P_ψ . Hence, for all but finitely many ψ , $P_\psi \notin \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$. In addition, by a standard counting argument, with overwhelming probability a Haar-random rank-one projector P_ψ requires exponential circuit size to implement [NC10, Gol08]. Moreover, [JW23] construct explicit families of states with exponential circuit lower bounds; thus the corresponding rank-one projectors are, eventually, not in $\mathbf{E}_n^{\text{eff}}$ (even up to inverse-polynomial approximation).

For QMA-complete local Hamiltonians, if one could (given the instance description) efficiently compile a polynomial-size circuit that implements a projector onto the ground space to constant precision, this would yield a BQP algorithm for a QMA-complete promise problem, implying $\text{QMA} \subseteq \text{BQP}$, which is considered unlikely [KKR06]. Finally, under standard pseudorandomness assumptions (against non-uniform quantum distinguishers), any measurement that separates pseudorandom states or unitaries from Haar-random distributions with non-negligible bias cannot be efficient—otherwise one could break the initial assumptions [JLS18, MPSY24, ABF⁺24].

3.5. Computational Trace Norm and Trace Distance

Having constructed our bases and cones of efficiently implementable measurements one straightforwardly via Definition 2.42 gets corresponding distinguishability norms, which also appear in the computational quantum cryptographic setting, see e.g. [JLS18, ABF⁺24, BCQ23].

Without computational constraints, where every measurement is implementable, the best possible distinguishability advantage between two quantum states ρ and σ is related to the trace distance $\|\rho - \sigma\|_1$. However, if the measurements are instead constrained to be implementable with poly-size quantum operations, the information-theoretic trace distance loses its operational meaning. In such cases, one instead considers the *computational distance*. This distance can essentially be recovered by a suitable *distinguishability norm*, see Definition 2.42, which in this context we call a *computational trace norm*.

Let us recall that the convex hull of efficient effect operators $\{\overline{\mathbf{E}}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ is a closed, convex, and informationally complete set, which additionally contains the trivial measurement $\mathbb{1} \in \overline{\mathbf{E}}_n^{\text{eff}}$ and satisfies the condition

$$E \in \overline{\mathbf{E}}_n^{\text{eff}} \implies \mathbb{1} - E \in \overline{\mathbf{E}}_n^{\text{eff}} \quad \forall E \in \overline{\mathbf{E}}_n^{\text{eff}} \quad (3.5.1)$$

for all $n \in \mathbb{N}$. For such sets, it follows from [RKW11, Section V] that

$$\|v\|_{\overline{\mathbf{M}}_n^{\text{eff}}} := \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \langle 2E - \mathbf{1}, v \rangle \quad (3.5.2)$$

defines a norm on the vector space of Hermitian operators $v \in \mathcal{B}^\dagger(\mathcal{H}_n)$ for any $n \in \mathbb{N}$. This naturally allows one to define a *computational trace norm*, where $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, $v = \rho - \sigma$, and $\text{Tr}[v] = 0$.

Definition 3.33 (Computational Trace Norm). Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$ and a family of informationally complete polynomially generated convex set of binary POVMs $\{\overline{\mathbf{M}}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, the *computational trace norm* between these states is given by

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} := 2 \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] . \quad (3.5.3)$$

For general $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$, one gets via Eq. (3.5.2), the more general

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} = 2 \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] - \text{Tr}[\rho - \sigma] . \quad (3.5.4)$$

Remark 3.34. It was shown in [MWW09] (and further discussed in [RKW11]), that norms defined via the theory of proper cones (such as the one in Eq. (3.5.2)) naturally satisfy the triangle inequality, positive definiteness and are absolute homogeneous, i.e.

- Triangle Inequality: for all $\rho, \sigma, \tau \in \mathcal{S}(\mathcal{H}_n)$

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} \leq \|\rho - \tau\|_{\overline{\mathbf{M}}_n^{\text{eff}}} + \|\tau - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} . \quad (3.5.5)$$

- Absolute Homogeneity: for all $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$ and $\lambda \in \mathbb{R}$

$$\|\lambda(\rho - \sigma)\|_{\overline{\mathbf{M}}_n^{\text{eff}}} = |\lambda| \|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} . \quad (3.5.6)$$

- Positive Definiteness: for all $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} = 0 \implies \rho = \sigma . \quad (3.5.7)$$

One straightforward property of the computational trace norm is that the optimization appearing on the RHS of Eq. (3.5.3) can be restricted to the set of efficient measurements $\mathbf{E}_n^{\text{eff}}$. In other words, the optimal distinguishing measurement lies in $\mathbf{M}_n^{\text{eff}}$.

Lemma 3.35. *Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, the computational trace norm can alternatively be expressed as:*

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} = \begin{cases} 2 \max_{E \in \mathbf{E}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] , \\ \max_{M=(E_1, E_2) \in \mathbf{M}_n^{\text{eff}}} \sum_{i=1}^2 |\text{Tr}[E_i(\rho - \sigma)]| , \\ \max_{M \in \mathbf{M}_n^{\text{eff}}} \|\mathcal{M}_M(\rho) - \mathcal{M}_M(\sigma)\|_1 \end{cases} \quad (3.5.8)$$

Proof. This follows from well-known arguments; for completeness we have included the proof in Section A.1.2. $\square$

In quantum information literature, the trace distance is defined via the trace norm; see e.g. [KW24, Section 6.1]. In the computational setting, one takes the same approach.

Definition 3.36 (Computational Trace Distance). Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$ and a family of informationally complete polynomially generated convex set of binary POVMs $\{\overline{\mathbf{M}}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, the *computational trace distance* between these states is given by

$$\hat{\Delta}(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}}. \quad (3.5.9)$$

As discussed in Chapter 2 the minimum error probability for discriminating between two quantum states via efficient measurements can be related to the computational trace distance. In particular, the minimum error probability using only computationally efficient binary measurements from $\overline{\mathbf{M}}_n^{\text{eff}}$ is given by

$$\hat{\mathcal{P}}_{\text{err}}(\rho, \sigma) = \frac{1}{2} (1 - \hat{\Delta}(\rho, \sigma)) = \frac{1}{2} - \frac{1}{4} \|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}}. \quad (3.5.10)$$

Remark 3.37 (Computational trace distance and cryptography). To further relate this to computational quantum cryptography, let us consider the following. Two families of quantum states $\{\rho_n\}_{n \in \mathbb{N}}, \{\sigma_n\}_{n \in \mathbb{N}} \in \{\mathcal{S}(\mathbb{C}^{2^n})\}_{n \in \mathbb{N}}$ are said to be *computationally indistinguishable* if for every quantum polynomial-time distinguisher $\mathcal{D}$ (possibly non-uniform, with quantum advice¹ α_n),

$$|\Pr[\mathcal{D}^{\alpha_n}(\rho_n) = 1] - \Pr[\mathcal{D}^{\alpha_n}(\sigma_n) = 1]| \leq \text{negl}(n). \quad (3.5.11)$$

This expression is known as the *computational distance* and, for every $n \in \mathbb{N}$, can be related to the computational trace distance via

$$\hat{\Delta}(\rho_n, \sigma_n) = \sup_{\mathcal{D} \in \overline{\mathbf{M}}_n^{\text{eff}}} |\Pr[\mathcal{D}^{\alpha_n}(\rho_n) = 1] - \Pr[\mathcal{D}^{\alpha_n}(\sigma_n) = 1]|. \quad (3.5.12)$$

Under standard quantum cryptographic assumptions (e.g., the existence of pseudorandom quantum states and pseudorandom unitaries), there are families of states that are indistinguishable by any QPT adversary. Therefore, explicit constructions of these cryptographic primitives, such as pseudorandom and pseudoentangled quantum states [JLS18, ABF⁺24, ABV23, LREJ25], are computationally indistinguishable within our framework under the corresponding cryptographic assumptions.

¹If one is given quantum advice, i.e., additional quantum side-information, then this should be incorporated into the states ρ_n, σ_n .

It was shown in [GE24, Lemma 2.5] that the computational distance is approximately sub-additive. In the following lemma, we express their result in terms of $\|\cdot\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}}$.

Lemma 3.38 (Sub-additivity of $\|\cdot\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}}$ [GE24]). *Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, the computational trace norm is approximately sub-additive under tensor products, i.e.,*

$$\|\rho^{\otimes m} - \sigma^{\otimes m}\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} \leq m \max_{i \in \{1, \dots, m\}} \|\rho_i - \sigma_i\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}}, \quad (3.5.13)$$

where $\rho_i = \rho^{\otimes(i-1)} \otimes \rho \otimes \sigma^{\otimes(m-i)}$, $\sigma_i = \rho^{\otimes(i-1)} \otimes \sigma \otimes \sigma^{\otimes(m-i)}$.

Proof. This follows directly from the triangle inequality property of the computational trace norm. For completeness we provide the proof in Section A.1.3, using the same steps as [GE24, Lemma 2.5]. $\square$

The states ρ_i, σ_i cannot be viewed as simply holding a single copy of ρ, σ . For both states one has the same additional registers which hold multiple copies of ρ and σ . These states have a computational cost associated with them, so there is a difference between having access to them for free and needing to first generate them. The triangle inequality can also be used to prove

$$\|\rho_1 \otimes \rho_2 - \sigma_1 \otimes \sigma_2\|_{\overline{\mathbf{M}}_{n_1+n_2}^{\text{eff}}} \leq \min \left\{ \begin{aligned} &\|\rho_1 \otimes (\rho_2 - \sigma_2)\|_{\overline{\mathbf{M}}_{n_1+n_2}^{\text{eff}}} + \|(\rho_1 - \sigma_1) \otimes \sigma_2\|_{\overline{\mathbf{M}}_{n_1+n_2}^{\text{eff}}} \\ &\|\sigma_1 \otimes (\rho_2 - \sigma_2)\|_{\overline{\mathbf{M}}_{n_1+n_2}^{\text{eff}}} + \|(\rho_1 - \sigma_1) \otimes \rho_2\|_{\overline{\mathbf{M}}_{n_1+n_2}^{\text{eff}}} \end{aligned} \right.$$

Chapter 4

Computational Divergences and Entropies

In the previous chapter, we introduced how complexity constraints reshape the space of quantum transformations and measurements available to bounded observers. This restriction induces different notions of order and distance between quantum states, but its consequences go beyond the cases considered there. This chapter develops complexity-constrained divergences and entropies that mirror their information-theoretic counterparts, building on the formalism introduced in Chapter 3. We first focus on single systems in Section 4.1, where restricting the available measurements leads to computational divergences, their regularized variants, and a corresponding notion of computational fidelity. We then turn to bipartite systems in Section 4.2, where restricting to efficient quantum channels induces a computational conditional min-entropy, its operational interpretation, and its non-conditional expression. Moreover, we show how the single-system setting can be seen as a special case of the latter one.

4.1. Computational Quantum Divergences

We introduce the computational divergences induced by efficient measurements and give them operational meaning, as sketched in Figure 4.1. In Section 4.1.1, we first define a computational max-divergence through the cone of efficient binary effects constructed in Chapter 3. This mirrors the information-theoretic max-divergence, which can be written as an optimization over the cone of positive semidefinite operators and is recovered when the computational restriction is removed. In Section 4.1.2, we then derive a family of computational measured Rényi divergences. Measured Rényi divergences are becoming well-studied objects in the quantum information literature, see e.g. [FFF26, RSB25, MH23]. In [FFF26] they were, among others, critical to deriving a generalized quantum asymptotic equipartition theorem (QAEF). This generalized QAEF can be used to derive a variety of statements, including a generalized Quantum Stein’s lemma. Our computational measured divergences can naturally be viewed as computational versions of these information-theoretic measured divergences.

Importantly, in Theorem 4.13, we recover the computational max-divergence as a special case ($\alpha \rightarrow \infty$) of the computational measured Rényi divergences, which can be viewed as one of the main results of this section. This gives the conic max-divergence an operational interpretation and naturally mimics the information-theoretic result of [MH23, Eq. III.65],

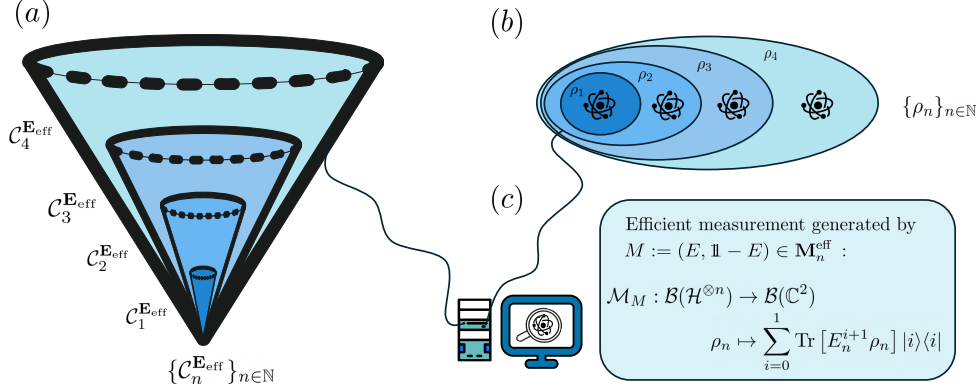


Figure 4.1.: Schematic representation of the single-system framework. **(a)** The family of cones induced by the set of efficiently generated quantum effect operators $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$. **(b)** A family of states $\{\rho_n\}_{n \in \mathbb{N}}$ with growing system size. **(c)** The measurement maps $\mathcal{M}_n^{\text{eff}}$ acting on ρ_n ; for instance, $p = \text{Tr}[E\rho_n]$ for $E \in \mathbf{E}_n^{\text{eff}}$. The computational distances and divergences are then defined by optimizing over these restricted measurement maps.

where it is shown that

$$D_{\max}^{\mathbf{M}}(\rho \parallel \sigma) = D_{\max}(\rho \parallel \sigma) \quad (4.1.1)$$

if $\mathbf{M}$ is the set of all binary POVMs.¹ Lastly, following the approach of [RKW11], we consider in Section 4.1.4 a *computational (binary) fidelity* and study its relation to the other quantities, including a straightforward equivalence to the computational measured Rényi divergence at $\alpha = 1/2$ and a computational analogue of the Fuchs-van der Graaf inequality.

4.1.1. Computational Max-Divergence

In the rest of this chapter, let $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ be an informationally complete polynomially generated set of effect operators, $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ its corresponding informationally complete polynomially generated set of binary POVMs, and its associated family of proper cones $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$. For each proper cone of efficient effect operators $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$, one can define a corresponding max-divergence using Definition 2.40. For any such cone, we call the corresponding divergence the *computational max-divergence*. We recall that this family of cones implicitly depends on a polynomial $n \mapsto p(n)$, which we sometimes state explicitly when it becomes relevant for clarity, however, most of the time we omit it to simplify notation.

¹It is also closely related to [MO14, Appendix A], where they show a slightly less general result, in that $\mathbf{M}$ is now the set of all POVMs, not just binary measurements.

Definition 4.1 (Computational Max-Divergence). Given a family of proper cones $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$ and any two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$, we denote the corresponding *computational max-divergence* by:

$$\mathring{D}_{\max}^{p(n)}(\rho \parallel \sigma) := \log \inf \{ \lambda \in \mathbb{R} \mid \rho \leq_{\mathcal{C}_n^{\mathbf{S}^{\text{eff}}}} \lambda \sigma \} = \log \sup_{v \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}} \frac{\text{Tr}[v\rho]}{\text{Tr}[v\sigma]}, \quad (4.1.2)$$

where we recall that $\mathcal{C}_n^{\mathbf{S}^{\text{eff}}}$ is the dual cone to $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ and thus also proper and $p(n)$ is the underlying polynomial that is used in the definition of the family $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$. If there does not exist any $\lambda \in \mathbb{R}$ such that $\rho \leq_{\mathcal{C}_n^{\mathbf{S}^{\text{eff}}}} \lambda \sigma$, then $\mathring{D}_{\max}^{p(n)}(\rho \parallel \sigma) = \infty$. We will occasionally drop the superscript and denote the computational max-divergence just by $\mathring{D}_{\max}$, when the underlying polynomial is clear from context or fixed in all expressions appearing in the same equation.

The equality in Eq. (4.1.2) holds whenever $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ is a proper cone; see [RKW11, Eqs. 4-6] and [GC24, Proposition 10]. Throughout, ratios of the form $\text{Tr}[v\rho]/\text{Tr}[v\sigma]$ are understood to be $+\infty$ if the denominator is zero and the numerator is strictly positive, while terms for which both are zero are omitted from the supremum.

Proposition 4.2 (Triangle Inequality for $\mathring{D}_{\max}$). *The computational max-divergence $\mathring{D}_{\max}(\rho \parallel \sigma)$ satisfies the triangle inequality, i.e. for any ρ, σ, ω such that the following quantities are finite,*

$$\mathring{D}_{\max}(\rho \parallel \sigma) \leq \mathring{D}_{\max}(\rho \parallel \omega) + \mathring{D}_{\max}(\omega \parallel \sigma) \quad (4.1.3)$$

This is a standard property of max-divergences defined as in Definition 2.40, since proper cones are in one-to-one correspondence with (transitive) semi-orders.

Remark 4.3. Since we construct the cone of efficient effects $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ rather implicitly, depending on some fixed polynomial and approximately universal gate set, existence of a tractable membership or linear-minimization oracle is unclear. Projection or Frank–Wolfe–type methods would require such an oracle to solve the membership problem. Hence we cannot provide a general algorithm for approximating the computational divergences that are based on these cones and introduced in the following section. Designing relaxations or hierarchies that admit verifiable oracles is an interesting open direction.

Super- and Sub-additivity Properties

Recall that the elements of $\mathbf{E}_n^{\text{eff}}$ are given by *all* POVM elements which can be generated with circuit size of (at most) $p(n)$ and possible polynomial classical post-processing with NOT and OR operations. So if we assume this polynomial to be strictly super-additive, i.e. $p(n+m) \geq p(n) + p(m) + 1$, then the family $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ and the family of its induced cones $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$ are stable under tensor products, in the sense that if $E_i \in \mathbf{E}_{n_i}^{\text{eff}}$, then $E_1 \otimes E_2 \in \mathbf{E}_{n_1+n_2}^{\text{eff}}$ and the same holds for $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$. With this in mind we are now ready to

study some of the super-additivity and sub-additivity properties of the computational max-divergence, where we follow the proof techniques from [RLW24].

Lemma 4.4 (Super-additivity of $\mathring{D}_{\max}$). *The computational max-divergence $\mathring{D}_{\max}(\rho\|\sigma)$ is super-additive, i.e., for any $\rho = \rho_1 \otimes \rho_2$, $\sigma = \sigma_1 \otimes \sigma_2$ such that $\rho_i, \sigma_i \in \text{Pos}(\mathcal{H}_{n_i})$,*

$$\mathring{D}_{\max}^{p(n_1+n_2)}(\rho\|\sigma) \geq \mathring{D}_{\max}^{p(n_1)}(\rho_1\|\sigma_1) + \mathring{D}_{\max}^{p(n_2)}(\rho_2\|\sigma_2). \quad (4.1.4)$$

Proof. By construction and super-additivity of p we have for $E_i \in \mathbf{E}_{n_i}^{\text{eff}}$ (for $i \in \{1, 2\}$), that $E_1 \otimes E_2 \in \mathbf{E}_{n_1+n_2}^{\text{eff}}$. Due to convexity, it then naturally follows that if $E_i \in \overline{\mathbf{E}}_{n_i}^{\text{eff}}$ (for $i \in \{1, 2\}$), then $E_1 \otimes E_2 \in \overline{\mathbf{E}}_{n_1+n_2}^{\text{eff}}$. We refer to Remark 3.12 for a more detailed discussion on this.

For any $v_i \in \mathcal{C}_{n_i}^{\text{eff}}$, there exists a λ_i and $E_i \in \overline{\mathbf{E}}_{n_i}^{\text{eff}}$ such that $v_i = \lambda_i E_i$. Since $E_1 \otimes E_2 \in \overline{\mathbf{E}}_{n_1+n_2}^{\text{eff}}$, it must hold that $v_1 \otimes v_2 \in \mathcal{C}_{n_1+n_2}^{\text{eff}}$. As such

$$\begin{aligned} \mathring{D}_{\max}^{p(n_1+n_2)}(\rho\|\sigma) &= \log \sup_{v \in \mathcal{C}_{n_1+n_2}^{\text{eff}}} \frac{\text{Tr}[v(\rho_1 \otimes \rho_2)]}{\text{Tr}[v(\sigma_1 \otimes \sigma_2)]} \geq \log \sup_{v_i \in \mathcal{C}_{n_i}^{\text{eff}}} \frac{\text{Tr}[(v_1 \otimes v_2)(\rho_1 \otimes \rho_2)]}{\text{Tr}[(v_1 \otimes v_2)(\sigma_1 \otimes \sigma_2)]} \\ &= \log \sup_{v_1 \in \mathcal{C}_{n_1}^{\text{eff}}} \frac{\text{Tr}[v_1 \rho_1]}{\text{Tr}[v_1 \sigma_1]} + \log \sup_{v_2 \in \mathcal{C}_{n_2}^{\text{eff}}} \frac{\text{Tr}[v_2 \rho_2]}{\text{Tr}[v_2 \sigma_2]} \\ &= \mathring{D}_{\max}^{p(n_1)}(\rho_1\|\sigma_1) + \mathring{D}_{\max}^{p(n_2)}(\rho_2\|\sigma_2). \end{aligned}$$

□

The same proof can be extended to operators of the form $\rho^{\otimes m}, \sigma^{\otimes m} \in \text{Pos}(\mathcal{H}_{n \cdot m})$. Here, one simply finds that

$$\mathring{D}_{\max}^{p(n \cdot m)}(\rho^{\otimes m}\|\sigma^{\otimes m}) \geq m \mathring{D}_{\max}^{p(n)}(\rho\|\sigma). \quad (4.1.5)$$

In other words, the super-additivity property follows from the observation that, informally, tensor products of efficient measurements are also an efficient measurement. This measurement can efficiently be reduced to a binary measurement, using logical OR gates if $m \in \text{poly}(n)$.

Sub-additivity cannot be proven via the same techniques. Nevertheless, acting on quantum states, it is possible to prove that the computational max-divergence must also be approximately sub-additive.

Lemma 4.5 (Sub-additivity of $\mathring{D}_{\max}$). *The computational max-divergence $\mathring{D}_{\max}(\rho\|\sigma)$ is approximately sub-additive for quantum states. For any $\rho = \rho_1 \otimes \rho_2$, $\sigma = \sigma_1 \otimes \sigma_2$ such that $\rho_i, \sigma_i \in \mathcal{S}(\mathcal{H}_{n_i})$ with cost $C(\rho_i)$, $C(\sigma_i)$,² it holds that*

$$\mathring{D}_{\max}^{p(n_1+n_2)}(\rho\|\sigma) \leq \mathring{D}_{\max}^{p(n_1+n_2)+C}(\rho_1\|\sigma_1) + \mathring{D}_{\max}^{p(n_1+n_2)+C}(\rho_2\|\sigma_2), \quad (4.1.6)$$

²Here, the cost is the minimal circuit size required to prepare them using gates from the prior fixed gate set $\mathcal{G}$.

where $C := \max\{C(\rho_1), C(\rho_2), C(\sigma_1), C(\sigma_2)\}$.

Proof. The proof technique is again inspired by [RLW24]. For any $\lambda_1, \lambda_2 \in \mathbb{R}$, it holds that

$$(\lambda_1 \cdot \lambda_2)\sigma_1 \otimes \sigma_2 - \rho_1 \otimes \rho_2 = (\lambda_1\sigma_1 - \rho_1) \otimes \lambda_2\sigma_2 + \rho_1 \otimes (\lambda_2\sigma_2 - \rho_2).$$

Let us consider any $\lambda = \lambda_1 \cdot \lambda_2 \geq 0$ for which it holds that

$$\text{Tr}[(\lambda\sigma - \rho)E] \geq 0, \quad \forall E \in \mathbf{E}_{n_1+n_2}^{\text{eff}}. \quad (4.1.7)$$

By construction it must similarly hold that

$$\text{Tr}[(\lambda\sigma - \rho)v] \geq 0, \quad \forall v \in \mathcal{C}_{n_1+n_2}^{\mathbf{E}^{\text{eff}}},$$

and, by Lemma 2.41, one then finds that

$$\bar{D}_{\max}^{c p(n_1+n_2)}(\rho\|\sigma) \leq \log \lambda.$$

Our goal is to show that $\lambda = \lambda_1 \cdot \lambda_2$ satisfies Eq. (4.1.7), if one sets

$$\log \lambda_i = \bar{D}_{\max}^{c p(n_1+n_2)+C}(\rho_i\|\sigma_i). \quad (4.1.8)$$

Note that, in particular, Eq. (4.1.8) implies $\lambda_i \geq 0$ for $i \in \{1, 2\}$. Since it takes at most C extra gates to generate σ_2 , for any $E \in \mathbf{E}_{n_1+n_2}^{\text{eff}}$ acting on $\mathcal{H}_{n_1} \otimes \mathcal{H}_{n_2}$ there exists another effect operator E' on $\mathcal{H}_{n_1}$ that first additionally prepares σ_2 and then implements E . By construction, it must hold that

$$\text{Tr}[(\lambda_1\sigma_1 - \rho_1) \otimes \sigma_2 E] = \text{Tr}[(\lambda_1\sigma_1 - \rho_1)E'].$$

Moreover, since E' uses at most $p(n_1 + n_2) + C$ gates, it follows from the definition of λ_1 that

$$\text{Tr}[(\lambda_1\sigma_1 - \rho_1) \otimes \sigma_2 E] = \text{Tr}[(\lambda_1\sigma_1 - \rho_1)E'] \geq 0.$$

Similarly, it must hold for any $E \in \mathbf{E}_{n_1+n_2}^{\text{eff}}$ that

$$\text{Tr}[\rho_1 \otimes (\lambda_2\sigma_2 - \rho_2)E] \geq 0. \quad (4.1.9)$$

Combining this, we find for $\lambda = \lambda_1 \cdot \lambda_2$ and any $E \in \mathbf{E}_{n_1+n_2}^{\text{eff}}$ that

$$\text{Tr}[(\lambda\sigma - \rho)E] = \lambda_2 \text{Tr}[(\lambda_1\sigma_1 - \rho_1) \otimes \sigma_2 E] + \text{Tr}[\rho_1 \otimes (\lambda_2\sigma_2 - \rho_2)E] \geq 0,$$

thus satisfying Eq. (4.1.7). As a result, it must hold that

$$\bar{D}_{\max}^{c p(n_1+n_2)}(\rho\|\sigma) \leq \log \lambda = \log \lambda_1 + \log \lambda_2 = \bar{D}_{\max}^{c p(n_1+n_2)+C}(\rho_1\|\sigma_1) + \bar{D}_{\max}^{c p(n_1+n_2)+C}(\rho_2\|\sigma_2).$$

□

Remark 4.6. It generally makes sense to consider quantum states ρ and σ that are efficiently preparable, i.e., where C scales polynomially in n_1, n_2 . This way, the max-divergences that appear in Lemma 4.5 are all related to cones generated by efficient effect operators. Moreover, the proof can straightforwardly be modified to hold for positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H}_{n_1} \otimes \mathcal{H}_{n_2})$. However, for any $\rho_i \in \text{Pos}(\mathcal{H}_{n_i})$, $C(\rho_i)$ would then be the gate complexity required to generate the corresponding normalized quantum state $\frac{\rho_i}{\text{Tr}[\rho_i]}$ and analogously for σ_i .

For states of the form $\rho^{\otimes m}, \sigma^{\otimes m} \in \mathcal{S}(\mathcal{H}_{n \cdot m})$, the same proof can be slightly modified, using [RLW24]

$$\lambda^m \sigma^{\otimes m} - \rho^{\otimes m} = \sum_{j=1}^m \lambda^{j-1} \rho^{\otimes m-j} \otimes (\lambda \sigma - \rho) \otimes \sigma^{\otimes j-1}. \quad (4.1.10)$$

One then retrieves the bound

$$\overset{c}{D}_{\max}^{p(n \cdot m)}(\rho^{\otimes m} \| \sigma^{\otimes m}) \leq m \overset{c}{D}_{\max}^{p(n \cdot m) + (m-1)C}(\rho \| \sigma), \quad (4.1.11)$$

where $C := \max\{C(\rho), C(\sigma)\}$. For fixed ρ, σ , one should expect the RHS of Eq. (4.1.11) to converge to the information-theoretic max-divergence in the limit $m \rightarrow \infty$. This is due to the fact that while the states are fixed, the gate complexity of the allowed measurement is increasing as a function of m . This inequality is not unexpected, as it always holds that

$$\overset{c}{D}_{\max}(\rho^{\otimes m} \| \sigma^{\otimes m}) \leq D_{\max}(\rho^{\otimes m} \| \sigma^{\otimes m}) = m D_{\max}(\rho \| \sigma). \quad (4.1.12)$$

Lastly, let us relate the computational max-divergence with the computational trace distance.

Lemma 4.7 (Computational Max-divergence and Computational Trace Distance). *Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, we have that*

$$\overset{c}{D}_{\max}(\rho \| \sigma) \geq \log \left(1 + \overset{c}{\Delta}(\rho, \sigma) \right). \quad (4.1.13)$$

Proof. The proof is straightforward and can be found in Section A.2.1. $\square$

4.1.2. Computational Measured Rényi Divergences

This section is devoted to computational measured Rényi divergences. Here, we consider an informationally complete polynomially generated set of binary POVMs $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, whose underlying polynomial is strictly super-additive. For each set $\mathbf{M}_n^{\text{eff}}$, we will define a measured Rényi divergence.

Definition 4.8 (Computational Measured Rényi Divergence). Given any two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$ and an element $\mathbf{M}_n^{\text{eff}}$ of an informationally complete polynomially generated set of binary POVMs $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, we denote the corresponding *computational measured Rényi divergence* for any $\alpha \in (0, 1) \cup (1, \infty)$ by:

$$\overset{\text{c}}{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) := \sup_{M \in \mathbf{M}_n^{\text{eff}}} D_\alpha(\mathcal{M}_M(\rho) \parallel \mathcal{M}_M(\sigma)), \quad (4.1.14)$$

where $D_\alpha(\rho \parallel \sigma)$ is the sandwiched Rényi divergence (Definition 2.15) and $\mathcal{M}(\cdot)$ is the measurement channel corresponding to the POVM M .

Moreover, we use $\overset{\text{c}}{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma)$ to denote

$$\overset{\text{c}}{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) := \begin{cases} \inf_{M \in \mathbf{M}_n^{\text{eff}}} Q_\alpha(\mathcal{M}(\rho) \parallel \mathcal{M}(\sigma)) & \text{if } \alpha \in (0, 1) \\ \sup_{M \in \mathbf{M}_n^{\text{eff}}} Q_\alpha(\mathcal{M}(\rho) \parallel \mathcal{M}(\sigma)) & \text{if } \alpha \in (1, \infty) \end{cases} \quad (4.1.15)$$

Definition 4.9 (Computational Measured Relative Entropy & Max-divergence). Given any two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$ with $\text{Tr}[\rho] > 0$ and $\rho \ll \sigma$ and an element $\mathbf{M}_n^{\text{eff}}$ of an informationally complete polynomially generated set of binary POVMs $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, the *computational measured relative entropy* and *max-divergence* between ρ, σ are, respectively, given by:

$$\overset{\text{c}}{D}^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) := \sup_{M \in \mathbf{M}_n^{\text{eff}}} \frac{\text{Tr}[\mathcal{M}(\rho) (\log \mathcal{M}(\rho) - \log \mathcal{M}(\sigma))]}{\text{Tr}[\mathcal{M}(\rho)]} = \lim_{\alpha \nearrow 1} \overset{\text{c}}{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma), \quad (4.1.16)$$

$$\overset{\text{c}}{D}_{\max}^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) := \sup_{M \in \mathbf{M}_n^{\text{eff}}} \log \inf\{\lambda \in \mathbb{R} \mid \mathcal{M}(\rho) \leq \lambda \mathcal{M}(\sigma)\} = \lim_{\alpha \nearrow \infty} \overset{\text{c}}{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma). \quad (4.1.17)$$

By taking the appropriate limits, one can retrieve the computational measured relative entropy and max-divergence from the computational measured Rényi divergences; see, e.g., [RSB25, Lemma 2]. We note that a priori one could have also defined the computational measured Rényi divergences using $\overline{\mathbf{M}}_n^{\text{eff}}$ instead of $\mathbf{M}_n^{\text{eff}}$. We show in Lemma 4.10 below, however, that these definitions are equivalent.

Lemma 4.10. For any $\mathbb{D} \in \{D_\alpha, D, D_{\max}\}$, where $\alpha \in (0, 1) \cup (1, \infty)$, it holds that

$$\overset{\text{c}}{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) = \overset{\text{c}}{\mathbb{D}}^{\overline{\mathbf{M}}_n^{\text{eff}}}(\rho \parallel \sigma). \quad (4.1.18)$$

Proof. Since $\mathbf{M}_n^{\text{eff}} \subset \overline{\mathbf{M}}_n^{\text{eff}}$, it always holds that

$$\overset{\text{c}}{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) \leq \overset{\text{c}}{\mathbb{D}}^{\overline{\mathbf{M}}_n^{\text{eff}}}(\rho \parallel \sigma). \quad (4.1.19)$$

To prove the reverse inequality, we first note that the relative entropy D and D_α are jointly convex on classical states in the range $\alpha \in (0, 1)$; the max-divergence $D_{\max}$ and D_α

are jointly quasi-convex for $\alpha \in [1, \infty)$ [Tom16]. Denoting the extremal points of $\overline{\mathbf{M}}_n^{\text{eff}}$ by $\text{extr}(\overline{\mathbf{M}}_n^{\text{eff}})$, a consequence of the (quasi-) convexity is that

$$\mathbb{D}^{\overline{\mathbf{M}}_n^{\text{eff}}}(\rho\|\sigma) := \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} \mathbb{D}(\mathcal{M}(\rho)\|\mathcal{M}(\sigma)) = \sup_{M \in \text{extr}(\overline{\mathbf{M}}_n^{\text{eff}})} \mathbb{D}(\mathcal{M}(\rho)\|\mathcal{M}(\sigma)).$$

Since the extremal points of $\overline{\mathbf{M}}_n^{\text{eff}}$ satisfy $\text{extr}(\overline{\mathbf{M}}_n^{\text{eff}}) \subset \mathbf{M}_n^{\text{eff}}$ by construction of $\overline{\mathbf{M}}_n^{\text{eff}}$, it holds for all $\mathbb{D} \in \{D, D_\alpha, D_{\max}\}$ that

$$\mathbb{D}^{\overline{\mathbf{M}}_n^{\text{eff}}}(\rho\|\sigma) = \sup_{M \in \text{extr}(\overline{\mathbf{M}}_n^{\text{eff}})} \mathbb{D}(\mathcal{M}(\rho)\|\mathcal{M}(\sigma)) \leq \sup_{M \in \mathbf{M}_n^{\text{eff}}} \mathbb{D}(\mathcal{M}(\rho)\|\mathcal{M}(\sigma)) = \mathbb{D}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma).$$

□

Properties of Measured Rényi Divergences

In [RSB25], properties of generic measured Rényi divergences are derived. For completeness, in this section we rephrase the results of [RSB25, Lemmas 2 & 5] in our setting of efficient measurements.

Lemma 4.11. *For any element $\mathbf{M}_n^{\text{eff}}$ of an informationally complete polynomially generated set of binary POVMs $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ and $\alpha, \beta \in (0, 1) \cup (1, \infty)$, the corresponding computational measured Rényi divergences satisfy*

1. (Non-negativity): For any two normalized quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$,

$$D_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) \geq 0. \quad (4.1.20)$$

2. (Monotonicity in α): For any $\alpha \leq \beta$ and $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$,

$$D_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) \leq D_\beta^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma). \quad (4.1.21)$$

3. (Joint (Quasi-) Convexity): For any two normalized quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, we have that $Q_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma)$ is jointly concave in ρ, σ if $\alpha \in (0, 1)$ and jointly convex if $\alpha \in (1, \infty)$. Moreover, $\tilde{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma)$ is jointly convex if $\alpha \in (0, 1)$ and jointly quasi-convex if $\alpha \in (1, \infty)$.

4. (Data Processing Inequality): For any $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$, let $\mathbf{E}_n^{\text{eff}}$ and $\mathbf{E}'_n^{\text{eff}}$ be two sets of (efficient) effect operators on $\mathcal{H}_n$ and $\mathcal{H}'_n$, respectively, with corresponding POVM sets $\mathbf{M}_n^{\text{eff}}$ and $\mathbf{M}'_n^{\text{eff}}$, and let $\mathcal{E} \in \text{CPTP}(\mathcal{H}_n, \mathcal{H}'_n)$ be a completely positive trace-preserving map such that for any $E \in \mathbf{E}_n^{\text{eff}}$, $\mathcal{E}^*(E) \in \mathbf{E}'_n^{\text{eff}}$. Then

$$\tilde{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) \geq \tilde{D}_\alpha^{\mathbf{M}'_n^{\text{eff}}}(\mathcal{E}(\rho)\|\mathcal{E}(\sigma)). \quad (4.1.22)$$

The first three properties are directly inherited from the information-theoretic sandwiched Rényi divergence. The last property can be understood as stating that we only consider $\mathcal{E}$ such that for every efficient measurement M on $\mathcal{H}'_n$, there exists another measurement $\bar{M} \in \bar{\mathbf{M}}_n^{\text{eff}}$ which satisfies $\bar{\mathcal{M}}(\rho) = \mathcal{M} \circ \mathcal{E}(\rho)$ for every ρ . In this case, optimizing over $M \circ \mathcal{E}$ is more restrictive than simply optimizing over efficient measurements on $\mathcal{H}_n$.

Proof. The proof for Lemma 4.11 directly follows from [RSB25]. However, for completeness, we include it in Section A.2.2. $\square$

Next, let us prove a Pinsker inequality for the computational measured Rényi divergences.

Lemma 4.12 (Computational Pinsker inequality). *Given two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, we have that for $\alpha \in (0, \infty)$,*

$$\hat{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) \geq \frac{\min\{1, \alpha\}}{2 \ln 2} \|\rho - \sigma\|_{\mathbf{M}_n^{\text{eff}}}^2. \quad (4.1.23)$$

Proof. As discussed in [RSB25], the bound for $\alpha \in (0, 1]$ follows directly from the classical result [Gil10]. Additionally, we convert the logarithm to base 2 and take the supremum over all efficient measurements. The bound for $\alpha \in (1, \infty)$ follows from the monotonicity in α property of the computational measured Rényi divergence (see Lemma 4.11). $\square$

Equivalence Between Conic and Measured Max-Divergence

We are now ready to prove one of the main results of this section, namely that the initially geometrically defined computational max-divergence ($\hat{D}_{\max}$ from Definition 4.1) and via the measured Rényi divergences constructed computational measured max-divergence ($\hat{D}_{\max}^{\mathbf{M}_n^{\text{eff}}}$ from Definition 4.9) are equivalent in our setting. The fact that these two divergences are actually equivalent is a priori not clear and this result highlights the strength of applying conic techniques to the computational setting.

This also naturally complements the known information-theoretic result, which states that there exists an optimal measurement which leaves the max-divergence invariant [MH23, MO14].

Theorem 4.13 (Max-Divergence Equivalence). *Let the families $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$, and $\{\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}\}_{n \in \mathbb{N}}$ be as in Chapter 3 and informationally complete and define $\hat{D}_{\max}$ and $\hat{D}_{\max}^{\mathbf{M}_n^{\text{eff}}}$ w.r.t. these as in Section 4.1.1 and Section 4.1.2, respectively. Then for any $n \in \mathbb{N}$ and two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$, it holds that*

$$\hat{D}_{\max}(\rho \parallel \sigma) = \hat{D}_{\max}^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma) =: \lim_{\alpha \rightarrow \infty} \hat{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma). \quad (4.1.24)$$

Proof. Using the approach of [RSB25], we first note that the computational max-divergence can alternatively be expressed as

$$\dot{D}_{\max}(\rho||\sigma) = \log \sup_{v \in \mathcal{C}_n^{\mathbf{E}^{\text{eff}}}} \frac{\text{Tr}[v\rho]}{\text{Tr}[v\sigma]}. \quad (4.1.25)$$

By construction, for every v , there exists a $\lambda \geq 0$ and an element $E \in \overline{\mathbf{E}}_n^{\text{eff}}$ such that $v = \lambda E$. Since the RHS of Eq. (4.1.25) is invariant under the scaling of λ , it must also hold that

$$\dot{D}_{\max}(\rho||\sigma) = \log \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \frac{\text{Tr}[E\rho]}{\text{Tr}[E\sigma]} = \sup_{M=(E^1, E^2) \in \overline{\mathbf{M}}_n^{\text{eff}}} \log \max_{i \in \{1,2\}} \frac{\text{Tr}[E^i \rho]}{\text{Tr}[E^i \sigma]}.$$

The last equality holds due to the condition that $E \in \overline{\mathbf{E}}_n^{\text{eff}}$ implies $1 - E \in \overline{\mathbf{E}}_n^{\text{eff}}$. We will now derive the same expression for the computational measured max-divergence. Let $M = (E^1, E^2) \in \overline{\mathbf{M}}_n^{\text{eff}}$ be a binary measurement with associated measurement map $\mathcal{M}$, then it is easy to check that the condition $\mathcal{M}(\rho) \leq \lambda \mathcal{M}(\sigma)$ is equivalent to

$$\lambda \geq \max_{i \in \{1,2\}} \frac{\text{Tr}[E^i \rho]}{\text{Tr}[E^i \sigma]}.$$

Hence it follows that

$$\inf\{\lambda \in \mathbb{R} | \mathcal{M}(\rho) \leq \lambda \mathcal{M}(\sigma)\} = \max_{i \in \{1,2\}} \frac{\text{Tr}[E^i \rho]}{\text{Tr}[E^i \sigma]}.$$

Combining these observations with Lemma 4.10, we directly find that

$$\begin{aligned} \dot{D}_{\max}^{\mathbf{M}_n^{\text{eff}}}(\rho||\sigma) &= \dot{D}_{\max}^{\overline{\mathbf{M}}_n^{\text{eff}}}(\rho||\sigma) \\ &= \sup_{M=(E^1, E^2) \in \overline{\mathbf{M}}_n^{\text{eff}}} \log \inf\{\lambda \in \mathbb{R} | \mathcal{M}(\rho) \leq \lambda \mathcal{M}(\sigma)\} \\ &= \sup_{M=(E^1, E^2) \in \overline{\mathbf{M}}_n^{\text{eff}}} \log \max_{i \in \{1,2\}} \frac{\text{Tr}[E^i \rho]}{\text{Tr}[E^i \sigma]} \\ &= \dot{D}_{\max}(\rho||\sigma), \end{aligned}$$

which concludes the proof. $\square$

One interesting consequence of Theorem 4.13 can be stated as follows. If two sets of measurements generate the same cone, then their corresponding measured max-divergences are also equal to each other. As we discussed in Chapter 3 the cone $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ may contain POVM elements which are on their own not efficiently preparable, however, due to the above argument, they do not aid in distinguishability when using the max-divergence as a distance measure.

Remark 4.14. That these two divergences —one defined via the geometric construction of cones of efficient effect operators, and the other via the optimization over measured $\alpha = \infty$ -Rényi divergences— are equivalent is a priori not clear in general. Let us consider a slightly more general setting where one starts with a family of admissible (not necessarily binary) POVMs $\mathbf{M}$ and generates from this the cone $\mathcal{C}^E$ as the smallest cone that contains all effect operators (POVM elements) appearing in all $M \in \mathbf{M}$. This is, among others, the setting of [RSB25]. Then, if one defines these two max-divergences w.r.t. these objects, one can show that the measured max-divergence is always upper bounded by the conical max-divergence; see [RSB25, Proposition 13]. In other words, the “ $\leq$ ” inequality of (4.1.24) is always true. Intuitively, this is because every POVM element is also an element of the cone $\mathcal{C}^E$. The other inequality and thus equality, however, may not always be true. This direction requires an extra condition. Up to rescaling, every element of the cone $\mathcal{C}^E$ needs to either be an effect operator that appears in some *binary* POVM in $\mathbf{M}$ or needs to be expressible as a convex mixture of such binary measurements. In our case this is not an issue, since we consider only binary POVMs. However, if one, e.g., considers $\mathbf{M}$ to be the set of LO measurements, which are necessarily generally non-binary, then the above proof of equality generally breaks down [RSB25]. This is because generating the required two-output measurements would require classical communication, which is not allowed.

4.1.3. Regularized Computational Measured Rényi Divergences

Let us define the regularized versions of the computational max-divergence and the computational measured Rényi divergences. We now pass from finite-block, per-instance quantities to *asymptotic rates*. In computational settings, an efficient tester is constrained not only in circuit size but also in the number of samples; accordingly, we adopt a *poly-sample* regime in which the number of copies satisfies $m(k) \in \text{poly}(k)$. Since we are studying asymptotic behavior, we henceforth consider families of states $\{\rho_k\}_{k \in \mathbb{N}}$ and $\{\sigma_k\}_{k \in \mathbb{N}}$. Again, we fix families of efficient measurements $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$. While the single-instance measures are evaluated at each fixed k , the regularized quantities are meaningful primarily at the family level.

Definition 4.15 (Regularized Computational Max-divergence). Given two families of states $\{\rho_k\}_{k \in \mathbb{N}}$ and $\{\sigma_k\}_{k \in \mathbb{N}}$, with $\rho_k, \sigma_k \in \mathcal{S}(\mathcal{H}_k)$, their *regularized computational max-divergence* when $m(k) \in \text{poly}(k)$ is given by,

$$D_{\max}^{\infty}(\{\rho_k\}_k \| \{\sigma_k\}_k) := \limsup_{k \rightarrow \infty} \frac{D_{\max}^{cp(k \cdot m(k))}(\rho_k^{\otimes m(k)} \| \sigma_k^{\otimes m(k)})}{m(k)}. \quad (4.1.26)$$

For the simple case in which $\rho_k = \rho$ ($\sigma_k = \sigma$), this regularization (denoted by $D_{\max}^{\infty}(\rho \| \sigma)$) admits a convenient equivalent form.

Lemma 4.16. *Given two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, their regularized computational max-divergence can be expressed as*

$$\overset{c}{D}_{\max}^{\infty}(\rho \parallel \sigma) = \lim_{m \rightarrow \infty} \frac{1}{m} \overset{c}{D}_{\max}(\rho^{\otimes m} \parallel \sigma^{\otimes m}) = \sup_{m \in \mathbb{N}} \frac{1}{m} \overset{c}{D}_{\max}(\rho^{\otimes m} \parallel \sigma^{\otimes m}). \quad (4.1.27)$$

Proof. The proof follows from the super-additivity property of the computational max-divergence (see Lemma 4.4) together with Fekete's lemma. $\square$

Definition 4.17 (Regularized Computational Measured Rényi Divergences). Given two families of states $\{\rho_k\}_{k \in \mathbb{N}}$ and $\{\sigma_k\}_{k \in \mathbb{N}}$, with $\rho_k, \sigma_k \in \mathcal{S}(\mathcal{H}_k)$, for any $\alpha \in (0, 1) \cup (1, \infty)$ their *regularized computational measured Rényi divergence* when $m(k) \in \text{poly}(k)$ is given by

$$\overset{c}{D}_{\alpha}^{\text{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k) := \limsup_{k \rightarrow \infty} \frac{\overset{c}{D}_{\alpha}^{\text{M}_{k \cdot m(k)}^{\text{eff}}}(\rho_k^{\otimes m(k)} \parallel \sigma_k^{\otimes m(k)})}{m(k)}. \quad (4.1.28)$$

Definition 4.18 (Regularized Computational Measured Relative Entropy). Given two families of quantum states $\{\rho_k\}_{k \in \mathbb{N}}$ and $\{\sigma_k\}_{k \in \mathbb{N}}$, with $\rho_k, \sigma_k \in \mathcal{S}(\mathcal{H}_k)$, their *regularized computational measured relative entropy* is given by,

$$\overset{c}{D}^{\text{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k) := \limsup_{k \rightarrow \infty} \frac{\overset{c}{D}^{\text{M}_{k \cdot m(k)}^{\text{eff}}}(\rho_k^{\otimes m(k)} \parallel \sigma_k^{\otimes m(k)})}{m(k)}. \quad (4.1.29)$$

Remark 4.19. In fact, for binary measurements super-additivity fails for general $\alpha \neq \infty$ [MH23]. It remains an interesting open question whether the regularized computational measured relative entropy can be equivalently expressed as a supremum, similar to the computational max-divergence.

We conclude by relating the various regularized divergences to each other.

Lemma 4.20. *For any two families of quantum states $\{\rho_k\}_{k \in \mathbb{N}}, \{\sigma_k\}_{k \in \mathbb{N}} \in \{\mathcal{S}(\mathcal{H}_k)\}_{k \in \mathbb{N}}$, $\alpha \in (0, 1)$, and $\beta \in (1, \infty)$, it holds that*

$$\begin{aligned} \overset{c}{D}_{\alpha}^{\text{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k) &\leq \overset{c}{D}^{\text{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k) \\ &\leq \overset{c}{D}_{\beta}^{\text{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k) \\ &\leq \overset{c}{D}_{\max}^{\infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k). \end{aligned}$$

Proof. This follows directly from the monotonicity property provided in Lemma 4.11. $\square$

4.1.4. Computational Fidelity

In this section we define the *computational (binary) fidelity* using the approach from [RKW11]. This is the computational analog to the *fidelity* [Tom16, Uhl76, Fuc96] between two post-measurement single-bit outputs and measures the similarity between two quantum states when restricted to binary efficient measurements. Additionally, we show that the resulting fidelity is effectively the computational measured Rényi divergence for $\alpha = \frac{1}{2}$. Lastly, we relate it to the *computational trace distance* and the *computational max-divergence*, following the techniques from [RKW11].

Definition 4.21 (Computational (Binary) Fidelity). Given any two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$ the *computational (binary) fidelity* is given by

$$\mathring{F}(\rho, \sigma) := \inf_{M=(E_1, E_2) \in \overline{\mathbf{M}}_n^{\text{eff}}} \left(\sum_{i=1}^2 \sqrt{\text{Tr}[E_i \rho] \text{Tr}[E_i \sigma]} \right)^2. \quad (4.1.30)$$

Lemma 4.22 (Computational Fidelity and Computational Measured Rényi Divergence). Given two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, it follows that

$$\mathring{D}_{1/2}^{\mathbf{M}_n^{\text{eff}}}(\rho \| \sigma) = -\log \mathring{F}(\rho, \sigma). \quad (4.1.31)$$

Proof. The proof can be found in Section A.2.3 □

Since the computational fidelity is related to the computational measured Rényi divergence for $\alpha = \frac{1}{2}$, due to Lemma 4.10 it automatically follows that

$$\inf_{M=(E_1, E_2) \in \overline{\mathbf{M}}_n^{\text{eff}}} \left(\sum_{i=1}^2 \sqrt{\text{Tr}[E_i \rho] \text{Tr}[E_i \sigma]} \right)^2 = \inf_{M=(E_1, E_2) \in \mathbf{M}_n^{\text{eff}}} \left(\sum_{i=1}^2 \sqrt{\text{Tr}[E_i \rho] \text{Tr}[E_i \sigma]} \right)^2, \quad (4.1.32)$$

meaning that one could have equivalently defined the computational fidelity by solely optimizing over measurements $\mathbf{M}_n^{\text{eff}}$. The computational fidelity and computational trace distance additionally straightforwardly satisfy a Fuchs-van de Graaf type inequality.

Lemma 4.23 (Computational Fuchs-van de Graaf). Given two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, it holds that

$$1 - \sqrt{\mathring{F}(\rho, \sigma)} \leq \mathring{\Delta}(\rho, \sigma) \leq \sqrt{1 - \mathring{F}(\rho, \sigma)}. \quad (4.1.33)$$

Proof. These inequalities follow directly from [RKW11, Proposition 19] applied to our family of informationally complete polynomially generated set of binary POVMs $\mathbf{M}_n^{\text{eff}}$. For completeness, we include a proof in Section A.2.4. □

We conclude this section by relating the fidelity to the computational max-divergence. Applying the computational Fuchs-van de Graaf inequality to this bound then yields a result similar to Lemma 4.7.

Lemma 4.24. *Given two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, we have that*

$$\tilde{F}(\rho, \sigma) \geq 2^{-\tilde{D}_{\max}(\rho\|\sigma)}. \quad (4.1.34)$$

Proof. By monotonicity of the measured Rényi α divergences, see Lemma 4.11, and Theorem 4.13 it follows that

$$\tilde{D}_{1/2}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) \leq \tilde{D}_{\max}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) = \tilde{D}_{\max}(\rho\|\sigma).$$

Applying Lemma 4.22, one then finds that

$$\tilde{F}(\rho, \sigma) \geq 2^{-\tilde{D}_{\max}(\rho\|\sigma)}.$$

□

The previous bounds allow us to derive another relation between the computational max-divergence and the computational trace distance.

Corollary 4.25. *Given two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_n)$, we have that*

$$\tilde{D}_{\max}(\rho\|\sigma) \geq -\log \left(1 - \left(\tilde{\Delta}(\rho, \sigma) \right)^2 \right). \quad (4.1.35)$$

Proof. This follows from Lemma 4.24 together with the computational Fuchs-van de Graaf inequality provided in Lemma 4.23. □

We provide a similar bound in Lemma 4.7. Whenever $\tilde{\Delta}(\rho, \sigma) \approx 0$, Lemma 4.7 is tighter. Conversely, for $\tilde{\Delta}(\rho, \sigma) \approx 1$, we find that Corollary 4.25 provides a substantially tighter bound.

4.2. Computational Conditional Min-Entropy

In this section, we first introduce the (*bipartite*) *computational max-divergence*, defined with respect to the cone of efficiently implementable Choi operators, and study its basic properties as well as its relation to other operationally meaningful quantities, such as a corresponding computational hypothesis-testing divergence. We then show, in Section 4.2.1, that the computational conditional min-entropy admits the same operational interpretation as in the information-theoretic setting: it characterizes the largest fidelity with a maximally entangled state achievable by a recovery channel acting on B . Next, in Section 4.2.2, we consider the special case of bipartite classical-quantum states and prove that the

corresponding operational quantity is the maximal guessing probability under efficiently implementable measurements. Finally, in Section 4.2.5, we turn to the non-conditional case and derive a single-letter expression mirroring the informational one.

In Section 4.1.1, the computational max-divergence is induced by the cone generated by efficiently implementable binary measurements. Here, we consider the analogous max-divergence induced by efficiently implementable completely positive unital (CPU) maps, represented as operators through the Choi-Jamiołkowski isomorphism. This viewpoint strictly generalizes the measurement-based approach of the previous section: by Lemma 3.16, every complexity-constrained effect operator from Definition 3.11 arises from an efficiently implementable CPU map. Moreover, in Section 4.2.3 we show that the max-divergence defined in Definition 4.1 is recovered as a special case of the construction introduced below.

Definition 4.26 (Bipartite Computational Max-Divergence). Given a family of proper cones $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ and two positive semidefinite operators $\rho_{AB}, \sigma_{AB} \in \text{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, we denote its corresponding computational max-divergence by:

$$\overset{c}{D}_{\max}^{A|B}(\rho_{AB} \parallel \sigma_{AB}) := \log \inf \{ \lambda \in \mathbb{R} \mid \rho_{AB} \leq_{\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}} \lambda \sigma_{AB} \} = \log \sup_{F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}} \frac{\text{Tr}[\rho_{AB} F_{AB}]}{\text{Tr}[\sigma_{AB} F_{AB}]} . \quad (4.2.1)$$

If there does not exist any $\lambda \in \mathbb{R}$ such that $\rho_{AB} \leq_{\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}} \lambda \sigma_{AB}$, then $\overset{c}{D}_{\max}^{A|B}(\rho_{AB} \parallel \sigma_{AB}) = \infty$.

Note that $\overset{c}{D}_{\max}^{A|B}$ is defined using the partial order from Eq.(3.3.10). Compared to Definition 4.1, the order here is induced by the cone generated by efficiently implementable CPU maps (via their Choi operators), rather than directly by efficiently implementable effect operators on $\mathcal{H}_n^A \otimes \mathcal{H}_n^B$.

Similarly to the POVM case (see Section 3.2), let us assume that the polynomial defining the family of sets of complexity-constrained Choi operators $\{\mathbf{J}_{n,A|B}^{p(n)}\}_{n \in \mathbb{N}}$ is strictly super additive, i.e., $p(n+m) \geq p(n) + p(m) + 1$. Then, the following lemma holds.

Lemma 4.27 (Super-additivity of the $\overset{c}{D}_{\max}^{A|B}$). *The computational max-divergence $\overset{c}{D}_{\max}^{A|B}(\rho_{AB} \parallel \sigma_{AB})$ is super-additive, i.e., for any $\rho_{AB} = \rho_{A_1 B_1}^1 \otimes \rho_{A_2 B_2}^2$ and $\sigma_{AB} = \sigma_{A_1 B_1}^1 \otimes \sigma_{A_2 B_2}^2$ such that $\rho_{A_i B_i}^i, \sigma_{A_i B_i}^i \in \text{Pos}(\mathcal{H}_{n_i}^A \otimes \mathcal{H}_{n_i}^B)$,*

$$\overset{c}{D}_{\max}^{A|B, p(n_1+n_2)}(\rho_{AB} \parallel \sigma_{AB}) \geq \overset{c}{D}_{\max}^{A|B, p(n_1)}(\rho_{A_1 B_1}^1 \parallel \sigma_{A_1 B_1}^1) + \overset{c}{D}_{\max}^{A|B, p(n_2)}(\rho_{A_2 B_2}^2 \parallel \sigma_{A_2 B_2}^2) .$$

Proof. The proof follows by arguments similar to those in Lemma 4.4. $\square$

Moreover, the structure of the cone of Choi operators allows us to prove the following lemma.

Lemma 4.28 (σ_B independence of the $\dot{D}_{\max}^{A|B}(\rho||\mathbb{1} \otimes \sigma_B)$). *Given a quantum state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, the computational max divergence satisfies*

$$\dot{D}_{\max}^{A|B}(\rho_{AB}||\mathbb{1}_A \otimes \sigma_B) = \dot{D}_{\max}^{A|B}(\rho_{AB}||\mathbb{1}_A \otimes \tau_B)$$

for any quantum states $\sigma_B, \tau_B \in \mathcal{S}(\mathcal{H}_n^B)$ that satisfy $\rho_{AB} \ll \mathbb{1}_A \otimes \sigma_B$, $\rho_{AB} \ll \mathbb{1}_A \otimes \tau_B$.

Proof. Let us recall that the computational max-divergence can be expressed as

$$\dot{D}_{\max}^{A|B}(\rho_{AB}||\mathbb{1}_A \otimes \sigma_B) = \log \sup_{F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}} \frac{\text{Tr}[\rho_{AB} F_{AB}]}{\text{Tr}[(\mathbb{1}_A \otimes \sigma_B) F_{AB}]}.$$

Every non-zero $F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ satisfies $\text{tr}_A[F_{AB}] = \lambda(F_{AB}) \mathbb{1}_B$ for some $\lambda(F_{AB}) > 0$. Hence, for $\text{Tr}[\sigma_B] = 1$,

$$\text{Tr}[(\mathbb{1}_A \otimes \sigma_B) F_{AB}] = \text{Tr}[\sigma_B \text{tr}_A[F_{AB}]] = \lambda(F_{AB}) \text{Tr}[\sigma_B] = \lambda(F_{AB}),$$

which is independent of σ_B . Therefore, each ratio in the supremum is independent of σ_B . Replacing σ_B by τ_B yields the claim. $\square$

This special structure of the Choi cone also allows us to identify the computational max divergence with a hypothesis-testing quantity.

Definition 4.29 (Computational hypothesis-testing Choi divergence). Fix n and a complexity bound $p(n)$, and let $\mathbf{E}_{n,A|B}^{\text{eff}}$ denote the corresponding set of efficiently generated Choi effect operators. For states $\rho_{AB}, \sigma_{AB} \in \text{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ and $\eta \in (0, 1]$, define

$$\dot{D}_H^{p(n),\eta}(\rho_{AB}||\sigma_{AB}) := -\log \inf_{\substack{E \in \mathbf{E}_{n,A|B}^{\text{eff}} \\ \text{Tr}[E\rho_{AB}] \geq \eta}} \frac{\text{Tr}[E\sigma_{AB}]}{\text{Tr}[E\rho_{AB}]}. \quad (4.2.2)$$

If the feasible set is empty, we set the corresponding divergence to ∞ .

Unlike the measurement-restricted computational divergences of Section 4.1.2 and the computational hypothesis-testing relative entropies introduced in [MKH⁺25, MRR⁺25], the divergence in Definition 4.29 is restricted to efficiently generated Choi effect operators. Such effects can be interpreted as the action of a CPTP map on subsystem B , followed by the projective measurement onto the maximally entangled state $|\Omega\rangle$. In particular, they do not span the set of all efficiently implementable measurements on AB .

Lemma 4.30 (Equivalence of hypothesis-testing and max divergence). *Let $\rho_{AB} \in \text{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ be a state, and let $\rho_B = \text{tr}_A[\rho_{AB}]$. Then*

$$\dot{D}_H^{p(n),\eta}(\rho_{AB}||\mathbb{1}_A \otimes \rho_B) = \dot{D}_{\max}^{A|B,p(n)}(\rho_{AB}||\mathbb{1}_A \otimes \rho_B)$$

for every feasible η , i.e., for every $0 < \eta \leq \sup_{E \in \mathbf{E}_{n,A|B}^{\text{eff}}} \text{Tr}[E\rho_{AB}]$.

Proof. Let $E_{AB} \in \mathbf{E}_{n,A|B}^{\text{eff}}$. By definition of the normalized Choi slice,

$$\text{tr}_A [E_{AB}] = \frac{1}{d_A} \mathbb{1}_B. \quad (4.2.3)$$

Hence

$$\text{Tr}[E_{AB}(\mathbb{1}_A \otimes \rho_B)] = \text{Tr}[\rho_B \text{tr}_A [E_{AB}]] = \frac{1}{d_A} \text{Tr}[\rho_B] = \frac{1}{d_A}. \quad (4.2.4)$$

Therefore, for every feasible E_{AB} ,

$$\frac{\text{Tr}[E_{AB}(\mathbb{1}_A \otimes \rho_B)]}{\text{Tr}[E_{AB}\rho_{AB}]} = \frac{1}{d_A \text{Tr}[E_{AB}\rho_{AB}]}. \quad (4.2.5)$$

It follows that

$$\overset{c}{D}_H^{p(n),\eta}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B) = -\log \inf_{\substack{E_{AB} \in \mathbf{E}_{n,A|B}^{\text{eff}} \\ \text{Tr}[E_{AB}\rho_{AB}] \geq \eta}} \frac{1}{d_A \text{Tr}[E_{AB}\rho_{AB}]} \quad (4.2.6)$$

$$= \log \left(d_A \sup_{\substack{E_{AB} \in \mathbf{E}_{n,A|B}^{\text{eff}} \\ \text{Tr}[E_{AB}\rho_{AB}] \geq \eta}} \text{Tr}[E_{AB}\rho_{AB}] \right). \quad (4.2.7)$$

Since η is feasible, the constraint $\text{Tr}[E_{AB}\rho_{AB}] \geq \eta$ does not affect the maximum, and thus

$$\overset{c}{D}_H^{p(n),\eta}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B) = \log \left(d_A \max_{E_{AB} \in \mathbf{E}_{n,A|B}^{\text{eff}}} \text{Tr}[E_{AB}\rho_{AB}] \right). \quad (4.2.8)$$

By the Choi-cone dual formulation of the computational max-divergence, equivalently by Lemma 4.28, this is exactly

$$\overset{c}{D}_{\max}^{A|B, p(n)}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B), \quad (4.2.9)$$

which proves the claim. $\square$

Similar to the informational setting, the (conditional) min-entropy is the entropy measure canonically associated with the max-divergence order: it quantifies the smallest scalar by which a reference operator of the form $\mathbb{1}_A \otimes \sigma_B$ must be scaled to dominate ρ_{AB} in the relevant partial order. In our complexity-constrained framework, the same construction applies with the cone-induced order $\leq_{\mathcal{C}_{n,A|B}^{\text{eff}}}$. Consequently, and in direct parallel with the informational case [KRS09], we define the computational conditional min-entropy from the computational max-divergence.

Definition 4.31. Given a family of proper cones $\{\mathcal{C}_{n,A|B}^{\text{eff}}\}_{n \in \mathbb{N}}$, and any quantum state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, we denote the corresponding *computational min-entropy* of A conditioned on B by:

$$\overset{c}{H}_{\min}(A|B)_\rho := - \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B]=1}} \overset{c}{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \sigma_B). \quad (4.2.10)$$

The expression of the computational min-entropy can be simplified as follows.

Corollary 4.32. *Given a quantum state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, its computational conditional min-entropy satisfies*

$$\mathring{H}_{\min}(A|B)_\rho = -\mathring{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B)$$

for $\rho_B = \text{tr}_A[\rho_{AB}]$.

Proof. It directly follows from the fact that the computational max divergence is independent of the σ_B state as proven in Lemma 4.28. $\square$

In the information-theoretic setting, one distinguishes between the two conditional min-entropies

$$\begin{aligned} H_{\min}(A|B)_\rho &:= H_\infty^\uparrow(A|B)_\rho = - \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B]=1}} D_{\max}(\rho_{AB} \parallel \mathbb{1}_A \otimes \sigma_B), \\ H_\infty^\downarrow(A|B)_\rho &:= -D_{\max}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B), \end{aligned}$$

where $\rho_B = \text{tr}_A[\rho_{AB}]$.

In general, these two quantities do not coincide, i.e., there is a non-zero gap between them in the information-theoretic setting [Tom16]. In our complexity-constrained framework, the computational conditional min-entropy defined above is the direct analogue of the up-arrow quantity, i.e.,

$$\mathring{H}_{\min}(A|B)_\rho = - \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B]=1}} \mathring{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \sigma_B).$$

One may also attempt to introduce the corresponding down-arrow quantity

$$\mathring{H}_\infty^\downarrow(A|B)_\rho := -\mathring{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B).$$

However, by Lemma 4.28, the quantity $\mathring{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \sigma_B)$ is independent of the choice of σ_B .

Remark 4.33 (Computational up- and down-arrow equivalence). For any state ρ_{AB} it holds that

$$\mathring{H}_{\min}(A|B)_\rho = \mathring{H}_\infty^\downarrow(A|B)_\rho = -\mathring{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \rho_B).$$

Hence, in our computational framework, the up-arrow and down-arrow definitions coincide. This is a special consequence of the Choi cone structure, where we exclusively optimize over efficient effect operators F_{AB} whose partial trace $\text{tr}_A[F_{AB}]$ are proportional to the identity. We highlight that a consequence of Theorem 4.36 is that we recover the computational min-entropy of [AHRA25]. That computational entropy is equivalent to the information-theoretic min-entropy $H_{\min}(A|B)_\rho$ when the computational restrictions are removed [AHRA25, Lemma 32]. As such, our conic structure naturally encodes the optimization over σ_B and does not lead to a distinct computational notion of $H_\infty^\downarrow(A|B)_\rho$.

Lemma 4.34 (Sub-additivity of the $\dot{H}_{\min}(A|B)_\rho$). *The computational min-entropy is sub-additive, i.e., for any $\rho_{A_1B_1} \in \text{Pos}(\mathcal{H}_n^{A_1} \otimes \mathcal{H}_n^{B_1})$ and $\rho_{A_2B_2} \in \text{Pos}(\mathcal{H}_n^{A_2} \otimes \mathcal{H}_n^{B_2})$, with $\rho_{A_1A_2B_1B_2} = \rho_{A_1B_1} \otimes \rho_{A_2B_2}$, it holds that*

$$\dot{H}_{\min}(A_1A_2|B_1B_2)_{\rho_{A_1A_2B_1B_2}} \leq \dot{H}_{\min}(A_1|B_1)_{\rho_{A_1B_1}} + \dot{H}_{\min}(A_2|B_2)_{\rho_{A_2B_2}}.$$

Proof. It follows from Corollary 4.32 together with the fact of the computational max-divergence being super-additive (Lemma 4.27). $\square$

Up to this point, we have introduced the computational max-divergence and the induced conditional min-entropy, and established the structural properties needed for its analysis. We now show that this entropy retains the standard operational interpretation, with the unrestricted recovery map replaced by an efficiently implementable channel.

4.2.1. The Operational Meaning of the Computational Min-Entropy

Let us now prove that, in the complexity-constrained setting, $\dot{H}_{\min}(A|B)_\rho$ admits an operational interpretation analogous to the standard (information-theoretic) conditional min-entropy [KRS09]. In particular, we recover the corresponding notion considered in [AHRA25]. Let us first prove the following lemma.

Lemma 4.35. *Let $\mathcal{H}_n^A$ and $\mathcal{H}_n^B$ be finite-dimensional Hilbert spaces. Let $\{\mathbf{J}_{n,A|B}^{p(n)}\}_{n \in \mathbb{N}}$ be the family of Choi operators of CPU maps generated by at most $p(n)$ given a gate-set $\mathcal{G}$ and $\mathcal{C}_{n,A|B}^{\mathbf{S}_{\text{eff}}}$ its corresponding dual cone. Then, for any $\rho_{AB} \in \text{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, it holds that*

$$\min_{\substack{\sigma_B \geq 0 \\ \mathbb{1}_A \otimes \sigma_B \geq_{\mathcal{C}_{n,A|B}^{\mathbf{S}_{\text{eff}}}} \rho_{AB}}} \text{Tr}[\sigma_B] = \max_{\substack{F_{AB} \geq_{\mathcal{C}_{n,A|B}^{\mathbf{E}_{\text{eff}}}} 0 \\ \text{tr}_A[F_{AB}] \leq \mathbb{1}_B}} \text{Tr}[\rho_{AB} F_{AB}]. \quad (4.2.11)$$

Proof. The argument is inspired by the proof of [KRS09, Lemma 4], but its implementation in our setting requires several nontrivial modifications in order to accommodate the complexity constraints.

Let us first exploit the special structure of the cone of efficiently generated Choi operators to reduce the problem to a dual program.

For each efficiently generated Choi operator F_{AB} we have that $\text{tr}_A[F_{AB}] = \mathbb{1}_B$. Therefore, for the cone of efficient Choi operators $\mathcal{C}_{n,A|B}^{\mathbf{E}_{\text{eff}}}$ defined in Definition 3.17, any element $F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}_{\text{eff}}}$ can be written as

$$F_{AB} = \sum_j \alpha_j J_{AB}^{(j)}, \quad \alpha_j \geq 0,$$

with $J_{AB}^{(j)}$ a (channel-normalized) Choi operator, i.e., $\text{tr}_A[J_{AB}^{(j)}] = \mathbb{1}_B$. In consequence, it also satisfies

$$\text{tr}_A[F_{AB}] = \left(\sum_j \alpha_j \right) \mathbb{1}_B =: t \mathbb{1}_B$$

for some scalar $t \geq 0$. If $t = 0$, then $F_{AB} = 0$ and the objective is zero. For $0 < t \leq 1$, define the rescaled operator

$$\hat{F}_{AB} := \frac{1}{t} F_{AB}.$$

Since $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ is a cone, then $\hat{F}_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$, and $\text{tr}_A[\hat{F}_{AB}] = \mathbb{1}_B$. Moreover, for any state $\rho_{AB} \geq 0$,

$$\text{Tr}[\rho_{AB} \hat{F}_{AB}] = \frac{1}{t} \text{Tr}[\rho_{AB} F_{AB}] \geq \text{Tr}[\rho_{AB} F_{AB}].$$

Thus, since the supremum in the r.h.s. of Eq.(4.2.11) is attained on elements with $\text{tr}_A[F_{AB}] = \mathbb{1}_B$, we can extend the maximization to all operators F_{AB} such that $\text{tr}_A[F_{AB}] \leq \mathbb{1}_B$. Therefore, we will prove the following claim:

$$\min_{\substack{\sigma_B \geq 0 \\ \mathbb{1}_A \otimes \sigma_B \in \mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}} \\ \rho_{AB}}} \text{Tr}[\sigma_B] = \max_{\substack{F_{AB} \geq 0 \\ F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} \\ \text{tr}_A[F_{AB}] \leq \mathbb{1}_B}} \text{Tr}[\rho_{AB} F_{AB}]. \quad (4.2.12)$$

We formulate Eq. (4.2.12) as a conic program using the tools introduced in Section 2.6 on the spaces

$$\mathcal{V}_1 := \text{Herm}(\mathcal{H}_n^B), \quad \mathcal{V}_2 := \mathcal{V}_{n,A|B}^*,$$

with cones

$$\mathcal{C}_1 := \text{Pos}(\mathcal{H}_n^B), \quad \mathcal{C}_2 := \mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}},$$

and identifying the elements of Eq.(2.6.6) and Eq.(2.6.7) as $c := \mathbb{1}_B$ and $b := \pi(\rho_{AB})$, and the linear map

$$\Phi : \mathcal{V}_1 \rightarrow \mathcal{V}_2, \quad \Phi(\sigma_B) := \pi(\mathbb{1}_A \otimes \sigma_B).$$

Then the primal problem is exactly

$$\gamma_{\text{primal}} = \inf_{\sigma_B \in \text{Herm}(\mathcal{H}_n^B)} \left\{ \text{Tr}[\sigma_B] : \sigma_B \in \text{Pos}(\mathcal{H}_n^B), \Phi(\sigma_B) - \pi(\rho_{AB}) \in \mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}} \right\}. \quad (4.2.13)$$

Since the pairing between $\mathcal{V}_{n,A|B}^*$ and $\mathcal{V}_{n,A|B}$ is given by

$$\langle \pi(Y), F_{AB} \rangle = \text{Tr}[Y F_{AB}], \quad Y \in \text{Herm}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B), \quad F_{AB} \in \mathcal{V}_{n,A|B},$$

the adjoint map $\Phi^* : \mathcal{V}_{n,A|B} \rightarrow \text{Herm}(\mathcal{H}_n^B)$ is

$$\Phi^*(F_{AB}) = \text{tr}_A[F_{AB}],$$

because

$$\langle \Phi(\sigma_B), F_{AB} \rangle = \text{Tr}[(\mathbb{1}_A \otimes \sigma_B)F_{AB}] = \text{Tr}[\sigma_B \text{tr}_A[F_{AB}]].$$

Hence, the dual problem is given by

$$\gamma_{\text{dual}} = \sup_{F_{AB} \in \mathcal{V}_{n,A|B}} \left\{ \text{Tr}[\rho_{AB} F_{AB}] : F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}, \mathbb{1}_B - \text{tr}_A[F_{AB}] \in \text{Pos}(\mathcal{H}_n^B) \right\},$$

which can be simplified as,

$$\gamma_{\text{dual}} = \sup_{F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}} \{ \text{Tr}[\rho_{AB} F_{AB}] : \text{tr}_A[F_{AB}] \leq \mathbb{1}_B \}. \quad (4.2.14)$$

Therefore, by weak duality for feasible primal-dual conic programs,

$$\min_{\substack{\sigma_B \geq 0 \\ \pi(\mathbb{1}_A \otimes \sigma_B) - \pi(\rho_{AB}) \in \mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}}}} \text{Tr}[\sigma_B] \geq \max_{\substack{F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} \\ \text{tr}_A[F_{AB}] \leq \mathbb{1}_B}} \text{Tr}[\rho_{AB} F_{AB}]. \quad (4.2.15)$$

To prove equality, it remains to show that there is no duality gap. For this, we verify Slater's condition for the primal problem. Since $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ is a proper cone in $\mathcal{V}_{n,A|B}$, its dual cone $\mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}}$ is a proper cone in $\mathcal{V}_{n,A|B}^*$. Thus it suffices to find $\sigma_B \in \text{ri}(\text{Pos}(\mathcal{H}_n^B))$ such that

$$\Phi(\sigma_B) - \pi(\rho_{AB}) = \pi(\mathbb{1}_A \otimes \sigma_B - \rho_{AB}) \in \text{ri}(\mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}}).$$

By taking $\sigma_B := 2\lambda_{\max}(\rho_{AB})\mathbb{1}_B$, then $\sigma_B \in \text{ri}(\text{Pos}(\mathcal{H}_n^B))$. Moreover, for every $F_{AB} \in \mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}} \setminus \{0\}$,

$$\begin{aligned} \langle \pi(\mathbb{1}_A \otimes \sigma_B - \rho_{AB}), F_{AB} \rangle &= \text{Tr}[(\mathbb{1}_A \otimes \sigma_B - \rho_{AB})F_{AB}] \\ &= 2\lambda_{\max}(\rho_{AB}) \text{Tr}[F_{AB}] - \text{Tr}[\rho_{AB} F_{AB}] \\ &\geq \lambda_{\max}(\rho_{AB}) \text{Tr}[F_{AB}] > 0, \end{aligned}$$

where the second to last inequality is Hölder inequality and the last strict inequality follows since $\text{Tr}[F_{AB}] = \|F_{AB}\|_1 > 0$ as by assumption $F_{AB} \neq 0$ and ρ_{AB} is a finite dimensional normalized quantum state, hence $\lambda_{\max}(\rho_{AB}) > 0$. Hence, by the standard characterization of the interior of the dual cone,

$$\pi(\mathbb{1}_A \otimes \sigma_B - \rho_{AB}) \in \text{ri}(\mathcal{C}_{n,A|B}^{\mathbf{S}^{\text{eff}}}).$$

Therefore, Slater's condition holds, and the primal and dual optimal values coincide. $\square$

Let us now prove the main theorem of this section.

Theorem 4.36. *Given a quantum state $\rho_{AB} \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, its computational min-entropy can be expressed as*

$$\hat{H}_{\min}(A|B)_\rho = -\log \hat{q}_{\text{corr}}(A|B)_\rho, \quad (4.2.16)$$

where

$$\hat{q}_{\text{corr}}(A|B)_\rho := d_A \cdot \max_T F((\text{Id}_A \otimes T)(\rho_{AB}), |\Omega_{AA'}\rangle\langle\Omega_{AA'}|) \quad (4.2.17)$$

is the maximal fidelity with a maximally entangled state achievable by channel $T : \mathcal{B}_1(\mathcal{H}_n^B) \rightarrow \mathcal{B}_1(\mathcal{H}_n^{A'})$, with $\mathcal{H}_n^{A'} \simeq \mathcal{H}_n^A$ and $|\Omega_{AA'}\rangle := \frac{1}{\sqrt{d_A}} \sum_{x \in [d_A]} |x\rangle_A \otimes |x\rangle_{A'}$.

Proof. By definition of the computational max-divergence,

$$\begin{aligned} \hat{H}_{\min}(A|B)_\rho &= - \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B]=1}} \hat{D}_{\max}^{A|B}(\rho_{AB} \parallel \mathbb{1}_A \otimes \sigma_B) \\ &= -\log \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B]=1}} \inf \left\{ \lambda \geq 0 : \rho_{AB} \leq_{\mathcal{C}_{n,A|B}^{\text{S}_{\text{eff}}}} \lambda(\mathbb{1}_A \otimes \sigma_B) \right\}. \end{aligned}$$

Introduce the unnormalised variable $\tilde{\sigma}_B := \lambda \sigma_B \geq 0$. Then the order constraint can be written as

$$\rho_{AB} \leq_{\mathcal{C}_{n,A|B}^{\text{S}_{\text{eff}}}} \mathbb{1}_A \otimes \tilde{\sigma}_B \iff \mathbb{1}_A \otimes \tilde{\sigma}_B - \rho_{AB} \in \mathcal{C}_{n,A|B}^{\text{S}_{\text{eff}}},$$

and $\text{Tr}[\tilde{\sigma}_B] = \lambda$. Hence

$$2^{-\hat{H}_{\min}(A|B)_\rho} = \inf_{\tilde{\sigma}_B \geq 0} \left\{ \text{Tr}[\tilde{\sigma}_B] : \mathbb{1}_A \otimes \tilde{\sigma}_B \geq_{\mathcal{C}_{n,A|B}^{\text{S}_{\text{eff}}}} \rho_{AB} \right\}.$$

This is exactly the primal conic program of Lemma 4.35, then it follows that

$$2^{-\hat{H}_{\min}(A|B)_\rho} = \max_{\substack{F_{AB} \in \mathcal{C}_{n,A|B}^{\text{E}_{\text{eff}} \\ \text{tr}_A[F_{AB}] = \mathbb{1}_B}} \text{Tr}[\rho_{AB} F_{AB}]. \quad (4.2.18)$$

Let us now prove the following equivalence:

$$\log \max_{\substack{F_{AB} \in \mathcal{C}_{n,A|B}^{\text{E}_{\text{eff}} \\ \text{tr}_A[F_{AB}] = \mathbb{1}_B}} \text{Tr}[\rho_{AB} F_{AB}] = \log \left(d_A \cdot \max_T F((\text{Id}_A \otimes T)(\rho_{AB}), |\Omega_{AA'}\rangle\langle\Omega_{AA'}|) \right).$$

By construction, $F_{AB} \in \mathcal{B}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ is a nonnegative operator such that $\text{tr}_A[F_{AB}] = \mathbb{1}_B$. Then, due to the Choi-Jamiołkowski isomorphism, it corresponds to the Choi operator of a CPU map T^* . Therefore,

$$\begin{aligned} \text{Tr}[\rho_{AB} F_{AB}] &= d_A \text{Tr}[\rho_{AB} (\text{Id}_A \otimes T^*) |\Omega_{AA'}\rangle\langle\Omega_{AA'}|] \\ &= d_A \text{Tr}[(\text{Id}_A \otimes T)(\rho_{AB})(|\Omega_{AA'}\rangle\langle\Omega_{AA'}|)] \\ &= d_A \langle\Omega_{AA'}| (\text{Id}_A \otimes T)(\rho_{AB}) |\Omega_{AA'}\rangle \\ &= d_A F((\text{Id}_A \otimes T)(\rho_{AB}), |\Omega_{AA'}\rangle\langle\Omega_{AA'}|), \end{aligned}$$

where the first equality follows from duality, with no complexity overhead as proven in Lemma 3.4, and the last one follows by the definition of the fidelity $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2 = \left(\text{Tr}\left[\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right]\right)^2$ taking into account that $|\Omega_{AA'}\rangle$ is a pure state.

Then, since the maximization is done for a linear function w.r.t. a convex set, it suffices to take the extremal point of the convex set, i.e., $\mathbf{J}_{n,A|B}^{p(n)}$. Lastly, due to the Choi-Jamiołkowski isomorphism to Lemma 3.4, this corresponds to the maximization w.r.t. the set of efficiently generated CPTP maps. $\square$

Therefore, the computational min-entropy admits the same operational interpretation as in the information-theoretic setting: it is the (negative logarithm of the) optimal fidelity with respect to a maximally entangled state achieved by acting on the conditioning system when the quantum channel is restricted to be efficiently implementable. This mirrors the characterization of [KRS09] and, under our complexity constraints, recovers the computational conditional min-entropy considered in [AHRA25]. It directly follows that dropping the computational restrictions, we recover the information-theoretic min-entropy; see also [AHRA25, Lemma 32].

4.2.2. The Classical-Quantum Case

Let us now study the case in which one of the registers is classical. This case can be seen as a reduction of the previous one, in which both subsystems $X \equiv A$ and $X' \equiv A'$ are classical, however, B may not be. Then, as in the information-theoretic case, we recover the interpretation as proportional to the optimal guessing probability under computational constraints on the allowed measurements, analogous to [AHRA25, Lemma 32].

Theorem 4.37. *Let $\rho_{XB} = \sum_{i=1}^{|X|} p_i |i\rangle\langle i|_X \otimes \rho_B^i$ be a cq state. Then*

$$\hat{H}_{\min}(X|B)_\rho = -\log \hat{p}_{\text{guess}}(X|B)_\rho \quad (4.2.19)$$

where

$$\hat{p}_{\text{guess}}(X|B)_\rho := \max_{\{E_B^i\}_{i \in X} \in \mathbf{M}_{n,|X|}^{p(n)}} \sum_{i \in X} p_i \text{Tr}[E_B^i \rho_B^i] \quad (4.2.20)$$

is the optimal guessing probability when restricted to efficient $|X|$ -outcome POVMs on B .

Proof. Recall Lemma 4.35. Then, for a given cq state $\rho_{XB} = \sum_i p_i |i\rangle\langle i|_X \otimes \rho_B^i$, it follows that

$$2^{-\tilde{H}_{\min}(X|B)_\rho} = \max_{\substack{E_{XB} \in \mathcal{C}_{n,X|B}^{\mathbf{E}^{\text{eff}}} \\ \text{tr}_X[E_{XB}] = \mathbb{1}_B}} \text{Tr}[\rho_{XB} E_{XB}] \quad (4.2.21)$$

$$= \max_{J_{XB} \in \mathbf{J}_{n,X|B}} \text{Tr}[\rho_{XB} J_{XB}] \quad (4.2.22)$$

$$= \max_{\substack{T^*: X' \rightarrow B \\ C(T^*) \leq p(n)}} \text{Tr} \left[\left(\sum_{i=1}^{|X|} p_i |i\rangle\langle i|_X \otimes \rho_B^i \right) d_A(\text{Id}_X \otimes T^*(|\Omega\rangle\langle\Omega|_{XX'})) \right] \quad (4.2.23)$$

$$= \max_{\substack{T^*: X' \rightarrow B \\ C(T^*) \leq p(n)}} \text{Tr} \left[\sum_{i=1}^{|X|} p_i \rho_B^i T^*(|i\rangle\langle i|_{X'}) \right] \quad (4.2.24)$$

$$= \max_{\{E_B^i\}_{i \in X} \in \mathbf{M}_{n,|X|}^{p(n)}} \sum_{i \in X} p_i \text{Tr}[E_B^i \rho_B^i], \quad (4.2.25)$$

where the second equality follows because the constraint $\text{tr}_X[E_{XB}] = \mathbb{1}_B$ restricts the cone $\mathcal{C}_{n,X|B}^{\mathbf{E}^{\text{eff}}}$ to its channel-normalized slice $\text{conv}(\mathbf{J}_{n,X|B}^{p(n)})$. Since the objective is linear, its maximum over this convex hull is attained on an element of $\mathbf{J}_{n,X|B}^{p(n)}$. The third equality follows from the definition of $\mathbf{J}_{n,X|B}$ and the fourth from

$$\text{Tr} \left[\left(\sum_{i=1}^{|X|} p_i |i\rangle\langle i|_X \otimes \rho_B^i \right) d_A(\text{Id}_X \otimes T^*(|\Omega\rangle\langle\Omega|_{XX'})) \right] \quad (4.2.26)$$

$$= \sum_{i=1}^{|X|} p_i \text{Tr} \left[(|i\rangle\langle i|_X \otimes \rho_B^i) \sum_{y,z} |y\rangle\langle z|_X \otimes T^*(|y\rangle\langle z|_{X'}) \right] \quad (4.2.27)$$

$$= \sum_{i=1}^{|X|} p_i \text{Tr} \left[\rho_B^i T^*(|i\rangle\langle i|_{X'}) \right]. \quad (4.2.28)$$

Finally the last equality is denoting the POVM elements $T^*(|i\rangle\langle i|_{X'}) =: E_B^i$, which by construction via the efficient map T^* form an efficient $|X|$ -outcome POVM $\{E_B^i\}_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)}$ and an analogous argument as for the second equality. Taking the logarithm and a minus sign concludes the proof. $\square$

Note that in the c-q setting we may also effectively replace the optimization over $\mathcal{C}_{n,A|B}^{\mathbf{E}^{\text{eff}}}$ by an optimization over efficient quantum-classical maps, which is also equivalent to pinched versions of the efficient quantum-quantum maps T^* , and the above optimization over efficient POVMs.

4.2.3. Measurement Channels for CQ-States

For classical-quantum states, we show that restricting to efficient Choi operations preserves the classical-quantum structure. Let ρ_{XB} be a classical-quantum state, i.e. $\rho_{XB} =$

$\sum_{i=1}^{|X|} |i\rangle\langle i|_X \otimes \rho_B^i$ is block diagonal in the computational basis of X . Then for every $F_{XB} \in \text{Herm}(\mathcal{H}_X \otimes \mathcal{H}_B)$ one has

$$\text{Tr}(\rho_{XB} F_{XB}) = \text{Tr}(\rho_{XB} \Delta_X(F_{XB})), \quad (4.2.29)$$

where $\Delta_X(F) = \sum_i (|i\rangle\langle i|_X \otimes \mathbb{1}_B) F (|i\rangle\langle i|_X \otimes \mathbb{1}_B)$ is the pinching map in the computational basis. Hence, in any cone program whose objective is of the form $F \mapsto \text{Tr}(\rho_{XB} F)$, one may w.l.o.g. restrict the optimization to X -diagonal operators.

In particular, the optimization over the cone of efficient Choi-operators in the computational min-entropy can be restricted by an optimization over the cone of efficient *flagged POVMs*

$$\tilde{\mathcal{C}}_{n,X|B}^{\mathbf{E}^{\text{eff}}} := \Delta_X(\mathcal{C}_{n,X|B}^{\mathbf{E}^{\text{eff}}}) = \text{cone} \left\{ \sum_{i=1}^{|X|} |i\rangle\langle i|_X \otimes E_B^i \mid \{E_B^i\}_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)} \right\}, \quad (4.2.30)$$

where $\mathbf{M}_{n,|X|}^{p(n)}$ denotes the family of $|X|$ -outcome POVMs on B implementable with complexity $p(n)$. The second equality is easily verified by explicit computation.

Equivalently, one might see the cone $\tilde{\mathcal{C}}_{n,X|B}^{\mathbf{E}^{\text{eff}}}$ as the cone of efficiently generated quantum-classical (q-c) measurement channels. Given a POVM $\{E_i\}_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)}$, its corresponding *quantum-classical measurement channel* $\mathcal{M} : \mathcal{B}(\mathcal{H}_n^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{X'})$ is given by

$$\mathcal{M}(\rho_B) := \sum_{i=1}^{|X|} \text{tr}(E_B^i \rho_B) |i\rangle\langle i|_{X'}. \quad (4.2.31)$$

Its adjoint $\mathcal{M}^* : \mathcal{B}(\mathcal{H}_n^{X'}) \rightarrow \mathcal{B}(\mathcal{H}_n^B)$ is a CPU map and satisfies

$$\mathcal{M}^*(|i\rangle\langle i|_{X'}) = E_B^i, \quad \mathcal{M}^*(|i\rangle\langle j|_{X'}) = 0 \quad (i \neq j), \quad (4.2.32)$$

so that its Choi operator

$$J(\mathcal{M}^*) = \sum_{i=1}^{|X|} |i\rangle\langle i|_X \otimes E_B^i, \quad (4.2.33)$$

is diagonal in X . Therefore, the optimization over $\tilde{\mathcal{C}}_{n,X|B}^{\mathbf{E}^{\text{eff}}}$ is equivalently an optimization over efficient q-c measurement channels $\mathcal{M}$.

When $|X| = 2$, every operator in the set that generates the cone $\tilde{\mathcal{C}}_{n,X|B}^{\mathbf{E}^{\text{eff}}}$ has the form

$$|0\rangle\langle 0|_X \otimes E_B + |1\rangle\langle 1|_X \otimes (\mathbb{1}_B - E_B), \quad (4.2.34)$$

where $\{E_B, \mathbb{1}_B - E_B\} \in \mathbf{M}_{n,2}^{p(n)}$ is an efficient binary POVM. Thus, in the binary c-q case, optimization over flagged POVMs is equivalently parametrized by the choice of an efficient effect operator E_B (the first POVM element), with the second outcome fixed by complement. In particular, this is the same binary-measurement model as in Definition 3.11 and section 3.4: the polynomially generated set of efficient effect operators for single systems and its induced cone of efficient effects describe exactly the admissible binary tests underlying the above flagged form.

Remark 4.38 (Efficient effect operators versus efficient Choi operators and the information-theoretic max-divergence). In the previous subsection, we considered the cone generated by the flagged Choi operators

$$\mathbf{J}_{n,X|B}^{p(n)} := \left\{ \sum_x |x\rangle\langle x|_X \otimes E_B^x \mid E_B^x = T^*(|x\rangle\langle x|) \text{ for a single efficient CPU map } T^* \right\}.$$

This construction implies that every effect E_B^x is individually efficient. More importantly, it requires the complete family $\{E_B^x\}_x$ to arise jointly from a single efficient CPU map.

One may instead consider families $\{E_B^x\}_x$ for which every effect belongs to some efficient binary POVM, without requiring a single efficient map implementing the complete family. These two notions coincide in the binary setting considered in Definition 3.11 and Section 3.4, but they need not coincide when the number of outcomes grows with the system size. In this case, individual efficient implementability does not imply joint efficient implementability. In other words, not every flagged POVM composed of individually efficient effects is itself efficiently implementable.

To illustrate this distinction, for each $n \in \mathbb{N}$ choose a permutation

$$\pi_n : \{0, 1\}^n \longrightarrow \{0, 1\}^n$$

whose inverse cannot be implemented by polynomial-size circuits. Such a family exists by a counting argument. Indeed, the number of circuits of size at most $2^{n/2}$ over a fixed finite gate set is at most $2^{O(n2^{n/2})}$, whereas the number of permutations of $\{0, 1\}^n$ is

$$(2^n)! = 2^{\Theta(n2^n)}.$$

Hence, one can choose $\{\pi_n\}_n$ such that the circuit complexity of π_n^{-1} is superpolynomial.

Now consider

$$F_{XB}^{(n)} := \sum_{x \in \{0,1\}^n} |x\rangle\langle x|_X \otimes |\pi_n(x)\rangle\langle \pi_n(x)|_B = \sum_{x \in \{0,1\}^n} |x\rangle\langle x|_X \otimes E_B^{x,n}.$$

Information-theoretically, $F_{XB}^{(n)}$ is the Choi operator of the q-c measurement channel T_{π_n} associated with the POVM $\{E_B^{x,n}\}_x$. Its dual is

$$T_{\pi_n}^*(Z) = \sum_x \langle x|Z|x\rangle |\pi_n(x)\rangle\langle \pi_n(x)|.$$

For every fixed $x \in \{0, 1\}^n$, the binary POVM

$$\{E_B^{x,n}, \mathbb{1}_B - E_B^{x,n}\}$$

is efficiently implementable. Indeed, for each fixed x , the value $\pi_n(x)$ is fixed in advance. The POVM can therefore be implemented by measuring in the computational basis and outputting 1 exactly when the observed string is $\pi_n(x)$, which requires only $O(n)$ gates. However, an implementation of the complete POVM $\{E_B^{x,n}\}_x$ would output $\pi_n^{-1}(y)$ on the

computational-basis input $|y\rangle$. It would therefore provide a polynomial-size implementation of π_n^{-1} , contradicting its construction. Thus, every effect $E_B^{x,n}$ is individually efficient, while the complete POVM is not efficiently implementable in the sense of Definition 3.8.

In the fully quantum setting, the corresponding distinction is between the cone generated by Choi operators of efficiently implementable CPU maps and a cone constructed from efficiently implementable global effects satisfying the Choi normalization condition. Although every positive operator F_{AB} satisfying

$$\mathrm{tr}_A[F_{AB}] = \mathbb{1}_B$$

is the Choi operator of a CPU map, an efficient implementation of F_{AB} as a measurement effect does not, by itself, provide an efficient implementation of the associated map. Indeed, the measurement only needs to produce a classical binary outcome, whereas implementing the map requires reproducing its complete action on arbitrary quantum inputs, including coherent superpositions.

Consequently, the computational max-divergence $\bar{D}_{\max}^{cA|B}$ considered here differs from the computational max-divergence induced by efficient global measurements. When computational restrictions are removed, the latter recovers the ordinary information-theoretic max-divergence. By contrast, the former remains induced by the cone

$$\{F_{AB} \geq 0 \mid \mathrm{tr}_A[F_{AB}] = \lambda \mathbb{1}_B \text{ for some } \lambda \geq 0\},$$

which is generally a strict subcone of $\mathrm{Pos}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$. Therefore, $\bar{D}_{\max}^{cA|B}$ does not generally reduce to the ordinary max-divergence, even though the conditional min-entropy induced by it recovers the information-theoretic conditional min-entropy when the complexity restrictions are removed.

4.2.4. Computational Max-Divergence and Measured Max-Divergence Relations

This classical-quantum reduction lets us extend Theorem 4.13 beyond the binary setting. In particular, we show that the computational max-divergence induced by the cone of efficiently implementable q-c measurement channels (equivalently, efficiently generated flagged POVMs) coincides with the computational measured Rényi divergence of order ∞ from Definition 4.9, for arbitrary outcome alphabets X . We do this via an embedding

$$\rho_B \mapsto \hat{\rho}_{XB} := |1\rangle\langle 1|_X \otimes \rho_B$$

storing the state in the B register while X holds an uncorrelated ancillary system.

Theorem 4.39 (Computational max-divergence and measured max-divergence equivalence). *Given a finite gate-set $\mathcal{G}$, let $\tilde{\mathcal{C}}_{n,X|B}^{\mathbf{E}^{\mathrm{eff}}}$ be the corresponding cone of efficiently generated flagged POVMs induced by $\mathbf{M}_{n,|X|}^{p(n)}$ as in Section 4.2.3. Then for any $\rho_B, \sigma_B \in \mathrm{Pos}(\mathcal{H}_n^B)$,*

$$\bar{D}_{\max}^{cX|B}(\hat{\rho}_{XB} \parallel \hat{\sigma}_{XB}) = \bar{D}_{\infty}^{\mathbf{M}_{n,|X|}^{p(n)}}(\rho_B \parallel \sigma_B),$$

where $\bar{D}_{\max}^{cX|B}(\hat{\rho}_{XB} \parallel \hat{\sigma}_{XB}) := \bar{D}_{\max}^{cX|B}(|1\rangle\langle 1|_X \otimes \rho_B \parallel |1\rangle\langle 1|_X \otimes \sigma_B)$ is the computational max divergence of Definition 4.26 with $X = A$ and

$$\bar{D}_{\infty}^{\mathbf{M}_{n,|X|}^{p(n)}}(\rho_B \parallel \sigma_B) := \sup_{M \in \mathbf{M}_{n,|X|}^{p(n)}} \log \inf \{ \lambda \in \mathbb{R} : \mathcal{M}_M(\rho_B) \leq \lambda \mathcal{M}_M(\sigma_B) \},$$

is the computational measured max-divergence of Definition 4.9 where $\mathcal{M}_M : \mathcal{B}(\mathcal{H}_n^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{X'})$ denotes the q -c measurement channel corresponding to M .

Proof. Let $M = \{E_B^i\}_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)}$ and recall that its q -c measurement channel is

$$\mathcal{M}_M(\tau) = \sum_{i=1}^{|X|} \text{Tr}(E_B^i \tau) |i\rangle\langle i|_{X'}.$$

Set $p_i := \text{Tr}(E_B^i \rho_B)$ and $q_i := \text{Tr}(E_B^i \sigma_B)$. Since $\mathcal{M}_M(\rho_B)$ and $\mathcal{M}_M(\sigma_B)$ are diagonal in the computational basis of X' , the operator inequality $\mathcal{M}_M(\rho_B) \leq \lambda \mathcal{M}_M(\sigma_B)$ holds if and only if $p_i \leq \lambda q_i$ for all i . Hence

$$\inf \{ \lambda : \mathcal{M}_M(\rho_B) \leq \lambda \mathcal{M}_M(\sigma_B) \} = \max_{i: q_i > 0} \frac{p_i}{q_i},$$

with the convention that if $q_i = 0$ and $p_i > 0$ for some i then the infimum equals $+\infty$. Therefore,

$$\bar{D}_{\infty}^{\mathbf{M}_{n,|X|}^{p(n)}}(\rho_B \parallel \sigma_B) = \sup_{M \in \mathbf{M}_{n,|X|}^{p(n)}} \log \max_{i \in [|X|]} \frac{\text{Tr}(E_B^i \rho_B)}{\text{Tr}(E_B^i \sigma_B)}. \quad (4.2.35)$$

Let $M = (E_i)_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)}$ be the POVM with the effect operator E_{i^*} that attains the maximum in Eq. (4.2.35). Consider the polynomial-time classical relabelling that maps the outcome i^* to 1. In this way we obtain a POVM $\{F_j\}_{j=1}^{|X|}$ such that

$$F_1 = E_{i^*}.$$

The resulting POVM is still efficiently implementable, up to polynomial classical post-processing. Hence

$$\max_{i \in [|X|]} \frac{\text{Tr}(E_i \rho_B)}{\text{Tr}(E_i \sigma_B)} = \frac{\text{Tr}(F_1 \rho_B)}{\text{Tr}(F_1 \sigma_B)}.$$

Taking the supremum over the relabelled POVMs gives

$$\bar{D}_{\infty}^{\mathbf{M}_{n,|X|}^{p(n)}}(\rho_B \parallel \sigma_B) = \sup_{\{F_i\}_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)}} \log \frac{\text{Tr}(F_1 \rho_B)}{\text{Tr}(F_1 \sigma_B)}. \quad (4.2.36)$$

Moreover, as proven by [RSB25, Lemma 2], the measured max-divergence is jointly quasi-convex. Then, for the convex hull of $\mathbf{M}_{n,|X|}^{p(n)}$, which we denote as $\overline{\mathbf{M}}_{n,|X|}^{p(n)}$, we have that

$$\tilde{D}_{\infty}^{\mathbf{M}_{n,|X|}^{p(n)}}(\rho_B \parallel \sigma_B) = \sup_{\{E_B^i\}_{i=1}^{|X|} \in \mathbf{M}_{n,|X|}^{p(n)}} \log \frac{\text{Tr}(E_B^1 \rho_B)}{\text{Tr}(E_B^1 \sigma_B)} = \sup_{\{E_B^i\}_{i=1}^{|X|} \in \overline{\mathbf{M}}_{n,|X|}^{p(n)}} \log \frac{\text{Tr}(E_B^1 \rho_B)}{\text{Tr}(E_B^1 \sigma_B)},$$

by the same convexity argument as in Lemma 4.10. On the other hand, by the definition of $\tilde{D}_{\max}^{A|B}(\rho_{AB} \parallel \sigma_{AB})$ in the c-q case,

$$\tilde{D}_{\max}^{X|B}(\hat{\rho}_{XB} \parallel \hat{\sigma}_{XB}) = \log \sup_{F_{XB} \in \mathcal{C}_{n,|X|B}^{\mathbf{E}_{\text{eff}}}} \frac{\text{Tr}(\hat{\rho}_{XB} F_{XB})}{\text{Tr}(\hat{\sigma}_{XB} F_{XB})}, \quad (4.2.37)$$

where $\hat{\rho}_{XB} = |1\rangle\langle 1|_X \otimes \rho_B$, $\hat{\sigma}_{XB} = |1\rangle\langle 1|_X \otimes \sigma_B$. Since $\hat{\rho}_{XB}$ and $\hat{\sigma}_{XB}$ are block-diagonal in the X basis, Section 4.2.3 implies that we may w.l.o.g. restrict to X -diagonal optimizers. By construction, for every $F_{XB} \in \mathcal{C}_{n,|X|B}^{\mathbf{E}_{\text{eff}}}$, there exists an $E_{XB} = \sum_{i=1}^{|X|} |i\rangle\langle i|_X \otimes E_B^i$ for some $\{E_B^i\}_{i=1}^{|X|} \in \overline{\mathbf{M}}_{n,|X|}^{p(n)}$ such that $F_{XB} = \lambda \cdot E_{XB}$. Then, since the RHS of Eq. (4.2.37) is scaling invariant and due to the form of the states $\hat{\rho}_{XB}$ and $\hat{\sigma}_{XB}$, we have

$$\frac{\text{Tr}(\hat{\rho}_{XB} F_{XB})}{\text{Tr}(\hat{\sigma}_{XB} F_{XB})} = \frac{\text{Tr}(E_B^1 \rho_B)}{\text{Tr}(E_B^1 \sigma_B)}.$$

Then,

$$\tilde{D}_{\max}^{X|B}(\hat{\rho}_{XB} \parallel \hat{\sigma}_{XB}) = \sup_{\{E_B^i\}_{i=1}^{|X|} \in \overline{\mathbf{M}}_{n,|X|}^{p(n)}} \log \frac{\text{Tr}(E_B^1 \rho_B)}{\text{Tr}(E_B^1 \sigma_B)}.$$

□

Mirroring the information-theoretic setting, computational indistinguishability and computational unpredictability are captured here by the computational max-divergence of Definition 4.1 and the computational conditional min-entropy [AHRA25], respectively. Both quantities arise from the same order induced by efficiently implementable channels, placing them within a single framework.

4.2.5. Non-conditional Computational Min-Entropy

We now consider the case in which there is no side information. Following [AHRA25, Definition 26], we obtain a single-system quantity by adjoining a trivial conditioning register. This is the non-conditional specialization of $\tilde{H}_{\min}(A|B)$ in our framework.

It is important to distinguish this construction from the flagged c-q reduction of Theorem 4.39. In the latter, one adjoins a classical flag on the first register and obtains a formulation in terms of efficient POVMs. In the present case, however, the conditioning register is one-dimensional, and the corresponding cone is generated by efficient state-preparation maps. Thus, the resulting quantity is not, in general, the same as the measured divergence of Theorem 4.39.

Definition 4.40 (Non-conditional computational min-entropy). Let B be a one-dimensional Hilbert space spanned by $|1\rangle_B$. For any state $\rho_A \in \mathcal{S}(\mathcal{H}_n^A)$, define

$$\hat{\rho}_{AB} := \rho_A \otimes |1\rangle\langle 1|_B .$$

The non-conditional computational min-entropy of ρ_A is

$$\mathring{H}_{\min}(A)_\rho := \mathring{H}_{\min}(A|B)_{\hat{\rho}} .$$

Since the conditioning register is trivial, the infimum over conditioning states disappears. The next lemma rewrites the resulting quantity in the form naturally associated with the $A|B$ cone. Concretely, the optimization is over the convex hull of efficiently preparable states on A .

Lemma 4.41. *Let B be a one-dimensional Hilbert space spanned by $|1\rangle_B$, and let*

$$\begin{aligned} \hat{\rho}_{AB} &:= \rho_A \otimes |1\rangle\langle 1|_B , \\ \hat{\mathbb{1}}_{AB} &:= \mathbb{1}_A \otimes |1\rangle\langle 1|_B . \end{aligned}$$

Let $\mathcal{H}_n^{A'} \simeq \mathcal{H}_n^A$, and define the set of efficiently preparable states on A' by

$$\mathcal{S}_{n,A}^{p(n)} := \left\{ \tau_A \in \mathcal{S}(\mathcal{H}_n^A) \mid \tau_A = \Lambda(|1\rangle\langle 1|_B) \text{ for some efficient CPTP map } \Lambda : \mathcal{B}(\mathcal{H}^B) \rightarrow \mathcal{B}(\mathcal{H}_n^A) \right\} .$$

Then

$$\mathring{H}_{\min}(A)_\rho = -\log \sup_{\tau_A \in \mathcal{S}_{n,A}^{p(n)}} \text{Tr} \left[\rho_A \tau_A^\top \right] ,$$

where the transpose is taken with respect to the computational basis used to define $|\Omega_{AA'}\rangle$.

Proof. By definition of the non-conditional computational min-entropy,

$$\mathring{H}_{\min}(A)_\rho = \mathring{H}_{\min}(A|B)_{\hat{\rho}} = - \inf_{\substack{\sigma_B \geq 0 \\ \text{Tr}[\sigma_B] = 1}} \mathring{D}_{\max}^{A|B}(\hat{\rho}_{AB} \| \mathbb{1}_A \otimes \sigma_B) .$$

Since B is one-dimensional, the only state on B is $|1\rangle\langle 1|_B$. Hence

$$\mathring{H}_{\min}(A)_\rho = -\mathring{D}_{\max}^{A|B}(\hat{\rho}_{AB} \| \hat{\mathbb{1}}_{AB}) .$$

Now apply Theorem 4.36 to the bipartite state $\hat{\rho}_{AB}$. This gives

$$\mathring{H}_{\min}(A)_\rho = -\log \left(d_A \sup_{\Lambda} F((\text{Id}_A \otimes \Lambda)(\hat{\rho}_{AB}), |\Omega\rangle\langle \Omega|_{AA'}) \right) ,$$

where the supremum is over efficient CPTP maps $\Lambda : \mathcal{B}(\mathcal{H}^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'})$. Since B is one-dimensional, every such Λ is a state-preparation map which can be associated with its unique image state $\tau_{A'} = \Lambda(|1\rangle\langle 1|_B) \in \mathcal{S}(\mathcal{H}_n^{A'})$ and so

$$(\text{Id}_A \otimes \Lambda)(\hat{\rho}_{AB}) = (\text{Id}_A \otimes \Lambda)(\rho_A \otimes |1\rangle\langle 1|_B) = \rho_A \otimes \tau_{A'}.$$

Therefore,

$$\check{H}_{\min}(A)_\rho = -\log \left(d_A \sup_{\tau_{A'} \in \mathcal{S}_{n,A'}^{p(n)}} F(\rho_A \otimes \tau_{A'}, |\Omega\rangle\langle \Omega|_{AA'}) \right).$$

Since one argument of the fidelity is pure,

$$F(\rho_A \otimes \tau_{A'}, |\Omega\rangle\langle \Omega|_{AA'}) = \langle \Omega_{AA'} | (\rho_A \otimes \tau_{A'}) | \Omega_{AA'} \rangle.$$

Using the identity

$$d_A \langle \Omega_{AA'} | (X_A \otimes Y_{A'}) | \Omega_{AA'} \rangle = \text{Tr}[X_A Y_A^\top],$$

valid for all compatible operators X_A and $Y_{A'}$, we obtain

$$d_A \langle \Omega_{AA'} | (\rho_A \otimes \tau_{A'}) | \Omega_{AA'} \rangle = \text{Tr}[\rho_A \tau_{A'}^\top].$$

Substituting this into the previous expression proves the claim. $\square$

This expression mirrors the ordinary non-conditional min-entropy obtained after dropping all computational constraints:

$$H_{\min}(A)_\rho = -\log \|\rho_A\|_\infty = -\log \sup_{\tau \in \mathcal{S}(\mathcal{H}_n^A)} \text{Tr}[\rho_A \tau]. \quad (4.2.38)$$

The only difference is that the unrestricted optimization over all states is replaced here by an optimization over efficiently preparable ones.

Moreover, Lemma 4.41 shows that the non-conditional min-entropy is determined by the maximal overlap with the transpose of an efficiently preparable state. For positive operators, it therefore defines a computational analogue of the largest eigenvalue. This motivates the associated computational operator norm on Hermitian operators, defined by taking the absolute value of the overlap. When the transposed efficiently preparable states span $\text{Herm}(\mathcal{H}_n^A)$, this norm coincides with the order-unit norm associated with the dual cone of $\text{cone}(\mathcal{S}_{n,A}^{\top, p(n)})$; see [RKW11, Jen14].

Corollary 4.42 (Operational norm in the non-conditional setting). *Let*

$$\mathcal{S}_{n,A}^{\top, p(n)} := \left\{ \tau_A^\top : \tau_A \in \mathcal{S}_{n,A}^{p(n)} \right\}, \quad \mathcal{C}_{n,A}^{\mathcal{S}} := \text{cone}(\mathcal{S}_{n,A}^{\top, p(n)}) \subseteq \text{Herm}(\mathcal{H}_n^A),$$

where the transpose is taken with respect to the computational basis. For $X_A \in \text{Herm}(\mathcal{H}_n^A)$, define

$$\|X_A\|_{\infty, \text{comp}} := \sup_{\omega_A \in \mathcal{S}_{n,A}^{\top, p(n)}} |\text{Tr}[X_A \omega_A]|.$$

Then $\|\cdot\|_{\infty, \text{comp}}$ is a seminorm on $\text{Herm}(\mathcal{H}_n^A)$. If further

$$\text{span}_{\mathbb{R}}(\mathcal{S}_{n,A}^{\top, p(n)}) = \text{Herm}(\mathcal{H}_n^A),$$

then $\|\cdot\|_{\infty, \text{comp}}$ is a norm. In this case,

$$\|X_A\|_{\infty, \text{comp}} = \inf \left\{ t \geq 0 : -t \mathbb{1}_A \leq_{(C^{\mathcal{S}_{n,A}^{\top}})^*} X_A \leq_{(C^{\mathcal{S}_{n,A}^{\top}})^*} t \mathbb{1}_A \right\}.$$

For $X_A \geq 0$ this is given by

$$\|X_A\|_{\infty, \text{comp}} = \sup_{\tau_A \in \mathcal{S}_{n,A}^{p(n)}} \text{Tr}[X_A \tau_A^{\top}].$$

Proof. Let us first prove that $\|\cdot\|_{\infty, \text{comp}}$ defines a seminorm. For every $\alpha \in \mathbb{R}$ and $X_A, Y_A \in \text{Herm}(\mathcal{H}_n^A)$, one has

$$\|\alpha X_A\|_{\infty, \text{comp}} = \sup_{\omega_A \in \mathcal{S}_{n,A}^{\top, p(n)}} |\text{Tr}[\alpha X_A \omega_A]| = |\alpha| \|X_A\|_{\infty, \text{comp}},$$

and

$$\begin{aligned} \|X_A + Y_A\|_{\infty, \text{comp}} &= \sup_{\omega_A \in \mathcal{S}_{n,A}^{\top, p(n)}} |\text{Tr}[(X_A + Y_A) \omega_A]| \\ &\leq \sup_{\omega_A \in \mathcal{S}_{n,A}^{\top, p(n)}} (|\text{Tr}[X_A \omega_A]| + |\text{Tr}[Y_A \omega_A]|) \\ &\leq \|X_A\|_{\infty, \text{comp}} + \|Y_A\|_{\infty, \text{comp}}. \end{aligned}$$

Hence $\|\cdot\|_{\infty, \text{comp}}$ is a seminorm on $\text{Herm}(\mathcal{H}_n^A)$.

Assume now that

$$\text{span}_{\mathbb{R}}(\mathcal{S}_{n,A}^{\top, p(n)}) = \text{Herm}(\mathcal{H}_n^A).$$

If $\|X_A\|_{\infty, \text{comp}} = 0$, then

$$\text{Tr}[X_A \omega_A] = 0 \quad \forall \omega_A \in \mathcal{S}_{n,A}^{\top, p(n)}.$$

By linearity, the same holds for every element of $\text{span}_{\mathbb{R}}(\mathcal{S}_{n,A}^{\top, p(n)}) = \text{Herm}(\mathcal{H}_n^A)$. Since the Hilbert–Schmidt inner product is non-degenerate on $\text{Herm}(\mathcal{H}_n^A)$, it follows that $X_A = 0$. Therefore $\|\cdot\|_{\infty, \text{comp}}$ is a norm.

Under the same assumption, the cone $\mathcal{C}_{n,A}^{\mathcal{S}^\top}$ is proper, and the corresponding dual order is well defined. Let $t \geq 0$. Then

$$-t \mathbb{1}_A \leq_{(\mathcal{C}_{n,A}^{\mathcal{S}^\top})^*} X_A \leq_{(\mathcal{C}_{n,A}^{\mathcal{S}^\top})^*} t \mathbb{1}_A \iff \text{Tr}[(t \mathbb{1}_A \pm X_A)\eta_A] \geq 0 \quad \forall \eta_A \in \mathcal{C}_{n,A}^{\mathcal{S}^\top}.$$

Since $\mathcal{C}_{n,A}^{\mathcal{S}^\top} = \text{cone}(\mathcal{S}_{n,A}^{\top,p(n)})$ and every $\omega_A \in \mathcal{S}_{n,A}^{\top,p(n)}$ has trace one, every nonzero $\eta_A \in \mathcal{C}_{n,A}^{\mathcal{S}^\top}$ can be written as

$$\eta_A = \lambda \bar{\eta}_A$$

for some $\lambda = \text{Tr}[\eta_A] > 0$ and some

$$\bar{\eta}_A \in \left\{ \zeta_A \in \mathcal{C}_{n,A}^{\mathcal{S}^\top} : \text{Tr}[\zeta_A] = 1 \right\} = \text{conv}(\mathcal{S}_{n,A}^{\top,p(n)}).$$

Hence, the preceding condition is equivalent to

$$\text{Tr}[(t \mathbb{1}_A \pm X_A)\bar{\eta}_A] \geq 0 \quad \forall \bar{\eta}_A \in \text{conv}(\mathcal{S}_{n,A}^{\top,p(n)}),$$

or, equivalently,

$$|\text{Tr}[X_A \bar{\eta}_A]| \leq t \quad \forall \bar{\eta}_A \in \text{conv}(\mathcal{S}_{n,A}^{\top,p(n)}).$$

This, in turn, is equivalent to

$$\sup_{\omega_A \in \mathcal{S}_{n,A}^{\top,p(n)}} |\text{Tr}[X_A \omega_A]| \leq t.$$

Indeed, one implication is immediate since $\mathcal{S}_{n,A}^{\top,p(n)} \subseteq \text{conv}(\mathcal{S}_{n,A}^{\top,p(n)})$. Conversely, if $\bar{\eta}_A = \sum_i p_i \omega_A^{(i)}$ is a convex combination of elements of $\mathcal{S}_{n,A}^{\top,p(n)}$, then

$$|\text{Tr}[X_A \bar{\eta}_A]| = \left| \sum_i p_i \text{Tr}[X_A \omega_A^{(i)}] \right| \leq \sum_i p_i |\text{Tr}[X_A \omega_A^{(i)}]| \leq \sup_{\omega_A \in \mathcal{S}_{n,A}^{\top,p(n)}} |\text{Tr}[X_A \omega_A]|.$$

Taking the infimum over all $t \geq 0$ proves the order-unit formula.

If $X_A \geq 0$, then since every $\omega_A \in \mathcal{S}_{n,A}^{\top,p(n)}$ is positive semidefinite the absolute value is redundant since $\text{Tr}[X_A \omega_A] \geq 0$.

□

Therefore, Lemma 4.41 together with Corollary 4.42 implies that,

$$\hat{H}_{\min}(A)_\rho = -\log \|\rho_A\|_{\infty, \text{comp}}, \quad (4.2.39)$$

mirroring the information-theoretic case.

Finally, the transpose in the preceding formulas is an artifact of the Choi representation used in the reduction to the non-conditional setting, rather than an intrinsic feature of the quantity. If the efficient state family is closed under computational-basis transpose up to polynomial overhead, then there exists a polynomial q such that

$$\mathcal{S}_{n,A}^{\top,p(n)} \subseteq \mathcal{S}_{n,A}^{q(n)} ,$$

so that, after replacing $p(n)$ by $q(n)$, the same optimizations may be written directly in terms of efficiently preparable states, without an explicit transpose.

Taken together, these observations show that the non-conditional computational min-entropy is governed by a computational analogue of the operator norm. For positive operators, it is given by the largest overlap with an efficiently preparable state, while under the spanning assumption, it extends to a genuine norm on $\text{Herm}(\mathcal{H}_n^A)$ through the associated order-unit structure. In this sense, the non-conditional reduction preserves the familiar interpretation of min-entropy as the logarithm of an operator norm, with the usual operator norm replaced by its complexity-constrained counterpart.

PART II

**COMPUTATIONAL SEPARATIONS
AND APPLICATIONS**

Chapter 5

Computational and Informational Separations

Once we have introduced the computational divergences and entropies, and shown that their operational meaning mirrors their information-theoretic counterparts, we can study how far the two settings can be from each other. In this chapter, we provide separations between the information-theoretic and computationally constrained quantities introduced in Chapter 4. We first focus on the single-system setting in Section 5.1, where a cryptographic construction gives a direct separation for computational divergences. We then move to bipartite states and study separations for the computational conditional min-entropy, first for unrestricted state constructions in Section 5.2, and then for efficiently generated states in Section 5.3, where the separations are obtained from an explicit LWE-based construction of scalable pseudorandom states and pseudorandom density matrices.

5.1. An Explicit Cryptographic Separation

We now exhibit an explicit pair of states that witnesses a sharp separation between the *computational* and *informational* divergences. The same calculations can be extended to other cryptographic constructions which exhibit a gap between the computational trace distance and the informational counterpart. Nevertheless, due to its simplicity, we focus on the pseudoentanglement construction proposed by [GE24]. Let

$$\psi_n^{AB} := \frac{1}{4} \left(|\Phi^+ \rangle \langle \Phi^+|^{AB} + |\Phi^- \rangle \langle \Phi^-|^{AB} \right) \otimes (\rho_{0,n}^A + \rho_{1,n}^A), \quad (5.1.1)$$

$$\phi_n^{AB} := \frac{1}{2} \left(|\Phi^+ \rangle \langle \Phi^+|^{AB} \otimes \rho_{0,n}^A + |\Phi^- \rangle \langle \Phi^-|^{AB} \otimes \rho_{1,n}^A \right), \quad (5.1.2)$$

where $|\Phi^\pm\rangle^{AB} = \frac{1}{\sqrt{2}}(|00\rangle^{AB} \pm |11\rangle^{AB})$ and $\{\rho_{0,n}^A, \rho_{1,n}^A\}_{n \in \mathbb{N}}$ are EFI pairs¹.

Assuming EFI pairs exist, [GE24] proves *poly-copy* computational indistinguishability of the two families, which in our notation implies

$$\hat{\Delta}(\psi_n^{AB}, \phi_n^{AB}) \leq \text{negl}(n).$$

¹An EFI pair [BCQ23] consists of two efficiently generated families of quantum states which are far in information-theoretic trace distance ($\Delta(\rho_{0,n}, \rho_{1,n}) \approx 1$), which are indistinguishable by computationally bounded adversaries given $m \in \text{poly}(n)$ copies, i.e., $|\Pr[\mathcal{D}(\rho_{n,0}^{\otimes m}) = 1] - \Pr[\mathcal{D}(\rho_{n,1}^{\otimes m}) = 1]| \leq \nu(n)$.

Therefore, by the computational Fuchs–van de Graaf inequality (Lemma 4.23) we obtain

$$1 - \text{negl}(n) \leq \tilde{F}(\psi_n^{AB}, \phi_n^{AB}) \leq 1. \quad (5.1.3)$$

This allows us to calculate the following upper bound for the computational measured Rényi divergence for $\alpha \in (0, 1/2)$,

$$\tilde{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\psi_n^{AB}, \phi_n^{AB}) \leq -\log(1 - \text{negl}(n)), \quad (5.1.4)$$

which follows from Lemma 4.22 together with the monotonicity property of the measured Rényi divergences (Lemma 4.11). Nevertheless, in the information-theoretic setting one has, by the Pinsker inequality, e.g., [Gil10], that

$$D_\alpha(\psi_n^{AB}, \phi_n^{AB}) \gtrsim \frac{\min\{1, \alpha\}}{8 \ln 2},$$

since the information-theoretic trace distance satisfies $\Delta(\psi_n^{AB}, \phi_n^{AB}) \approx 1/2$ and the fidelity is bounded away from 1:

$$\frac{1}{4} \leq F(\psi_n^{AB}, \phi_n^{AB}) \leq \frac{3}{4}. \quad (5.1.5)$$

Therefore, under standard hardness assumptions, the computational measures collapse ($\tilde{\Delta}(\psi_n^{AB}, \phi_n^{AB}) \leq \text{negl}(n)$, $1 - \text{negl}(n) \leq \tilde{F}(\psi_n^{AB}, \phi_n^{AB}) \leq 1$, and $\tilde{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\psi_n^{AB}, \phi_n^{AB}) \leq \text{negl}(n)$ for $\alpha \leq \frac{1}{2}$), whereas their informational counterparts are $\Theta(1)$.²

5.2. Separations Between Computational and Information-Theoretic Notions of Min-Entropy: the Unconstrained Setting

Building on Section 4.2, we use the operational characterization of $\tilde{H}_{\min}(A|B)$ as the *maximal achievable fidelity with the maximally entangled state under complexity-constrained operations* from Theorem 4.36 to derive explicit, quantitative bounds on $\tilde{H}_{\min}(A|B)$ for specific families of states. In particular, we focus on fully quantum states and analyze two complementary scenarios, as summarized in Figure 5.1.

In Section 5.2.1, we study achievable gaps using pure states. By adapting the Hayashi–Schur entanglement concentration protocol [MH07] to our channel model, we obtain an explicit upper bound on $\tilde{H}_{\min}(A|B)$ from a single efficient, state-independent distillation protocol [LREJ25]. As was observed in [LREJ25], this analysis can be reduced to two regimes. If $H_{\min}(A)_\rho \leq O(\log n)$, then the computational min-entropy is guaranteed to scale in a similar fashion to the information-theoretic min-entropy. Conversely, for larger

²Indeed, the gap can be polynomially amplified by taking poly-many copies of pseudoentangled states, as proven in [GE24, Lemma 5.1].

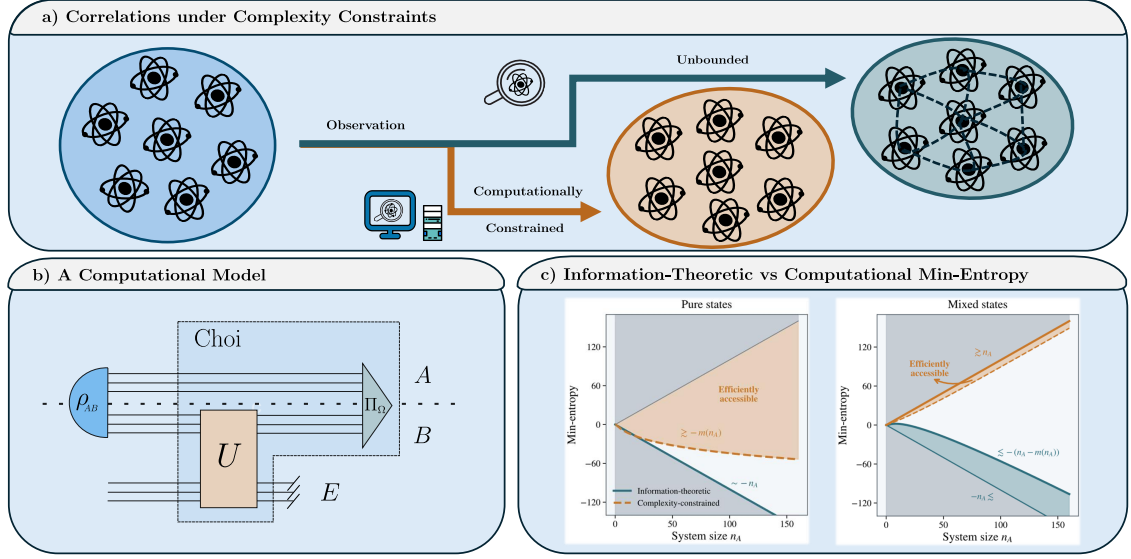


Figure 5.1.: Overview of the bipartite separations studied in this chapter. **(a)** A bipartite state ρ_{AB} may contain entanglement or quantum side information that is present information-theoretically but only partially available to a computationally bounded observer. **(b)** To model this restriction, we use the complexity-constrained channel framework introduced in Section 3.1. **(c)** The resulting computational min-entropy, introduced in Section 4.2, can exhibit a gap with respect to its information-theoretic counterpart.

$H_{\min}(A)_{|\psi\rangle}$, the same universal protocol saturates and $\tilde{H}_{\min}(A|B)$ scales logarithmically with respect to the number of copies. We then show that this saturation is essentially unavoidable in the worst case by constructing families of highly entangled pure states for which polynomial-time operations fail to access the underlying $\Theta(n)$ entanglement, leaving only a polylogarithmic amount of efficiently distillable entanglement.

Afterwards, in Section 5.2.2, we show that mixed states can hide correlations even more effectively. Via the generalized Hilbert–Schmidt ensemble [HLW06, CN15, Hal98, ZPNC11, BMB⁺25], we construct mixed states with highly negative information-theoretic conditional min-entropy for which $\tilde{H}_{\min}(A|B)$ is nearly maximal. This means that for any efficient observer holding B , the state essentially appears maximally mixed.

5.2.1. Separation for Pure States

We start by proving an upper bound on the computational min-entropy for pure states. The bound is achieved by constructing an efficient Choi operator that takes as input k copies of an arbitrary pure state $|\psi_{AB}\rangle$. We note that the Choi operator is independent of the state $|\psi_{AB}\rangle$. In this sense, we say the operator is state agnostic or that the state is unknown to the protocol.

Given a pure bipartite quantum state $|\psi_{AB}\rangle \in \mathcal{H}_n^A \otimes \mathcal{H}_n^B$, it follows from Theorem 4.36 that the computational min-entropy of the state $|\psi_{AB}\rangle^{\otimes k}$ is given by

$$\tilde{H}_{\min}(A^k|B^k)_{|\psi\rangle^{\otimes k}} = -\log \left(d_{A_k} \cdot \max_T \left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes T)(|\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k}) \right| \Omega_{A_k A'_k} \right\rangle \right). \quad (5.2.1)$$

We first show that there exists a single efficient channel T , independent of the input pure state, which yields the desired upper bound on $\tilde{H}_{\min}(A^k|B^k)$. The procedure is inspired by [LREJ25, Algorithm 1], using the entanglement distillation protocol proposed in [MH07]. This protocol is state-agnostic: Alice and Bob perform local Schur sampling described in Section 2.7 on k copies, getting the state the following decomposition:

$$|\psi_{AB}\rangle^{\otimes k} = \sum_{\lambda \in \mathcal{I}(k,d)} \sqrt{\text{Pr}(\lambda)} \left| \Phi_{AB}^\lambda \right\rangle \otimes \left(U_{\mathcal{V}_\lambda}^A \otimes U_{\mathcal{V}_\lambda}^B \right) \left| \phi_{AB}^+ \right\rangle,$$

where $\left| \Phi_{AB}^\lambda \right\rangle \in \mathcal{U}_\lambda^A \otimes \mathcal{U}_\lambda^B$, and $\left| \phi_{AB}^+ \right\rangle \in \mathcal{V}_\lambda^A \otimes \mathcal{V}_\lambda^B$ is a maximally entangled state of Schmidt rank $\dim \mathcal{V}_\lambda$, as discussed in Lemma 2.47. Then, both keep the multiplicity registers $\mathcal{V}_\lambda^A, \mathcal{V}_\lambda^B$, i.e., the ones corresponding to the irreps of the symmetric group and discard the irrep registers $\mathcal{U}_\lambda^A, \mathcal{U}_\lambda^B$, i.e., the ones corresponding to the unitary group. Finally, they apply the inverse Schur transform on $\mathcal{V}_\lambda$ to return to the computational basis. The overall procedure is efficient whenever the Schur transform can be implemented efficiently. In particular, it is efficient for $k = \text{poly}(n)$ [LREJ25]. In order to apply these results to bound the computational min-entropy, we first prove that this protocol can be adapted to the channel structure underpinning the maps T .

Lemma 5.1 (Channel reduction for the Hayashi–Schur concentration map of [MH07]). *Fix the computational basis on $\mathcal{H}_n^{A_k} \simeq \mathcal{H}_n^{A'_k}$ and let*

$$|\Omega\rangle_{AA'} = \frac{1}{\sqrt{d_{A_k}}} \sum_{x=1}^{d_{A_k}} |x\rangle_A |x\rangle_{A'} \quad (5.2.2)$$

be the maximally entangled state, where $d_{A_k} := \dim \mathcal{H}_n^{A_k}$. Let $\Lambda : \mathcal{B}(\mathcal{H}_n^{A_k} \otimes \mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}(\mathcal{H}_n^{A_k} \otimes \mathcal{H}_n^{A'_k})$ be the CPTP map implementing the above described protocol. Assume moreover that the Schur transform, the inverse Schur transform, and their transposes in the fixed computational basis admit polynomial-size implementations. Then there exists an efficiently implementable CPTP map $T : \mathcal{B}_1(\mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}_1(\mathcal{H}_n^{A'_k})$ such that for every bipartite pure state $|\psi_{AB}\rangle \in \mathcal{H}_n^A \otimes \mathcal{H}_n^B$,

$$\langle \Omega_{AA'} | (\text{Id}_{A_k} \otimes T) \left(|\psi\rangle\langle\psi|_{AB}^{\otimes k} \right) | \Omega_{AA'} \rangle = \langle \Omega_{AA'} | \Lambda \left(|\psi\rangle\langle\psi|_{AB}^{\otimes k} \right) | \Omega_{AA'} \rangle. \quad (5.2.3)$$

Proof. Let us first provide the CPTP description of the algorithm that first implements the Schur sampling, subsequently discards $\mathcal{U}_\lambda$, and lastly applies the inverse Schur transform on $\mathcal{V}_\lambda$ described by [MH07, LREJ25]. For each partition $\lambda \vdash k$, let Π_λ^A and Π_λ^B denote the

local Schur–Weyl projectors on $\mathcal{H}_n^{A_k}$ and $\mathcal{H}_n^{B_k}$. Fix the orthonormal bases $\{|u\rangle\}_u$ of $\mathcal{H}_{U_\lambda^A}$ and $\{|v\rangle\}_v$ of $\mathcal{H}_{U_\lambda^B}$. Let

$$W_\lambda^A : \mathcal{H}_{V_\lambda^A} \rightarrow \mathcal{H}_n^{A_k} \quad \text{and} \quad W_\lambda^{A'} : \mathcal{H}_{V_\lambda^B} \rightarrow \mathcal{H}_n^{A'_k}$$

be the isometries implementing the inverse Schur transform on V_λ extended by 0 outside the λ -block. Let us define Kraus operators

$$A_{\lambda,u} := W_\lambda^A (\langle u|_{\mathcal{U}_\lambda^A} \otimes \mathbb{1}_{V_\lambda^A}) \Pi_\lambda^A, \quad B_{\mu,v} := W_\mu^{A'} (\langle v|_{\mathcal{U}_\mu^B} \otimes \mathbb{1}_{V_\mu^B}) \Pi_\mu^B.$$

Then, the action of the channel on a state $\rho_{AB} \in \mathcal{B}(\mathcal{H}_n^{A_k} \otimes \mathcal{H}_n^{B_k})$ can be written as

$$\Lambda(\rho_{AB}) = \sum_{\lambda,u} \sum_{\mu,v} (A_{\lambda,u} \otimes B_{\mu,v}) \rho_{AB} (A_{\lambda,u} \otimes B_{\mu,v})^*. \quad (5.2.4)$$

It follows that

$$\begin{aligned} \langle \Omega | \Lambda(\rho_{AB}) | \Omega \rangle &= \sum_{\lambda,\mu,u,v} \langle \Omega | (A_{\lambda,u} \otimes B_{\mu,v}) \rho_{AB} (A_{\lambda,u} \otimes B_{\mu,v})^* | \Omega \rangle \\ &= \sum_{\lambda,\mu,u,v} \langle \Omega | (\mathbb{1} \otimes A_{\lambda,u}^\top B_{\mu,v}) \rho_{AB} (\mathbb{1} \otimes (A_{\lambda,u}^\top B_{\mu,v})^*) | \Omega \rangle. \end{aligned}$$

where the second equality follows from the ‘transpose trick’ [KW24, Eq. 2.2.40], i.e., for all $A \in \mathbb{C}^{d \times d}$, $(A \otimes \mathbb{1}) |\Omega_{AA'}\rangle = (\mathbb{1} \otimes A^\top) |\Omega_{AA'}\rangle$. Denoting

$$C_{\lambda,\mu,u,v} := A_{\lambda,u}^\top B_{\mu,v}, \quad \text{and} \quad T(X) := \sum_{\lambda,\mu,u,v} C_{\lambda,\mu,u,v} X C_{\lambda,\mu,u,v}^*,$$

we see that

$$\begin{aligned} \langle \Omega | \Lambda(\rho_{AB}) | \Omega \rangle &= \sum_{\lambda,\mu,u,v} \langle \Omega | (\mathbb{1} \otimes C_{\lambda,\mu,u,v}) \rho_{AB} (\mathbb{1} \otimes C_{\lambda,\mu,u,v}^*) | \Omega \rangle \\ &= \langle \Omega | (\text{Id} \otimes T)(\rho_{AB}) | \Omega \rangle, \end{aligned}$$

where the channel $T(X)$ is CP by construction. Next, we prove that this operation is trace preserving. First, using $\sum_u |u\rangle \langle u| = \mathbb{1}_{\mathcal{U}_\lambda^A}$ and that W_λ^A is an isometry, we have

$$\sum_u A_{\lambda,u}^* A_{\lambda,u} = \Pi_\lambda^A.$$

Hence, $\sum_{\lambda,u} A_{\lambda,u}^* A_{\lambda,u} = \sum_\lambda \Pi_\lambda^A = \mathbb{1}_{\mathcal{H}_n^{A_k}}$. A similar argument shows that $\sum_{\mu,v} B_{\mu,v}^* B_{\mu,v} = \mathbb{1}_{\mathcal{H}_n^{B_k}}$. Therefore,

$$\begin{aligned} \sum_{\lambda,\mu,u,v} C_{\lambda,\mu,u,v}^* C_{\lambda,\mu,u,v} &= \sum_{\mu,v} B_{\mu,v}^* \left(\sum_{\lambda,u} A_{\lambda,u}^* A_{\lambda,u} \right)^\top B_{\mu,v} \\ &= \sum_{\mu,v} B_{\mu,v}^* \mathbb{1}_{\mathcal{H}_n^{A'_k}} B_{\mu,v} = \sum_{\mu,v} B_{\mu,v}^* B_{\mu,v} = \mathbb{1}_{\mathcal{H}_n^{B_k}}. \end{aligned}$$

This proves that T is CPTP. Moreover, by the assumption that the Schur transform, the inverse Schur transform, and their transposes admit polynomial-size implementations, T is efficiently implementable. □

Moreover, the aforementioned algorithm can be further simplified. Let us now consider a fixed state $\rho_{AB} := |\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k}$. By [LREJ25, Lemma 12], it holds that

$$(\Pi_\lambda^A \otimes \Pi_\mu^B) \rho_{AB} (\Pi_\lambda^A \otimes \Pi_\mu^B) = 0 \quad \text{for all } \lambda \neq \mu. \quad (5.2.5)$$

In particular, when evaluating $\langle\Omega| \Lambda(\rho_{AB}) |\Omega\rangle$ we may therefore restrict to the diagonal $\mu = \lambda$ terms in Eq. (5.2.4). Given k copies of an unknown bipartite pure state $|\psi_{AB}\rangle^{\otimes k}$, there exists a state-agnostic CPTP map $\Lambda : \mathcal{B}(\mathcal{H}_n^{A_k} \otimes \mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}(\mathcal{H}_n^{A_k} \otimes \mathcal{H}_n^{A'_k})$ implementing the Hayashi–Schur entanglement concentration protocol. In particular, its output on $|\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k}$ is block-diagonal in the Schur label and can be written as

$$\Lambda \left(|\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k} \right) = \sum_{\lambda \vdash k} \Pr(\lambda) \left| \phi_\lambda^+ \right\rangle \left\langle \phi_\lambda^+ \right|, \quad (5.2.6)$$

where $|\phi_\lambda^+\rangle$ is a maximally entangled state supported on the multiplicity subspace $\mathcal{V}_\lambda^A \otimes \mathcal{V}_\lambda^{A'}$ (hence corresponding to $\log \dim \mathcal{V}_\lambda$ ebits), the summands have pairwise orthogonal support for distinct λ , and $\Pr(\lambda) := \text{tr}[\Pi_\lambda^A \rho_A^{\otimes k}]$ with $\rho_A := \text{tr}_B |\psi_{AB}\rangle\langle\psi_{AB}|$. Moreover, by Lemma 5.1, the overlap with $|\Omega_{AA'}\rangle$ satisfies

$$\langle\Omega_{AA'}| \Lambda \left(|\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k} \right) |\Omega_{AA'}\rangle = \langle\Omega_{AA'}| (\text{Id}_{A_k} \otimes T) \left(|\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k} \right) |\Omega_{AA'}\rangle$$

for a CPTP map $T : \mathcal{B}_1(\mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}_1(\mathcal{H}_n^{A'_k})$ acting only on Bob’s system. Moreover, as proven in [CHW07, Kro19], the Schur transform can be approximated on a quantum computer to precision δ in time $O(\text{poly}(k, n, \log(1/\delta)))$; see Theorem 2.46. If, in addition, the chosen gate set $\mathcal{G}$ is closed under transpose in the fixed computational basis³ (or more generally closed under transpose up to constant overhead), then the transpose of such an implementation has the same asymptotic gate complexity. Therefore the transpose operations appearing in Lemma 5.1 do not change the efficiency of the resulting channel T , which can be implemented in polynomial time.

Theorem 5.2 (Upper bound on $\hat{H}_{\min}^c$ for pure states). *Given a bipartite state $|\psi_{AB}\rangle \in \mathcal{H}_n^A \otimes \mathcal{H}_n^B$ with min-entropy $H_{\min}(A)_\rho = -\log \|\rho_A\|_\infty$, where $\rho_A := \text{tr}_B |\psi_{AB}\rangle\langle\psi_{AB}|$, then its computational conditional min-entropy obeys*

$$\hat{H}_{\min}^c(A^k|B^k)_{|\psi\rangle^{\otimes k}} \leq -\frac{k}{4} \min\{H_{\min}(A)_\rho, \log k\} + \log \frac{3}{2} + \text{negl}(n), \quad (5.2.7)$$

where $k = \text{poly}(n)$ is the number of input copies.

Proof. The proof is given in Section A.3.1. □

³This holds, for example, for the standard gate set $\{H, S, CNOT\}$.

Let us now study the converse bound. To do so, let us first introduce two families of states that are statistically indistinguishable given polynomially many copies but for which there exists a gap in their information-theoretic conditional min-entropy:

$$\begin{aligned}\mathcal{E}_{\text{Haar}} &= \{|\psi_{AB}\rangle, \quad |\psi_{AB}\rangle \sim \text{Haar}(n)\}, \\ \mathcal{E}_m &= \{U_\pi|0\rangle^{n-m}|\phi_m\rangle : U_\pi \in P, |\phi_m\rangle \sim \text{Haar}(m)\},\end{aligned}\tag{5.2.8}$$

where $\mathcal{E}_m$ is a Haar-subsystem state ensemble, with P is the set of unitaries implementing a random permutation on n -bit strings $\pi : \{0, 1\}^n \rightarrow \{0, 1\}^n$, $U_\pi|x\rangle = |\pi(x)\rangle$. Then, given a function $m : \mathbb{N} \rightarrow \mathbb{N}$ such that $m = \omega(\log n_A)$ and $\mathcal{H}_A \simeq \mathcal{H}_B$, where $n = n_A + n_B$, both ensembles are statistically indistinguishable given access to $k = \text{poly}(n)$ many copies, as proven in [LREJ25, Lemma 16].

Let us now calculate their corresponding values of the (information-theoretic) min-entropy.

Lemma 5.3 (Min-entropy of a Haar-random state). *Let $k \in \text{poly}(n)$. For a state $|\psi_{AB}\rangle \in \mathcal{H}_n^A \otimes \mathcal{H}_n^B$ drawn from the Haar random ensemble $\mathcal{E}_{\text{Haar}}$, one has, almost surely,*

$$H_{\min}(A^k | B^k)_{|\psi\rangle^{\otimes k}} = -k(n_A - O(1)).\tag{5.2.9}$$

Proof. We defer the proof, which follows from the asymptotic spectral distribution of Haar-random reduced states, to Section A.3.2. $\square$

Let us now bound the case in which the state is given by the Haar-subsystem state ensemble.

Lemma 5.4 (Min-entropy of a Haar-subsystem random state). *Let $k \in \text{poly}(n)$. For a state $|\psi_{AB}\rangle \in \mathcal{H}_n^A \otimes \mathcal{H}_n^B$ drawn from the Haar-subsystem random ensemble $\mathcal{E}_{m(n_A)}$ of Eq. (5.2.8), one has, almost surely,*

$$H_{\min}(A^k | B^k)_{|\psi\rangle^{\otimes k}} \geq -k m(n_A).\tag{5.2.10}$$

Proof. Similarly to the proof of Lemma 5.3, let us express the conditional min-entropy of a pure state in the following way:

$$H_{\min}(A|B)_{|\psi\rangle} = -2 \log(\text{Tr} \sqrt{\rho_A}),$$

where $\rho_A = \text{tr}_B(|\psi_{AB}\rangle\langle\psi_{AB}|)$. On the other hand, when $|\psi_{AB}\rangle \sim \mathcal{E}_m$, we use the deterministic bound

$$\text{Tr} \sqrt{\psi_A} \leq \sqrt{\text{rank}(\psi_A)} \leq 2^{m/2},$$

and therefore,

$$H_{\min}(A|B)_{|\psi\rangle \sim \mathcal{E}_m} \geq -2 \log(2^{m/2}) = -m.\tag{5.2.11}$$

Lastly, since in the informational case the conditional min-entropy is additive, one has that

$$H_{\min}(A^k|B^k)_{|\psi\rangle^{\otimes k}} = k \cdot H_{\min}(A|B)_{|\psi\rangle}.$$

□

Let us now prove the main theorem of this section. We characterize the computational min-entropy for a Haar random state (see Eq.(5.2.8)) and compare it with the information-theoretic one. This yields a large separation between the two entropies. The separation holds unconditionally when one is restricted to polynomially many copies and polynomially generated quantum channels.

Theorem 5.5 (Lower bound on the computational min-entropy for pure states). *Fix an $A|B$ bipartition with $n_A = n_B = \frac{n}{2}$, a function $m : \mathbb{N} \rightarrow \mathbb{N}$ such that $m(n_A) = \omega(\log n_A)$ and let $k \in \text{poly}(n)$ be the number of copies of a given state. Then, there exists a family of bipartite pure states $\{|\psi_{AB,n}\rangle\}_n$ such that*

$$\begin{aligned} H_{\min}(A^k|B^k)_{|\psi_n\rangle^{\otimes k}} &= -k(n_A - O(1)), \\ \overset{c}{H}_{\min}(A^k|B^k)_{|\psi_n\rangle^{\otimes k}} &\geq -k m(n_A). \end{aligned}$$

Proof. We prove this in Section A.3.3 by combining the moment comparison between $\mathcal{E}_{\text{Haar}}$ and $\mathcal{E}_m$ with the information-theoretic min-entropy estimates of Lemmas 5.3 and 5.4. □

In the following, we summarize bounds of this form by writing

$$\overset{c}{H}_{\min}(A^k|B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} \geq -k \cdot \omega(\log n_A),$$

meaning that the above inequality holds for every function $m(n_A) = \omega(\log n_A)$. Similarly, an estimate of the form

$$\overset{c}{H}_{\min}(A^k|B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} \leq -k \cdot \Omega(\log k)$$

means that there exists a constant $C > 0$ such that

$$\overset{c}{H}_{\min}(A^k|B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} \leq -C \cdot k \cdot \log k$$

for all sufficiently large n_A . Then, given $k = \text{poly}(n)$ copies, there exists a family of states $\{|\psi_n\rangle\}_n$ whose computational conditional min-entropy is given by

$$-k \cdot \omega(\log n) \leq \overset{c}{H}_{\min}(A^k|B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} \leq -k \cdot \Omega(\log k) \quad (5.2.12)$$

Therefore, since k is polynomial in n we can write

$$\overset{c}{H}_{\min}(A^k|B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} = -\Theta(k \cdot \log k), \quad (5.2.13)$$

while its informational one satisfies

$$H_{\min}(A^k|B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} = -k \cdot (n_A - O(1)).$$

By using the subadditivity property of the $\mathring{H}_{\min}(A|B)$ one gets that the computational min-entropy per copy is given by

$$\mathring{H}_{\min}(A|B)_{|\psi_{AB,n}\rangle} \geq -\omega(\log n_A), \quad (5.2.14)$$

while, in the informational case, one has that

$$H_{\min}(A|B)_{|\psi_{AB,n}\rangle} = -n_A + O(1). \quad (5.2.15)$$

5.2.2. Separation for Mixed States

In this section we turn to mixed states. We begin by extending our notion of a random ensemble beyond pure states, using the canonical construction obtained by partially tracing out m qubits from a Haar-random $(n+m)$ -qubit state. The resulting induced ensemble, also known as the GHSE ensemble [HLW06, CN15, Hal98, ZPNC11, BMB⁺25], will be our main object of study.

Definition 5.6 (Generalized Hilbert-Schmidt ensemble (GHSE) [BMB⁺25]). Let $\text{Haar}(n+m)$ be the Haar measure on pure states of $n+m$ qubits. Then, the *GHSE* ensemble is defined as

$$\eta_{n,m} := \{\text{tr}_m(|\psi\rangle\langle\psi|)\}_{|\psi\rangle \sim \text{Haar}(n+m)}.$$

Similarly to Theorem 5.5, one can explicitly characterize the computational min-entropy of the GHSE ensemble. For mixed states, however, one can obtain a larger gap. We obtain that when restricted to a polynomial number of copies and to polynomially implementable quantum operations, the corresponding computational min-entropy unconditionally scales as that of a maximally mixed state (up to negligible error) while its information-theoretic min-entropy is upper bounded by $-k \cdot (n_A - \omega(\log n))$, where k is the number of copies. Therefore, under complexity-constrained restrictions, one can get an almost maximal gap between the computational and the information-theoretic notion of conditional min-entropy.

Theorem 5.7 (Lower bound on the computational min-entropy for mixed states). *Fix an $A|B$ bipartition with $n_A = n_B$, a function $m : \mathbb{N} \rightarrow \mathbb{N}$ such that $m(n_A) = \omega(\log n_A)$ and let $k \in \text{poly}(n)$ be the number of copies of a given state. Then, there exists a family of bipartite mixed states $\{\rho_{n,m}\}_n$ such that*

$$\begin{aligned} H_{\min}(A^k|B^k)_{\rho_{n,m}^{\otimes k}} &\leq -k \cdot (n_A - m(n_A)) + \text{negl}(n), \\ \mathring{H}_{\min}(A^k|B^k)_{\rho_{n,m}^{\otimes k}} &\geq k \cdot n_A - \text{negl}(n). \end{aligned}$$

Proof. The proof can be found in Section A.3.4. It uses the GHSE moment decomposition of [BMB⁺25, Theorem D.1] to bound the computational term, while the information-theoretic one follows from entropy estimates on the purification of Definition 5.6. $\square$

Then, given $k = \text{poly}(n)$ copies, there exists a family of mixed states $\{\rho_{n,m}\}_n$ whose computational conditional min-entropy is asymptotically maximal:

$$\tilde{H}_{\min}(A^k|B^k)_{\rho_{n,m}^{\otimes k}} \geq k \cdot n_A - \text{negl}(n),$$

while its information-theoretic conditional min-entropy remains highly negative:

$$H_{\min}(A^k|B^k)_{\rho_{n,m}^{\otimes k}} \leq -k \cdot (n_A - m(n_A)) + \text{negl}(n).$$

By subadditivity of $\tilde{H}_{\min}(A|B)$ and additivity of the ordinary conditional min-entropy, this yields the corresponding per-copy bounds

$$\tilde{H}_{\min}(A|B)_{\rho_{n,m}} \geq n_A - \text{negl}(n), \quad H_{\min}(A|B)_{\rho_{n,m}} \leq -(n_A - m(n_A)) + \text{negl}(n).$$

Thus, in the mixed-state regime, the computational conditional min-entropy is nearly maximal even though the information-theoretic conditional min-entropy remains highly negative, yielding an almost maximal separation between operationally accessible and information-theoretic side information.

5.3. Separations Between Computational and Information-Theoretic Notions of Min-Entropy: the Efficient Setting

The gap between the information-theoretic and the computational conditional min-entropies can also be extended to the case where the states witnessing the separation are efficiently generated. This will hold under the assumption that the Learning With Errors (LWE) problem is hard for quantum polynomial-time algorithms. More specifically, we will work with Asymptotically Random States (ARS) [BS19]: an ARS generator assumes quantum oracle access to a random function and produces states that are statistically indistinguishable from Haar-random states. These generators allow us to construct *scalable* pseudorandom states (PRS), for which the security parameter is independent of the size of the generated state [BS20, BCJZ26]. Then, by replacing the random function by a quantum-secure pseudorandom function (PRF), scalable PRS can be efficiently generated.

In Section 5.3.1, we introduce the cryptographic background necessary for the construction of these pseudorandom states. Moreover, in Lemma 5.15, we provide a concrete instantiation of this functionality based on a specific hardness problem: subexponential LWE. While the same gap could be obtained by assuming the coarse-grained hardness of the PRS functionality, we give this explicit instantiation for completeness.

Lastly, in Section 5.3.2, we prove the main results of this section: the separations for efficiently generated states in both the pure- and mixed-state settings.

5.3.1. Pseudorandom States and Learning with Errors

We start by defining the basic cryptographic primitive that allows us to provide the efficient gap: (scalable) pseudorandom quantum states.

Definition 5.8 (Scalable Pseudorandom State (PRS) Generator [BS20, BCJZ26]). Let $\lambda \in \mathbb{N}$ be the security parameter. We say that a pair of polynomial-time quantum algorithms (K, G) is a scalable pseudorandom-state (PRS) generator if, for any $n \in \mathbb{N}$, the following holds:

- **Key generation:** $K(1^n, 1^\lambda)$ is a polynomial algorithm which outputs a classical key s .
- **State generation:** Given s in the support of $K(1^n, 1^\lambda)$, the algorithm $G(1^n, 1^\lambda, s)$ outputs an n -qubit quantum state.
- **Quantum pseudorandomness:** For any non-uniform QPT distinguisher $\mathcal{D}$ and any $k \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda)$ such that

$$\left| \Pr_{\substack{s \leftarrow K(1^n, 1^\lambda) \\ \psi \leftarrow G(1^n, 1^\lambda, s)}} [\mathcal{D}(\psi^{\otimes k}) = 1] - \Pr_{\psi \sim \mathcal{E}_{\text{Haar}}} [\mathcal{D}(\psi^{\otimes k}) = 1] \right| \leq \nu(\lambda). \quad (5.3.1)$$

The classical analogue of pseudorandom states is given by pseudorandom functions (PRFs). Since our distinguishers are quantum, we will need the corresponding notion of quantum-secure pseudorandom functions.

Definition 5.9 (Quantum-Secure Pseudorandom Function (QPRF) [BS20]). Let $L : \mathbb{N} \rightarrow \mathbb{N}$ be polynomially bounded, let $\mathcal{K} = \{\mathcal{K}_\lambda\}_{\lambda \in \mathbb{N}}$ be an efficiently samplable family of key distributions, and let

$$\text{PRF}_\lambda : \mathcal{K}_\lambda \times \{0, 1\}^\lambda \longrightarrow \{0, 1\}^{L(\lambda)}$$

be efficiently computable. We say that PRF is a quantum-secure pseudorandom function if, for every non-uniform QPT distinguisher $\mathcal{D} = \{\mathcal{D}_\lambda\}_{\lambda \in \mathbb{N}}$ with quantum oracle access, there exists a negligible function $\nu(\lambda)$ such that, for every $\lambda \in \mathbb{N}$,

$$\left| \Pr_{s \leftarrow \mathcal{K}_\lambda} [\mathcal{D}_\lambda^{\text{PRF}_\lambda(s, \cdot)}(1^\lambda) = 1] - \Pr_{f \leftarrow (\{0, 1\}^{L(\lambda)})^{\{0, 1\}^\lambda}} [\mathcal{D}_\lambda^f(1^\lambda) = 1] \right| \leq \nu(\lambda). \quad (5.3.2)$$

We call a QPRF *scalable* if its input and output lengths can be chosen as arbitrary polynomially bounded functions of the security parameter [BCJZ26, Definition 3.27]. Such scalable QPRFs exist assuming quantum-secure one-way functions [BCJZ26, Fact 3.29].

Moreover, PRS can be instantiated from quantum-secure PRFs. In this chapter, we will focus on the construction based on Asymptotically Random State (ARS) generators.

Definition 5.10 (Asymptotically Random State (ARS) Generator [BS20]). A QPT algorithm Gen with input $(1^n, 1^\lambda)$ for $n, \lambda \in \mathbb{N}$ and quantum oracle access to $U_f : |x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$ for $f : \{0, 1\}^n \rightarrow \{0, 1\}^{\text{poly}(n, \lambda)}$, is an ARS generator if there exists a negligible function $\nu(\lambda)$ such that for every $k \in \text{poly}(\lambda)$, for all natural numbers n, λ ,

$$\Delta(\mathbb{E}_{\substack{|\psi\rangle \leftarrow \text{Gen}^{U_f(1^n, 1^\lambda)} \\ f \leftarrow (\{0, 1\}^{\text{poly}(n, \lambda)})^{\{0, 1\}^n}}} [|\psi\rangle\langle\psi|^{\otimes k}], \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} [|\psi\rangle\langle\psi|^{\otimes k}]) \leq \nu(\lambda).$$

We use the scalable ARS construction of [BCJZ26, Theorem 4.7]. More precisely, this result shows that, for any algorithm making at most k oracle calls to the state-preparation isometry, the ARS output is within trace distance

$$\varepsilon_{\text{ARS}}(n, \lambda, k) \leq (1 + \sqrt{2\pi}) k \sqrt{n} 2^{-\lambda/2} + 2\pi k 2^{-\lambda} + 2^{-\lambda} \quad (5.3.3)$$

from k copies of a Haar-random n -qubit state. Replacing the random function by a (scalable) quantum-secure PRF gives a scalable PRS generator by [BCJZ26, Theorem 5.1]. In the sequel we use this construction.

For the concrete LWE-based instantiation used below, the security bound satisfies the following estimate.

Lemma 5.11 (QPRF instantiation of the scalable ARS generator of [BCJZ26]). *Suppose the scalable ARS generator of [BCJZ26, Theorem 4.7] is instantiated, as in [BCJZ26, Theorem 5.1], with a scalable quantum-secure PRF. Let $\varepsilon_{\text{PRS}}(n, \lambda, k)$ denote the distinguishing advantage of a QPT distinguisher between k copies of the resulting PRS output and k copies of a Haar-random n -qubit state. Then*

$$\varepsilon_{\text{PRS}}(n, \lambda, k) \leq (1 + \sqrt{2\pi}) k \sqrt{n} 2^{-\lambda/2} + 2\pi k 2^{-\lambda} + 2^{-\lambda} + \varepsilon_{\text{QPRF}}(n, \lambda, k), \quad (5.3.4)$$

where $\varepsilon_{\text{QPRF}}$ is the distinguishing advantage for the quantum-secure pseudorandom function as in Eq. (5.3.2).

Proof. Let us prove it by a hybrid argument. Let H_0 be the experiment in which the distinguisher receives k copies generated by the PRS, namely the ARS generator run with oracle access to PRF_s . Let H_1 be the same experiment with the PRF oracle replaced by a truly random function. Let H_2 be k copies of a Haar-random n -qubit state.

By the triangle inequality,

$$|\Pr[H_0 = 1] - \Pr[H_2 = 1]| \leq |\Pr[H_0 = 1] - \Pr[H_1 = 1]| + |\Pr[H_1 = 1] - \Pr[H_2 = 1]|.$$

The second term is bounded by [BCJZ26, Theorem 4.7]. For the first term, a qPRF adversary simulates the k executions of the ARS generator using its oracle, then runs the final distinguisher; see Eq. (5.3.2). $\square$

We now state the concrete post-quantum hardness assumption used in this section. We first recall the relevant LWE distributions.

Definition 5.12 (Matrix distributions). Let $q \geq 2$ be an integer. We write $U_q^{r \times s}$ for the uniform distribution over $\mathbb{Z}_q^{r \times s}$. For $\sigma > 0$, let $D_{q,\sigma}$ denote the discretised Gaussian distribution over $\mathbb{Z}_q$ with width σ , and let $D_{q,\sigma}^{r \times s}$ denote the product distribution over $\mathbb{Z}_q^{r \times s}$ obtained by sampling all entries independently from $D_{q,\sigma}$.

For parameters m, ℓ, q, σ , define the noisy rank- ℓ matrix distribution $L_{q,\ell,\sigma}^{m \times m}$ over $\mathbb{Z}_q^{m \times m}$ as follows:

1. sample $B, C \leftarrow U_q^{\ell \times m}$ independently;
2. sample $E \leftarrow D_{q,\sigma}^{m \times m}$;
3. output $B^T C + E \bmod q$.

Definition 5.13 (Decision LWE). For parameters n, m, q, σ , the problem $\text{LWE}_{n,m,q,\sigma}$ is to distinguish the following two distributions over $\mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$:

$$(A, u), \quad A \leftarrow U_q^{m \times n}, \quad u \leftarrow U_q^m,$$

and

$$(A, As + e), \quad A \leftarrow U_q^{m \times n}, \quad s \leftarrow U_q^n, \quad e \leftarrow D_{q,\sigma}^m.$$

The advantage of an algorithm is the absolute difference between its acceptance probabilities on these two distributions.

We use the following fine-grained, post-quantum LWE assumption. It is a subexponential-strength variant of the standard LWE hardness assumptions; see, e.g., [BLP⁺13, Pei16] for background on LWE parameters and worst-case hardness evidence.

Assumption 5.14 (Subexponential-time LWE). *There exist constants $\alpha > 0$, $\beta > 1$ and $\delta > 0$ such that every quantum algorithm of size $\text{poly}(m)$ has advantage at most*

$$2^{-\Omega(\ell^\alpha)}$$

in solving $\text{LWE}_{\ell,m,q,\sigma}$ for parameters satisfying

$$\begin{aligned} m &= \ell^\beta, \\ q &= \text{poly}(m), \\ 2\sqrt{m} &\leq \sigma \leq q^{1-\delta}. \end{aligned}$$

Moreover, Zhandry proved that the GGM construction of PRFs [GGM86] is quantum-secure when instantiated with a length-doubling PRG secure against quantum distinguishers [Zha21, Theorem 5.5]. Together with the construction of quantum-secure PRGs from quantum-secure one-way functions, this yields quantum-secure PRFs from quantum-secure one-way functions [Zha21, Corollary 1.3].

For the quantitative bounds needed here, however, we avoid this generic route and instead directly construct a length-doubling PRG from Assumption 5.14.

Lemma 5.15 (Length-doubling PRG from subexponential-time LWE). *Assume Assumption 5.14. Then there exists a family of pseudorandom generators (PRG), indexed by a public matrix,*

$$G_{\ell,A} : \{0,1\}^{d_\ell} \rightarrow \{0,1\}^{2d_\ell}, \quad A \in \mathbb{Z}_q^{m \times \ell},$$

where $d_\ell = \ell \log q + m \log \sigma$, such that, for $A \leftarrow U_q^{m \times \ell}$, every QPT distinguisher $D = \{D_\ell\}_{\ell \in \mathbb{N}}$ has advantage at most

$$\left| \Pr_{\substack{A \leftarrow U_q^{m \times \ell} \\ r \leftarrow U_{d_\ell}}} [D_\ell(A, G_{\ell,A}(r)) = 1] - \Pr_{\substack{A \leftarrow U_q^{m \times \ell} \\ u \leftarrow U_{2d_\ell}}} [D_\ell(A, u) = 1] \right| \leq 2^{-\Omega(\ell^\alpha)}.$$

Proof. The proof is given in Section A.3.5. $\square$

Moreover, Zhandry's proof of the quantum security of the GGM construction [Zha21, Theorem 5.5] shows that any quantum-query adversary breaking the GGM PRF on n -bit inputs with advantage ε yields, with only polynomial overhead and a factor n loss in advantage, an oracle-security distinguisher for the underlying length-doubling PRG. By the security conversion for PRGs in [Zha21, Corollary 5.4], which is based on [Zha21, Theorem 4.5], the length-doubling PRG of Lemma 5.15 therefore gives a quantum-secure GGM PRF. Consequently, for n and query complexity $\text{poly}(d_\ell)$, the QPRF distinguishing advantage is bounded, up to the polynomial losses in Zhandry's reduction, by

$$\text{poly}(d_\ell, n) \cdot 2^{-\Omega(\ell^\alpha)} = 2^{-\Omega(\ell^\alpha)},$$

since $d_\ell = \text{poly}(\ell)$.

5.3.2. Separation for Efficiently Generated States

We now prove the main results of this section. Under the hardness assumption introduced in Assumption 5.14, the unconstrained separations of Theorems 5.5 and 5.7 can be reproduced with efficiently generated keyed ensembles. For pure states, Theorem 5.16 replaces Haar-random states by scalable pseudorandom states. For mixed states, Theorem 5.18 uses a scalable pseudorandom density matrix obtained by tracing out part of a scalable PRS, inspired by the construction of [BMB⁺25].

Theorem 5.16 (Lower bound on the computational min-entropy for efficiently generated pure states). *Fix an $A|B$ bipartition with $n_A = n_B = \frac{n}{2}$, a function $m : \mathbb{N} \rightarrow \mathbb{N}$ such that $m(n_A) = \omega(\log n_A)$ and let $k \in \text{poly}(n)$ be the number of copies of a given state. Then, assuming the hardness of subexponential-time LWE (Assumption 5.14) there exists an efficiently generated keyed ensemble of bipartite pure states $\{|\psi_{s,n}\rangle\}_{s,n}$ such that, with probability $1 - \text{negl}(n)$ over the key s ,*

$$\begin{aligned} H_{\min}(A^k | B^k)_{|\psi_{s,n}\rangle^{\otimes k}} &= -k(n_A - O(1)), \\ \tilde{H}_{\min}(A^k | B^k)_{|\psi_{s,n}\rangle^{\otimes k}} &\geq -k m(n_A). \end{aligned}$$

Proof. The argument in Section A.3.6 follows the strategy of Theorem 5.5. The main difference is that the Haar-random state is replaced by an efficiently generated scalable pseudorandom state instantiated with Assumption 5.14. $\square$

The previous theorem provides an efficiently generated analogue of Theorem 5.5. With overwhelming probability over the key, the generated state is nearly maximally entangled from the information-theoretic point of view, while its computational conditional min-entropy remains lower bounded by $-k m(n_A)$. Therefore, under Assumption 5.14, the separation obtained for Haar-random states persists for efficiently generated pseudorandom states. This remains true even when the recovery channel is chosen using the classical description (G, s) of the state-preparation circuit, since such a choice is included among the efficient channels considered in the theorem. This is analogous to the state-aware results of [LREJ25], although here the obstruction concerns the one-sided recovery task defining $\tilde{H}_{\min}(A|B)$ rather than LOCC distillation or dilution.

Similarly to the pure-state case, the separation obtained in Theorem 5.7 for the unconstrained mixed-state construction also persists for efficiently generated states, under the hardness assumption of Assumption 5.14. In this case, the relevant cryptographic primitive is a pseudorandom density matrix (PRDM), which generalizes pseudorandom states to mixed states by requiring computational indistinguishability from the generalized Hilbert–Schmidt ensemble [BMB⁺25]. We use the following scalable version of this cryptographic primitive.

Definition 5.17 (Scalable pseudorandom density-matrix generator). Let $\lambda \in \mathbb{N}$ be the security parameter and let $m = m(n)$ be the mixedness parameter. We say that a pair of polynomial-time quantum algorithms (K, G) is a scalable pseudorandom density-matrix (PRDM) generator if, for any $n, m \in \mathbb{N}$, the following holds:

- **Key generation:** $K(1^n, 1^m, 1^\lambda)$ outputs a classical key s .
- **State generation:** Given s in the support of $K(1^n, 1^m, 1^\lambda)$, the algorithm $G(1^n, 1^m, 1^\lambda, s)$ outputs an n -qubit density matrix $\rho_{s,n,m}$.
- **Quantum pseudorandomness:** For any non-uniform QPT distinguisher $\mathcal{D}$ and any $k \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda)$ such that

$$\left| \Pr_{s \leftarrow K(1^n, 1^m, 1^\lambda)} [\mathcal{D}(\rho_{s,n,m}^{\otimes k}) = 1] - \Pr_{\rho \sim \eta_{n,m}} [\mathcal{D}(\rho^{\otimes k}) = 1] \right| \leq \nu(\lambda), \quad (5.3.5)$$

where $\eta_{n,m}$ denotes the generalized Hilbert–Schmidt ensemble obtained by tracing out m qubits from a Haar-random $(n + m)$ -qubit pure state.

Let us highlight that, unlike the fixed-parameter PRDMs introduced in [BMB⁺25], the security parameter λ in the previous definition is independent of the dimension of the pseudorandom density matrix. This construction allows us to prove the following separation.

Theorem 5.18 (Efficiently generated mixed-state separation). *Fix an $A|B$ bipartition with $n_A = n_B = n/2$, let $m(n_A) = \omega(\log n_A)$ with $m(n_A) \leq n_A$, and let $k \in \text{poly}(n)$ be the number of copies. Fix also a polynomial circuit-size bound $p(n)$ and a constant $c > 0$. Assuming Assumption 5.14, there exists an efficiently generated keyed ensemble of bipartite mixed states $\{\rho_{s,AB,n}\}_{s,n}$ such that, with probability $1 - \text{negl}(n)$ over the key s ,*

$$\begin{aligned} H_{\min}(A^k|B^k)_{\rho_{s,n}^{\otimes k}} &\leq -k(n_A - m(n_A)) + \text{negl}(n), \\ \tilde{H}_{\min}(A^k|B^k)_{\rho_{s,n}^{\otimes k}} &\geq kn_A - O(n^{-c}). \end{aligned}$$

Proof. In Section A.3.7, we adapt the unconstrained mixed-state argument of Theorem 5.7. The use of Assumption 5.14 is the same as in Theorem 5.16: it provides an efficient pseudorandom construction, while tracing out the last registers transfers the GHSE estimates to efficiently generated mixed states. $\square$

Thus, in the efficiently generated setting, we obtain that, with overwhelming probability over the key, the generated states have information-theoretic conditional min-entropy of order $-k(n_A - m(n_A))$, whereas their computational conditional min-entropy is almost maximal. The computational bound is slightly weaker than in the unconstrained mixed-state construction, since the error is $O(n^{-c})$ rather than negligible. This loss comes from the way the high-moment tail bound is applied, where we compare the overlap with $(1 + n^{-c})/d_n^k$. Since $c > 0$ is arbitrary, the error can be made smaller than any prescribed inverse polynomial. Thus the efficiently generated mixed states still exhibit an almost maximal separation between information-theoretic and computationally accessible correlations.

Chapter 6

Applications of Efficient Quantum Measurements

In this chapter we develop applications of the computational divergences introduced in Section 4.1. In Section 6.1 we revisit the quantum Stein's lemma under computational constraints (Theorem 6.2) and show that the optimal one-sided error exponent is upper bounded by the regularized computational measured relative entropy, thereby giving it an operational meaning (see Definition 4.9). In Section 6.2 we define computational quantum resources based on our computational divergences and establish an asymptotic continuity bound for the so-called computational measured relative entropy of a resource (Theorem 6.7). Finally, further focusing on entanglement (Section 6.3), we introduce the *computational measured relative entropy of entanglement* and relate it to the previously proposed computational entanglement measures from [ABV23] in Lemma 6.14.

6.1. Computational Hypothesis Testing

Let us consider the problem of binary hypothesis testing, where one is tasked with discriminating between two input states. The main result of this section is a rigorous study of the optimal type II error decay rate in asymmetric hypothesis testing, commonly known as the Stein's exponent, when restricted to efficient binary measurements [HP91, ON00].

In binary quantum hypothesis testing, the goal is to discriminate between two quantum states; one of the states, ρ , is called the *null hypothesis* H_0 and the other state, σ , is known as the *alternate hypothesis* H_1 . While in the information-theoretic setting one typically studies two fixed states ρ and σ , in our complexity constrained framework the studied objects are families of states $\{\rho_k\}_{k \in \mathbb{N}}$ and $\{\sigma_k\}_{k \in \mathbb{N}}$ of which we are given $m(k) \in \text{poly}(k)$ polynomially many copies.

Thus, we study the distinguishing rate in the asymptotic limit, i.e., when $k \rightarrow \infty$ given $m(k)$ copies of $\rho_k, \sigma_k \in \mathcal{S}(\mathcal{H}_k)$ under the restriction that the observer is constrained to implement efficient binary measurements. We say that the observer determines the system to be in state ρ_k , instead of σ_k if the performed measurement M outputs 0. Conversely, if the output is 1, the observer assumes that they initially held $m(k)$ copies of σ_k .

For any $k, m(k) \in \mathbb{N}$, let $\mathbf{M}_{k \cdot m(k)}^{\text{eff}}$ denote the set of efficient binary measurements acting on $\mathcal{H}_{k \cdot m(k)}$. For any feasible binary measurement $M = (E_{k \cdot m(k)}, \mathbb{1} - E_{k \cdot m(k)}) \in \mathbf{M}_{k \cdot m(k)}^{\text{eff}}$, there are two types of errors that arise:

- **Type-I error:** the probability of incorrectly rejecting the null hypothesis (i.e., mistakenly inferring σ_k when the state is actually ρ_k). It is hence defined as

$$\alpha_{m(k)}(M) := \text{Tr} \left[\rho_k^{\otimes m(k)} (\mathbb{1} - E_{k \cdot m(k)}) \right]. \quad (6.1.1)$$

- **Type-II error:** the probability of incorrectly accepting the null hypothesis (i.e., mistakenly inferring ρ_k when the state is actually σ_k). It is given by

$$\beta_{m(k)}(M) := \text{Tr} \left[\sigma_k^{\otimes m(k)} E_{k \cdot m(k)} \right]. \quad (6.1.2)$$

We are primarily interested in the *asymmetric* setting, where the aim is to minimize the Type-II error while the Type-I error is constrained to not exceed a prescribed threshold $\varepsilon \in (0, 1)$ [HP91, ON00]. Formally, one defines

$$\beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}) := \min_{M \in \mathbf{M}_{k \cdot m(k)}^{\text{eff}}} \{ \beta_{m(k)}(M) \mid \alpha_{m(k)}(M) \leq \varepsilon \}, \quad (6.1.3)$$

to be the optimal type-II error given a type-I error bound by ε . As mentioned above, our goal is to characterize the scaling of $\beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}})$ in the asymptotic regime. In this setting, the optimal Type-II error probability is expected to decay exponentially with the number of copies $m(k) = \mathcal{O}(\text{poly}(n))$, i.e., $\beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}) \sim 2^{-\xi^{\text{eff}} m(k) + o(m(k))}$, where the optimal (asymptotic) error exponent can be expressed as,

$$\xi^{\text{eff}} = \lim_{\varepsilon \rightarrow 0} \limsup_{k \rightarrow \infty} -\frac{1}{m(k)} \log \beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}). \quad (6.1.4)$$

If one adopts the approach from [YHKH⁺22, MKH⁺25], then this exponent can be expressed via the complexity relative entropy, which we define below for completeness.

Definition 6.1 (Complexity Relative Entropy [MKH⁺25]). Let $\mathbf{E}_n^{\text{eff}}$ be the set of efficient effect operators on $\mathcal{H}_n$ that can be generated using at most $p(n)$ elementary gates. Then for any two positive semi-definite operators $\rho, \sigma \in \text{Pos}(\mathcal{H}_n)$ and any $\eta \geq 0$, the *complexity relative entropy* is given by:

$$D_{\text{H}}^{p(n), \eta}(\rho \parallel \sigma) := -\log \inf_{\substack{E \in \mathbf{E}_n^{\text{eff}} \\ \text{Tr}[E\rho] \geq \eta}} \frac{\text{Tr}[E\sigma]}{\text{Tr}[E\rho]}. \quad (6.1.5)$$

It then directly follows from [MKH⁺25, Proposition 1] that the optimal type II error is expressed as

$$\beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}) = (1 - \varepsilon) 2^{-D_{\text{H}}^{p(k \cdot m(k)), 1-\varepsilon}(\rho_k^{\otimes m(k)} \parallel \sigma_k^{\otimes m(k)})}.$$

This yields an optimal error exponent of

$$\begin{aligned}\xi^{\text{eff}} &= \lim_{\varepsilon \rightarrow 0} \limsup_{k \rightarrow \infty} -\frac{\log(1 - \varepsilon)}{m(k)} + \frac{D_{\text{H}}^{p(k \cdot m(k)), 1 - \varepsilon}(\rho_k^{\otimes m(k)} \parallel \sigma_k^{\otimes m(k)})}{m(k)} \\ &= \lim_{\varepsilon \rightarrow 0} \limsup_{k \rightarrow \infty} \frac{D_{\text{H}}^{p(k \cdot m(k)), 1 - \varepsilon}(\rho_k^{\otimes m(k)} \parallel \sigma_k^{\otimes m(k)})}{m(k)}.\end{aligned}\quad (6.1.6)$$

Here, our main contribution is Theorem 6.2, which shows that the optimal error exponent is bounded by the regularized computational measured relative entropy. In doing so, we also establish a connection between the computational measured Rényi divergences of this work and the complexity relative entropy from [MKH⁺25].

Theorem 6.2 (One-Sided Computational Stein’s Lemma). *Let $\{\rho_k\}_k$ and $\{\sigma_k\}_k$ be two families of quantum states with $\rho_k, \sigma_k \in \mathcal{S}(\mathcal{H}_k)$. Let $\{\mathbf{M}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ be informationally complete and polynomially generated. For any $0 < \varepsilon < 1$, the optimal type II error exponent $\xi^{\text{eff}}(\{\rho_k\}_k \parallel \{\sigma_k\}_k)$ for testing $H_0 : \rho_k^{\otimes m(k)}$ against $H_1 : \sigma_k^{\otimes m(k)}$, with $m(k) = \mathcal{O}(\text{poly}(k))$, using efficient measurements satisfies:*

$$\xi^{\text{eff}}(\{\rho_k\}_k \parallel \{\sigma_k\}_k) := \lim_{\varepsilon \rightarrow 0} \limsup_{k \rightarrow \infty} -\frac{1}{m(k)} \log \beta_{m(k)}^{\varepsilon}(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}) \leq \dot{D}^{\text{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \parallel \{\sigma_k\}_k), \quad (6.1.7)$$

where $\beta_{m(k)}^{\varepsilon}(\mathbf{M}_{k \cdot m(k)}^{\text{eff}})$ is the minimal type II error given the type I error $\leq \varepsilon$, and $\dot{D}^{\text{M}^{\text{eff}}, \infty}$ is the regularized computational measured relative entropy.

Proof. To prove this result, we follow the proof strategy of [BaHLP14, Theorem 16], adapting it to the polynomially constrained setting. The proof is included in Section A.4.1 for completeness. $\square$

In the simple case, where ρ, σ are fixed and one distinguishes $\rho^{\otimes m}$ from $\sigma^{\otimes m}$, this reduces to

$$\xi^{\text{eff}}(\rho \parallel \sigma) := \lim_{\varepsilon \rightarrow 0} \limsup_{m \rightarrow \infty} -\frac{1}{m} \log \beta_m^{\varepsilon}(\mathbf{M}_m^{\text{eff}}) \leq \dot{D}^{\text{M}^{\text{eff}}, \infty}(\rho \parallel \sigma).$$

This bound is tight for the original quantum Stein’s lemma [ON00], which one recovers by removing the computational constraints on $\mathbf{M}_k^{\text{eff}}$. Moreover, Lemma 4.20 implies that the regularized computational max-divergence is an upper bound on $\xi^{\text{eff}}(\{\rho_k\}_k \parallel \{\sigma_k\}_k)$.

6.2. Computational Resource Theory

Properties of quantum systems such as entanglement or magic can be studied systematically by *quantum resource theories* [CG19]. As defined in Definition 2.31, a resource theory is a pair $\mathcal{R} = (\mathcal{F}, \mathcal{O})$ where, for any Hilbert spaces $\mathcal{H}, \mathcal{H}_1, \mathcal{H}_2$,

- $\mathcal{F}(\mathcal{H}) \subseteq \mathcal{S}(\mathcal{H})$ is the set of *free states* on $\mathcal{H}$, i.e., those that lack the resource and
- $\mathcal{O}(\mathcal{H}_1, \mathcal{H}_2) \subseteq \text{CPTP}(\mathcal{H}_1, \mathcal{H}_2)$ is the set of *free operations*.

A free operation, $\Lambda \in \mathcal{O}(\mathcal{H}_1, \mathcal{H}_2)$, must minimally satisfy $\Lambda(\sigma) \in \mathcal{F}(\mathcal{H}_2)$ for every $\sigma \in \mathcal{F}(\mathcal{H}_1)$. When studying, e.g., entanglement as a resource, one chooses $\mathcal{F} = \text{SEP}$ (separable states) and $\mathcal{O} = \text{LOCC}$ (local operations + classical communication).

When restricted to polynomially implementable operations, families of states with high resource content can become computationally indistinguishable from low-resource families. This phenomenon, known as pseudoresourced quantum states [HBK25, BMB⁺25, GY26] has a direct consequence: when constrained to polynomially implementable operations, information-theoretic resource measures will generally not be operationally meaningful. For entanglement there exist, e.g., exponential separations between information-theoretic and computational measures [ABF⁺24, ABV23, LREJ25]. Nevertheless, to the best of our knowledge, no general measure for polynomially-constrained resources has been proposed in the literature.

There exist several measures for quantifying resources in quantum information theory, ranging from geometric to witness-based approaches. Typically, these measures assess the resource content of a state by evaluating its “distance” from the set of free states. In this section, we focus on entropic measures, defining the computational counterpart of the *measured relative entropy of resource* and the *max-relative entropy of resource*. Throughout this section, $\mathcal{O}_n(\mathcal{H}_1, \mathcal{H}_2)$ will denote the free operations realized by poly-size circuits, i.e., the “efficient” free operations from the Hilbert space $\mathcal{H}_1$ to $\mathcal{H}_2$. In the same way, $\mathcal{F}_n \equiv \mathcal{F}(\mathcal{H}_n)$ represents the set of free states, i.e. the ones without the resource.

Definition 6.3 (Set of Efficient Free Quantum Operations). Given $\{\mathbf{E}_n^{\text{eff},(i)}\}_{n \in \mathbb{N}}$, where

$$\mathbf{E}_n^{\text{eff},(i)} \subset \mathcal{B}(\mathcal{H}_n^i)$$

are families of informationally complete polynomially generated sets of effect operators on systems $i = 1, 2$, with possibly different polynomials. On the sets of free states $\mathcal{F}(\mathcal{H}_n^1)$ and $\mathcal{F}(\mathcal{H}_n^2)$, a CPTP map $\Lambda_n \in \mathcal{O}(\mathcal{H}_n^1, \mathcal{H}_n^2)$ is said to be an *efficient free quantum operation* if the following two conditions hold.

1. **State condition:** $\Lambda_n(\sigma) \in \mathcal{F}(\mathcal{H}_n^2)$ for all $\sigma \in \mathcal{F}(\mathcal{H}_n^1)$.
2. **Test stability:** $\Lambda_n^*(\mathbf{E}_n^{\text{eff}}(\mathcal{H}_n^2)) \subseteq \overline{\mathbf{E}}_n^{\text{eff}}(\mathcal{H}_n^1)$.

Therefore, the set of free operations can be seen as quantum channels that map free states into free states, not generating the studied resource. The main difference with respect to the information-theoretic case lies in the added complexity constraint: these transformations cannot generate more complexity either. Let us now define computational resource distance measures through our proposed computational divergences.

Definition 6.4 (Computational Measured Relative Entropy of Resource). Given a quantum state $\rho \in \mathcal{S}(\mathcal{H}_n)$, its *computational measured relative entropy of resource* is given by,

$$\mathring{D}_{\mathcal{F}_n}(\rho) := \min_{\sigma \in \mathcal{F}_n} \mathring{D}^{\mathbf{M}_n^{\text{eff}}}(\rho \parallel \sigma). \quad (6.2.1)$$

Definition 6.5 (Computational Max-Relative Entropy of Resource). Given a quantum state $\rho \in \mathcal{S}(\mathcal{H}_n)$, its *computational max-relative entropy of resource* is given by,

$$\mathring{D}_{\max, \mathcal{F}_n}(\rho) := \min_{\sigma \in \mathcal{F}_n} \mathring{D}_{\max}(\rho \parallel \sigma). \quad (6.2.2)$$

In the information-theoretic setting, the (unmeasured) relative entropy of resource and the smoothed max-relative entropy have standard operational roles: the former captures asymptotic hypothesis-testing/interconversion rates, while the latter characterizes one-shot formation/distillation (e.g., entanglement cost and distillable entanglement) under non-entangling or catalytic transformations [Dat09b, BD11, LBT19, RFWG19, RW20]. In the computational setting, a full operational characterization remains open. Nevertheless, by parallelism with the unconstrained case, we adopt their measured and restricted counterparts. Moreover, they satisfy the crucial properties of resource monotones under our allowed maps, as we now prove.

Lemma 6.6. *Assuming that the set of free states $\mathcal{F}_n = \mathcal{F}(\mathcal{H}_n)$ is closed, the computational resource measures $\mathring{\mathbb{D}} \in \{\mathring{D}, \mathring{D}_{\max}\}$ that we have defined satisfy the following properties,*

- **Faithfulness:** *Given a quantum state $\rho \in \mathcal{S}(\mathcal{H}_n)$, its resource measure $\mathring{D}_{\mathcal{F}_n}(\rho)$ is zero iff $\rho \in \mathcal{F}(\mathcal{H}_n)$.*
- **Monotonicity:** *For any free operation $\Lambda_n \in \mathcal{O}_n(\mathcal{H}_n^1, \mathcal{H}_n^2)$,*

$$\mathring{\mathbb{D}}_{\mathcal{F}(\mathcal{H}_n^2)}(\Lambda_n(\rho)) \leq \mathring{\mathbb{D}}_{\mathcal{F}(\mathcal{H}_n^1)}(\rho). \quad (6.2.3)$$

Proof. On the one hand, if $\rho \in \mathcal{F}(\mathcal{H}_n)$, then it naturally holds that

$$\mathring{\mathbb{D}}_{\mathcal{F}_n}(\rho) = 0. \quad (6.2.4)$$

The other direction which is necessary to prove faithfulness then directly follows for $\mathring{D}_{\max}$ from Lemma 4.7 and for the measured relative entropy measures from Lemma 4.12 together with the fact that the computational trace distance $\mathring{\Delta}$ is a norm and the set $\mathcal{F}_n(\mathcal{H}_n)$ is closed. For the monotonicity property, it follows from the state condition of Definition 6.3 that,

$$\begin{aligned} \mathring{\mathbb{D}}_{\mathcal{F}(\mathcal{H}_n^2)}(\Lambda_n(\rho)) &= \min_{\tau \in \mathcal{F}(\mathcal{H}_n^2)} \mathring{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\Lambda_n(\rho) \parallel \tau) \leq \min_{\sigma \in \mathcal{F}(\mathcal{H}_n^1)} \mathring{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\Lambda_n(\rho) \parallel \Lambda_n(\sigma)) \\ &\leq \mathring{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\Lambda_n(\rho) \parallel \Lambda_n(\sigma)), \end{aligned}$$

for $\mathring{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}} \in \{\mathring{D}^{\mathbf{M}_n^{\text{eff}}}, \mathring{D}_{\max}\}$ and any $\sigma \in \mathcal{F}(\mathcal{H}_n^1)$, since $\Lambda_n(\mathcal{F}(\mathcal{H}_n^1)) \subseteq \mathcal{F}(\mathcal{H}_n^2)$. Then, by the test stability property of Definition 6.3 together with Property 4 of Lemma 4.11 (see also [RSB25, Lemma 5]) it follows that, for all $\sigma \in \mathcal{F}(\mathcal{H}_n^1)$,

$$\mathring{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\Lambda_n(\rho) \|\Lambda_n(\sigma)) \leq \mathring{\mathbb{D}}^{\mathbf{M}_n^{\text{eff}}}(\rho \|\sigma).$$

The proof then concludes by taking the minimum. $\square$

For the case of the computational measured relative entropy, it follows from [SW23, Theorem 11] that it satisfies asymptotic continuity with respect to the trace norm. Concretely, for any convex set $\mathcal{F}_n$ and assuming $\frac{1}{2}\|\rho - \rho'\|_1 \leq \varepsilon$ it then holds that

$$\left| \mathring{D}_{\mathcal{F}_n}(\rho) - \mathring{D}_{\mathcal{F}_n}(\rho') \right| \leq g(\varepsilon) + \varepsilon \sup_{M \in \mathbf{M}_n^{\text{eff}}} \kappa_M, \quad (6.2.5)$$

where $\kappa_M = \sup_{\mu, \nu} \left| D_{\mathcal{F}_n}^M(\mu) - D_{\mathcal{F}_n}^M(\nu) \right|$ with $D_{\mathcal{F}_n}^M(\rho) := \inf_{\sigma \in \mathcal{F}_n} D(\mathcal{M}_M(\rho) \|\mathcal{M}_M(\sigma))$ is the maximal absolute difference of the measured relative entropy of resource between any two states and $g(\varepsilon) := (1 + \varepsilon)h\left(\frac{\varepsilon}{1+\varepsilon}\right)$ and $h(\cdot)$ is the binary entropy.

Since we demonstrate an exponential separation between the information-theoretic trace distance and its computational counterpart, continuity bounds phrased in the computational metric can be exponentially sharper. This makes the above bound in practice less useful and motivates a tighter relation between our computational divergences and the corresponding restricted trace distance—so that the right-hand side of the inequality depends only on a computational notion of distinguishability. Theorem 6.7 provides exactly this: an asymptotic continuity bound for the computational measured relative entropy of resource in terms of the computational trace distance, furnishing a direct analytic bridge with clear operational meaning. Such a bound can essentially be derived via [BLT25, Corollary 14]. However, for completeness and convenience of the reader, a novel and self-contained proof of Theorem 6.7 is provided in Section A.4.2.¹

Theorem 6.7 (Asymptotic Continuity Bound for the Computational Measured Relative Entropy of Resource). *Let $\mathcal{F}_n \subset \mathcal{S}(\mathcal{H}_n)$ be closed, convex, and bounded, and suppose that it contains a full rank state σ^* . For $\rho, \rho' \in \mathcal{S}(\mathcal{H}_n)$ and $\mathring{\Delta}(\rho, \rho') = \frac{1}{2}\|\rho - \rho'\|_{\mathbf{M}_n^{\text{eff}}} \leq \varepsilon$ it holds that,*

$$\left| \mathring{D}_{\mathcal{F}_n}(\rho) - \mathring{D}_{\mathcal{F}_n}(\rho') \right| \leq (1 + \varepsilon)h\left(\frac{\varepsilon}{1+\varepsilon}\right) + \varepsilon \left(\kappa + \log\left(\frac{2}{\varepsilon}\right) \right), \quad (6.2.6)$$

where $h(\cdot)$ is the binary entropy and $\kappa := -\log(\lambda_{\min}(\sigma^*)) > 0$ is the negative logarithm of the smallest eigenvalue of the full rank state σ^* .

Proof. The proof is provided in Section A.4.2. $\square$

¹We were made aware after submitting this manuscript that an asymptotic continuity bound similar to Theorem V.7 was independently established in [BLT25, Corollary 14], in a somewhat more general and abstract setting, and using different techniques.

Moreover, as discussed in [Win16], κ is guaranteed to be finite since $\mathcal{F}_n$ contains at least one full-rank state. For example, in the case of the entanglement resource, $\mathcal{F}_n = \text{SEP}_n \ni \frac{\mathbb{1}}{2^n}$ and so $\kappa = n$. Therefore, for the *computational measured relative entropy of resource* another property is fulfilled: *computational faithfulness*.

Corollary 6.8. *Let $\mathcal{F}_n \subset \mathcal{S}(\mathcal{H}_n)$ be closed and convex, and suppose that it contains a full-rank state σ_n^* such that $\kappa_n := -\log \lambda_{\min}(\sigma_n^*)$ is polynomially bounded in n . Given two families of states $\{\rho_n\}_{n \in \mathbb{N}}$ and $\{\rho'_n\}_{n \in \mathbb{N}}$ which are computationally indistinguishable, then,*

$$|\mathring{D}_{\mathcal{F}_n}(\rho_n) - \mathring{D}_{\mathcal{F}_n}(\rho'_n)| \leq \text{negl}(n). \quad (6.2.7)$$

Proof. Computational indistinguishability implies $\mathring{\Delta}(\rho_n, \rho'_n) \leq \varepsilon(n)$ for some negligible function ε . The result then follows from Theorem 6.7, since κ_n is polynomially bounded and hence

$$\varepsilon(n) \left(\kappa_n + \log \frac{2}{\varepsilon(n)} \right) = \text{negl}(n).$$

□

The operational implications of this equation are as follows: if a family of states $\{\rho_n\}_{n \in \mathbb{N}}$ with high (information-theoretic) resource content is computationally indistinguishable from a family of free states $\{\sigma_n\}_{n \in \mathbb{N}}$ with $\sigma_n \in \mathcal{F}_n$, then the computational measured relative entropy of resource of $\{\rho_n\}_{n \in \mathbb{N}}$ is negligible, i.e., $\mathring{D}_{\mathcal{F}_n}(\rho_n) = \text{negl}(n)$. This phenomenon has been studied before in the context of pseudoentanglement [ABF⁺24, ABV23, LREJ25] and other pseudoresources [HBK25, BMB⁺25, GY26] from a more heuristic perspective: if a computational distinguisher is able to distill more resource from one of the families, then it can distinguish them. Here, we provide an analytical expression that quantifies this for every resource.

6.3. Computational Entanglement Measures

Let us now focus on the resource of entanglement. Analogously to the information-theoretic scenario in, e.g. [Pia09], we define the computational measured relative entropy of entanglement as

$$\mathring{E}_R(\rho) := \mathring{D}_{\text{SEP}_n}(\rho) = \min_{\sigma \in \text{SEP}_n} \mathring{D}^{\text{M}_n^{\text{eff}}}(\rho \| \sigma), \quad (6.3.1)$$

where we take $\mathcal{F}_n = \text{SEP}_n$ to be the set of separable states, i.e.,

$$\text{SEP}_n := \text{conv}\{\rho_A^n \otimes \rho_B^n : \rho_A^n \in \mathcal{S}(\mathcal{H}_n^A), \rho_B^n \in \mathcal{S}(\mathcal{H}_n^B)\}. \quad (6.3.2)$$

Let us again show an explicit separation between computational and information-theoretic measures. Using the explicit cryptographic construction of Section 5.1,

$$\begin{aligned}\psi_n^{AB} &:= \frac{1}{4} \left(|\Phi^+\rangle\langle\Phi^+|^{AB} + |\Phi^-\rangle\langle\Phi^-|^{AB} \right) \otimes (\rho_{0,n}^A + \rho_{1,n}^A), \\ \phi_n^{AB} &:= \frac{1}{2} \left(|\Phi^+\rangle\langle\Phi^+|^{AB} \otimes \rho_{0,n}^A + |\Phi^-\rangle\langle\Phi^-|^{AB} \otimes \rho_{1,n}^A \right),\end{aligned}$$

where $|\Phi^+\rangle^{AB} = \frac{1}{\sqrt{2}}(|00\rangle^{AB} + |11\rangle^{AB})$, $|\Phi^-\rangle^{AB} = \frac{1}{\sqrt{2}}(|00\rangle^{AB} - |11\rangle^{AB})$ and $\{\rho_{0,n}^A, \rho_{1,n}^A\}_{n \in \mathbb{N}}$ are EFI pairs. One can easily see that $E_R(\phi_n^{AB}) \approx 1$ by construction. On the other hand, as stated in Section 5.1, $\hat{\Delta}(\psi_n^{AB}, \phi_n^{AB}) \leq \text{negl}(n)$. Then, by Theorem 6.7, it follows that $\hat{E}_R(\phi_n^{AB}) \leq \text{negl}(n)$. Moreover, this explicit gap can again be (polynomially) amplified as proven in [GE24, Lemma 5.1].

To relate the *computational measured relative entropy of entanglement* to the *(one-shot) computational entanglement cost* and the *(one-shot) computational distillable entanglement* defined by [ABV23], we first fix the class of efficient free operations we work with. In entanglement theory, local operations and classical communication (LOCC) form the canonical class of free maps: they cannot increase entanglement, and they directly model a practical two-party communication setting.

Definition 6.9 (LOCC Map [ABV23]). A quantum channel is said to be an LOCC map

$$\Gamma : \mathcal{B}_1(\mathcal{H}_A \otimes \mathcal{H}_B) \mapsto \mathcal{B}_1(\mathcal{H}_{A'} \otimes \mathcal{H}_{B'})$$

if it can be implemented by a two-party interactive protocol where each party can implement arbitrary local quantum computations and the two parties can exchange arbitrary classical communication.

We now consider their complexity-constrained counterparts.

Definition 6.10 (Circuit Description of an LOCC Map [ABV23]). Given an LOCC map Γ , its circuit description is given by two families of circuits $\{\mathcal{C}_{A,i}\}_{i \in \{1, \dots, r\}}$ and $\{\mathcal{C}_{B,i}\}_{i \in \{1, \dots, r\}}$, each of them acting on $n_A + t_A + c$ and $n_B + t_B + c$ qubits, respectively, such that the following procedure implements the map Γ on an arbitrary input $\varphi_{AB} \in (\mathbb{C}^2)^{\otimes n_A} \otimes (\mathbb{C}^2)^{\otimes n_B}$:

1. Registers A and B of n_A and n_B qubits are initialized in the state φ_{AB} . Ancilla registers A' and B' of t_A and t_B qubits are initialized in the $|0\rangle$ state. Communication register C is also initialized in the $|0\rangle$ state.
2. For $i = 1, \dots, r$, the circuit $\mathcal{C}_{A,i}$ is applied to registers A , A' and C . Then, register C is measured in the computational basis. Next, the circuit $\mathcal{C}_{B,i}$ is applied to registers B , B' , and C . Lastly, register C is measured in the computational basis.
3. The final output of the LOCC map in the subspace $\mathcal{H}_{A'}$ corresponds to the state in the registers A and A' . In the same way, the state in the registers B and B' is the final output on $\mathcal{H}_{B'}$.

For the remainder of this section we fix a polynomial $c(n)$. We say that an LOCC map Γ_n is efficient if Γ_n has a circuit description whose total number of gates is at most $c(n)$ [ABV23]. We want to view the class of LOCC operations on $\mathcal{H}_{n_A(n)} \otimes \mathcal{H}_{n_B(n)}$ with circuit size at most $c(n)$ as a set of free operations, where we set $\dim(\mathcal{H}) = 2$.

It is easy to see that, by construction, this set satisfies the *state condition*. This is because LOCC operations map separable states to separable states [CG19, HHHH09]. To satisfy *test stability*, we impose the following straightforward property. Every efficient LOCC channel in our set can be viewed as a mapping

$$\Gamma_n : \mathcal{H}_{n_A(n)} \otimes \mathcal{H}_{n_B(n)} \mapsto \mathcal{H}_{\bar{A}} \otimes \mathcal{H}_{\bar{B}}, \quad (6.3.3)$$

where $\mathcal{H}_{\bar{A}} \otimes \mathcal{H}_{\bar{B}}$ contains at most polynomially many qubits. If the maximal allowed circuit size for binary measurements on $\mathcal{H}_{n_A(n)} \otimes \mathcal{H}_{n_B(n)}$ is given by $q(n)$, then we restrict the corresponding circuit size on $\mathcal{H}_{\bar{A}} \otimes \mathcal{H}_{\bar{B}}$ to be bounded by a (fixed) polynomial $p(n)$ such that

$$q(n) \geq p(n) + c(n). \quad (6.3.4)$$

This guarantees test stability, as it ensures that for any efficient binary measurement $\mathcal{M}$ that acts on $\Gamma_n(\rho)$, $\mathcal{M} \circ \Gamma_n$ is a *valid* efficient binary measurement on $\mathcal{H}_{n_A(n)} \otimes \mathcal{H}_{n_B(n)}$. As such, the resulting computational measured relative entropy of entanglement satisfies monotonicity under efficient LOCC operations.

Let us take the class of efficient LOCC as the set of free operations as in Definition 6.3. It is easy to see that, by construction, it satisfies the state condition: LOCC operations map separable states to separable states.

With efficient LOCC fixed as our class of free operations, we are now ready to introduce the computational entanglement measures defined by [ABV23]. Operationally, they encode how many Bell states $\Phi := |\Phi_1\rangle\langle\Phi_1|$ one can distill from or, respectively, are required to produce a given state using only efficient LOCC operations, where the Bell state is defined as $|\Phi_1\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Moreover, it satisfies that $\Phi^{\otimes n} = |\Phi_n\rangle\langle\Phi_n|$, where $|\Phi_n\rangle := \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} |i\rangle|i\rangle$.

Definition 6.11 (Computational One-Shot Entanglement Cost [ABV23]). Let $\varepsilon : \mathbb{N}_+ \rightarrow [0, 1]$ and $n \in \mathbb{N}_+$. Fix polynomial functions $n_A, n_B : \mathbb{N}_+ \rightarrow \mathbb{N}_+$. Let $\{\rho_{AB}^n\}_n$ be a family of quantum states such that, for any $n \geq 1$, $\rho_{AB}^n \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ is a bipartite state on $n_A(n) + n_B(n)$. The function $k : \mathbb{N} \rightarrow \mathbb{N}$ is an upper bound on the computational entanglement cost of the family $\{\rho_{AB}^n\}_n$, i.e. $\hat{E}_C^\varepsilon(\{\rho_{AB}^n\}_n) \leq k(n)$, if there exists an efficient LOCC map family $\{\Gamma_n\}_n$ such that, for each $n \geq 1$, Γ_n takes as input $k(n)$ maximally entangled d -dimensional states, and

$$1 - F(\rho_{AB}^n, \Gamma_n(\Phi^{\otimes k(n)})) \leq \varepsilon(n).$$

Definition 6.12 (Computational One-Shot Distillable Entanglement [ABV23]). Let $\varepsilon : \mathbb{N}_+ \rightarrow [0, 1]$ and $n \in \mathbb{N}_+$. Fix polynomial functions $n_A, n_B : \mathbb{N}_+ \rightarrow \mathbb{N}_+$. Let $\{\rho_{AB}^n\}_n$ be

a family of quantum states such that, for any $n \geq 1$, $\rho_{AB}^n \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$ is a bipartite state on $n_A(n) + n_B(n)$. The function $m : \mathbb{N} \rightarrow \mathbb{N}$ is a lower bound on the computational distillable entanglement of the family $\{\rho_{AB}^n\}_n$, i.e. $\hat{E}_D^\varepsilon(\{\rho_{AB}^n\}_n) \geq m(n)$, if there exists an efficient LOCC map family $\{\Gamma_n\}_n$ such that, for each $n \geq 1$, Γ_n outputs a $2m(n)$ -qubit state, and

$$1 - F(\Gamma_n(\rho_{AB}^n), \Phi^{\otimes m(n)}) \leq \varepsilon(n).$$

For a fixed state ρ_{AB}^n and $\varepsilon := \varepsilon(n)$, we say that

$$\hat{E}_C^\varepsilon(\rho_{AB}^n) := \min_k k(n), \quad \hat{E}_D^\varepsilon(\rho_{AB}^n) := \max_m m(n). \quad (6.3.5)$$

This is well-defined for any fixed polynomial $c(n)$ which determines the set of efficient LOCC channels. We now evaluate $\hat{E}_R$ on the maximally entangled state Φ_n before comparing it with these computational entanglement measures.

Lemma 6.13. *Assume that the Bell projector $\Phi_n = |\Phi_n\rangle\langle\Phi_n| = \Phi_1^{\otimes n} \in \mathbf{E}_{2n}^{\text{eff}}$ is efficiently generated.² Then it follows that*

$$\hat{E}_R(\Phi^{\otimes n}) = n,$$

where $\Phi^{\otimes n} := |\Phi_n\rangle\langle\Phi_n| \in \mathcal{B}((\mathbb{C}_A^2)^{\otimes n} \otimes (\mathbb{C}_B^2)^{\otimes n})$.

Note that the condition $\Phi^{\otimes n} \in \mathbf{E}_{2n}^{\text{eff}}$ is typically implied by the much simpler one of $\Phi \in \mathbf{E}_2^{\text{eff}}$ whenever the polynomial in the construction of the family $\{\mathbf{E}_n^{\text{eff}}\}_{n \in \mathbb{N}}$ is (super)-additive.

Proof. We first show that $\hat{E}_R(\Phi^{\otimes n}) \geq n$. Since Φ can be efficiently generated, the measurement $M_{\Phi_n} = (\Phi_n, \mathbb{1} - \Phi_n) \in \mathbf{M}_{2n}^{\text{eff}}$. Then, since $\langle\Phi_n|\sigma|\Phi_n\rangle \leq \frac{1}{2^n}$ for any $\sigma \in \text{SEP}_n$, we have $D(\mathcal{M}_{M_{\Phi_n}}(\Phi_n) \parallel \mathcal{M}_{M_{\Phi_n}}(\sigma)) \geq n$ for any $\sigma \in \text{SEP}_n$. Let us prove this in detail for completeness. Due to convexity of the map $\sigma \mapsto \langle\Phi_n|\sigma|\Phi_n\rangle$ it suffices to consider product states $\sigma = \rho_A \otimes \rho_B$. Now by a standard Choi argument we have

$$\begin{aligned} \langle\Phi_n|\rho_A \otimes \rho_B|\Phi_n\rangle &= 2^{-n} \text{Tr} \left[\left(\sum_{i,j=0}^{2^n-1} |i\rangle\langle j|_A \otimes |i\rangle\langle j|_B \right) (\rho_A \otimes \rho_B) \right] \\ &= 2^{-n} \text{Tr} [\rho_A \rho_B^T] \leq 2^{-n} \|\rho_A\|_2 \|\rho_B^T\|_2 \leq 2^{-n}, \end{aligned} \quad (6.3.6)$$

where in the last line we used Cauchy-Schwartz inequality and normalization of the states.

$$\begin{aligned} \hat{E}_R(\Phi_n) &= \min_{\sigma \in \text{SEP}_n} \sup_{M \in \mathbf{M}_n^{\text{eff}}} \hat{D}(\mathcal{M}_M(\Phi_n) \parallel \mathcal{M}_M(\sigma)) \\ &\geq \min_{\sigma \in \text{SEP}_n} D(\mathcal{M}_{M_{\Phi_n}}(\Phi_n) \parallel \mathcal{M}_{M_{\Phi_n}}(\sigma)) \\ &= \min_{\sigma \in \text{SEP}_n} (-\log \langle\Phi_n|\sigma|\Phi_n\rangle) \geq n. \end{aligned}$$

²For example, any set of efficient effects $\mathbf{E}_n^{\text{eff}}$ which is generated by the gateset $\mathcal{G} \supset \{H, \text{CNOT}\}$ and the super-additive polynomial $p(n) = n^2$.

We now show $\mathring{E}_R(\Phi_n) \leq n$. By definition,

$$\mathring{E}_R(\Phi_n) = \min_{\sigma \in \text{SEP}_n} \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} \mathring{D}(\mathcal{M}_M(\Phi_n) \| \mathcal{M}_M(\sigma)) \leq \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} \mathring{D}(\mathcal{M}_M(\Phi_n) \| \mathcal{M}_M(\sigma^*)),$$

where $\sigma_n^* = 2^{-n} \sum_{x \in \{0,1\}^n} |x\rangle\langle x|_A \otimes |x\rangle\langle x|_B$ which is separable. Lastly, by data processing,

$$\sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} \mathring{D}(\mathcal{M}_M(\Phi_n) \| \mathcal{M}_M(\sigma^*)) \leq D(\Phi_n \| \sigma^*) = -\langle \Phi_n | \log \sigma^* | \Phi_n \rangle = n.$$

□

Therefore, in this simple case of effectively not assuming computational restrictions, our measure attains the expected maximum.

Let us now also relate $\mathring{E}_R$ to the computational entanglement cost and computational distillable entanglement proposed by the authors of [ABV23]. However, first we note that since we are only considering efficient LOCC protocols, one can without loss of generality assume that every $k(n)$ and $m(n)$ appearing in Eq (6.3.5) is bounded by some fixed polynomial. Since the desired Bell projector can be efficiently prepared, we also assume that it is efficient to first run an efficient LOCC mapping and then apply the binary measurement corresponding to the desired Bell projector.

Lemma 6.14. *Assume that the Bell projector is efficient in the sense we describe above. Then given polynomials n_A, n_B and a family of states $\{\rho_{AB}^n\}_n$, such that for all $n \in \mathbb{N}$ $\rho_{AB}^n \in \mathcal{S}(\mathcal{H}_n^A \otimes \mathcal{H}_n^B)$, the following inequalities must hold:*

$$\hat{E}_D^\varepsilon(\rho_{AB}^n) \leq \mathring{E}_R(\rho_{AB}^n) + g_1\left(\sqrt{\varepsilon(n)}, n, \rho_{AB}^n\right), \quad (6.3.7)$$

$$\hat{E}_C^\varepsilon(\rho_{AB}^n) \geq \mathring{E}_R(\rho_{AB}^n) - g_2\left(\sqrt{\varepsilon(n)}, n\right), \quad (6.3.8)$$

where $\mathring{E}_D(\rho_{AB}^n)$ and $\mathring{E}_C(\rho_{AB}^n)$ are, respectively, the computational distillable entanglement (Definition 6.12) and the computational entanglement cost (Definition 6.11), and

$$g_1\left(\sqrt{\varepsilon(n)}, n, \rho_{AB}^n\right) := \left(1 + \sqrt{\varepsilon(n)}\right) h\left(\frac{\sqrt{\varepsilon(n)}}{1 + \sqrt{\varepsilon(n)}}\right) + \sqrt{\varepsilon(n)} \left(2\hat{E}_D^\varepsilon(\rho_{AB}^n) + \log \frac{2}{\sqrt{\varepsilon(n)}}\right),$$

$$g_2\left(\sqrt{\varepsilon(n)}, n\right) := \left(1 + \sqrt{\varepsilon(n)}\right) h\left(\frac{\sqrt{\varepsilon(n)}}{1 + \sqrt{\varepsilon(n)}}\right) + \sqrt{\varepsilon(n)} \left(n_A + n_B + \log \frac{2}{\sqrt{\varepsilon(n)}}\right).$$

Proof. We first derive Eq. (6.3.7), which relates the computational measured entropy of resource to the computational one-shot distillable entanglement. By Definition 6.12, we say that $\mathring{E}_D(\rho_{AB}^n) \geq m(n)$ if there exists an efficient LOCC map Γ_n of circuit size $c(n)$ such that

$$1 - F\left(\eta_{AB}, \Phi^{\otimes m(n)}\right) \leq \varepsilon(n), \quad \text{where } \eta_{AB} := \Gamma_n(\rho_{AB}^n). \quad (6.3.9)$$

Using the (information-theoretic) Fuchs–van de Graaf inequality [FvdG99] and the fact that the quantum fidelity is non-decreasing under quantum channels [Tom16, Proposition 3.2], this also implies that $\dot{\Delta}(\eta_{AB}, \Phi^{\otimes m(n)}) \leq \sqrt{\varepsilon(n)}$. We then apply Theorem 6.7 and choose the full rank state to be the maximally mixed state to achieve the bound.

$$\begin{aligned} |\dot{E}_R(\Phi^{\otimes m(n)}) - \dot{E}_R(\eta_{AB})| &\leq g' \left(\sqrt{\varepsilon(n)}, n, m(n) \right) \\ &:= \left(1 + \sqrt{\varepsilon(n)} \right) h \left(\frac{\sqrt{\varepsilon(n)}}{1 + \sqrt{\varepsilon(n)}} \right) \\ &\quad + \sqrt{\varepsilon(n)} \left(2m(n) + \log \frac{2}{\sqrt{\varepsilon(n)}} \right). \end{aligned} \quad (6.3.10)$$

We note that this holds as long as $M_{\Phi_m} := (\Phi_m, \mathbb{1} - \Phi_m)$ is an efficient measurement on $\mathcal{H}_{m(n)} \otimes \mathcal{H}_{m(n)}$. We denote the underlying polynomial of $\dot{E}_R(\eta_{AB})$ by $q(n)$ and the underlying polynomial of $\dot{E}_R(\rho_{AB}^n)$ by $p(n)$. We choose $q(n)$ to be large enough to satisfy the above property and small enough to satisfy

$$p(n) \geq q(n) + c(n). \quad (6.3.11)$$

Roughly speaking, this is meant to capture the fact that for any given state ρ_{AB}^n , every measurement of the form “efficient LOCC operation Γ_n + applying the binary measurement M_{Φ_m} ” is a valid efficient measurement on the input state.

By Lemma 6.14, it holds that $\dot{E}_R(\Phi^{\otimes m(n)}) = m(n)$. Moreover, $\dot{E}_R(\eta_{AB}) \leq \dot{E}_R(\rho_{AB}^n)$ holds by the monotonicity property. We therefore obtain the inequality

$$m(n) \leq \dot{E}_R(\rho_{AB}^n) + g' \left(\sqrt{\varepsilon(n)}, n, m(n) \right). \quad (6.3.12)$$

Taking the supremum over all achievable $m(n)$ yields the stated bound on $\dot{E}_D$.

We derive Eq. (6.3.8) in a similar fashion. By Definition 6.11, we have that the computational one-shot entanglement cost satisfies $\dot{E}_C(\rho_{AB}^n) \leq k(n)$ if there exists an efficient LOCC map Γ_n of circuit size $c(n)$ such that

$$1 - F(\rho_{AB}^n, \eta_{AB}) \leq \varepsilon(n), \quad \text{where } \eta_{AB} := \Gamma_n(\Phi^{\otimes k(n)}). \quad (6.3.13)$$

Again, by the Fuchs–van de Graaf inequality [FvdG99] and the fact that the quantum fidelity is non-decreasing under quantum channels [Tom16, Proposition 3.2], it must hold that $\dot{\Delta}(\rho_{AB}^n, \eta_{AB}) \leq \sqrt{\varepsilon(n)}$. Then, by Theorem 6.7,

$$|\dot{E}_R(\eta_{AB}) - \dot{E}_R(\rho_{AB}^n)| \leq g_2 \left(\sqrt{\varepsilon(n)}, n \right). \quad (6.3.14)$$

Additionally, it must hold that

$$\dot{E}_R(\eta_{AB}) \leq E_R(\eta_{AB}) \leq E_R(\Phi^{\otimes k(n)}) = k(n). \quad (6.3.15)$$

Here, E_R is the unmeasured relative entropy of entanglement.³ The first inequality holds due to the data processing inequality of the relative entropy; see, e.g. [Tom16, Section 4]. The second inequality holds due to the monotonicity property of E_R [VP98]. The last equality follows from [VP98, Theorem 3]. Therefore, it follows that

$$k(n) \geq \mathring{E}_R(\rho_{AB}^n) - g_2\left(\sqrt{\varepsilon(n)}, n\right). \quad (6.3.16)$$

Taking the infimum over all feasible $k(n)$ yields the desired inequality. $\square$

These bounds are conceptually very similar to [Win16, Corollary 4]. Moreover, for reasonable choices of parameters, one typically expects $2\hat{E}_D^\varepsilon(\rho_{AB}^n) \leq n_A + n_B$ to hold. This is because the output of an entanglement distillation protocol is expected to have a smaller system size than the original input. Whenever this is the case, g_1 can be replaced with g_2 in Eq. (6.3.7) to achieve a looser, but simpler, bound.

A consequence of Lemma 6.14 is that $\mathring{E}_R$ is sandwiched between the computational distillable entanglement and the computational entanglement cost, up to continuity terms. This parallels the information-theoretic hierarchy $E_D \leq E_R \leq E_C$ and confirms that $\mathring{E}_R$ behaves as an entanglement monotone in the computational setting.

Remark 6.15. For the computational max-relative entropy of entanglement defined in Definition 6.5, a direct comparison with the one-shot measures of [ABV23] is not straightforward. In the unconstrained setting, the max-relative entropy of entanglement (equivalently, the log-robustness) attains a one-shot operational meaning after smoothing with respect to non-entangling operations, a class that strictly contains LOCC [Dat09a, BD11]. Analogously, smoothing the computational max-relative entropy we define and taking the efficient free operations to be (efficient) non-entangling maps may clarify its relation to the measures in [ABV23].

³See [VPRK97, VP98] for a rigorous definition.

Chapter 7

Quantum Pseudoresources and Cryptographic Implications

This chapter is based on [GY26]. In the classical setting, one-way functions play the role of a minimal assumption for computational cryptography, while in the quantum setting weaker primitives such as pseudorandom states and EFI pairs can already support cryptographic tasks. The minimal assumption for quantum cryptography remains open, and recent results show that pseudoentanglement is necessary for EFI pairs. We study this question through the broader notion of quantum pseudoresources: pairs of state families with a gap in their resource content which remain indistinguishable to efficient observers. We first introduce EPFI pairs and show how they yield canonical quantum commitments in Section 7.1. Then, in Section 7.2, we prove that pseudoresources imply EPFI pairs. Finally, Section 7.3 specializes this construction to pure and mixed pseudoentanglement, and ends with computationally locked entanglement as a functionality intrinsically tied to hidden entanglement.

7.1. Quantum Commitments and EPFI Pairs

The cryptographic primitive of EFI pairs (Efficiently generated, statistically far and indistinguishable states) was first defined in [BCQ23], and it consists of a pair of states ρ and σ that are far in trace distance but cannot be efficiently distinguished by polynomial-time algorithms. In [BCQ23], they showed that EFI pairs are equivalent to quantum commitments, more specifically, to the statistically binding variant of canonical quantum commitments introduced in [Yan22]. Consequently, EFI pairs serve as a fundamental assumption for the existence of commitments, oblivious transfer, multiparty computation, and computational zero-knowledge proofs for non-trivial languages.

Definition 7.1 (EFI pair). A pair of mixed states $(\rho_{0,\lambda}, \rho_{1,\lambda})$ is an EFI pair, if

- **Efficient generation:** there exists a QPT algorithm A that on input $(1^\lambda, b)$ outputs $\rho_{b,\lambda}$.
- **Statistical distance:** $\Delta(\rho_{0,\lambda}, \rho_{1,\lambda}) \geq \Omega\left(\frac{1}{\text{poly}(\lambda)}\right)$.
- **Computational indistinguishability:** for any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr\left[\mathcal{D}(\sigma_\lambda, \rho_{\lambda,0}^{\otimes m}) = 1\right] - \Pr\left[\mathcal{D}(\sigma_\lambda, \rho_{\lambda,1}^{\otimes m}) = 1\right] \right| \leq \nu(\lambda).$$

As mentioned above, EFI pairs of states are equivalent to quantum commitments as proven by [BCQ23], more specifically to *canonical quantum commitments*, which is defined as follows.

Definition 7.2 (Canonical quantum commitment [Yan22]). Given an ensemble of polynomial-time uniformly generated pairs of quantum circuits $\{(Q_{\lambda,0}, Q_{\lambda,1})\}_\lambda$, a *canonical quantum commitment* scheme is defined by the following stages:

- **Commit stage:** the committer chooses the committed bit $b \in \{0, 1\}$ and performs the quantum circuit $Q_{\lambda,b}$ to the register pair (C, R) , initialized in the $|0\rangle$ state¹. Then the committer sends the register C to the receiver, i.e., the state $\rho_{\lambda,b} := \text{Tr}_R(Q_{\lambda,b} |0\rangle\langle 0|_{CR} Q_{\lambda,b}^\dagger)$.
- **Reveal stage:** the committer sends the bit b and the register R to the receiver. The receiver performs $Q_{\lambda,b}^\dagger$ on (C, R) and aborts if the measured registers are not in the $|0\rangle$ state.

Definition 7.3 (Computational hiding). Given a canonical quantum commitment scheme in which the committed states are given by the families of mixed states $\{\rho_{\lambda,0}\}_\lambda$ and $\{\rho_{\lambda,1}\}_\lambda$, the scheme is computationally hiding if, for any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr[\mathcal{D}(\sigma_\lambda, \rho_{\lambda,0}^{\otimes m}) = 1] - \Pr[\mathcal{D}(\sigma_\lambda, \rho_{\lambda,1}^{\otimes m}) = 1] \right| \leq \nu(\lambda).$$

Definition 7.4 (Honest statistical binding [Yan22]). A canonical quantum commitment scheme satisfies honest statistical binding if, for any auxiliary state $|\psi\rangle$ and any unitary $U \in \mathcal{U}(\mathcal{H})$ there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left\| \left(Q_{\lambda,1} \otimes \mathbb{1}_Z |0\rangle\langle 0|_{CR} Q_{\lambda,1}^\dagger \otimes \mathbb{1}_Z \right) \left(\mathbb{1}_C \otimes U_{RZ} \right) \left(Q_{\lambda,0} |0\rangle_{CR} \otimes |\psi\rangle_Z \right) \right\|_2 \leq \nu(\lambda). \quad (7.1.1)$$

Informally, the honest-binding property allows the receiver to cheat only in the reveal stage of the commitment functionality. As proven by [Yan22], honest binding for canonical quantum commitments is equivalent to the notion of sum-binding [Unr16]. Moreover, since sum-binding is equivalent to AQY binding [AQY22, MY22], oblivious transfer and multiparty computing can be constructed from it [BCKM21, GLSV21]. From now on, we will refer to the honest statistical binding commitments as statistical binding commitments.

We note that the construction of [Yan22] can be modified with commitments generated by a pair of uniformly generated families of quantum circuits $(\{Q_0^{k(\lambda)}\}_{k \in \{0,1\}^{\kappa(\lambda)}}, \{Q_1^{k'(\lambda)}\}_{k' \in \{0,1\}^{\kappa(\lambda)}})$.

¹For simplicity we write the tensor product of k registers in the $|0\rangle$ state as $|0\rangle$ (instead of $|0\rangle^{\otimes k}$), when it is clear from the context.

To commit to a bit $b \in \{0, 1\}$, the committer first chooses a secret key k (or k') uniformly at random and then applies the corresponding circuit $Q_b^{k(\lambda)}$, preparing the state $|\Psi_b^{k(\lambda)}\rangle_{CR} = Q_b^{k(\lambda)} |0\rangle_{CR}$. Then, they send the commitment register C as in Definition 7.2. In the reveal stage, the committer sends to the receiver the chosen key together with b and the register R .²

The statistical binding property in this setting requires that for all $k, k' \in \{0, 1\}^{\kappa(\lambda)}$,

$$\left\| \left(Q_1^{k'(\lambda)} \otimes \mathbb{1}_Z |0\rangle \langle 0|_{CR} Q_1^{\dagger k'(\lambda)} \otimes \mathbb{1}_Z \right) \left(\mathbb{1}_C \otimes U_{RZ} \right) \left(Q_0^{k(\lambda)} |0\rangle_{CR} \otimes |\psi\rangle_Z \right) \right\|_2 \leq \nu(\lambda), \quad (7.1.2)$$

Likewise, the computational hiding property in this case is given by,

$$\left| \Pr_{k(\lambda)} \left[\mathcal{D}(\sigma_\lambda, \rho_{k(\lambda),0}^{\otimes m}) = 1 \right] - \Pr_{k'(\lambda)} \left[\mathcal{D}(\sigma_\lambda, \rho_{k'(\lambda),1}^{\otimes m}) = 1 \right] \right| \leq \nu(\lambda). \quad (7.1.3)$$

When the commitment procedure is instantiated by a *family* of circuits rather than a single pair, we adopt a stronger honest statistical binding requirement: the honest committing states must be pairwise far for every choice of keys. Formally, Eq. (7.1.2) requires the negligible overlap bound for each pair of states indexed by $(k, k') \in \{0, 1\}^\kappa$. In particular, Eq. (7.1.2) reduces to Def. 7.4 when both families degenerate to a singleton.

By contrast, the hiding property must hold on the receiver's commit-phase view. In our canonical commitment scheme the key k is written into the register R and revealed only during the reveal stage, so the receiver's commit-phase view is the *mixture* over k . Accordingly, the hiding property is stated on the mixtures as in Eq. (7.1.3).

The construction of quantum commitments [Yan22] from EFI pairs given in [BCQ23][Lemma 4.7] is similar in spirit to Theorem 7.7: given an EFI pair, each efficient generator defines a distinct committing state. The statistical binding and computational hiding properties follow directly from the statistical fairness and computational indistinguishability of the two EFI states. Conversely, to show that quantum commitments imply EFI pairs, [BCQ23][Lemma 4.10] derives EFI pairs from semi-honest oblivious transfer, which in turn can be constructed from commitments as shown in [BCQ23][Lemma 4.9]. In this work, we define EPFI pairs by refining the requirements for EFI pairs.

7.1.1. EPFI Pairs

In order to achieve our results, we need to slightly modify the functionality of Definition 7.1 as follows. We consider two efficiently generated keyed families of states $\{\rho_k\}_k$ and $\{\sigma_{k'}\}_{k'}$, and we require:

- **Pairwise statistical fairness.** For every $k, k' \in \{0, 1\}^{\kappa(\lambda)}$,

$$\Delta(\rho_k, \sigma_{k'}) \approx \Omega(1/\text{poly}).$$

²The key k can be stored, without loss of generality, in the register R .

- **Computational indistinguishability.** For every polynomial $m(\lambda)$, the mixed states,

$$\tilde{\rho}^{(m)} := \mathbb{E}_{k \leftarrow \{0,1\}^{\kappa(\lambda)}} [\rho_k^{\otimes m(\lambda)}], \quad \tilde{\sigma}^{(m)} := \mathbb{E}_{k' \leftarrow \{0,1\}^{\kappa(\lambda)}} [\sigma_{k'}^{\otimes m(\lambda)}]$$

are computationally indistinguishable for any quantum polynomial-time distinguisher.

We call such a pair an EPFI pair (Efficiently generated, Pairwise Statistically Far and Computationally Indistinguishable).³ Conceptually, the EPFI pair primitive can be seen as the minimal cryptographic primitive that simultaneously supports indistinguishability by averaging over the ensemble and pairwise statistical fairness for every element of it.

Definition 7.5 (EPFI pair). A pair of ensembles of mixed states $(\{\psi_{k(\lambda)}\}_{k(\lambda)}, \{\phi_{k'(\lambda)}\}_{k'(\lambda)})$ indexed by $k, k' \in \{0,1\}^{\kappa(\lambda)}$ is an **Efficiently generated, pairwise far** and computational indistinguishable pair (EPFI pair), if

- **Efficient generation:** Given $k(\lambda)$ (or $k'(\lambda)$), there exists a QPT algorithm A that on input $(1^\lambda, k, b)$ (or $(1^\lambda, k', b)$) outputs $\psi_{k(\lambda)}$ (or $\phi_{k'(\lambda)}$).
- **Pairwise statistically far:** For all $k, k' \in \{0,1\}^{\kappa(\lambda)}$,

$$\Delta(\psi_{k(\lambda)}, \phi_{k'(\lambda)}) \geq \Omega\left(\frac{1}{\text{poly}(\lambda)}\right).$$

- **Computational indistinguishability:** for any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr_{k(\lambda)} [\mathcal{D}(\sigma_\lambda, \psi_{k(\lambda)}^{\otimes m}) = 1] - \Pr_{k'(\lambda)} [\mathcal{D}(\sigma_\lambda, \phi_{k'(\lambda)}^{\otimes m}) = 1] \right| \leq \nu(\lambda).$$

The main difference between the two primitives is that each element of each ensemble has to be statistically far in terms of trace distance from every element of the other ensemble, while computational indistinguishability holds for the ensemble itself.

We note that an EFI pair is a special case of an EPFI pair where each family has a single element. However, an EFI pair cannot be constructed directly from an EPFI pair instance: even if all pairs $(\rho_k, \sigma_{k'})$ are far, their corresponding mixtures $\tilde{\rho}^{(m)}$ and $\tilde{\sigma}^{(m)}$ may be statistically *close* by joint convexity of trace distance.

Remark 7.6 (No key-averaging collapse to EFI). For any distributions over keys, joint convexity of the trace distance [Ras07] yields

$$\Delta(\tilde{\phi}^{(m)}, \tilde{\psi}^{(m)}) \leq \mathbb{E}_{k,k'} [\Delta(\phi_k^{\otimes m}, \psi_{k'}^{\otimes m})],$$

where $\tilde{\psi}^{(m)} := \mathbb{E}_{k \leftarrow \{0,1\}^\kappa} \{\psi_{k(\lambda)}^{\otimes m}\}$ and $\tilde{\phi}^{(m)} := \mathbb{E}_{k' \leftarrow \{0,1\}^\kappa} \{\phi_{k'(\lambda)}^{\otimes m}\}$. Thus, EPFI's pairwise gaps do not imply $\Delta(\tilde{\phi}^{(m)}, \tilde{\psi}^{(m)}) \geq \Omega(1/\text{poly})$ in general. Therefore, by just averaging the

³Note that the trace distance condition is stated for *each pair of keys* k, k' , whereas computational indistinguishability is defined for the *ensembles* themselves.

keys the resulting pair of states does not need to produce an EFI pair. Moreover, due to data processing inequality, any CPTP map acting on both ensembles cannot increase the trace distance,

$$\Delta(\Lambda(\tilde{\phi}^{(m)}), \Lambda(\tilde{\psi}^{(m)})) \leq \Delta(\tilde{\phi}^{(m)}, \tilde{\psi}^{(m)}).$$

Note that the quantum channel Λ is agnostic to the key since it acts on both states. A simple toy example shows how averaging over ensembles can potentially decrease the trace distance. Consider two ensembles $\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$ and $\{|+\rangle\langle +|, |-\rangle\langle -|\}$. One can easily see that every pairwise combination leads to a $\Delta = 1/\sqrt{2}$ yet the mixtures coincide: $1/2(|0\rangle\langle 0| + |1\rangle\langle 1|) = 1/2(|+\rangle\langle +| + |-\rangle\langle -|)$. While this pair of ensembles is not an EPFI pair due to the indistinguishability condition, it gives a clear insight of why averaging can decrease the trace distance.

7.1.2. EPFI Pairs Imply Quantum Commitments

We can now prove the main result of this section, which is the construction of a canonical quantum commitment from EPFI. As with EFI pairs, EPFI pairs naturally lead to the construction of canonical quantum commitments.

- **Commit stage:** The committer commits to a bit b by generating a bipartite state $|\psi^b\rangle_{CR}$ and sending the register C to the receiver.
- **Reveal stage:** The committer discloses register R along with the bit b , allowing the receiver to verify the commitment by projecting onto $|\psi^b\rangle_{CR}$. If the verification fails, the receiver aborts.

In our construction, we introduce a classical key to accommodate the use of state ensembles. This modification does not alter the protocol since the key is also revealed in the opening stage. Our approach closely follows the honest statistical binding and computational hiding canonical quantum commitments construction from EFI pairs of [BCQ23]. Here, the committed register C contains a state sampled from one of the EPFI pairs. In the reveal stage, the committer provides the receiver with the purification of the sampled state, the secret key associated with the state, and the committed bit.

Honest binding follows from the statistical distance between any state sampled from one family of the EPFI pair and all states from the opposing ensemble together with Uhlmann's theorem [Uhl76]. Similarly, computational hiding follows from the computational indistinguishability of EPFI pairs. Thus, just as EFI pairs yield a statistically binding canonical quantum commitment when restricted to two algorithms, EPFI pairs provide the same commitment structure when generalized to two families of algorithms for two possible committed values. This new functionality, EPFI pairs, will play a central role in the construction of cryptography from different quantum pseudoresources, as sketched in Figure 7.1.

Theorem 7.7. *Assuming the existence of EPFI pairs of mixed states, there exist statistically binding and computationally hiding canonical quantum commitments.*

Proof. Let $\{Q_\psi^k\}_k$ and $\{Q_\phi^{k'}\}_{k'}$ be the two families of algorithms that generate the corresponding families of states $\{\psi_k\}_k$ and $\{\phi_{k'}\}_{k'}$.⁴ The construction of the canonical quantum commitment is as follows. Let the algorithms $\{Q_{b,\lambda}^k\}_k$ with $b \in \{0, 1\}$ correspond to the aforementioned $\{Q_\psi^k\}_k$ and $\{Q_\phi^{k'}\}_{k'}$, i.e., each family of states corresponds to one value of b . As in [BCQ23], the committer generates $\text{poly}(\lambda)$ copies of the state by applying $\bigotimes_{i=1}^{\text{poly}(\lambda)} (Q_{b,\lambda}^k)_i$, sending $\text{poly}(\lambda)$ copies to the receiver. Then, the committed state defined in Definition 7.2 is given by $\rho_{C,b}^{\otimes \text{poly}(\lambda)}$, where $\rho_{C,b} = \text{Tr}_R(Q_b^k |0\rangle\langle 0|_{CR} Q_b^{k\dagger})$ is a state from the EPFI ensemble (Definition 7.5). For the opening, the committer sends the $\text{poly}(\lambda)$ registers R , together with the key k ⁵ and the committed bit b .

Let us first prove honest statistical binding. Given the two states ψ_k and $\phi_{k'}$, $\Delta(\psi_k, \phi_{k'}) \geq \Omega(1/\text{poly}(\lambda))$ by definition of the EPFI. Lastly, by taking polynomially many copies of the states, $\Delta(\psi_k^{\otimes \text{poly}(\lambda)}, \phi_{k'}^{\otimes \text{poly}(\lambda)}) \geq 1 - \text{negl}(\lambda)$ by Theorem 2.18. Moreover, due to the Fuchs–van de Graaf inequality for the squared fidelity convention used in this thesis,

$$F(\psi_k^{\otimes \text{poly}(\lambda)}, \phi_{k'}^{\otimes \text{poly}(\lambda)}) \leq 1 - \left(\Delta(\psi_k^{\otimes \text{poly}(\lambda)}, \phi_{k'}^{\otimes \text{poly}(\lambda)}) \right)^2 \leq \text{negl}(\lambda). \quad (7.1.4)$$

Therefore, by Uhlmann’s theorem (Theorem 2.19), the norm in Eq. 7.1.2 is bounded by the square root of the fidelity above, and the scheme satisfies honest binding.

Computational hiding follows from the computational indistinguishability property of the EPFI pairs: without the key, a distinguisher cannot infer the ensemble from which the state has been sampled, satisfying Eq. 7.1.3.

□

Remark 7.8 (EFI is a special case of EPFI). Let $\{\rho_{0,\lambda}\}_\lambda$ and $\{\rho_{1,\lambda}\}_\lambda$ be an EFI pair as in Definition 7.1. Define EPFI families $(\{\psi_{k,\lambda}\}_{k \in \mathcal{K}_\lambda}, \{\phi_{k',\lambda}\}_{k' \in \mathcal{K}'_\lambda})$ as in Definition 7.5 with singleton key spaces $\mathcal{K}_\lambda = \mathcal{K}'_\lambda = \{0\}$ and set $\psi_{0,\lambda} := \rho_{0,\lambda}$ and $\phi_{0,\lambda} := \rho_{1,\lambda}$. In this case both families are singletons, and all EPFI requirements reduce to the corresponding EFI properties: efficient generation, pairwise statistical fairness, and computational indistinguishability.

7.2. Pseudoresources and Quantum Cryptography

In this section, we establish a foundational connection between computational-based quantum cryptography and resource theories. We begin by briefly recalling the formalism

⁴We omit the security parameter λ for simplicity.

⁵Please note that the construction can be extended to quantum keys, i.e., the key is a quantum state, in the subspace of the registers R .

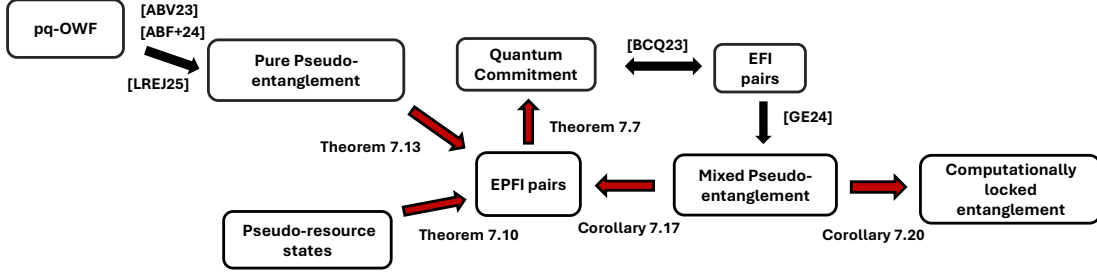


Figure 7.1.: Summary of results and their relation to previously defined primitives.

of resource theories (see Definition 2.31), define the concept of *pseudoresource*, and finally discuss how the existence of pseudoresources implies the existence of EFI pairs of ensembles and, thus, computationally based cryptography. The relation with the primitives considered in this chapter is summarized in Figure 7.1.

7.2.1. Resource Theory and Pseudoresources

We use the resource-theoretic notation introduced in Definitions 2.31 and 2.32. Thus, a resource theory is denoted by $\mathcal{R} = (\mathcal{F}, \mathcal{O})$, where $\mathcal{F}(\mathcal{H}) \subseteq \mathcal{S}(\mathcal{H})$ is the set of free states and $\mathcal{O}$ is the corresponding set of free operations. In the case of entanglement, the free states are separable states and the free operations can be chosen as LOCC maps, as recalled in Definition 2.25. Throughout this section, the set $\mathcal{F}(\mathcal{H})$ is assumed to be convex and closed, to contain a full-rank state, and all systems are finite-dimensional, as required for the continuity bound of Lemma 2.33. We quantify resources by the relative entropy of resource R_{rel} defined in Definition 2.32.

When taking into account computationally bounded operations, families of states that can be distinguished in the asymptotic regime might become indistinguishable against polynomially bounded quantum adversaries. This property becomes even more interesting when these states have very different properties. We focus on the case in which both families have a substantial difference in terms of resources.

Definition 7.9 (η -gap pseudoresource). Let $\lambda \in \mathbb{N}_+$ and $\eta : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be arbitrary. A pair of families $\{\psi_{k(\lambda)}\}_{k(\lambda)}$ and $\{\phi_{k'(\lambda)}\}_{k'(\lambda)}$ of (potentially mixed) states indexed by $k(\lambda), k'(\lambda) \in \{0, 1\}^{\kappa(\lambda)}$ is said to form an η -gap pseudoresourced for $\mathcal{R}$ if,

1. **Efficient generation:** Given $k(\lambda)$ (or $k'(\lambda)$), there exists a QPT algorithm A that on input $(1^\lambda, k)$ (or $(1^\lambda, k')$) outputs $\psi_{k(\lambda)}$ (or $\phi_{k'(\lambda)}$).
2. **Resource gap:** For all $k, k' \in \{0, 1\}^{\kappa(\lambda)}$,

$$|R_{\text{rel}}(\psi_{k(\lambda)}) - R_{\text{rel}}(\phi_{k'(\lambda)})| \geq \eta,$$

where $R_{\text{rel}}(\rho)$ is the relative entropy of resource $\mathcal{R}$.

3. **Computational indistinguishability:** For any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr_{k(\lambda)}[\mathcal{D}(\sigma_\lambda, \psi_{k(\lambda)}^{\otimes m}) = 1] - \Pr_{k'(\lambda)}[\mathcal{D}(\sigma_\lambda, \phi_{k'(\lambda)}^{\otimes m}) = 1] \right| \leq \nu(\lambda).$$

Under computational restrictions, resource properties can therefore be hidden from efficient observers. This phenomenon has been studied for entanglement [ABF⁺24, ABV23, GE24, LREJ25], magic [GLG⁺24], and coherence [HBK25]. More generally, pseudoresources can be defined for arbitrary resource monotones [HBK25, BMB⁺25]; here we focus on the relative entropy of resource, which is the entropic measure naturally associated with the resource theory $\mathcal{R}$.

7.2.2. Pseudoresources Imply EPFI Pairs

We prove now the main technical contribution of our result.

Theorem 7.10 (Pseudoresource implies EPFI pairs). *For any $\eta \geq 2 + 1/\text{poly}(n)$, assuming the existence of η -gap pseudoresource with $K_{\mathcal{R}} := \sup_{\tau, \tau'} |R_{\text{rel}}(\tau) - R_{\text{rel}}(\tau')| = \text{polylog}(d)$, EPFI pairs exist.*

Proof. The construction of an EPFI pair given a pair of pseudoresourced families of states is straightforward: each family of the pseudoresourced states corresponds to an EPFI ensemble. The efficient generation of the EPFI pairs together with the property of computational indistinguishability follow from the definition of η -gap pseudoresource.

To prove statistical distance, it follows from Equation (2.5.11) that, for every $k, k' \in \{0, 1\}^{\kappa(\lambda)}$

$$\Delta(\psi_{k(\lambda)}, \phi_{k'(\lambda)}) \geq \frac{|R_{\text{rel}}(\psi_{k(\lambda)}) - R_{\text{rel}}(\phi_{k'(\lambda)})| - 2}{K_{\mathcal{R}}}.$$

Moreover, by taking into account that $\eta \geq 2 + 1/\text{poly}(n)$ and $K_{\mathcal{R}} = \text{polylog}(d)$,

$$\Delta(\psi_{k(\lambda)}, \phi_{k'(\lambda)}) \geq \Omega\left(\frac{1}{\text{poly}(n)}\right) \quad \forall k, k' \in \{0, 1\}^{\kappa(\lambda)}.$$

□

Remark 7.11. The construction of EPFI pairs from pseudorandom families of states is given by the asymptotic continuity of the relative entropy of resource. Nevertheless, a similar construction holds for any resource monotone which is asymptotically continuous.

Therefore, just assuming the existence of pseudoresource families whose relative entropy of resource presents a gap of at least $2 + 1/\text{poly}(n)$, EPFI pairs and thus, quantum commitments and all the primitives that follow from it such as oblivious transfer and multiparty computing can be constructed. Constructions such as the one of pseudoresources from pseudo-random density matrices [BMB⁺25] hold⁶. A similar construction was also proposed by [GLG⁺24] for the specific case of pseudomagic⁷. Nevertheless, we generalize the result for any pseudoresource.

7.3. Cryptography from Pseudoentanglement

Since the proposal of the notion of pseudoentanglement [ABF⁺24], different definitions of the same phenomenon have been proposed [ABV23, GE24, LREJ25], extending the definition to mixed states and using different measures of entanglement that allow one to obtain a maximal separation in terms of entanglement between both families of states. Nevertheless, all definitions capture the same phenomenon: given two families of states that are efficiently generated, both families present a gap in the entanglement, yet they are computationally indistinguishable.

Although the existence of mixed-state pseudoentanglement was proven to be a minimal assumption for computationally based cryptography [GE24], no cryptographic primitives have been constructed from it to the best of our knowledge. The goal of this section is to establish a direct connection between the different definitions of pseudoentanglement and cryptography. The first subsection studies the construction of EPFI pairs from pure state pseudoentanglement, while the second extends this result to the case of mixed states. Lastly, we propose a new functionality that is intrinsically dependent on the resource of entanglement.

7.3.1. Pure State Pseudoentanglement Implies EPFI Pairs

The first formalization of pseudoentanglement, introduced in [ABF⁺24], concerns families of pure states. This is natural for that construction, but also for the chosen measure, since the entropy of entanglement is defined for pure bipartite states. Later works, such as [ABV23, LREJ25], consider more general entanglement measures, although their constructions still produce pseudoentangled families of pure states.

Let us first define the notion of pure state pseudoentanglement [ABF⁺24].

Definition 7.12 (Pure η -gap pseudoentanglement). Let $\lambda \in \mathbb{N}_+$ and $\eta : \mathbb{N}_+ \rightarrow \mathbb{N}_+$ be arbitrary. A pair of ensembles of bipartite pure states $\{|\psi^{k(\lambda)}\rangle_{AB}\}_{k(\lambda)}$, $\{|\phi^{k'(\lambda)}\rangle_{AB}\}_{k'(\lambda)}$ indexed by $k, k' \in \{0, 1\}^{\kappa(\lambda)}$ is said to have pure η -pseudoentanglement if,

⁶Note that in [BMB⁺25] EFI pairs are constructed assuming the existence of pseudo-random density matrices.

⁷The construction of pseudomagic states of [GLG⁺24] actually implies our proposed primitive of EPFI pairs, as it is proven in SM V.B of [GLG⁺24].

1. **Efficient generation:** Given $k(\lambda)$ (or $k'(\lambda)$), there exists a QPT algorithm A that on input $(1^\lambda, k, b)$ (or $(1^\lambda, k', b)$) outputs $|\psi^{k(\lambda)}\rangle_{AB}$ (or $|\phi^{k'(\lambda)}\rangle_{AB}$).

2. **Entanglement gap:** For all $k, k' \in \{0, 1\}^{\kappa(\lambda)}$,

$$|E(|\psi^{k(\lambda)}\rangle_{AB}) - E(|\phi^{k'(\lambda)}\rangle_{AB})| \geq \eta,$$

where $E(|\psi\rangle_{AB}) := E_{A/B}(|\psi\rangle\langle\psi|_{AB})$ is the entanglement entropy from Definition 2.26.

3. **Computational indistinguishability:** For any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr_{|\varphi\rangle \leftarrow \{|\psi^{k(\lambda)}\rangle_{AB}\}_{k(\lambda)}} [\mathcal{D}(\sigma_\lambda, |\varphi\rangle^{\otimes m}) = 1] - \Pr_{|\varphi\rangle \leftarrow \{|\phi^{k'(\lambda)}\rangle_{AB}\}_{k'(\lambda)}} [\mathcal{D}(\sigma_\lambda, |\varphi\rangle^{\otimes m}) = 1] \right| \leq \nu(\lambda).$$

Having introduced the definition of pseudoentanglement, we can prove the main statement of this section.

Theorem 7.13 (Pure pseudoentanglement implies EPFI pairs). *For any $\eta \geq 1/2e + 1/\text{poly}(n)$ and assuming the existence of pure η -gap pseudoentanglement, then EPFI pairs exist.*

Proof. The construction of an EPFI pair given a pair of pseudoentangled pair of families of pure states is slightly different from that of Theorem 7.10. We define the two families with the reduced density matrices of the pseudoentangled states:

$$\begin{aligned} \{\rho_{A,\psi}^{k(\lambda)}\}_{k(\lambda)} &= \{ \text{Tr}_B (|\psi^{k(\lambda)}\rangle\langle\psi^{k(\lambda)}|_{AB}) \}_{k(\lambda)}, \\ \{\sigma_{A,\phi}^{k'(\lambda)}\}_{k'(\lambda)} &= \{ \text{Tr}_B (|\phi^{k'(\lambda)}\rangle\langle\phi^{k'(\lambda)}|_{AB}) \}_{k'(\lambda)}, \end{aligned}$$

where the entanglement is measured across the cut $(A : B)$. The properties of efficient generation and computational indistinguishability of the EPFI pairs follow from the definition of η -gap pure pseudoentanglement.

We now prove that for all $k, k' \in \{0, 1\}^{\kappa(\lambda)}$, $\rho_{A,\psi}^{k(\lambda)}$ and $\sigma_{A,\phi}^{k'(\lambda)}$ are statistically far. We have that

$$\Delta(\rho_{A,\psi}^{k(\lambda)}, \sigma_{A,\phi}^{k'(\lambda)}) \geq \frac{|S(\rho_{A,\psi}^{k(\lambda)}) - S(\sigma_{A,\phi}^{k'(\lambda)})| - c}{2n(\lambda)} = \frac{|E(|\psi^{k(\lambda)}\rangle_{AB}) - E(|\phi^{k'(\lambda)}\rangle_{AB})| - c}{2n(\lambda)},$$

where the first inequality follows from Lemma 2.23, and the last equality uses the definition of the entanglement entropy. Therefore, using the fact that for all $k, k' \in \{0, 1\}^{\kappa(\lambda)}$ the entanglement entropies of pairwise sampled states from the ensembles is $\eta \geq 1/2e + 1/\text{poly}(n)$,

$$\Delta(\rho_{A,\psi}^{k(\lambda)}, \sigma_{A,\phi}^{k'(\lambda)}) \geq \Omega\left(\frac{1}{\text{poly}(\lambda)}\right) \quad \forall k, k' \in \{0, 1\}^{\kappa(\lambda)}.$$

□

Every known construction of pseudoentanglement from pure state ensembles [ABF⁺24, ABV23, LREJ25] exhibits an entanglement entropy gap of at least $1/2e + 1/\text{poly}(n)$, making them suitable for constructing EPFI pairs. While the definitions in [ABV23, LREJ25] allow for a larger gap when considering computationally efficient entanglement measures, it is the information-theoretic measure of entanglement entropy that determines the relationship between the gap and trace distance, and thus enables the construction of EPFI pairs. The connection between information-theoretic and computationally meaningful entanglement measures, along with its cryptographic implications, is explored in more detail in the following section.

Remark 7.14. We notice that in the proof of Theorem 7.13, we do not need indistinguishability between the pure states, but only of subsystem A (or B). In this case, we can also achieve EPFI pairs under a weaker notion of pure-state pseudoentanglement.

The use of entanglement entropy as a measure of entanglement in the case of pure states requires a lower bound of $\omega(\log n)$ for the low entangled family. However, when computational measures of entanglement are taken into account, the entanglement gap can be even larger for pure states, i.e., $\Omega(n)$ vs. $o(1)$ for other entanglement measures as proven in [LREJ25]. While it is true that entanglement entropy loses its operational meaning when it comes to quantifying the distillable entanglement (or entanglement cost) taking into account computational efficiency, it is relevant for the construction of EPFI pairs.

7.3.2. Mixed State Pseudoentanglement Implies EPFI Pairs

Unlike pure state pseudoentanglement, the entanglement in pseudoentangled mixed states cannot be quantified using the entanglement entropy. The necessity of an operationally meaningful entanglement monotone motivated [ABV23] to use the entanglement cost, E_C , and the distillable entanglement, E_D , together with their computationally meaningful counterparts, in which the generation or distillation of entanglement has to be implementable by a polynomial-time algorithm (see Definitions 6.11 and 6.12). The definition of pseudoentanglement with these entanglement monotones allowed [GE24] to construct a pseudoentangled mixed-state family from EFI pairs or, equivalently, to prove mixed pseudoentanglement as a new minimal assumption for the existence of computationally based cryptography, as sketched in Figure 7.1. Moreover, the use of other entanglement monotones also allows the existence of a maximal gap in the pseudoentangled families of $\Theta(n)$ vs 0 .⁸ We now use this perspective to construct EPFI pairs from pseudoentangled mixed states, starting with our proposed definition.

Definition 7.15 (Mixed η -gap pseudoentanglement). Let $\lambda \in \mathbb{N}_+$ and $\eta : \mathbb{N}_+ \rightarrow \mathbb{N}_+$ be arbitrary. A pair of families of mixed bipartite states $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ and $\{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}$ indexed by $k, k' \in \{0, 1\}^{\kappa(\lambda)}$ is said to have *mixed η -gap pseudoentanglement* if:

⁸In contrast to pure pseudoentanglement, where the maximal gap is given by $\Theta(n)$ vs $\omega(\log(n))$.

1. **Efficient generation:** Given $k(\lambda)$ (or $k'(\lambda)$), there exists a QPT algorithm A that on input $(1^\lambda, k, b)$ (or $(1^\lambda, k', b)$) outputs $\psi_{AB}^{k(\lambda)}$ (or $\phi_{AB}^{k'(\lambda)}$).
2. **Entanglement gap:** For all $k, k' \in \{0, 1\}^{\kappa(\lambda)}$,

$$\left| E_R^\infty(\psi_{AB}^{k(\lambda)}) - E_R^\infty(\phi_{AB}^{k'(\lambda)}) \right| \geq \eta.$$

3. **Computational indistinguishability:** For any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr_{\rho \leftarrow \{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}} [\mathcal{D}(\sigma_\lambda, \rho^{\otimes m}) = 1] - \Pr_{\rho \leftarrow \{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}} [\mathcal{D}(\sigma_\lambda, \rho^{\otimes m}) = 1] \right| \leq \nu(\lambda).$$

Our definition is based on the ones proposed by [ABV23, GE24] but with two modifications. Foremost, both families of states have to be efficiently generated, contrary to previous definitions in which only the low entangled family was the efficiently generated one. The other major difference is that we consider the (regularised) relative entropy of entanglement, instead of the computational distillable entanglement or the computational entanglement cost, which was used in those results.

Unlike in [ABV23, GE24], our proposed definition of pseudoentanglement is defined in the asymptotic IID setting without taking into account computational entanglement measures. As proposed in [ABV23], their pseudoentanglement construction can be extended to the computational asymptotic IID setting by considering the following measures. Moreover, we adapt the one-shot computational entanglement measures introduced in Definitions 6.11 and 6.12 to the case where, for each security parameter $\lambda \in \mathbb{N}$, the possible states of size λ are indexed by a classical key $k \in \{0, 1\}^{\kappa(\lambda)}$, with $\kappa : \mathbb{N}_+ \rightarrow \mathbb{N}_+$. We will write such a keyed family as $\{\rho_{AB}^k\}_k$, keeping the security parameter implicit in the key size. In the distillation task, the efficient LOCC map, understood through the circuit description of Definition 6.10, has access to the key and acts as $\Gamma(k, \rho^k)$, with input $|k\rangle\langle k|_{A'} \otimes \rho_{AB}^k \otimes |k\rangle\langle k|_{B'}$ across the bipartition $A'A : B'B$. The same convention applies to the cost of generating all the states associated with all possible keys. The corresponding keys can also be quantum, more specifically EFI pairs, as proven in [GE24].

Definition 7.16 (Asymptotic IID computational entanglement measures). Given a keyed family of bipartite states $\{\rho_{AB}^k\}_{k(\lambda)}$, its asymptotic IID computational entanglement cost and distillable entanglement are defined as

$$\begin{aligned} \hat{E}_C^\infty(\{k, \rho_{AB}^k\}) &:= \lim_{\epsilon \rightarrow 0} \limsup_{t \rightarrow \infty} \frac{1}{t} \hat{E}_C^\epsilon(\{k, (\rho_{AB}^k)^{\otimes t}\}), \\ \hat{E}_D^\infty(\{k, \rho_{AB}^k\}) &:= \lim_{\epsilon \rightarrow 0} \liminf_{t \rightarrow \infty} \frac{1}{t} \hat{E}_D^\epsilon(\{k, (\rho_{AB}^k)^{\otimes t}\}), \end{aligned}$$

where $\hat{E}_C^\epsilon$ and $\hat{E}_D^\epsilon$ are defined in Definition 6.11 and Definition 6.12.

Let us now show that our definition is a relaxation in the condition of the entanglement gap with respect to the ones of [ABV23, GE24] in the asymptotic IID setting.

Given a ‘highly entangled’ family $\{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}$ such that $\hat{E}_D^\infty(\{k', \phi_{AB}^{k'}\}) \geq d(\lambda)$ and a ‘low entangled’ family $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ such that $\hat{E}_C^\infty(\{k, \psi_{AB}^k\}) \leq c(\lambda)$ with $c(\lambda) < d(\lambda)$, then, given two states $\phi_{AB}^{k'(\lambda)}$ and $\psi_{AB}^{k(\lambda)}$,

$$d(\lambda) - c(\lambda) \leq \hat{E}_D^\infty(\{k', \phi_{AB}^{k'}\}) - \hat{E}_C^\infty(\{k, \psi_{AB}^k\}) \leq E_R^\infty(\phi_{AB}^{k'(\lambda)}) - E_R^\infty(\psi_{AB}^{k(\lambda)})$$

for all k and k' , since for any family $\hat{E}_D^\infty \leq E_D^\infty \leq E_R^\infty \leq E_C^\infty \leq \hat{E}_C^\infty$ under LOCC operations.

Corollary 7.17 (Mixed pseudoentanglement implies EPFI pairs). *For $\eta \geq 2 + 1/\text{poly}(n)$ and assuming the existence of mixed η -pseudoentanglement, EPFI pairs exist.*

Please note that the construction is similar to the one of Theorem 7.10, but taking into account that in this case the studied resource is the entanglement. In the case of entanglement, the regularised relative entropy of resource is equivalent to the regularised relative entropy of entanglement, whose asymptotic continuity bound is given by Lemma 2.30.

Therefore, every state that has a large pseudoentangled gap $d(\lambda) - c(\lambda) \geq 2 + 1/\text{poly}(n)$ is eligible for building EPFI pairs. On the other hand, there potentially exist pairs of families of mixed states which do not present a gap in the computational measures of entanglement, i.e., they are not pseudoentangled following the definition of [ABV23, GE24], but from which EPFI pairs can be constructed.

7.3.3. Beyond EPFI: Computationally Locked Entanglement

Previous definitions of pseudoentanglement focus on efficiently generated states with low entanglement that are computationally indistinguishable from highly entangled states. This is an analogue of the ‘classical’ concept of pseudorandomness: a simple object (such as a pseudorandom string or pseudoentangled state) that can replace a complex one (resp. random string or highly entangled state).

On the other hand, given how central entanglement is in quantum information, and the plethora of applications that require highly entangled states, we can flip this question and ask for efficiently generated states with high entanglement that are computationally indistinguishable from low entangled ones (that may be efficiently generated or not). We denote this as *computationally locked entanglement*, and it can be used to conceal entanglement that could be distilled only with the help of a secret key. We notice that pseudoentangled states where both low- and high-entangled families are efficiently generated (as in our definition of pseudoentanglement) exhibit computationally locked entanglement.

Definition 7.18 (Computationally locked entanglement). Let $\lambda \in \mathbb{N}$, $n : \mathbb{N} \rightarrow \mathbb{N}$ be a polynomially bounded function, $\epsilon : \mathbb{N} \rightarrow [0, 1]$ and $c, d : \mathbb{N} \rightarrow \mathbb{N}$ with $c < d$. A family of $2n(\lambda)$ -qubit bipartite states $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ is said to have *computationally locked entanglement* (ϵ, c, d) if there is a family of $2n(\lambda)$ -qubit bipartite states $\{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}$, such that:

1. The computational entanglement cost of the family $\{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}$ is upper bounded as $\hat{E}_C^\epsilon(\{k', \phi_{AB}^{k'}\}) \leq c$.
2. Given $k(\lambda)$, there exists a QPT algorithm A that on input $(1^\lambda, k, b)$ outputs $\psi_{AB}^{k(\lambda)}$.
3. The computational distillable entanglement of the family $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ is lower bounded as $\hat{E}_D^\epsilon(\{k, \psi_{AB}^k\}) \geq d$.
4. For any non-uniform QPT distinguisher $\mathcal{D}$ with advice σ_λ and any $m \in \text{poly}(\lambda)$, there exists a negligible function $\nu(\lambda) > 0$ such that:

$$\left| \Pr_{\rho \leftarrow \{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}} [\mathcal{D}(\sigma_\lambda, \rho^{\otimes m}) = 1] - \Pr_{\rho \leftarrow \{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}} [\mathcal{D}(\sigma_\lambda, \rho^{\otimes m}) = 1] \right| \leq \nu(\lambda).$$

Computationally locked entanglement can be viewed as a “dual” notion to pseudoentanglement, as defined in [ABV23, GE24]. An example of computational locked entanglement is the construction of pseudorandom density matrices [BMB⁺25]. However, in this case, the “low-entangled” family is also efficiently generated. It would be interesting to investigate how pseudoentanglement and computationally locked states relate to each other, i.e., if one implies the other, or if they are incomparable.

We now prove the intuitive consequences of computationally locked states for parties that do not have access to the key.

Lemma 7.19. *Given a family of $2n(\lambda)$ -qubit bipartite states $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ that have computationally locked entanglement (ϵ, c, d) , where $c < d$, its computational distillable entanglement without access to the key k is upper bounded as $\hat{E}_D^\epsilon(\{\psi_{AB}^k\}) \leq c$.*

Proof. It can be proven by contradiction. Suppose that there exists a family of states $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ that have computationally locked entanglement as defined in Definition 7.18 such that, not given the keys k , $\hat{E}_D^\epsilon(\{\psi_{AB}^k\}) > c$. As proven in [ABV23], $\hat{E}_D^\epsilon < \hat{E}_C^\epsilon$. Moreover, $\hat{E}_C^\epsilon(\{k', \phi_{AB}^{k'(\lambda)}\}) < c$ by construction. Then, if there is a poly time algorithm that is able to distill $\hat{E}_D^\epsilon(\{\psi_{AB}^{k(\lambda)}\}) > c$, it would be able to distinguish between the pair of families $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ and $\{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}$ which is not possible by definition. $\square$

Corollary 7.20. *Given a $2n(\lambda)$ -qubit bipartite state family $\{\psi_{AB}^{k(\lambda)}\}_{k(\lambda)}$ which has computationally locked entanglement (ϵ, d) , i.e., w.r.t. a family of $2n(\lambda)$ -qubit separable bipartite states $\{\phi_{AB}^{k'(\lambda)}\}_{k'(\lambda)}$, no entanglement can be distilled without the respective keys k .*

An application of the computationally locked functionality is an authenticated quantum teleportation protocol. The scheme is similar to the original quantum teleportation [BBC⁺93], but here the receiver cannot access the teleported state without using a secret key in each interaction. In particular, for privacy against keyless adversaries, the classical correction string can be broadcast publicly, and we do not rely on a pre-authenticated classical channel. In the standard protocol, Alice and Bob share an entangled state, Alice performs a Bell measurement on the input and her share of the entanglement pair, and she sends the classical correction string to Bob, who applies the corresponding Pauli correction and recovers the unknown state.

Our goal is that Alice can distribute the entanglement during the protocol, and can publicly broadcast the teleportation corrections, while any party without Bob's key cannot efficiently recover the teleported state. This is conceptually different from assuming an authenticated and confidential classical channel: in our setting, the secrecy of the teleported state is achieved by keying the entangled resource itself. Moreover, under our computational assumption, the same family of states and its corresponding keys can be used a polynomial number of times, in contrast to one-time information-theoretic approaches such as Clifford-based encryption/twirling [DLT02, ABOE08].

Further applications of computationally locked entanglement may arise in quantum networks where entanglement must be distributed while preventing users from accessing it without authorization. Since routing in quantum networks relies on entanglement swapping [BDCZ98, Cal17], encoding network links or nodes with computationally locked entanglement may provide a way to certify or enforce routing policies under computational assumptions.

PART III

NEAR-TERM APPLICATIONS: QUANTUM CRYPTOGRAPHY BEYOND QKD

Chapter 8

A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions

Since the early stages of quantum cryptography, quantum protocols that achieve information-theoretical security (i.e., security against unbounded adversaries) have been known for primitives that are impossible in the classical setting [BB84]. However, it was soon realized that even quantumly, a very limited family of cryptographic primitives can achieve such a level of security [LC97, May97], and therefore, the use of computational assumptions is necessary to reach quantum advantage.

Multi-party computation (MPC) is a very versatile primitive that plays a central role in classical cryptography. In MPC, many parties want to collectively compute a function that depends on their private inputs, while keeping the inputs secret (even if many of these parties are malicious and deviate from the original protocol). This functionality can be constructed from another primitive: oblivious transfer (OT) [DFL⁺09]. However, it is not expected that OT (and thus MPC) can be constructed only from one-way functions (OWFs)¹².

On the other hand, [GLSV21, BCKM21] showed that OT *can* be built from OWFs and quantum resources. More precisely, [GLSV21, BCKM21] showed how to build equivocal and extractable commitment schemes, which were known to be sufficient to build quantum protocols for OT [DFL⁺09] and MPC [IPS08, Kil88]³.

One important feature of [GLSV21, BCKM21] is that the quantum resources required for such protocols are exactly the same as the ones used in some quantum key distribution (QKD) protocols: they only need to prepare, communicate, and measure one-qubit states in conjugate bases. Given the huge progress in the implementation of QKD in very different setups [AAB⁺23, GBR⁺23, PSC⁺23] [ZMM⁺24], it would be expected that one could easily implement such quantum protocols with current technology.

Unfortunately, this is not the case. While the protocols proposed by [GLSV21, BCKM21] are important to theoretically understand the power of quantum resources, their building blocks put serious barriers in their experimental implementation. More concretely, they pose the following difficulties:

¹One-way functions are functions that are easy to compute and hard to invert. These are considered the minimal computational assumption in classical cryptography.

²More formally, [IR90] shows that no MPC protocol can be built from OWF in a black-box way.

³We notice that later, it was shown that MPC can be built even from weaker computation assumptions such as pseudo-random states [MY22, AQY22].

- **Fragility against errors.** The protocols proposed by [GLSV21] and [BCKM21] do not tolerate experimental errors such as bit flips during the state distribution steps.
- **Practical hash functions and zero knowledge proofs.** Zero knowledge proofs used in [GLSV21] do not make black-box use of OWFs, and the practical implementation of the protocol would require the arithmetical description of the inner functioning of the one-way functions. It is currently unclear how to achieve such descriptions for widespread heuristic implementations of post-quantum OWFs like the secure hash standard (SHA), making their integration in the [GLSV21] scheme non-trivial.
- **Inefficiency.** The iterative structure of the protocol proposed by [BCKM21] allows the protocol to make a black-box use of the underlying components but requires a very large quantity of quantum states, which prevents it from being practically implementable. For example, one run of that protocol requires an order of 10^{13} BB84 states [BB84], which are provided by days of transmission given a state-of-the-art discrete-variable QKD setup.

The main contribution of this chapter is to provide a noise-tolerant protocol for OT, based on the structure from [BCKM21], while avoiding some of its bottlenecks and while making it efficiently implementable. In particular, with this new protocol, we expect that around 10^7 BB84 states would be sufficient instead of 10^{13} , improving the transmission time to the order of seconds. Moreover, we notice that with this amount of quantum resources, we can distill more than one OT at once, which will be crucial in future uses of this protocol to implement MPC.

8.1. Background and Our Results

As previously mentioned, the goal of [GLSV21, BCKM21] is to achieve a quantum protocol for oblivious transfer, since it can be used to construct MPC in a generic way [IPS08]. Oblivious transfer is a cryptographic primitive where Alice chooses two messages, m_0 and m_1 , Bob chooses a bit $b \in \{0, 1\}$, and Bob learns m_b (notice that Bob should not learn $m_{\bar{b}}$ and Alice should not learn b).

In the quantum oblivious transfer (QOT) protocol proposed in [CK90, BBCS92], Alice sends BB84 states to Bob. The encoded states are $\left(\left| x_i^{OT} \right\rangle_{\theta_i^{OT}} \right)_{i \in [2\lambda_{OT}]}$, where $x_i^{OT} \in \{0, 1\}$ is the value of the bit, $\theta_i^{OT} \in \{+, \times\}$ is the choice of encoding bases and λ_{OT} is a security parameter. Bob chooses measurement bases $\hat{\theta}_i^{OT} \in \{+, \times\}$ and obtains measurement results $\hat{x}_i^{OT} \in \{0, 1\}$. Since we cannot guarantee that a malicious Bob will follow the protocol (in particular, that he keeps these qubits in a quantum memory instead of measuring them), Alice and Bob perform a sub-protocol that will give guarantees to Alice that Bob did measure the quantum state. However, this sub-protocol comes with a cost, since it adds the need for bit commitment, and requires the added cost of cryptographic assumptions. More concretely, to ensure Bob's measurement, Bob *commits* to the measurement bases that he

chose and the outcomes resulting from the measurement. Alice can then choose a subset of these pairs of measurements/outcomes, and she asks Bob to *open* the corresponding commitments. Then, she can check if the outcomes are consistent with the BB84 states that she sent. If they are, it can be shown that there are strong guarantees that Bob measured (most of) the qubits sent by Alice [BF10]. The bit commitment sub-protocol is finished when Alice sends the bases of her original BB84 states. Bob then divides his indices into two different subsets $I_b = \{i : \theta_i^{OT} = \hat{\theta}_i^{OT}\}$ and $I_{\bar{b}} = \{i : \theta_i^{OT} \neq \hat{\theta}_i^{OT}\}$, which depend on the bit choice bit b . Bob sends the two sets I_0 and I_1 to Alice who can encode the messages m_0 and m_1 using $\mathbf{x}_0^{OT} = \{x_i^{OT} : i \in I_0\}$ and $\mathbf{x}_1^{OT} = \{x_i^{OT} : i \in I_1\}$ as encrypting keys. Bob receives the encoded messages and decodes m_b .

The subtleties here are the properties needed in the commitments to prove that the protocol is secure. In standard bit commitment schemes, there are two properties of interest: *hiding*, meaning that the receiver cannot learn the committed message before the opening, and *binding*, meaning that after committing, there is only one value that can be opened by the committer. However, these properties alone do not allow us to prove the security of the quantum OT protocol in the simulation-based setting (which is needed to use this building block in MPC protocols). To prove the simulation-security of the QOT protocol, a strengthening of these properties is needed, namely:

Equivocality: A bit commitment protocol is called equivocal if there is a simulator (also called equivocator) that can perform a “dummy commitment” that can be opened to any desired value. Moreover, such a dummy commitment cannot be distinguished from the real ones by any polynomial-time distinguisher.⁴

Extractability: A bit commitment protocol is called extractable if there is a simulator (also called extractor) that is able to extract the commitment message in the commitment phase of the protocol.⁵

In [DFL⁺09], they show that equivocation and extraction of the quantum bit commitment are sufficient properties to prove the simulation-security of the QOT protocols of [CK90, BBCS92]. Moreover, the technical contribution of [GLSV21, BCKM21] was to show a quantum protocol bit commitment that is equivocal and extractable. The main observation that allowed us to improve the parameters of such a protocol is that full extraction is not needed in order to prove the security of the QOT protocol. Instead, we show that *relaxed extractability* is sufficient, and we show a more efficient bit commitment protocol that achieves this property (and is still equivocal). We describe the details of our contributions in the next section.

⁴We notice that for this to be true, the simulator has to have some leverage (e.g., a trapdoor, or the ability to rewind), otherwise the protocol would not be secure.

⁵We have here the same considerations as in Footnote 4.

8.2. Technical Overview

We now explain the approach of [BCKM21] to achieve extractable and equivocal commitment, and discuss how we modify it. In their result, they propose two compilers: in the first one, they propose an equivocal commitment scheme based on any bit commitment scheme that satisfies only “vanilla” binding and hiding. Moreover, the resulting commitment is extractable if the original commitment was also extractable. In the second commitment, they show how to turn an equivocal commitment into an extractable one (while losing equivocality). We note that in this second step, even if the original bit commitment scheme is classical, the resulting one would be a quantum protocol. They achieve the equivocal and extractable bit commitment scheme by starting with a standard bit commitment scheme, and then applying the equivocal compiler, followed by the extractable compiler, and then finally the equivocal compiler again.

In the equivocal compiler, *EqCommitment*, the committer generates λ pairs of random bits (u_i^0, u_i^1) . For each i , the committer and the receiver proceed as follows, sequentially: the committer commits, using the base bit commitment scheme, to each one of them twice leading to the commitments $(c_{i,0}^0, c_{i,1}^0, c_{i,0}^1, c_{i,1}^1)$. Then, the receiver chooses a random bit γ_i and the committer opens $c_{i,0}^{\gamma_i}, c_{i,1}^{\gamma_i}$. The verifier aborts if such pairs of commitments open to different values. At the end of this interaction, the committer sends $e_i = b \oplus u_i^{\gamma_i}$ for every i . In the opening phase, the committer chooses bits d_i and opens $c_{i,d_i}^{\gamma_i}$ and the receiver accepts if all the e_i ’s decommit to the same value b .

In the equivocality proof, the equivocator guesses γ_i , and then commits to two different values on $c_{i,0}^{\gamma_i}, c_{i,1}^{\gamma_i}$. At each step, the equivocator uses Watrous’ rewinding [Wat06] to amplify the success probability to $1 - \text{negl}(\lambda)$. In this case, in the opening phase, the equivocator can choose which $c_{i,d_i}^{\gamma_i}$ to open, so that the decommitment will be the desired bit. The binding/equivocal property follows from the fact that, if many of the checks on γ_i ’s passed, then most of the pairs $c_{i,0}^{\gamma_i}, c_{i,1}^{\gamma_i}$ commit to the same value, and therefore the decommitted bit is fixed.

The modification that we propose to the equivocal compiler is simple but impactful. Instead of repeating the commitment/checking λ times sequentially, our protocol only performs it once. The proof of equivocality follows exactly as in [BCKM21], but the binding property completely breaks. In particular, a malicious committer can guess γ , and commit to two different values of $c_0^{\gamma}, c_1^{\gamma}$, and therefore, with probability $\frac{1}{2}$, the receiver’s checks pass and the committer can open to any value of their choice. However, we show that this protocol satisfies what we call *relaxed binding*. We defer the formal definition of this property to Section 8.5.1, but, intuitively, it says that, if we commit to m bits using our bit commitment scheme and all of the tests pass, then, with overwhelming probability, only a logarithmic small fraction of these bits are non-binding. As we discuss next, we show that this property is sufficient in the extractable compiler. Moreover, since this sub-protocol is repeated many times in later parts of the protocol, reducing its complexity has a big impact in the runtime of the classical post-processing of our final protocol.

We switch gears now to the extractable compiler of [BCKM21]. This compiler is heavily inspired by the structure of the quantum OT protocol of [CK90, BBCS92]. In this compiler, the committer generates 2λ BB84 states and sends them to the receiver. As in the quantum OT protocol, the receiver equivocally commits to measurement bases and outcomes, then the committer challenges the receiver to open a subset of size λ of such commitments and verify the consistency of the opened values with the committer's original encoded states. If such a test passes, the committer divides the remaining encoded states into $\sqrt{\lambda}$ strings $x_1, \dots, x_{\sqrt{\lambda}}$ of size $\sqrt{\lambda}$, and using $\sqrt{\lambda}$ hashes from a 2-universal hash function $h_1, \dots, h_{\sqrt{\lambda}}$, the committer sends $\hat{h}_i = h_i(x_i) \oplus b$

The opening in [BCKM21] is followed by the committer sending $x_1, \dots, x_{\sqrt{\lambda}}, b$, which is followed by the receiver checking if these values are consistent with their measurement outcomes and if $\hat{h}_i = h_i(x_i) \oplus b$.

The hiding property of the commitment comes from the properties of 2-universal hash functions along with the entropy of x_i from the receiver's perspective due to the uncertainty relations of measurement outcomes that we can achieve with the binding property of the commitment scheme. We notice that the relaxed binding of the base commitment instead of a full binding preserves the overall proof of hiding with minimal losses.

The extraction property is (roughly) proved as follows. Using the equivocality of the base bit commitment scheme, the extractor can delay the measurement until the committer reveals which subset of positions they will check. At this point, the extractor measures those positions on random bases, and then equivocates the opening to these values. Later in the protocol, whenever the committer sends the bases, the extractor is able to measure *all* of the qubits in the correct bases and find the (purported) values of $x_1, \dots, x_{\sqrt{\lambda}}$, and then extract the committed values from $\hat{h}_i$.

The first problem of this compiler is that it assumes that all of the parties have access to perfect (noiseless) devices. In particular, noise will make it impossible for the extractor to extract the correct values. A second problem appears while splitting the string in $\sqrt{\lambda}$ blocks, since it causes a quadratic loss in the security of the protocol. Finally, we notice that, in [BCKM21], this protocol is repeated multiple times: firstly in the equivocal compiler, and then to commit to many values in the final quantum OT protocol. This causes the huge overhead needed in the total number of qubits required for their protocol.

In our work, we solve all of these problems at once. First, in order to enable extraction even in the presence of noise, we send the syndrome of the encoded values according to a linear error correcting code, and, as in the QKD setting, this allows the extractor to correct the faulty positions. Second, we replace the use of hashes so that, with the same number of BB84 states, we can commit to many qubits with equivocality and some extraction properties. Our proposed Equivocal and Relaxed-Extractable commitment scheme (*ERE-Commitment*), can be divided into two main different subroutines: the generation of a set of random seeds that are relaxed-extractable and the use of these seeds as “keys” in a set of equivocal commitments. We show that the keys of almost all the equivocal commitments are extractable, which is sufficient for a simulator to open most of the messages. We notice that the structure of our scheme is conceptually different from

the one proposed by [BCKM21], allowing the drastic reductions of the needed quantum resources.

The challenge is to generate random seeds that are relaxed-extractable. To do so, we make use of quantum resources again. Given the distribution of BB84 states, the sender can distill random seeds from the encoded states. By using an equivocal commitment subroutine, these seeds cannot be distilled by a receiver that is forced to measure. However, an efficient simulator is able to extract them.

More concretely, we split the encoded string $\mathbf{x}_{EX}$ into $(\tilde{\mathbf{x}}^j)_{j \in [2k]}$ subsets of size m . Using privacy amplification techniques found in QKD [Ren08], we distill uniformly random strings $\mathbf{s}^j$ which are used as seeds of pseudo-random generators, i.e., we expand them to the pseudo-random strings $(\mathbf{p}^j)_{j \in [2k]}$. Each $\mathbf{p}^j$ represents a concatenation of $2w$ values $\mathbf{p}_i^j$ that are used for their randomness as needed in statistically binding bit commitment schemes used for equivocation.

ERE-Commitment allows us to achieve the independence of seeds required to open the different commitments, while avoiding the modular composition proposed by [BCKM21]. The reason is that we consider that the $2k$ seed families $(\mathbf{p}^j)_{j \in [2k]}$ are grouped in k pairs $(\mathbf{p}^1, \mathbf{p}^2), \dots, (\mathbf{p}^{2k-1}, \mathbf{p}^{2k})$, so each of the $r \in [k]$ pairs can be used for parallel commitment of w values. For the r -th pair, each of the $q \in [w]$ instances makes use of two seeds $\mathbf{p}_i^j$ from each family of the pair, in such a way that the final set of seeds for the q -th EqCommitment instance of the r -th family pair is $\left((\mathbf{p}_{i_{q,0}}^{r,0}, \mathbf{p}_{i_{q,1}}^{r,0}, \mathbf{p}_{i_{q,0}}^{r,1}, \mathbf{p}_{i_{q,1}}^{r,1})_{q \in [w]} \right)_{r \in [k]}$. Afterwards, Bob sends Alice the corresponding x_i^{EX} and θ_i^{EX} , then Alice checks that $x_i^{EX} = \hat{x}_i^{EX}$ whenever $\theta_i^{EX} = \hat{\theta}_i^{EX}$ and that the $\mathbf{p}^j$ were generated in an honest way. This ensures that the seeds $\mathbf{p}_i^j$ originate from the proper $\mathbf{x}^{EX}$ with probability $1 - \text{negl}(k)$. Bob can then commit to $((\hat{x}_i^{OT}, \hat{\theta}_i^{OT}))_{i \in [2\lambda_{OT}]}$ using $\mathbf{p}_i^j$.

To prove that ERE-Commitment is relaxed extractable, we make use of equivocality and quantum rewinding. A simulator commits to dummy values instead of $((\hat{x}_i^{EX}, \hat{\theta}_i^{EX}))_{i \in [4\lambda_{EX}]}$. After learning the challenged subset E , the simulator measures the corresponding states $(|x_i^{EX}\rangle_{\theta_i^{EX}})_{i \in E}$, and, applying Watrous' quantum rewinding, opens to the measured values. Once the simulator has passed the decommitment check, Bob announces the bases. The simulator will then measure $(|x_i^{EX}\rangle_{\theta_i^{EX}})_{i \in \bar{E}}$ in the correct bases obtaining $\mathbf{x}^{EX}$. It is in this step in which the simulator has to make use of the error correction code for obtaining the correct seeds $(\mathbf{p}^j)_{j \in [2k]}$ given the syndromes $(\text{Synd}_j)_{j \in [2k]}$. The simulator will obtain the seeds of the equivocal commitments, and thus extracts $|\bar{T}| - \omega(\log^2(k))$ of the non-challenged committed values $((\hat{x}_i^{OT}, \hat{\theta}_i^{OT}))_{i \in \bar{T}}$ with probability $1 - \text{negl}(k)$.

Despite these changes, we prove that the OT functionality remains simulation-based secure given access to an equivocal and χ -relaxed extractable bit commitment. With a practical implementation as our main goal, we explicitly take into account possible experimental errors and their correction, and we derive analytical expressions based on [BF10] that

Alice	Bob
<i>Input:</i> Messages (m_0, m_1)	<i>Input:</i> Bit $b \in \{0, 1\}$
<i>Assumptions:</i> ERE-Commitment PRG, and 2-universal hash functions h_j .	
1. Chooses $\mathbf{x} \leftarrow \{0, 1\}^{2\lambda_{\text{OT}}}$ and $\boldsymbol{\theta} \leftarrow \{+, \times\}^{2\lambda_{\text{OT}}}$, prepares $ x\rangle_{\boldsymbol{\theta}}$, and sends the $2\lambda_{\text{OT}}$ BB84 states.	1.a Receives the BB84 states.
2. Acts as receiver in the ERE-Commitment execution for $(\hat{x}_i, \hat{\theta}_i)_{i \in [2\lambda_{\text{OT}}]}$.	1.b Samples measurement bases $\hat{\boldsymbol{\theta}} \leftarrow \{+, \times\}^{2\lambda_{\text{OT}}}$ and measures to obtain $\hat{\mathbf{x}} \leftarrow \{0, 1\}^{2\lambda_{\text{OT}}}$
3. Chooses a random challenge subset $T \subset [2\lambda_{\text{OT}}]$ with $ T = \lambda_{\text{OT}}$ and sends T .	2. Executes ERE-Commitment to commit to $(\hat{x}_i, \hat{\theta}_i)_{i \in [2\lambda_{\text{OT}}]}$.
4. Runs ERE-Decommitment for each $i \in T$ and obtains $(\hat{x}_i, \hat{\theta}_i)_{i \in T}$.	3. Receives T .
5. Checks that $x_i = \hat{x}_i$ whenever $\theta_i = \hat{\theta}_i$ on T , up to experimental error α ; if the test fails, aborts.	4. Opens the corresponding commitments on all $i \in T$.
6. Discards T , reorders $(x_i, \theta_i)_{i \in \bar{T}}$, and sends $\boldsymbol{\theta}$ (on $\bar{T}$).	6. Discards T , reorders $(\hat{x}_i, \hat{\theta}_i)_{i \in \bar{T}}$, and receives $\boldsymbol{\theta}$.
7. Receives (I_0, I_1) .	7. Partitions $\bar{T}$ into $I_b = \{i : \theta_i = \hat{\theta}_i\}$ and $I_{\bar{b}} = \{i : \theta_i \neq \hat{\theta}_i\}$; sends (I_0, I_1) .
8. Defines $\mathbf{x}_0 = \{x_i : i \in I_0\}$, $\mathbf{x}_1 = \{x_i : i \in I_1\}$ and sends syndromes $\text{Synd}_0, \text{Synd}_1$ for correcting a proportion α of errors in $\mathbf{x}_0, \mathbf{x}_1$.	8. Receives $\text{Synd}_0, \text{Synd}_1$.
9. Samples seeds s_0, s_1 and sends $(s_0, \text{PRG}(h(s_0, \mathbf{x}_0)) \oplus m_0)$ and $(s_1, \text{PRG}(h(s_1, \mathbf{x}_1)) \oplus m_1)$	9. Receives the seeds and masked messages.
	10. Performs error correction on $\mathbf{x}_b$ using Synd_b and decrypt m_b using $\hat{\mathbf{x}}_b = \{\hat{x}_i : i \in I_b\}$. If error correction fails, chooses $\mathbf{x}_b$ as a random string in $\{0, 1\}^{\lambda_{\text{OT}}}$.

Protocol 8.1.: Our Proposed QOT protocol.

facilitate benchmarking of the required quantum resources. The resulting QOT protocol is given in Protocol 8.1.

Moreover, we propose a method to distill a number n_{OT} of QOT keys from a single run of the protocol, which is particularly useful when a large number of BB84 states is employed to achieve a smaller sampling error (and hence a tighter Δ -security bound).

8.3. Related Works

We compare and benchmark our proposed protocol with other composable secure QOT protocols based on one-way functions that have been proposed in the literature. In [ABKK23], the authors also focus on reducing the round complexity, but they work in the Random Oracle Model (ROM). In this setting, equivocality and extractability of commitments can be obtained from the random oracle, so no additional quantum resources are needed for the commitment layer itself. The cost of the quantum communication in [ABKK23] comes entirely from the OT core, which is structurally similar to CK88. Under our common benchmarking assumptions (target trace distance, security level, and an upper bound on the number of RO queries), the resulting number of BB84 states in their protocols is of the same order of magnitude as in our construction, as shown in Table 8.1 and detailed in Section A.5.2. However, these estimates for [ABKK23] do not take into account practical errors such as bit-flips, photon loss, or multi-photon events, whereas our benchmark explicitly includes such imperfections.

It is important to stress that the comparison in Table 8.1 concerns *only* the quantum communication cost (number of BB84 states) under a common concrete security target. In particular, we do not claim to match the classical efficiency of ROM-based commitments. Conceptually, our assumption is strictly weaker: our protocol is proved secure in the plain model under the minimal assumption of post-quantum one-way functions, without relying on a random oracle or heuristic instantiations thereof. In this sense, our construction achieves comparable BB84 cost while working under strictly milder computational assumptions than [ABKK23].

On the other hand, the number of BB84 states necessary to instantiate [BCKM21] has been obtained by adapting our security bounds to their protocol structure. A detailed explanation of how all entries in Table 8.1 are computed is provided in Section A.5.2.

Further proposals, such as [CCLY22], consider relaxed simulation notions, where the runtime of the simulator depends polynomially on the desired security level. In our work, we can achieve negligible distinguishing probability with polynomial-time simulators.

8.4. Notions in Quantum Cryptography

8.4.1. Notation for Cryptographic Protocols

We use the notation fixed in Chapter 2 for quantum states, trace distance, negligible functions, asymptotic notation, and the cryptographic conventions used throughout this

	[BCKM21]	[ABKK23]		Our protocol	
α	—	—	—	—	0.006
ϑ	—	—	—	—	0.001
q_{RO}	—	2^{64}	2^{64}	—	—
N_{BB84}	2.27×10^{13}	3.22×10^6	1.43×10^6	3.33×10^7	7.47×10^7
T_{acq}	8.6 months	3.22 s	1.43 s	33.3 s	74.7 s

Table 8.1.: Performance benchmark for a target trace distance $\Delta \leq 10^{-15}$. Here, α denotes the bit-flip probability, ϑ the fraction of leaked bits due to, for example, multi-photon events, q_{RO} the number of queries to the random oracle (when applicable), N_{BB84} the number of exchanged BB84 states, and T_{acq} the corresponding acquisition time assuming a rate of 1 MHz. From left to right, the columns correspond to: the protocol of [BCKM21], the 3-round and 4-round communication protocols of [ABKK23], and our protocol in the ideal (error-free) and realistic noisy settings.

thesis. In this chapter, the security parameters of the oblivious-transfer and extraction subprotocols are denoted by λ_{OT} and λ_{EX} , respectively. For the BB84 encoding, we write $|x\rangle_+ := |x\rangle$ and $|x\rangle_- := H|x\rangle$ for $x \in \{0, 1\}$, where H is the Hadamard gate, and extend this notation componentwise as $|\mathbf{x}\rangle_\theta := \bigotimes_i |x_i\rangle_{\theta_i}$.

8.4.2. Bit Commitment

A bit commitment is a two-party interactive functionality between a quantum committer C and a quantum receiver R . A commitment protocol is composed of two phases: the committing phase and the decommitting phase, where $C = \{C_{\text{com}}, C_{\text{dec}}\}$ and $R = \{R_{\text{com}}, R_{\text{dec}}\}$. During the committing phase, the committer C commits to a bit $b \in \{0, 1\}$, with $C_{\text{com}}(1^\lambda, b)$ representing this step for the committer and $R_{\text{com}}(1^\lambda)$, for the receiver. After the interaction, C_{com} outputs a state $\rho_{\text{com}}^C \in \mathcal{B}_1(\mathcal{H}_C)$ and $R_{\text{com}}, \rho_{\text{com}}^R \in \mathcal{B}_1(\mathcal{H}_R)$. All in all, the commitment phase is written as $(\rho_{\text{com}}^C, \rho_{\text{com}}^R) \leftarrow \langle C_{\text{com}}(1^\lambda, b), R_{\text{com}}(1^\lambda) \rangle$. Additionally, ρ_{com}^C and ρ_{com}^R may be entangled. During the decommitment phase, the interaction between $C_{\text{dec}}(\rho_{\text{com}}^C)$ and $R_{\text{dec}}(\rho_{\text{com}}^R)$ leads to an output of R_{dec} that is either b' or $\perp$.

Security Against Malicious Receiver

The two notions of security that we use against malicious receivers are as follows:

Definition 8.1 (Computational hiding). A bit commitment protocol (C, R) is computationally hiding if, given any polynomial-size interactive receiver $R_{\text{com}}^* = \{R_{\text{com}, \lambda}^*, \rho_\lambda\}$ with

$OUT_R\langle C(1^\lambda, b), R_{com,\lambda}^*(\rho_\lambda) \rangle$ being the output of the receiver after the commitment phase, there exists a negligible function $\nu(\cdot)$ such that:

$$\left| \Pr[OUT_R\langle C_{com}(1^\lambda, 0), R_{com}^*(\rho_\lambda) \rangle = 1] - \Pr[OUT_R\langle C_{com}(1^\lambda, 1), R_{com}^*(\rho_\lambda) \rangle = 1] \right| = \nu(\lambda).$$

Definition 8.2 (Equivocality). Given a set of auxiliary states $\{\rho_\lambda, \sigma_\lambda\}_{\lambda \in \mathbb{N}}$ with $\rho_\lambda, \sigma_\lambda \in \mathcal{B}_1(\mathbb{C}^n)$, a bit commitment protocol (C, R) is *equivocal* if, for any poly-size receiver $R^* = \{R_{com,\lambda}^*, R_{dec,\lambda}^*, \rho_\lambda\}$, there exists a non-uniform quantum polynomial time equivocator $\mathcal{Q}_{R^*} = \{\mathcal{Q}_{R^*,com}, \mathcal{Q}_{R^*,dec}\}$ such that, for any poly-size distinguisher $\mathcal{D} = \{\mathcal{D}_\lambda, \sigma_\lambda\}$, there exists a negligible function $\nu(\lambda) > 0$ for the committed bit $b \in \{0, 1\}$:

$$|\Pr[\mathcal{D}^*(\sigma_\lambda, Real_b) = 1] - \Pr[\mathcal{D}^*(\sigma_\lambda, Ideal_b) = 1]| \leq \nu(\lambda),$$

provided that:

Real_b: Represents the interaction between $R_{com}^*(\rho)$ and $C_{com}(1^\lambda, b)$ such that $(\rho_{com}^C, \rho_{com}^{R^*}) \leftarrow \langle C_{com}(b), R_{com}^*(\rho) \rangle$ and $\rho_{final}^{R^*} \leftarrow \langle C_{dec}(\rho_{com}^C), R_{dec}^*(\rho_{com}^{R^*}) \rangle$.

Ideal_b: The quantum simulator outputs $(\rho_{com}^C, \rho_{com}^{R^*}) \leftarrow \mathcal{Q}_{R^*,com}(\rho)$. Then, $\rho_{final}^{R^*} \leftarrow \langle \mathcal{Q}_{R^*,dec}(\rho_{com}^C, b), R_{dec}^*(\rho_{com}^{R^*}) \rangle$.

Security Against Malicious Sender

The security definitions of bit commitment against malicious sender are written as:

Definition 8.3 (Statistical binding). A bit commitment protocol (C, R) is statistically-binding if, for every unbounded-size committer $C^* = \{C_{com}^*, C_{dec}^*\}$ and $\lambda \in \mathbb{N}$, such that:

$$(\rho_{com}^{C^*}, \rho_{com}^R) \leftarrow (\langle C_{com}^*(1^\lambda, b), R_{com}(1^\lambda) \rangle),$$

there exists a bit $b \in \{0, 1\}$ and a negligible function $\nu(\cdot)$ such that with probability at least $1 - \nu(\lambda)$,

$$\Pr[OUT_R\langle C_{dec}^*(1^\lambda, \rho_{com}^{C^*}(\lambda)), R_{dec}(1^\lambda, \rho_{com}^R(\lambda)) \rangle \neq b] \leq \nu(\lambda).$$

In this work, we use a weaker notion of binding that we call relaxed statistical binding. Unlike the general definition of statistically-binding bit commitment, each individual commitment is not statistically binding by itself. However, committing to a collection of bits binds for a fraction $1 - \eta$ of the bits.

Definition 8.4 (η -relaxed statistical binding). Let (C, R) be a bit commitment protocol. For a sequence of m commitments $(C_i^* = (C_{com,i}^*, C_{dec,i}^*))_{i \in [m]}$ with $\lambda \in \mathbb{N}$ such that:

$$(\rho_{com}^{C^*}(\lambda), \rho_{com}^R(\lambda)) \leftarrow (\langle C_{com,i}^*(1^\lambda, b_i), R_{com,i}(1^\lambda) \rangle)_{i \in [m]}.$$

(C, R) is η -relaxed-binding if given the commitment there exists a set $I \subseteq [m]$, where $|I| \geq (1 - \eta)m$ and a negligible function $\nu(\cdot)$ such that with probability at least $1 - \nu(\lambda)$ over $(\rho_{com}^{C^*}(\lambda), \rho_{com}^R(\lambda))$, there exists a string $\mathbf{b} \in \{0, 1\}^{|I|}$ such that,

$$\Pr[OUT_R(\langle C_{dec}^*(1^\lambda, \rho_{com}^{C^*}(\lambda)), R_{dec}(1^\lambda, \rho_{com}^R(\lambda)) \rangle)_{|I|} \neq \mathbf{b}(\lambda)] \leq \nu(\eta).$$

Relaxed extractability works similarly to relaxed binding: after committing to many bits, most of them can be extracted.

Definition 8.5 (χ -relaxed extractability). Let (C, R) be a bit commitment protocol. Given a sequence of auxiliary states $(\rho_i, \sigma_i)_{i \in [m]}$ where $\rho_i, \sigma_i \in \mathcal{B}_1(\mathbb{C}^\lambda)$ and $\lambda \in \mathbb{N}$, m sequential repetitions of commitments $(\rho_{com,1}^{C^*}, \rho_{com,1}^R), \dots, (\rho_{com,m}^{C^*}, \rho_{com,m}^R)$ are χ -relaxed-extractable if, for any poly-size quantum malicious committer $C^* = \{(C_{com,i}^*(1^\lambda))_{i \in [m]}, (\rho_i)_{i \in [m]}\}$, there exists a quantum polynomial time extractor $\mathcal{Q}_{C^*} = \{\mathcal{Q}_{C^*,com}, \mathcal{Q}_{C^*,dec}\}$ such that, for any poly-size distinguisher $\mathcal{D}^* = \{(\mathcal{D}_i^*)_{i \in [m]}, (\sigma_i)_{i \in [m]}\}$ and for every polynomial-size opening strategy $(C_{dec,i}^*)_{i \in [m]}$, there is a negligible function $\nu(\chi) > 0$ that obeys:

$$|\Pr[\mathcal{D}^*(\sigma_\lambda, Real) = 1] - \Pr[\mathcal{D}^*(\sigma_\lambda, Ideal) = 1]| = \nu(\chi),$$

with:

Real: It represents the interaction between $R_{dec}(\rho_{com}^R)$ and $C_{dec}^*(\rho_{com}^{C^*})$ such that $(\rho_{final}^{C^*}, b_i) \leftarrow \langle C_{dec}^*(\rho_{com}^{C^*}), R_{dec}(\rho_{com}^R) \rangle_i$, with $b_i \in \{0, 1, \perp\}$. Then, the output after m sequential repetitions is given by $(\rho_{final}^{C^*}(m), \mathbf{b}(m)) \leftarrow (\langle C_{com,i}^*(1^\lambda, b_i), R_{com,i}(1^\lambda) \rangle)_{i \in [m]}$, where $\mathbf{b} \in \{0, 1\}^m$.

Ideal: The simulator computes $(\rho_{com}^C(m), \rho_{com}^R(m), \mathbf{b}^*(m)) \leftarrow (\mathcal{Q}_{C^*,com}(\rho_i))_{i \in \lambda}$, where $\mathbf{b}^* \in \{0, 1, ?\}^m$. Then, $(\rho_{final}^C(m), \mathbf{b}(m)) \leftarrow \langle C_{dec}^*(\rho_{com}^C(m)), R_{dec}(\rho_{com}^R(m)) \rangle$. The simulator outputs FAIL if :

- $|\{i : b_i^* = ?\}| \geq \chi m$, or
- if for any i such that $b_i^* \in \{0, 1\}$, $b_i \neq b_i^*$.

If the simulator does not output FAIL, it outputs $(\rho_{final}^C(m), \mathbf{b}_S(m))$, where $\mathbf{b}_S(j) = \mathbf{b}^*(j)$ every time $b_i^* \neq ?$, and replaces the ? cases by the values $\mathbf{b}(j)$ opened by $C_{dec}^*(\rho_{com}^C(j))$.

8.4.3. Leftover Hash Lemma

The conditional min-entropy was introduced in Definition 2.22. We recall the leftover hash lemma in the form needed in the security proofs below.

Lemma 8.6 (Leftover Hash Lemma with Quantum Side Information [Ren08]). *Let ρ_{XE} be a hybrid state and $h(r, x) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ a 2-universal hash function, with r uniformly distributed over $\{0, 1\}^m$. Then, $K = h(r, x)$ satisfies:*

$$\Delta(\rho_{K_b K_b E}, \frac{1}{2^\ell} \mathbb{1} \otimes \rho_{K_b E}) \leq \frac{1}{2} \sqrt{2^{\ell - H_{\min}(X|E)}}.$$

Lemma 8.7 (Conditional min-entropy [BF10]). *Given an n -qubit system A , let the state $|\psi_{AE}\rangle = \sum_{b \in \{0,1\}^n} |b\rangle \otimes |\psi_E^b\rangle \in \mathcal{H}_A \otimes \mathcal{H}_E$ have a relative Hamming weight δ -close to α , with $\delta + \alpha \leq 1/2$. Let X be the random variable obtained by measuring A in the bases $H^\theta \{|0\rangle, |1\rangle\}^{\otimes n}$ for $\theta \in \{0, 1\}^n$. Then:*

$$H_{\min}(X|E) \geq d_H(\theta, \hat{\theta}) - h_2(\delta + \alpha)n, \quad (8.4.1)$$

where $\hat{\theta} \in \{0, 1\}^n$ denotes the measurement bases committed by Bob, $h_2(\cdot) : [0, 1] \rightarrow [0, 1]$, the binary entropy function and $d_H(\cdot, \cdot) : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \mathbb{N}$, the Hamming distance function.

8.4.4. Watrous Rewinding Lemma

Lemma 8.8 (Rewinding lemma with small perturbations [Wat06]). *Let Q be an (n, k) -quantum circuit that outputs a classical bit b and m qubits. For any input state $|\psi\rangle \in \mathbb{C}^n$, let $p(\psi)$ be the probability of measuring the classical bit $b = 0$ such that the state after measuring the action of the circuit Q is $|\phi_0(\psi)\rangle \in \mathbb{C}^m$. Let $p_0 \in (0, 1)$, $\epsilon \in (0, 1/2)$ and $q \in (0, 1)$, where q is the probability that the quantum circuit acts like a unitary, be such that:*

1. $|p(\psi) - q| < \epsilon, \forall |\psi\rangle \in \mathbb{C}^n$,
2. $p_0 < p(\psi), \forall |\psi\rangle \in \mathbb{C}^n$,
3. $p_0(1 - p_0) \leq q(1 - q)$.

Then, there exists a general quantum circuit R of size $\mathcal{O}\left(\frac{\log(1/\epsilon)|Q|}{p_0(1-p_0)}\right)$, such that, for every input state $|\psi\rangle$, the output $R(|\psi\rangle)$ satisfies:

$$\|R(|\psi\rangle) - |\phi_0(\psi)\rangle\|_1 \leq 4\sqrt{\epsilon} \frac{\log(1/\epsilon)}{p_0(1-p_0)}.$$

8.4.5. Error Resilience

Handling error correction in our protocol involves more subtleties compared to, for example, QKD, since in our case Alice and Bob are not trusted. In quantum communication protocols, there are two primary sources of errors: noise and losses. In the case of losses, since the proposed protocol incorporates bit commitments for each BB84 state, the loss of a state will result in the absence of a commitment for that instance of the protocol. Discarding non-committed instances has no impact on the security analysis of the protocol.

The potential presence of malicious parties prevents them from cooperating to estimate the noise. Therefore, an error correction subroutine must be included in the protocol to handle errors without any collaboration between parties [DFSS08, ENG⁺14, LPAK23]. An $(\mathcal{M}, K, t)$ -error correction code can correct errors in a code C composed of K elements from a metric space $\mathcal{M}$, such that for every element $w \in \mathcal{M}$, there exists at most one codeword $c \in C$ within a ball of radius t around w .

We propose the use of a non-interactive (classical) error correction code, which is syndrome-based, such as Low-Density Parity-Check (LDPC) codes [Gal60]. The syndrome of an element w is its projection onto the orthogonal subspace of the code, capturing all the necessary information for decoding.

Finally, we consider the possibility of multiphoton events. While an ideal BB84 source generates single photons, experimental sources could accidentally generate more than one copy of a BB84 instance. We take into account this leakage of information for the privacy amplification of our proposed protocol. We present this as a general variable ϑ that accounts for the information leakage.

8.5. Equivocal Commitment

In this section, we present a compiler that yields an equivocal η -relaxed statistically binding bit commitment from a statistically binding computationally hiding bit commitment. As previously discussed in the Introduction, our compiler is directly based on [BCKM21], but we reduce the number of repetitions in order to improve efficiency at the cost of requiring relaxed binding properties. The protocol is described in Algorithm 1 and we prove the relaxed binding property in Section 8.5.1 and equivocality in Section 8.5.2.

8.5.1. Relaxed Statistical Binding

Given a statistically-binding and computationally-hiding commitment, we can build a commitment named *EqCommitment* that is η -relaxed statistically binding (Definition 8.4) and computationally hiding (Definition 8.1).

Theorem 8.9. *If Com is a statistically binding commitment scheme, then, for any function $\eta(n)$, *EqCommitment* is $(\eta, 2^{-n\eta})$ -relaxed statistically binding.*

Algorithm 1 EqCommitment: Equivocal commitment

Input of C: bit $b \in \{0, 1\}$, seeds $\mathbf{p}_0^0, \mathbf{p}_1^0, \mathbf{p}_0^1, \mathbf{p}_1^1 \in \{0, 1\}^{\lambda_{FQS}}$.

Assumptions: Bit commitment Com_r that is statistically binding and computationally hiding and which uses r as the randomness for the commitment.

Commit phase:

- 1: C chooses two random bits $u^0, u^1 \in \{0, 1\}$.
- 2: C and R commit to four values, by using the input seeds in the following way:
 - $c_0^0 = Com_{\mathbf{p}_0^0}(C(u^0), R)$
 - $c_1^0 = Com_{\mathbf{p}_1^0}(C(u^0), R)$
 - $c_0^1 = Com_{\mathbf{p}_0^1}(C(u^1), R)$
 - $c_1^1 = Com_{\mathbf{p}_1^1}(C(u^1), R)$
- 3: R sends a bit $\gamma \in \{0, 1\}$ to C
- 4: R and C decommit $u_0^\gamma = Decom_{\mathbf{p}_0^\gamma}(c_0^\gamma)$ and $u_1^\gamma = Decom_{\mathbf{p}_1^\gamma}(c_1^\gamma)$
- 5: R aborts if $u_0^\gamma \neq u_1^\gamma$.
- 6: C computes $e = b \oplus u^\gamma$ and sends it to R.

Decommit phase:

- 7: C chooses a random bit $\delta \in \{0, 1\}$ and sends both, δ and b to R.
- 8: C and R decommit c_δ^γ :

$$u_\delta^\gamma = Decom_{\mathbf{p}_\delta^\gamma}(c_\delta^\gamma)$$

- 9: R aborts if $u_\delta^\gamma \neq b \oplus e$.

Proof. Let us consider n sequential repetitions of the EqCommitment protocol. Let us denote $c_{\delta,i}^\gamma$ as the commitment c_δ^γ of the i -th run of the protocol. Notice that all $c_{\delta,i}^\gamma$ are statistically binding, so with overwhelming probability, we can define the values $u_{\delta,i}^\gamma$ such that the opening of $c_{\delta,i}^\gamma$ to a value that is different from $u_{\delta,i}^\gamma$ fails.

Let us define the set of $B_0 = \{i : \exists \gamma \ u_{0,i}^\gamma \neq u_{1,i}^\gamma\}$. Let us also define the event E_0 where the check on Step 5 passes on all runs of the protocol. Notice that for each $j \notin B_0$, the j -th run of the commitment is statistically binding, and the goal is now to connect $|B_0|$ and $\Pr[E_0]$. It can be shown that:

$$\Pr[E_0] \leq \frac{1}{2^{|B_0|}}, \quad (8.5.1)$$

which is equivalent to the fact that if $\Pr[E_0] \geq 2^{-n\eta}$, then $|B_0| \leq n\eta$, which shows that EqCommitment is $(\eta, 2^{-n\eta})$ -relaxed statistically binding. The proof of Equation (8.5.1) holds since for each $i \in B_0$, there exists one γ_i^* such that $u_{0,i}^{\gamma_i^*} \neq u_{1,i}^{\gamma_i^*}$, and in order for E_0 to be true, the challenged γ_i for that run is different than γ_i^* , which happens with probability $\frac{1}{2}$. Since all challenges are picked independently, the probability is:

$$\Pr[\forall i \in B_0 : \gamma_i \neq \gamma_i^*] = \prod_{i \in B_0} \Pr[\gamma_i \neq \gamma_i^*] \leq \frac{1}{2^{|B_0|}},$$

Algorithm 2 Equivocal simulator $\mathcal{Q}_{Eq}$

-
- 1: $\mathcal{Q}_{R^*}$ samples a random bit $\tilde{\gamma} \in \{0, 1\}$ and stores it in register P.
 - 2: C samples two random bits $\tilde{u}, \tilde{y} \in \{0, 1\}$
 - 3: Commitment subprotocol for the simulated committer $\tilde{C}$
 - If $\tilde{\gamma} = 0$, $\tilde{C}$ commits to:
 1. $\tilde{c}_0^0 = \text{Com}_{p_0^0}(\tilde{C}(\tilde{y}), R^*)$
 2. $\tilde{c}_1^0 = \text{Com}_{p_1^0}(\tilde{C}(\tilde{y}), R^*)$
 3. $\tilde{c}_0^1 = \text{Com}_{p_0^1}(\tilde{C}(\tilde{u}), R^*)$
 4. $\tilde{c}_1^1 = \text{Com}_{p_1^1}(\tilde{C}(1 - \tilde{u}), R^*)$
 - If $\tilde{\gamma} = 1$, $\tilde{C}$ commits to:
 1. $\tilde{c}_0^0 = \text{Com}_{p_0^0}(\tilde{C}(\tilde{u}), R^*)$
 2. $\tilde{c}_1^0 = \text{Com}_{p_1^0}(\tilde{C}(1 - \tilde{u}), R^*)$
 3. $\tilde{c}_0^1 = \text{Com}_{p_0^1}(\tilde{C}(\tilde{y}), R^*)$
 4. $\tilde{c}_1^1 = \text{Com}_{p_1^1}(\tilde{C}(\tilde{y}), R^*)$
 - Run commitment phase between C and R^* .
 - Measure γ in the register A. If $\tilde{\gamma} = \gamma$, the simulator was successful, output 0. Otherwise, output 1.
 - 4: Let $u = b \oplus e$
 - 5: $\mathcal{Q}_{R^*}$ and R^* decommits to:
 - If $\tilde{\gamma} = 0$, decommits to the $(u \oplus \tilde{u} + 2)^{th}$ committed bit.
 - If $\tilde{\gamma} = 1$, decommits to the $(u \oplus \tilde{u})^{th}$ committed bit.
-

where the inequality accounts for the trivial case where $\nexists \gamma_i$ such that $u_{0,i}^{\gamma_i} \neq u_{1,i}^{\gamma_i}$, resulting in $\Pr[E_0] = 0$. $\square$

8.5.2. Equivocality

Our proof of equivocality is the same as [BCKM21], and we provide it here for completeness. We apply Watrous' rewinding on the equivocal simulator in Algorithm 2 to pass the verification test while not having consistent commitments.

Theorem 8.10. *If Com is a computationally hiding commitment, then EqCommitment is an equivocal bit commitment*

Proof. In order to apply the rewinding lemma, we can consider a communication scheme as follows.

- C : the committer C commits and communicates with the malicious receiver R^* through the polynomial-size register M .
- R^* : the malicious receiver has three different registers: W , V and A . The first one, W , represents the quantum input. The registers V and A are the workspace of R^* and both are initialized in the zero state. While V is a polynomial-size register, A is a one qubit register. R^* measures this second register A for sampling $\gamma \in \{0, 1\}$. After the honesty check, R^* outputs the registers (W, V, A, M) .
- Q_{R^*} : the quantum simulator works with registers P and Z in addition to (W, V, A, M) . P is a one qubit register for the guess $\tilde{\gamma}$, and Z is the auxiliary register in which Algorithm 2 is implemented.

The condition to apply Watrous' lemma is that the probability for the simulator to guess the output has to be nearly independent of the probability distribution of the malicious receiver R^* 's choice. This is guaranteed by the computationally hiding property of the underlying bit commitment. Then, $|p(\psi) - 1/2| = \text{negl}(\lambda)$, for every input state $|\psi\rangle$ of R^* in $\mathcal{Q}_{Eq}$. Due to Lemma 8.8, there is a poly-size circuit R that outputs a state inverse-exponentially close to the final state conditioned to $\tilde{\gamma} = \gamma$ of the Algorithm 2.

Once the simulator $Q_{R^*, com}$ has measured the qubit register P and obtained γ , it proceeds to do the honesty check, as described in Algorithm 2 and outputs the registers (W, V, A, M) after tracing out the remaining registers.

If a quantum poly-time distinguisher $\mathcal{D}^* = \{\mathcal{D}_\lambda^*, \sigma_\lambda\}$ could distinguish $Real_b$ and $Ideal_b$, it would mean that $\mathcal{D}^*$ is able to discriminate the commitments of the simulator Q_{R^*} and the honest committer C . This would imply that the distinguisher is able to open the committed values which contradicts the definition of computational hiding of the base commitment scheme.

□

8.6. Equivocal and Relaxed-Extractable Commitment

We construct an equivocal and χ -relaxed extractable commitment (Definition 8.5), named *ERE-Commitment*, based on an equivocal and η -relaxed statistically binding commitment. We present our protocol in Algorithm 3 and then prove its equivocality and the relaxed extractability properties.

8.6.1. Equivocality

The proof of equivocality is very similar to Theorem 8.10. Nevertheless, we also provide in Corollary 8.12 the condition for the Naor's commitment subprotocol of the ERE-Commitment to be computationally hiding, giving in Lemma 8.22 an analytical expression that quantifies it.

Algorithm 3 ERE-Commitment: Equivocal and relaxed extractable commitment

Committer C Input: A sequence of bits $(b_i)_{i \in [wk]}$.

Assumptions: The parties use a $PRG : \{0, 1\}^{\lambda_{PQS}} \rightarrow \{0, 1\}^{2w \cdot \lambda_{PQS}}$, 2-universal hash functions $h_j(\cdot, \cdot) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^{\lambda_{PQS}}$, an EqCommitment that is equivocal and a statistically binding and computationally hiding commitment Com_r which uses r as the randomness of the commitment.

- 1: C samples $\mathbf{x} \leftarrow \{0, 1\}^{4\lambda_{EX}}$ and $\boldsymbol{\theta} \leftarrow \{+, \times\}^{4\lambda_{EX}}$ and sends $|\mathbf{x}\rangle_{\boldsymbol{\theta}}$ to R.
- 2: R chooses his measurement bases $\hat{\boldsymbol{\theta}} \leftarrow \{+, \times\}^{4\lambda_{EX}}$ and measures $\hat{\mathbf{x}} \leftarrow \{0, 1\}^{4\lambda_{EX}}$.
- 3: R sequentially commits to $\left((\hat{x}_i, \hat{\theta}_i)\right)_{i \in [4\lambda_{EX}]}$ using EqCommitment.
- 4: C challenges a random subset $E \subset [4\lambda_{EX}]$, such that $|E| = 2\lambda_{EX}$.
- 5: C and R execute EqDecommitment sequentially for every $i \in E$. Then, C obtains $\left((\hat{x}_i, \hat{\theta}_i)\right)_{i \in E}$. C aborts if the commitment fails to open.
- 6: C checks that $x_i = \hat{x}_i$ whenever $\theta_i = \hat{\theta}_i$, up to a fraction α due to the experimental error. If the test does not pass, C aborts.
- 7: C divides $\bar{E}$ into $2k$ consecutive disjoint subsets $M_j \subseteq \bar{E}$ of size m , with m given by the size of the preimage of a 2-universal hash function $\{h_j(\cdot, \cdot) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^{\lambda_{PQS}}\}_{j \in [2k]}$, where $k = \lfloor \lambda_{EX}/m \rfloor$.
- 8: For every $j \in [2k]$, C samples $\mathbf{r}^j \in \{0, 1\}^n$ and compute $\mathbf{s}^j = h_j(\mathbf{r}^j, \tilde{\mathbf{x}}^j)$ where $\tilde{\mathbf{x}}^j = \{x_i : i \in M_j\}$.
- 9: Given $PRG : \{0, 1\}^{\lambda_{PQS}} \rightarrow \{0, 1\}^{2w \cdot \lambda_{PQS}}$, C generates $\mathbf{p}^j = PRG(\mathbf{s}^j) = (\mathbf{p}_1^j, \dots, \mathbf{p}_{2w}^j)$ for every $j \in [2k]$, where $\mathbf{p}_i^j \in \{0, 1\}^{\lambda_{PQS}}$.
- 10: C sends $(M_j)_{j \in [2k]}$, $\boldsymbol{\theta}$ and $(\mathbf{r}^j)_{j \in [2k]}$ to R. C sends the syndromes $(Synd_j)_{j \in [2k]}$ to correct a proportion α of errors.
- 11: C and R sequentially execute k sessions of w parallel EqCommitment. For each $r \in [k]$:
 - 11.1: C commits in parallel to w pairs of random bits with $\left(\mathbf{p}_{i_{q,0}}^{j^{r,0}}, \mathbf{p}_{i_{q,1}}^{j^{r,0}}, \mathbf{p}_{i_{q,0}}^{j^{r,1}}, \mathbf{p}_{i_{q,1}}^{j^{r,1}}\right)_{q \in [w]} = \left(\mathbf{p}_{2q-1}^{2r-1}, \mathbf{p}_{2q}^{2r-1}, \mathbf{p}_{2q-1}^{2r}, \mathbf{p}_{2q}^{2r}\right)_{q \in [w]}$ as seeds.
 - 11.2: For the honesty check of the EqCommitment, R samples $\gamma_r \xleftarrow{\$} \{0, 1\}$ and sends it to C.
 - 11.3: C sends $\tilde{\mathbf{x}}^{j^r, \gamma_r}$ and $\mathbf{p}^{j^r, \gamma_r}$ to R. R checks if,
 - $\tilde{x}_i = \hat{x}_i$ whenever $\tilde{\theta}_i = \hat{\theta}_i$, and, if there is a proportion of $\tilde{x}_i \neq \hat{x}_i$ larger than α , it aborts.
 - $PRG(h_{j^r, \gamma_r}(\tilde{\mathbf{x}}^{j^r, \gamma_r})) = \mathbf{p}^{j^r, \gamma_r}$ and, if not, aborts.
 - $Decom_{\mathbf{p}_{i_{q,0}}^{j^r, \gamma_r}}(\mathbf{c}_{i_{q,0}}^{j^r, \gamma_r}) = Decom_{\mathbf{p}_{i_{q,1}}^{j^r, \gamma_r}}(\mathbf{c}_{i_{q,1}}^{j^r, \gamma_r})$ for every $q \in [w]$ and, if not, aborts.
 - 11.4: C commits to each $(b_i)_{i \in [w]}$ as described in EqCommitment.
- 12: When necessary, C and R open to any of the committed bits $(b_i)_{i \in [wk]}$ by executing EqDecommitment. In order to do so, C sends the seeds $\mathbf{p}_{i_{q,\delta}}^{j^r, \gamma}$ to R.

Theorem 8.11. *If Com is computationally hiding, then ERE-Commitment is an equivocal bit commitment.*

Proof. The main difference with respect to the proof for EqCommitment is that the equivocal simulator has to answer correctly the commitment of w equivocal boxes in parallel with single challenge bit γ_r . Given a computationally hiding bit commitment, $|p(\psi) - 1/2| = \text{negl}(\lambda)$ also for this case. Hence, all the properties of the Algorithm 2 and the application of Watrous' rewinding lemma are equivalent for this scheme. Therefore, from computational hiding, $Real_b$ and $Ideal_b$ of Def. 8.2 are indistinguishable. $\square$

Corollary 8.12. *If $Com_{\mathbf{p}_i^j}$ is implemented using Naor's bit commitment with input seeds $\mathbf{p}_i^j \in \{0,1\}^{\lambda_{PQS}}$, where $(\mathbf{p}_1^j, \dots, \mathbf{p}_{2w}^j) = PRG(\mathbf{s}^j)$ for $\mathbf{s}^j \in \{0,1\}^{\lambda_{PQS}}$, then ERE-Commitment is an equivocal bit commitment.*

Proof. We have to prove that the Naor's bit commitment scheme [Nao91] is computationally hiding giving the pseudorandom seeds $\mathbf{p}_i^j \in \{0,1\}^{\lambda_{PQS}}$. Naor's bit commitment is an interactive bit commitment in which the receiver sends $k \xleftarrow{\$} \{0,1\}^{3\lambda}$ to the committer, which commits to the bit b by sending $(b \cdot k) \oplus W(r)$, where $r \xleftarrow{\$} \{0,1\}^\lambda$ and $W : \{0,1\}^{\lambda_{PQS}} \rightarrow \{0,1\}^{3\lambda}$ is a pseudo-random generator. Thus, the computational hiding property is guaranteed by the pseudorandomness property of W . In this case, the seed $\mathbf{p}_i^j$ of the Naor's base commitment is not randomly sampled by the committer, but obtained as the output of a PRG which input is $(\mathbf{s}^j)_{j \in [2k]}$. $(\mathbf{s}^j)_{j \in [2k]}$ are generated by hashing the strings $(\tilde{\mathbf{x}}^j)_{j \in [2k]}$. Therefore, computational hiding would be fulfilled if the strings $(\mathbf{s}^j)_{j \in [2k]}$ are random strings. By applying the leftover hash lemma (Lemma 8.6), the obtained seeds are arbitrarily close to a random string, as proven in Theorem A.1 of Section A.5.1. $\square$

8.6.2. Relaxed Extractability

Before proving relaxed-extractability, we first show that our commitment scheme is relaxed-binding.

Theorem 8.13. *If Com is a statistically binding commitment scheme, then for any function $\eta(k)$ ERE-Commitment is η -relaxed statistically binding.*

Proof. The proof is similar to the one of Theorem 8.9. In ERE-Commitment, the seeds of the base commitments Com_r are used as described in Algorithm 3. Therefore, when the honesty check is done, the sampled population is not the number of sequential repetitions of the EqCommitment wk but the number of different families k since the EqCommitment instances which make use of seeds derived from the same family $\mathbf{p}^j$ are opened together. $\square$

To prove extractability, we will use the equivocality properties of EqCommitment. To prove that the simulator $\mathcal{Q}_{C^*}(\rho)$ defined in Algorithm 4 that represents the *Ideal* interaction from Definition 8.5 is indistinguishable from the *Real* one, we use the following sequence of hybrid arguments:

- **Hybrid 0.** Corresponds to the real protocol described in Algorithm 3.
- **Hybrid 1.** Similar to Hyb_0 except in Step 3, in which we execute the equivocal simulator $\mathcal{Q}_{R^*,com}$ for $i \in [4\lambda_{EX}]$, and in Step 5, to open the challenged commitments $i \in E$, we execute $\mathcal{Q}_{R^*,dec}$.
- **Hybrid 2.** Similar to Hyb_1 , except that $|x\rangle_\theta$ is not measured during Step 2 of 3, but rather during Step 3 of Algorithm 4, i.e., only the positions $i \in E$ are measured and only after obtaining the sequence of indices.
- **Hybrid 3.** Identical to Hyb_2 but the simulator performs the error correction as described in Step 7 of 4 and then, the honesty check as described in Step 8 of Algorithm 4.
- **Hybrid 4.** Identical to Hyb_3 but the simulator extracts the non-challenged commitments as described in Step 9 of 4.
- **Hybrid 5.** The *Ideal* distribution, i.e., the extractable simulator described in Algorithm 4. The difference with respect to Hyb_4 is that the simulator opens the non-extracted commitments to the value opened by the malicious committer, as described on Step 10.

Lemma 8.14. *There exists a negligible function $\nu(\lambda)$, such that:*

$$|\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1] - \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_0) = 1]| = \nu(\lambda).$$

Proof. Suppose that there exists a non-negligible function for which $\mathcal{D}^*$ can distinguish between Hyb_0 and Hyb_1 . This is equivalent to:

$$\left| \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_{0,i-1}) = 1] - \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_{0,i}) = 1] \right| \geq \frac{1}{\text{poly}(\lambda)4\lambda},$$

where $\text{Hyb}_{0,j}$ is the sub-hybrid that is identical to Hyb_0 until the committed bit $j \in [4\lambda_{EX}]$ and then identical to Hyb_1 . Therefore, $\mathcal{D}^*$ would be able to distinguish between the outputs ρ_{final}^{CO} of an honest and an equivocated commitment, contradicting Definition 8.2.

□

Lemma 8.15. $\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_2) = 1] = \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1].$

Proof. Since the states corresponding to the subset $E \subset [4\lambda_{EX}]$ are also measured with randomly sampled bases $(\hat{\theta}_i)_{i \in E}$, the two experiments have exactly the same output distribution. $\square$

Lemma 8.16. *There exists a negligible function $\nu(\lambda)$, such that,*

$$|\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_3) = 1] - \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_2) = 1]| = \nu(\lambda).$$

Proof. A quantum poly-time distinguisher $\mathcal{D}^*$ would be able to discriminate between both distributions if Hyb_2 and Hyb_3 had different probabilities for output $\perp$ during the honesty check. If $\delta > \alpha$, both Hybrids abort. Otherwise, as discussed in Section 8.4.5, an $(\mathcal{M}, 4\lambda_{EX}, 4\alpha\lambda_{EX})$ -error correction code can unambiguously correct up to a proportion of α errors. Therefore, for any $\delta \leq \alpha$, there is at most one codeword inside the radius of the error correction code. Therefore, both distributions are indistinguishable. $\square$

Lemma 8.17. *There exists a negligible function $\nu(\lambda)$ such that:*

$$|\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_4) = 1] - \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_3) = 1]| = \nu(\lambda).$$

Proof. The only difference between Hyb_3 and Hyb_4 comes from the fact that the *Ideal* distribution, given by Hyb_4 , outputs FAIL whenever $b_i^* \notin \{\perp, b_i, ?\}$ and the proportion of ? outputs is larger than $\chi = \eta + \zeta$, where ζ is the proportion of commitments seeded in a dishonest way.

In order to bound the probability of having more than a proportion ζ of dishonestly seeded commitments, let us consider again k sequential repetitions of w parallel EqCommitment protocols, as described in Algorithm 3. For the r -th instance, the q -th EqCommitment $(c_{i_{q,0}}^{j^{r,0}}, c_{i_{q,1}}^{j^{r,0}}, c_{i_{q,0}}^{j^{r,1}}, c_{i_{q,1}}^{j^{r,1}})$ of the w parallel commitments, uses as seeds $(\mathbf{p}_{i_{q,0}}^{j^{r,0}}, \mathbf{p}_{i_{q,1}}^{j^{r,0}}, \mathbf{p}_{i_{q,0}}^{j^{r,1}}, \mathbf{p}_{i_{q,1}}^{j^{r,1}})$.

Each $r \in [k]$ sequential honesty check consists of a sequence of w EqCommitment that use the seeds $\mathbf{p}^{j^{r,0}} \leftarrow \tilde{\mathbf{x}}^{j^{r,0}}$ and $\mathbf{p}^{j^{r,1}} \leftarrow \tilde{\mathbf{x}}^{j^{r,1}}$. Let us define B_1 as the set of r indices with at least one dishonestly seeded commitment $B_1 = \{r : \exists \gamma, \delta, q \text{ such that } \mathbf{p}_{i_{q,\delta}}^{j^{r,\gamma}} \text{ is not the seed of the commitment}\}$. Let us also define the event E_1 where the check on Step 11 of Algorithm 3 passes on all k sets. The goal is now to connect $|B_1|$ and $\Pr[E_1]$. It can be shown that:

$$\Pr[E_1] \leq \frac{1}{2^{|B_1|}}, \quad (8.6.1)$$

which is equivalent to the fact that, if $\Pr[E_1] \geq 2^{-\zeta k}$, then $|B_1| \leq \zeta k$. The proof of Equation (8.6.1) holds since, for each $r \in B_1$, there exists one set of indices $\{\gamma^*, \delta^*, q^*\}$ such that $\mathbf{p}_{i_{q^*,\delta^*}}^{j^{r,\gamma^*}}$ is not the seed of the commitment, and in order for E_1 to be true, the

Algorithm 4 Extractable simulator

-
- 1: $C^*(\rho)$ outputs Ψ .
 - 2: For $i \in [4\lambda_{EX}]$, execute sequentially the quantum equivocal simulator $\mathcal{Q}_{R^*}$ to generate a tuple of $((c_\delta^i)_{\delta=0,1}^i)_{i \in [4\lambda_{EX}]}$ dummy commitments. After each commitment session, $(\rho_i^S, \rho_{com,i}^y) \leftarrow \mathcal{Q}_{R^*,com}(\rho)$, where ρ_i^S is stored by $\mathcal{Q}_{C^*}(\rho)$, and $\rho_{com,i}^y$ is used by the malicious committer C^* as an input in the next session.
 - 3: Once C^* has announced the challenged subset $E \subset [4\lambda_{EX}]$, $\mathcal{Q}_{C^*}(\rho)$ samples $\hat{\theta} \leftarrow \{+, \times\}^{2\lambda_{EX}}$. $\mathcal{Q}_{C^*}$ measures the i^{th} qubit of Ψ , Ψ_i , for every $i \in E$ in the previous sampled bases, obtaining $(\hat{x}_i)_{i \in E}$.
 - 4: By applying the equivocal simulator $\mathcal{Q}_{Eq}$, $\mathcal{Q}_{C^*}$ opens the challenged bits $z_i \in E$ to the correct values $(\hat{x}_i, \hat{\theta}_i)_{i \in E}$. To do so, it executes for each $i \in E$: $\rho_{final}^x \leftarrow \langle \mathcal{Q}_{R^*,dec}(\hat{x}_i, \hat{\theta}_i, z_i, \rho_{com}^x), C_{dec}^*(\rho_{com}^y) \rangle$, where ρ_{com}^x is the current state of C^* , updated after each session.
 - 5: If C^* aborts at any point, $\mathcal{Q}_{C^*}(\rho)$ outputs $\perp$, otherwise continues.
 - 6: C^* and R discard the tested positions and both reorder their respective tuples $((x_i, \theta_i))_{i \in \bar{E}}$ and $((\hat{x}_i, \hat{\theta}_i))_{i \in \bar{E}}$. C^* sends θ , computes the $(p^j)_{j \in [2k]}$ as described in Algorithm 3 and sends $(r^j)_{j \in [2k]}$.
 - 7: $\mathcal{Q}_{C^*}(\rho)$ measures each qubit $\psi_i \in \bar{E}$ in the bases θ_i to obtain $\hat{x}_i = x_i$. $\mathcal{Q}_{C^*}(\rho)$ performs the error correction on $\hat{x}$ by using the syndrome $Synd$. Then, $\mathcal{Q}_{C^*}(\rho)$ computes the seeds $\left((\hat{p}_{i_{q,0}}^{j_{r,0}}, \hat{p}_{i_{q,1}}^{j_{r,0}}, \hat{p}_{i_{q,0}}^{j_{r,1}}, \hat{p}_{i_{q,1}}^{j_{r,1}})_{q \in [w]} \right)_{r \in [k]}$.
 - 8: $\mathcal{Q}_{C^*}(\rho)$ executes the honesty check as described in Step 11 of Algorithm 3.
 - 9: $\mathcal{Q}_{C^*}(\rho)$ extracts the non challenged commitments:
 - 9.1: $\mathcal{Q}_{C^*}(\rho)$ checks if $Decom_{\hat{p}_{i_{q,0}}^{j_{r,\gamma}}}(c_{i_{q,0}}^{j_{r,\gamma}}) = Decom_{\hat{p}_{i_{q,1}}^{j_{r,\gamma}}}(c_{i_{q,1}}^{j_{r,\gamma}})$ where $\hat{p}_{i_{q,\delta}}^{j_{r,\gamma}}$ are the ones obtained in Step 7.
 - 9.2: For the positions where this is the case, $b_{r,q}^* = Decom_{\hat{p}_{i_{q,0}}^{j_{r,\gamma}}}(c_{i_{q,0}}^{j_{r,\gamma}} \oplus e_{r,q})$, otherwise $b_{r,q}^* = ?$.
 - 9.3: If the proportion of $b_i^* = ?$ is larger than $\chi = \eta + \zeta$, with $\eta = \zeta = \omega(\log^2(k)/k)$, $\mathcal{Q}_{C^*}(\rho)$ outputs abort, where $i = (r-1) * w + q$ are the reordered indices.
 - 9.4: Otherwise, $\mathcal{Q}_{C^*}(\rho)$ outputs $(\rho_{com}^C, \rho_{com}^R, b^*)$, where $b^* \in \{0,1\}^{wk}$ is the string of bits in which $b_i^* = b_i^*$ for the positions in which the output was not ? and an equivocated dummy commitment for the remaining positions by executing the quantum equivocal simulator $\mathcal{Q}_{Eq}$. ρ_{com}^C is the resulting state of C^* and $\rho_{com}^R = (\theta, \hat{\theta}, \hat{x})$.
 - 10: $\mathcal{Q}_{C^*}(\rho)$ opens the equivocated commitments corresponding to ? to the value opened by C^* in these positions by applying the equivocal simulator $\mathcal{Q}_{Eq}$.
-

challenged family of commitments seeded by $\mathbf{p}^{j^{r,\gamma}}$ for the r -th set of w EqCommitment instances is different from $\mathbf{p}^{j^{r,\gamma^*}}$, which happens with probability $\frac{1}{2}$. Since all challenges are picked independently,

$$\Pr[\forall r \in B_1 : \gamma \neq \gamma^*] = \prod_{r \in B_1} \Pr[\gamma \neq \gamma^*] \leq \frac{1}{2^{|B_1|}}.$$

Hyb_3 and Hyb_4 would be distinguishable if the malicious committer $\mathbf{C}^*$ has committed to b_i with an equivocated commitment more than a proportion $\eta = \omega(\log^2(k)/k)$, or if the malicious committer $\mathbf{C}^*$ uses as seeds $\mathbf{p}^j \neq \text{PRG}(h(\tilde{\mathbf{x}}^j))$ in more than a proportion $\zeta = \omega(\log^2(k)/k)$ of the keys and Hyb_3 has not output $\perp$ in the honesty check phase. This happens with probability:

$$\begin{aligned} \Pr[\text{FAIL}|\text{Ideal}] &= \Pr[\text{proportion of non extractable commitments} \geq \eta + \zeta] \\ &\leq \Pr[\text{proportion of non extractable commitments} \geq \zeta]. \end{aligned}$$

Then, $\Pr[\text{FAIL}|\text{Ideal}] \leq \text{negl}(\lambda)$ given Theorem 8.13. $\square$

Lemma 8.18. $\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_5) = 1] \equiv \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_4) = 1]$.

Proof. Similar to the proof of Lemma 8.14. In the opening phase, since the opened value is the same as the one opened by the malicious committer $\mathbf{C}^*$, both distributions are indistinguishable. $\square$

Thus, ERE-Commitment is equivocal and $\omega(\log^2(k)/k)$ -relaxed-extractable in the sense of Definitions 8.2 and 8.5.

8.7. OT from Equivocal and Relaxed-Extractable Bit Commitment

In this section, we build an oblivious transfer functionality assuming an equivocal and χ -relaxed extractable bit commitment, with $\chi = \omega(\log^2(k)/k)$. The structure is similar to [DFL⁺09, BCKM21], but applying it to our setting. Lastly, we show how to optimize the number of runs of OT in Corollary 8.25.

The oblivious transfer functionality $\mathcal{F}_{OT}(\cdot, \cdot)$ is a quantum two-parties interactive functionality in which Alice, A , inputs two messages $m_0, m_1 \in \{0, 1\}^\lambda$, and Bob, B , inputs a choice bit $b \in \{0, 1\}$. The output after the interaction is (m_0, m_1) on Alice's side and (b, m_b) on Bob's side. More concretely, the OT functionality $\mathcal{F}_{OT}$ works as follows:

- $\mathcal{F}_{OT}((m_0, m_1), \cdot)$ takes an input b (or $\perp$), returns m_b to B and END (or $\perp$) to A .
- $\mathcal{F}_{OT}(\cdot, b)$ takes an input (m_0, m_1) (or $\perp$), returns END to A and m_b (or $\perp$) to B .

Algorithm 5 Oblivious transfer protocol**Alice A Input:** Messages $m_0, m_1 \in \{0, 1\}^\lambda$.**Bob B Input:** Bit $b \in \{0, 1\}$.**Assumptions:** The parties use an ERE-Commitment that is χ -relaxed extractable and equivocal, $PRG : \{0, 1\}^{\lambda_{PQS}} \rightarrow \{0, 1\}^w$ and 2-universal hash functions $h_j(\cdot, \cdot) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^{\lambda_{PQS}}$.

- 1: A chooses $\mathbf{x} \leftarrow \{0, 1\}^{2\lambda_{OT}}$ and $\boldsymbol{\theta} \leftarrow \{+, \times\}^{2\lambda_{OT}}$ and sends $|x\rangle_{\boldsymbol{\theta}}$ to B .
- 2: B samples his measurement bases $\hat{\boldsymbol{\theta}} \leftarrow \{+, \times\}^{2\lambda_{OT}}$ and measures $\hat{\mathbf{x}} \leftarrow \{0, 1\}^{2\lambda_{OT}}$. A and B execute ERE-Commitment to commit to $((\hat{x}_i, \hat{\theta}_i))_{i \in [2\lambda_{OT}]}$.
- 3: A challenges a random subset $T \subset [2\lambda_{OT}]$, such that $|T| = \lambda_{OT}$.
- 4: A and B execute ERE-Decommitment for every $i \in T$. Then, A obtains $((\hat{x}_i, \hat{\theta}_i))_{i \in T}$.
- 5: A checks that $x_i = \hat{x}_i$ whenever $\theta_i = \hat{\theta}_i$, up to an experimental error α . If the test does not pass, A aborts.
- 6: A and B discard the tested positions and both reorder their respective sequences $((x_i, \theta_i))_{i \in \bar{T}}$ and $((\hat{x}_i, \hat{\theta}_i))_{i \in \bar{T}}$. A sends $\boldsymbol{\theta}$ to B .
- 7: B partitions the set $\bar{T}$ into two different subsets: the subset in which both have measured in the same bases $I_b = \{i : \theta_i = \hat{\theta}_i\}$ and the subset in which they have not measured in the same bases $I_{\bar{b}} = \{i : \theta_i \neq \hat{\theta}_i\}$. B sends (I_0, I_1) to A .
- 8: A sends the syndromes $Synd_0$ and $Synd_1$ for correcting a proportion α of errors of the bit strings $\mathbf{x}_0$ and $\mathbf{x}_1$ to B , where $\mathbf{x}_0 = \{x_i : i \in I_0\}$ and $\mathbf{x}_1 = \{x_i : i \in I_1\}$.
- 9: A samples seeds $s_0, s_1 \leftarrow \{0, 1\}^{k(\lambda_{PQS})}$ and sends $(s_0, PRG(h(s_0, \mathbf{x}_0)) \oplus m_0, s_1, PRG(h(s_1, \mathbf{x}_1)) \oplus m_1)$.
- 10: B performs error correction on $\mathbf{x}_b$ using the syndrome $Synd_b$ and decrypts m_b by using $\hat{\mathbf{x}}_b$, with $\hat{\mathbf{x}}_b = \{\hat{x}_i : i \in I_b\}$. If the error correcting protocol fails, B chooses $\mathbf{x}_b$ to be a random string $s \leftarrow \{0, 1\}^{\lambda_{OT}/2}$.

Definition 8.19 (Secure OT functionality). A protocol $\langle A(m_0, m_1), B(b) \rangle$ that implements the ideal OT functionality $\mathcal{F}_{OT}(\cdot, \cdot)$ is said to be simulation-based secure if it satisfies:

- **Security against a malicious Alice.** Given the interaction between a quantum poly-time non-uniform malicious Alice, A^* , and an honest Bob, their final state will be $\rho_{OUT, A^*} \langle A^*(m_0, m_1), B(b) \rangle$ and $OUT_B \langle A^*(m_0, m_1), B(b) \rangle$. There exists a simulator Sim_{A^*} that interacts with the ideal functionality $\mathcal{F}_{OT}(\cdot, b)$ in the following way. For every non-uniform advice $\rho, \sigma \in \mathcal{B}_1(\mathcal{H})$, where ρ and σ may be entangled, $Sim_{A^*}(\rho)$ sends either (m_0, m_1) or $\perp$ to $\mathcal{F}_{OT}(\cdot, b)$ and outputs the final state ρ_{SIM, OUT, A^*} . The output of the ideal functionality to the Bob is given by OUT_B . Then, for any quantum poly-time non-uniform distinguisher $\mathcal{D}^*$:

$$\begin{aligned}
& |\Pr[\mathcal{D}^*(\sigma, (\rho_{SIM, OUT, A^*}, OUT_B)) = 1] \\
& \quad - \Pr[\mathcal{D}^*(\sigma, (\rho_{OUT, A^*} \langle A^*(m_0, m_1), B(b) \rangle, OUT_B \langle A^*(m_0, m_1), B(b) \rangle)) = 1]| \\
& = \text{negl}(\lambda).
\end{aligned} \tag{8.7.1}$$

- **Security against a malicious Bob.** Given the interaction between an honest Alice and a quantum poly-time non-uniform malicious Bob, B^* , their final state will be $OUT_B \langle A(m_0, m_1), B^*(b) \rangle$ and $\rho_{OUT, B^*} \langle A(m_0, m_1), B^*(b) \rangle$. There exists a simulator Sim_{B^*} that interacts with the ideal functionality $\mathcal{F}_{OT}((m_0, m_1), \cdot)$ in the following way. For every non-uniform advice $\rho, \sigma \in \mathcal{B}_1(\mathcal{H})$, where ρ and σ may be entangled, $Sim_{B^*}(\rho)$ sends either $b \in \{0, 1\}$ or $\perp$ to $\mathcal{F}_{OT}((m_0, m_1), \cdot)$ and outputs the final state ρ_{SIM, OUT, B^*} . The output of the ideal functionality to Bob is given by OUT_A . Then, for any quantum poly-time non-uniform distinguisher $\mathcal{D}^*$:

$$\begin{aligned}
& |\Pr[\mathcal{D}^*(\sigma, (OUT_A, \rho_{SIM, OUT, B^*}) = 1] \\
& \quad - \Pr[\mathcal{D}^*(\sigma, (OUT_A \langle A(m_0, m_1), B^*(b) \rangle, \rho_{OUT, B^*} \langle A(m_0, m_1), B^*(b) \rangle)) = 1]| \\
& = \text{negl}(\lambda).
\end{aligned} \tag{8.7.2}$$

8.7.1. Security Against a Malicious Alice

We will now establish the security of Bob against a quantum poly-time non-uniform malicious Alice, given Definition 8.19. To prove that Eq. 8.7.1 holds in the proposed scheme, the following hybrid argument is required:

- **Hybrid 0.** The *Real* distribution given by Algorithm 5.
- **Hybrid 1.** Similar to Hyb_0 except in Step 2, where we execute the equivocal simulator $\mathcal{Q}_{R^*, com}$ for $i \in [2\lambda_{OT}]$. The second difference with respect to Hyb_0 is that in Step 4, to open the challenged commitments $i \in T$, we execute $\mathcal{Q}_{R^*, dec}$.
- **Hybrid 2.** The *Ideal* distribution given by Algorithm 6.

Lemma 8.20. *There exists a negligible function $\nu(\lambda)$, such that:*

$$\begin{aligned}
& \left| \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1] \right. \\
& \quad \left. - \Pr[\mathcal{D}^*(\sigma, (\rho_{OUT, A^*} \langle A^*(\rho), B(b) \rangle, OUT_B \langle A^*(\rho), B(b) \rangle)) = 1] \right| = \nu(\lambda).
\end{aligned}$$

Algorithm 6 Simulator Sim_{A^*}

-
- 1: $A^*(\rho)$ outputs the message Ψ .
 - 2: Sim_{A^*} executes $2\lambda_{OT}$ sessions of ERE-Commitment to commit to dummy values by executing the equivocal simulator $\mathcal{Q}_{R^*}$.
 - 3: Sim_{A^*} obtains the challenged set $T \subset [2\lambda_{OT}]$ output by A^* .
 - 4: Sim_{A^*} samples $\hat{\theta} \leftarrow \{+, \times\}^{\lambda_{OT}}$. Sim_{A^*} measures the i^{th} qubit of Ψ , Ψ_i , for every $i \in T$ in the previous sampled bases, obtaining $\{\hat{x}_i\}_{i \in T}$. By applying Watrous' rewinding, Sim_{A^*} opens the challenged bits to the correct values $\{\hat{x}_i, \hat{\theta}_i\}_{i \in T}$.
 - 5: Sim_{A^*} discards the tested positions. A sends θ to B . Sim_{A^*} obtains θ and measures each of the qubits $\psi_i \in \bar{T}$ in the bases θ_i to obtain $\hat{x}_i = x_i$.
 - 6: Sim_{A^*} partitions the set $\bar{T}$ into two different subsets, by randomly sampling a bit $d_i \in \{0, 1\}$ for each $i \in \bar{T}$, to create the sequences $I_0 = \{i : d_i = 0\}$ and $I_1 = \{i : d_i = 1\}$. Sim_{A^*} sends (I_0, I_1) to A^* .
 - 7: Sim_{A^*} obtains $(s_0, \text{PRG}(h(s_0, \mathbf{x}_0)) \oplus m_0, s_1, \text{PRG}(h(s_1, \mathbf{x}_1)) \oplus m_1)$, and computes m_0 and m_1 . Both, $\mathbf{x}_0$ and $\mathbf{x}_1$ are the corrected measured strings $\hat{\mathbf{x}}_0$ and $\hat{\mathbf{x}}_1$ with the syndromes Synd_0 and Synd_1 . If the error correcting protocol fails, Sim_{A^*} chooses $\mathbf{x}_0$ or/and $\mathbf{x}_1$ to correspondingly be the random strings $s_0, s_1 \leftarrow \{0, 1\}^{\lambda_{OT}/2}$.
 - 8: If A^* aborts at any point, Sim_{A^*} sends $\perp$ to $\mathcal{F}_{OT}(\cdot, b)$. Otherwise, it sends (m_0, m_1) to $\mathcal{F}_{OT}(\cdot, b)$. The output is the final state of A^* .
-

Proof. It follows from the equivocality property of ERE-Commitment. □

Lemma 8.21. $\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1] \equiv \Pr[\mathcal{D}^*(\sigma, (\rho_{\text{SIM}, \text{OUT}, A^*}, \text{OUT}_B)) = 1]$.

Proof. It follows from the fact that the delay in the measurement of ψ_i with $i \in T$ cannot be noticed by A^* since, in both cases, the measurement bases are sampled at random, similar to Lemma 8.15. Moreover, in both cases the syndromes used for performing the error correction are the same ones. Therefore, if the syndromes sent by A^* are not correct, both hybrids would lead to the same random distribution. Then, both extracted messages m_0 and m_1 are obtained by using the same error correction syndromes as in Hyb_1 , leading to the same distribution. □

8.7.2. Security Against a Malicious Bob

In this section, we prove security against malicious Bob. We start by analyzing the number of qubits necessary in the protocol to achieve the desired level of security in Lemma 8.22. Then, we prove the security of the protocol in Theorem 8.23.

Lemma 8.22 (Distance bound for a malicious Bob in QOT). *Consider a QOT protocol between an honest Alice and a malicious Bob (Algorithm 5) with ERE-Commitment as the bit commitment subprotocol (Algorithm 3). Let $b \in \{0, 1\}$ be a random bit and $K_b = h(r, \mathbf{x}_{I_b}) \in \{0, 1\}^\ell$ and $K_{\bar{b}} = h(r, \mathbf{x}_{\bar{I}_b}) \in \{0, 1\}^\ell$, the keys. $h(r, x) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ is a*

2-universal hash function, and I_0 and I_1 are the two subsets in which $\bar{T} \subset [2\lambda_{OT}]$ the unchallenged commitments are divided. Then, given K_b , for any sampling errors $\xi, \delta > 0$, a bit-flip error $\alpha \geq 0$ and a proportion $\vartheta \geq 0$ of leaked bits:

$$\Delta(\rho_{K_b K_b E}, \frac{1}{2^\ell} \mathbb{1} \otimes \rho_{K_b E}) \leq \frac{1}{2} \cdot 2^{-\frac{1}{2}[(\frac{1}{2}-\xi-2\vartheta)\frac{\lambda_{OT}}{2} - h_2(\delta+\alpha+\chi)\frac{\lambda_{OT}}{2}(1-2\vartheta)-\ell-q]} + \sqrt{6} \exp(-\lambda_{OT}\delta^2/100) + 2 \exp(-\xi^2\lambda_{OT}/2), \quad (8.7.3)$$

where E denotes Bob's quantum state, k , the number of different seed families as described in Algorithm 3 and q , the length of the syndrome needed for performing the error correction. $\mathbb{1} \in \mathcal{B}(\mathbb{C}^{2^\ell})$ is the identity operator.

Proof. The opening and checking process, which is based on sampling, follows a structure akin to the proof provided in [BF10]. Interested readers can refer to it for a detailed analysis of the proof. This proof is done in the EPR setting in which Alice distributes half of an EPR pair to Bob in order for her to perform the sampling. Since Bob's and Alice's actions are equivalent in both settings, it suffices to prove the bound in the EPR setting. As the measurement bases are chosen randomly by Alice and Bob, the sampling strategy will be performed on a random subset $S \subseteq T$ defined as $S = \{i \in T : \theta_i = \hat{\theta}_i\}$. Alice will accept the opening phase of the bit commitment if $w(\mathbf{x}|_S, \hat{\mathbf{x}}|_S) \leq \alpha$, where $w(\cdot) = d_H(\cdot)/n$, with $n \in \mathbb{N}$ being the length of the bit-string.

Once the testing phase is accepted, the joint state $|\psi_{A_{\bar{T}}E}\rangle$ will be a superposition of states with relative Hamming weight δ close to α within $A_{\bar{T}}$ due to the statistical error of the sampling. This error given by the sampling strategy will be bounded by Hoeffding's bound:

$$\epsilon^\delta \leq \sqrt{6} \exp(-\lambda_{OT}\delta^2/100).$$

This bound is obtained in Appendix B.4 of [BF10]. Assuming that Alice chooses her bases following a uniform random distribution and for $\theta|_{\bar{T}}$ representing the subsets of bases that were not challenged, $d_H(\theta|_{\bar{T}}, \hat{\theta}|_{\bar{T}}) \geq (1/2 - \gamma)|\bar{T}| = (1/2 - \gamma)\lambda_{OT}$ with Hoeffding's bound:

$$\epsilon^\gamma \leq 2 \exp(-2\gamma^2\lambda_{OT}).$$

After the commitment and check subprotocols, the next step in the QOT protocol is when indices Alice sends the bases $\theta|_{\bar{T}}$ to Bob, and Bob partitions $\bar{T}$ into two sets, I_0 and I_1 . There exists a bit $b \in \{0, 1\}$ such that $d_H(\theta|_{I_b}, \hat{\theta}|_{I_b}) \geq (1/2 - 2\gamma - 2\vartheta)\lambda_{OT}/2$, regardless of how Bob splits the sets of indices and taking the maximum advantage of the leaked bits, i.e., all the leaked bits are used in the set I_b . Moreover, the effective size of the non-sampled subset $I_{\bar{b}}$ for which the binary entropy is bounded is $(1 - 2\vartheta)$. By applying

Algorithm 7 Simulator Sim_{B^*}

-
- 1: Sim_{B^*} chooses $\mathbf{x} \leftarrow \{0, 1\}^{2\lambda_{OT}}$ and $\boldsymbol{\theta} \leftarrow \{+, \times\}^{2\lambda_{OT}}$ and sends $|\mathbf{x}\rangle_{\boldsymbol{\theta}}$ to B^* .
 - 2: B^* and Sim_{B^*} execute $2\lambda_{OT}$ sessions of ERE-Commitment. Then, Sim_{B^*} extracts the committed values $(\hat{\theta}_i, \hat{x}_i)_{i \in [2\lambda_{OT}]}$ by applying the extractable simulator given by Algorithm 4 and aborts if the proportion of extracted commitments equal to ? is larger than $\chi = \eta + \zeta$.
 - 3: Sim_{B^*} challenges a random subset $T \subseteq [2\lambda_{OT}]$, such that $|T| = \lambda_{OT}$.
 - 4: Sim_{B^*} and B^* execute ERE-Decommitment for every $i \in T$. Then, Sim_{B^*} obtains the set $((\hat{x}_i, \hat{\theta}_i))_{i \in T}$ and checks that $x_i = \hat{x}_i$ whenever $\theta_i = \hat{\theta}_i$, up to an experimental error α . If the test does not pass, Sim_{B^*} aborts.
 - 5: Sim_{B^*} discards the tested positions and sends $\boldsymbol{\theta}$ to B^* .
 - 6: Sim_{B^*} obtains I_0 and I_1 . Let S be the sequence of indices in I_0 such that $\theta_i \neq \hat{\theta}_i$. If $|S| \geq (1/2 - \xi) \lambda_{OT}/2$ set $b = 1$, otherwise set $b = 0$.
 - 7: Sim_{B^*} obtains m_b from $\mathcal{F}_{OT}$ and sets $m_{\bar{b}} = 0$.
 - 8: Sim_{B^*} samples seeds $s_0, s_1 \leftarrow \{0, 1\}^{k(\lambda_{PQS})}$ and sends $(s_0, \text{PRG}(h(s_0, \mathbf{x}_0)) \oplus m_0, s_1, \text{PRG}(h(s_1, \mathbf{x}_1)) \oplus m_1)$, where $\mathbf{x}_0 = \{x_i : i \in I_0\}$ and $\mathbf{x}_1 = \{x_i : i \in I_1\}$. He also sends the syndromes Synd_0 and Synd_1 for correcting a proportion α of errors of the bit strings $\mathbf{x}_0$ and $\mathbf{x}_1$. Sim_{B^*} outputs the final state of B^* .
-

Equation (8.4.1) to our state $\rho_{X_{\bar{b}}A^bE}$, itself obtained from measuring the subsystem $A^{\bar{b}}$ on $|\psi_{A^{\bar{b}}A^bE}\rangle$, privacy amplification gives, for $\xi = 2\gamma$:

$$\begin{aligned} H_{\min}(X_{\bar{b}}|A^bE) &\geq d_H(\boldsymbol{\theta}|_{I_{\bar{b}}}, \hat{\boldsymbol{\theta}}|_{I_{\bar{b}}}) - h_2(\delta + \alpha + \eta + \zeta)|I_{\bar{b}}| \\ &\geq \left(\frac{1}{2} - \xi - 2\vartheta\right) \frac{\lambda_{OT}}{2} - h_2(\delta + \alpha + \chi) \frac{\lambda_{OT}}{2} (1 - 2\vartheta). \end{aligned}$$

Given a χ -relaxed extractable ERE-Commitment, with $\chi = \eta + \zeta = \omega(\log^2(k)/k)$, ERE-Commitment is relaxed extractable.

□

We now move on to prove the security of the protocol against a quantum poly-time malicious Bob and an honest Alice. The sequence of hybrid arguments for this case is:

- **Hybrid 0.** The *Real* distribution given by Algorithm 5.
- **Hybrid 1.** Similar to Hyb_0 except in Step 2, in which we execute the extractable simulator $\mathcal{Q}_{R^*, \text{com}}$ of Algorithm 4.
- **Hybrid 2.** The *Ideal* distribution given by Algorithm 7.

Theorem 8.23. *There exists a negligible function $\nu(\lambda)$, such that,*

$$\left| \Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1] - \Pr[\mathcal{D}^*(\sigma, (\text{OUT}_A \langle A(m_0, m_1), B^*(b) \rangle), \rho_{\text{OUT}, B^*} \langle A(m_0, m_1), B^*(b) \rangle) = 1] \right| = \nu(\lambda).$$

Proof. It follows from the $\omega(\log^2(k)/k)$ -relaxed-extractability definition of the bit commitment. $\square$

Lemma 8.24. $|\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1] - \Pr[\mathcal{D}^*(\sigma, (\text{OUT}_A, \rho_{\text{SIM}, \text{OUT}, B^*}) = 1]| = \nu(\lambda).$

Proof. We prove this by contradiction. Suppose that there exists a distinguisher such that:

$$|\Pr[\mathcal{D}^*(\sigma, \text{Hyb}_1) = 1] - \Pr[\mathcal{D}^*(\sigma, (\text{OUT}_A, \rho_{\text{SIM}, \text{OUT}, B^*}) = 1]| \geq \frac{1}{\text{poly}(\lambda)}.$$

This would mean that there is a non-uniform quantum poly-time distinguisher $\mathcal{D}^*$ that can distinguish between the messages $m_{\bar{b}}$ obtained in Hyb_1 and $m_{\bar{b}}$ obtained in Hyb_2 . Since $m_{\bar{b}}$ is encoded as $\text{PRG}(h(s_{\bar{b}}, \hat{x}_{\bar{b}})) \oplus m_{\bar{b}}$, this would be equivalent to distinguishing between the seeds $h(s_{\bar{b}}, \hat{x}_{\bar{b}})$ of Hyb_1 and Hyb_2 . This leads to a contradiction since $h(s_{\bar{b}}, \hat{x}_{\bar{b}})$ is statistically close to a random string due to privacy amplification, as proven in Lemma 8.22. Moreover, in both distributions, the applied syndromes for the error correction of the strings $\hat{x}_{\bar{b}}$ and $\hat{x}_b$ are the same ones, leading to the same corrected strings. $\square$

Corollary 8.25. *Given one implementation of the QOT protocol described in Algorithm 5 with the quantum distributed key $\mathbf{x} \in \{0, 1\}^{2\lambda_{\text{OT}}}$, a quantity n_{OT} of 1-out-of-2 oblivious transfers can be distilled, where $n_{\text{OT}} = \lfloor \lambda_{\text{OT}}/v \rfloor$ and where v is the size of the preimage of a hash function such that $h(r, x) : \{0, 1\}^{m(\lambda_{\text{PQS}})} \times \{0, 1\}^m \rightarrow \{0, 1\}^{\lambda_{\text{PQS}}}$. The parameter v is determined in such a way that:*

$$\begin{aligned} \Delta(\rho_{K_{\bar{b}}K_bE}, \frac{1}{2^\ell} \mathbb{1} \otimes \rho_{K_bE}) &\leq \frac{1}{2} \cdot 2^{-\frac{1}{2}[(\frac{1}{2}-\xi)\frac{v}{2}-\vartheta\lambda_{\text{OT}}-h_2(\delta+\alpha+\chi)(\frac{v}{2}-\vartheta\lambda_{\text{OT}})-\ell-q]} \\ &\quad + \sqrt{6} \exp(-\lambda_{\text{OT}}\delta^2/100) + 2 \exp(-\xi^2\lambda_{\text{OT}}/2), \end{aligned}$$

with $\ell = \lambda_{\text{PQS}}$.

Proof. The proof is similar to the one in Lemma 8.22. The sampling requires a large number of $\hat{x}_i$ and $\hat{\theta}_i$. On step 7 of Algorithm 5, Bob can choose subsets of length v of $\hat{\theta}_i$ and $\hat{x}_i$ to generate n_{OT} seeds $s_i \in \{0, 1\}^{\lambda_{\text{PQS}}}$. By selecting a large enough v , we avoid finite-set sampling errors, leading ξ and δ to be representative of the whole population as in Theorem 8.10, and thus:

$$H_{\min}(X_{\bar{b}}|A^b E) \geq \left(\frac{1}{2} - \xi\right) \frac{v}{2} - \vartheta \lambda_{OT} - h_2(\delta + \alpha + \chi) \left(\frac{v}{2} - \vartheta \lambda_{OT}\right). \quad (8.7.4)$$

In order to obtain n_{OT} 1-out-of-2 OTs, the first difference with respect to Algorithm 5 is that, in Step 7, Bob chooses the tuple of sets $(I_{b,j})_{j \in [n_{OT}]}$ and $(I_{\bar{b},j})_{j \in [n_{OT}]}$, such that $I_{b,j} = \{i : \theta_i = \hat{\theta}_i\}$, $I_{b,j} \cap I_{b,j'} = \emptyset$, $I_{\bar{b},j} = \{i : \theta_i \neq \hat{\theta}_i\}$ and $I_{\bar{b},j} \cap I_{\bar{b},j'} = \emptyset$. Then Bob sends $((I_0, I_1)_j)_{j \in [n_{OT}]}$ to Alice. The second difference is that, in Step 8, we define $\mathbf{x}_{0,j} = \{x_i : i \in I_{0,j}\}$ and $\mathbf{x}_{1,j} = \{x_i : i \in I_{1,j}\}$. Alice can then send $((m_0, m_1)_j)_{j \in [n_{OT}]}$ different encoded messages. For each encoded message, Alice chooses at random a pair $(I_0, I_1)_j$ over the entire tuple $((I_{0,j}, I_{1,j}))_{j \in [n_{OT}]}$ and shares it with Bob, who can decrypt m messages m_b by using the strings $(\hat{\mathbf{x}}_{b,j})_{j \in [n_{OT}]}$ with $\hat{\mathbf{x}}_{b,j} = \{\hat{x}_i : i \in I_{b,j}\}$. $\square$

DISCUSSION AND OPEN PROBLEMS

Discussion and Open Problems

The goal of this thesis is to develop a formalism that captures how computational constraints modify central concepts of quantum information theory and their operational consequences.

The first part of the thesis provides the basic framework for this point of view. In Chapter 3, we model the operations available to a computationally bounded observer by imposing polynomial bounds on the gate complexity of quantum channels and measurements. This leads to sets of efficiently implementable maps, POVMs, and effect operators, and allows us to use cone-theoretic tools to define computational analogues of the order structure underlying distinguishability. In particular, the resulting cones induce a computational partial order, a computational trace norm and trace distance, and the operator norms that define the entropic constructions.

In Chapter 4, this geometric framework is turned into computational versions of standard information-theoretic quantities. For single systems, the computational max-divergence arises from the cone-induced order, in the spirit of [RKW11, GC24], while polynomially generated binary POVMs lead to computational measured Rényi divergences, following the approach of [RSB25, MH23]. We show that these two constructions are compatible: in the limit $\alpha \rightarrow \infty$, the computational measured Rényi divergence coincides with the computational max-divergence (Theorem 4.13). This equivalence supports the consistency of the framework and illustrates the usefulness of cone theory in complexity-constrained quantum information. The same chapter also relates the resulting quantities to computational versions of fidelity and trace distance. In particular, we obtain a computational Pinsker inequality (Lemma 4.12) and a computational Fuchs–van de Graaf inequality (Lemma 4.23), showing that the usual relations between distinguishability measures survive after restricting the available measurements.

We then move to bipartite systems, where the cone of efficiently implementable Choi operators induces a different computational max-divergence and, from it, a computational conditional min-entropy. This entropy keeps the operational meaning of the information-theoretic conditional min-entropy as an optimal recovery fidelity [KRS09]: in the fully quantum setting, it quantifies the largest overlap with a maximally entangled state attainable by an efficient channel acting on the conditioning system (Theorem 4.36). For classical-quantum states, it reduces to the optimal probability of inferring a classical variable from quantum side information using efficient measurements (Theorem 4.37). When the conditioning system is trivial, it reduces to a single-system entropy determined by a computational analogue of the operator norm (Definition 4.40 and Corollary 4.42). This also unifies our proposed framework with previous computational min-entropy notions [AA26, AHRA25]. Finally, Section 4.2.4 shows that the single-system measured

max-divergence can be recovered from the bipartite construction through a suitable embedding, whereas Remark 4.38 explains why the underlying bipartite max-divergence itself should not be expected to reduce to the usual information-theoretic one when computational constraints are removed.

Overall, the proposed framework brings a wide range of computational quantum information measures within a common perspective. It suggests that complexity-constrained quantum information should not be viewed merely as a collection of restricted tasks, but as a modification of the core mathematical formalism from which information-theoretic quantities arise. The next step is to understand this structure systematically. Several notions have now appeared in this direction, including computational divergences, entropies, entanglement measures, and relative-entropy quantities discussed throughout the thesis. Nevertheless, their precise relationships remain only partially understood. A central question is therefore which familiar notions of quantum information survive under computational restrictions and, when they do not, how their computational counterparts should be formalized.

This structural programme becomes concrete through several technical questions. One starting point would be to formalize a computational mutual information as a notion of correlations accessible to efficient observers. A second direction is to compare the orders induced by different classes of admissible operations. A third direction is to extend the approach to more general Rényi-divergences [Tom16]. This is interesting not only as a formal generalization, but because different Rényi parameters control different operational regimes, such as one-shot bounds, strong-converse exponents, and regularized rates [ANSV08, MH23]. In the regularized setting, for binary measurements super-additivity fails for general $\alpha \neq \infty$ [MH23]; it remains open whether the regularized computational measured relative entropy can nevertheless be expressed through a supremum similar to the one defining the computational max-divergence. A further issue concerns smoothing. Computational notions of smoothing interact naturally with cryptographic objects, such as pseudorandom generators [CCL⁺17, MRR⁺25], while information-theoretic smoothing yields useful chain rules [AA26, AHRA25]. Finding a formulation that combines both features would give robust one-shot quantities, stable under computational indistinguishability and applicable to tasks such as randomness extraction, privacy amplification, compression, and resource conversion under efficient operations.

After establishing the mathematical framework and its operational meaning, we study how different the computational and information-theoretic regimes can be. In Chapter 5, we first consider the single-system setting and show that, assuming the existence of EFI pairs, explicit separations appear for the computational trace distance, computational fidelity, and measured Rényi divergences introduced in our framework (Section 5.1). We then turn to bipartite systems, where computational complexity can reshape the scaling of operationally accessible quantum correlations. This is captured by large gaps between $\tilde{H}_{\min}(A|B)$ and $H_{\min}(A|B)$, both in the unconstrained and in the efficiently generated settings (Sections 5.2 and 5.3). The distinction already appears for pure states, where one-shot processing of polynomially many copies leaves only a logarithmic amount of entanglement within efficient reach in the worst case. For mixed states, the effect is sharper:

we construct ensembles with highly negative information-theoretic conditional min-entropy but a complexity-constrained conditional min-entropy that is, up to negligible corrections, as large as that of an uncorrelated maximally mixed state. In this regime, computational complexity can render an extensive amount of entanglement or quantum side information inaccessible for any efficient observer.

An important open problem is to understand how explicit and physically natural this phenomenon can be made. Our maximal mixed-state separation is obtained from the generalized Hilbert–Schmidt ensemble [HLW06, CN15, Hal98, ZPNC11] in the unconstrained case, while the efficiently generated version relies on scalable pseudorandom states instantiated from subexponential LWE. A natural next step is to replace these constructions by families that arise in many-body physics, where the relevant observer is usually restricted to local measurements, low-depth dynamics, or polynomial-time post-processing. Possible candidates include low- or intermediate-temperature Gibbs states of interacting local Hamiltonians [Alh23, KAA21], states generated by shallow local dynamics that are pseudorandom to efficient observers [BHHP25], and other physically motivated pseudorandom or pseudoentangled many-body constructions. Resolving these questions would show whether almost-maximal separations between information-theoretic and operationally accessible side information are exceptional consequences of randomness, or a robust feature of complexity-constrained quantum many-body systems. It would also connect the formalism to thermodynamics, where the relevant question is often not what work or resource can be extracted in principle, but what can be extracted by an agent with efficient control [MKH⁺25, GJB⁺18].

In Chapter 6, the measurement-based quantities are applied in two directions. First, in asymmetric binary hypothesis testing, the regularized computational measured relative entropy has an operational meaning as an upper bound on the Stein exponent under computationally restricted measurements (Section 6.1 and theorem 6.2), directly paralleling the information-theoretic setting. Second, in resource theories, these divergences define computational resource measures, and the asymptotic continuity bound for the computational measured relative entropy (Section 6.2 and Theorem 6.7) implies that computationally indistinguishable families of states also have computationally indistinguishable resource contents. Specializing to entanglement, this lets us compare the induced computational measured relative entropy of entanglement $\hat{E}_R$ with the computational distillable entanglement and entanglement cost of [ABV23]: under efficient LOCC, up to a continuity term, one has $\hat{E}_D(\rho_{AB}^n) \leq \hat{E}_R(\rho_{AB}^n) \leq \hat{E}_C(\rho_{AB}^n)$, thereby reproducing the information-theoretic hierarchy.

On the technical side, a key open problem is to establish nontrivial lower bounds on the computational Stein exponent. Such bounds would turn the converse result of Theorem 6.2 into a sharper operational characterization of hypothesis testing under efficient measurements. From the resource-theoretic viewpoint, it would be valuable to clarify the role of our computational divergences in one-shot formation and distillation tasks under efficient free operations. Pursuing this direction, including smoothed variants of the computational max-divergence and efficient analogues of non-entangling maps, may

illuminate further connections between the measures introduced here and prior work in computational entanglement theory [ABV23, LREJ25].

The last chapter of this part studies the connection between complexity-constrained quantum resources and cryptography in the opposite direction. In Chapter 7, we show how quantum pseudoresources in general and pseudoentanglement in particular can lead to relevant quantum cryptographic primitives. The key step is the construction of EPFI pairs from pseudoresources: the resource gap gives pairwise statistical farness, while computational indistinguishability is inherited from the pseudoresource families, leading to canonical quantum commitments and the primitives that follow from them (Sections 7.1 to 7.3). This raises a broader question about the minimal assumptions behind quantum cryptography. According to [Kre21], $PP \neq BQP$ is necessary for the existence of PRS; however, such a condition is not known for EFI pairs, and therefore the primitives obtained from EPFI pairs and pseudoresources might be weaker. Understanding whether EPFI pairs, EFI pairs, PRS, and pseudoresources require the same complexity-theoretic assumptions would clarify whether MiniQCrypt has one underlying minimal world or several genuinely different ones. Beyond this complexity-theoretic question, one would also like to know which quantum resources are really responsible for cryptographic power. The relation between pseudomagic and pseudoentanglement has been studied for pure states [GOL25], but remains largely unexplored for mixed states. While the construction in [BMB⁺25] simultaneously exhibits pseudoentanglement and pseudomagic, it is unclear whether the two phenomena are independent in general. Constructing mixed pseudoentangled states without magic, or proving that this is impossible under natural assumptions, would clarify whether computational indistinguishability can hide entanglement independently of other resources.

Another direction is to make the cryptographic use of hidden resources more concrete. The computationally locked entanglement functionality introduced in Chapter 7 suggests that entanglement can be efficiently unlocked only with the appropriate key. Understanding whether such a functionality can be built from weaker assumptions, and whether it can support tasks such as authenticated teleportation or certified routing in quantum networks, would help turn pseudoresource phenomena into operational cryptographic primitives.

The last part of the thesis turns to a more concrete cryptographic application. In Chapter 8, we construct a practical quantum oblivious transfer protocol from one-way functions in the plain model. The protocol follows the structure of [BCKM21], but replaces full extractability by relaxed extractability, which is enough for simulation-based security and substantially reduces the quantum resources needed. This is formalized through the equivocal and relaxed-extractable commitment of Section 8.6 and the QOT protocol of Section 8.7, Protocol 8.1. The security analysis also incorporates experimental imperfections, such as bit-flip errors and multiphoton leakage, leading to explicit bounds against malicious receivers (Lemma 8.22 and theorem 8.23) and to the possibility of distilling several OT keys from one execution (Corollary 8.25). The resulting benchmark in Table 8.1 and section A.5.2 shows that the quantum communication cost is reduced from the order of 10^{13} BB84 states

in the previous plain-model construction to around 10^7 in the parameter regime considered here.

Several questions remain open in this direction. At the protocol level, a natural first step is to study analogous QOT constructions in the random-oracle or quantum random-oracle models, such as [ABKK23], while incorporating the same practical experimental errors considered here. This would allow for a fair comparison not only in terms of communication complexity and quantum resources, but also in terms of post-processing running time and robustness to realistic noise. From the composability point of view, the next step is to compile the proposed QOT functionality into quantum-secure MPC protocols [IPS08, Kil88, DFL⁺09], keeping track of the concrete parameters throughout the compilation. This is the point at which the resource savings of the protocol should be tested against the overheads of a full cryptographic application.

A more conceptual direction is to connect this near-term protocol back to the pseudoresource perspective. Commitments can be obtained from weaker assumptions such as pseudoresources or EPFI pairs (Theorems 7.7 and 7.10), but it remains unclear how to implement these functionalities in practice, or how to instantiate them directly from genuinely quantum primitives without relying on classical cryptographic assumptions such as one-way functions. Understanding this connection would bring together the two cryptographic directions of the thesis: near-term protocols based on minimal classical assumptions, and cryptography whose security comes from computationally hidden quantum resources.

Appendix A

Appendix Material

A.1. Proofs of Chapter 3

A.1.1. Proof of Corollary 3.26

Recall that we want to effectively place a bound on the diamond norm between measurement maps corresponding to binary POVMs, given the operator norm difference between their effect operators.

Proof of Corollary 3.26. By Proposition 3.24 there exists a binary measurement M' such that

$M = (E_1, E_2)$ and $M' = (F_1, F_2) = (\frac{1}{k} \sum_{i=1}^k E_{j_i}, \mathbb{1} - \frac{1}{k} \sum_{i=1}^k E_{j_i})$ satisfy

$$\|E_1 - F_1\|_\infty = \|E_2 - F_2\|_\infty \leq \varepsilon . \quad (\text{A.1.1})$$

Recall that the measurement maps they induce are given by

$$\mathcal{M}(\rho) = \sum_{i=0}^1 \text{Tr}[E_{i+1}\rho] |i\rangle\langle i| , \quad \mathcal{M}'(\rho) = \sum_{i=0}^1 \text{Tr}[F_{i+1}\rho] |i\rangle\langle i| ,$$

for any state $\rho \in \mathcal{S}(\mathcal{H}_n)$ (and some choice of n). Alternatively, for any state $\rho \in \mathcal{S}(\mathcal{H}_n \otimes \mathcal{H}')$ on the extended Hilbert space $\mathcal{H}'$, $\mathcal{M} \otimes \text{Id}$ and $\mathcal{M}' \otimes \text{Id}$ are mappings of the form

$$\mathcal{M} \otimes \text{Id}(\rho) = \sum_{i=0}^1 |i\rangle\langle i| \otimes \text{tr}_{\mathcal{H}_n}[E_{i+1}\rho] : \mathcal{S}(\mathcal{H}_n \otimes \mathcal{H}') \rightarrow \mathcal{S}(\mathbb{C}^2 \otimes \mathcal{H}') , \quad (\text{A.1.2})$$

where $\mathcal{M}' \otimes \text{Id}$ is analogous and tr stands for the partial trace. Hence, for these measure-

ments, it must hold that

$$\begin{aligned}
\|\mathcal{M} - \mathcal{M}'\|_{\diamond} &= \sup_{\rho \in \mathcal{S}(\mathcal{H}_n \otimes \mathcal{H}')} \|((\mathcal{M} - \mathcal{M}') \otimes \text{Id})(\rho)\|_1 \\
&= \sup_{\rho} \left\| \sum_{i=0}^1 |i\rangle\langle i| \otimes \text{tr}_{\mathcal{H}_n}[(E_{i+1} - F_{i+1})\rho] \right\|_1 \\
&= \sup_{\rho} \sum_{i=0}^1 \|\text{tr}_{\mathcal{H}_n}[(E_{i+1} - F_{i+1})\rho]\|_1 \\
&\leq \sup_{\rho} \sum_{i=0}^1 \|(E_{i+1} - F_{i+1})\rho\|_1 \\
&\leq \sup_{\rho} \sum_{i=0}^1 \|E_{i+1} - F_{i+1}\|_{\infty} \|\rho\|_1 \\
&\leq 2\varepsilon .
\end{aligned}$$

The first inequality follows from the fact that the partial trace is trace-norm contractive, and the second is an application of Hölder's inequality. The last inequality follows then from Eq. (A.1.1). $\square$

A.1.2. Proof of Lemma 3.35

Recall that we want to prove the following three equivalences

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} = \begin{cases} 2 \max_{E \in \mathbf{E}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] & (i) \\ \max_{M=(E_1, E_2) \in \mathbf{M}_n^{\text{eff}}} \sum_{i=1}^2 |\text{Tr}[E_i(\rho - \sigma)]| & (ii) \\ \max_{M \in \mathbf{M}_n^{\text{eff}}} \|\mathcal{M}_M(\rho) - \mathcal{M}_M(\sigma)\|_1 . & (iii) \end{cases} \quad (\text{A.1.3})$$

Let us prove them separately.

Proof of first equality (i). By definition of

$$\|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} = 2 \max_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] \geq 2 \max_{E \in \mathbf{E}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] ,$$

we get the first inequality. The other inequality follows since the functional $E \mapsto \text{Tr}[E(\rho - \sigma)]$ is linear and thus convex and so

$$\begin{aligned}
2 \max_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] &= 2 \max_{E \in \text{extr}(\overline{\mathbf{E}}_n^{\text{eff}})} \text{Tr}[E(\rho - \sigma)] \\
&\leq 2 \max_{E \in \mathbf{E}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] .
\end{aligned} \quad (\text{A.1.4})$$

Combining both inequalities proves the desired result. $\square$

Proof of second equality (ii). By the above proven first equivalence, it holds that

$$\begin{aligned}
& 2 \max_{E \in \mathbf{E}_n^{\text{eff}}} \text{Tr} [E (\rho - \sigma)] \\
&= \max_{E \in \mathbf{E}_n^{\text{eff}}} |\text{Tr} [E (\rho - \sigma)]| + |\text{Tr} [-E (\rho - \sigma)]| \\
&= \max_{E \in \mathbf{E}_n^{\text{eff}}} |\text{Tr} [E (\rho - \sigma)]| + |\text{Tr} [(1 - E) (\rho - \sigma)]| \\
&= \max_{M=(E_1, E_2) \in \mathbf{M}_n^{\text{eff}}} |\text{Tr} [E_1 (\rho - \sigma)]| + |\text{Tr} [E_2 (\rho - \sigma)]| \\
&= \max_{M=(E_1, E_2) \in \mathbf{M}_n^{\text{eff}}} \sum_{i=1}^2 |\text{Tr} [E_i (\rho - \sigma)]| .
\end{aligned}$$

□

Proof of third equality (iii). For any binary measurement $M = (E_1, E_2) \in \mathbf{M}_n^{\text{eff}}$, it further holds that

$$\begin{aligned}
& \sum_{i=1}^2 |\text{Tr} [E_i (\rho - \sigma)]| \\
&= \left\| \sum_{i=0}^1 \text{Tr} [E_{i+1} (\rho - \sigma)] |i\rangle\langle i| \right\|_1 \\
&= \left\| \sum_{i=0}^1 \text{Tr} [E_{i+1} \rho] |i\rangle\langle i| - \sum_{i=0}^1 \text{Tr} [E_{i+1} \sigma] |i\rangle\langle i| \right\|_1 \\
&= \|\mathcal{M}_M(\rho) - \mathcal{M}_M(\sigma)\|_1 ,
\end{aligned}$$

Optimizing over $M = (E_1, E_2) \in \mathbf{M}_n^{\text{eff}}$ then yields (iii). □

A.1.3. Proof of Lemma 3.38

Recall that we want to prove that $\|\cdot\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}}$ is sub-additive, i.e.

$$\|\rho^{\otimes m} - \sigma^{\otimes m}\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} \leq m \max_{i \in \{1, \dots, m\}} \|\rho_i - \sigma_i\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} ,$$

where $\rho_i = \rho^{\otimes(i-1)} \otimes \rho \otimes \sigma^{\otimes(m-i)}$, $\sigma_i = \rho^{\otimes(i-1)} \otimes \sigma \otimes \sigma^{\otimes(m-i)}$.

Proof of Lemma 3.38. Note that $\rho_{i-1} = \sigma_i$ for $i = 2, \dots, m$, and $\rho_m = \rho^{\otimes m}$, $\sigma_1 = \sigma^{\otimes m}$. As such

$$\begin{aligned}
\rho^{\otimes m} - \sigma^{\otimes m} &= \rho_m - \sigma_1 = \sum_{j=1}^m \rho_j - \sum_{j=1}^{m-1} \rho_j - \sigma_1 \\
&= \sum_{j=1}^m (\rho_j - \sigma_j) .
\end{aligned}$$

Using this and the triangle inequality repeatedly now yields the claim

$$\begin{aligned}
\|\rho^{\otimes m} - \sigma^{\otimes m}\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} &\leq \sum_{j=1}^m \|\rho_j - \sigma_j\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} \\
&\leq \sum_{j=1}^m \max_{i \in \{1, \dots, m\}} \|\rho_i - \sigma_i\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} \\
&\leq m \max_{i \in \{1, \dots, m\}} \|\rho_i - \sigma_i\|_{\overline{\mathbf{M}}_{n \cdot m}^{\text{eff}}} .
\end{aligned}$$

□

A.2. Proofs of Section 4.1

A.2.1. Proof of Lemma 4.7

Recall, we want to prove that

$$\dot{D}_{\max}(\rho \|\sigma) \geq \log \left(1 + \dot{\Delta}(\rho, \sigma) \right) .$$

Proof of Lemma 4.7. If $\dot{D}_{\max}(\rho \|\sigma) = \infty$, then this trivially holds. Assume that $\dot{D}_{\max}(\rho \|\sigma) < \infty$. Hence there exists some λ such that $\rho \leq_{\mathcal{C}_n^{\text{S}_{\text{eff}}}} \lambda \sigma$. Then, for any such λ , we have

$$\begin{aligned}
\frac{1}{2} \|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} &= \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E(\rho - \sigma)] \\
&\leq \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E(\lambda \sigma - \sigma)] \\
&= (\lambda - 1) \sup_{E \in \overline{\mathbf{E}}_n^{\text{eff}}} \text{Tr}[E\sigma] \\
&\leq \lambda - 1 ,
\end{aligned}$$

where for the inequality we are using that the partial order $\leq_{\mathcal{C}_n^{\text{S}_{\text{eff}}}}$ is the one induced by the dual cone to $\mathcal{C}_n^{\mathbf{E}^{\text{eff}}}$ to which $\overline{\mathbf{E}}_n^{\text{eff}}$ contains a base. Rearranging gives, $\lambda \geq 1 + \frac{1}{2} \|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}}$ and taking the infimum over all such suitable λ yields the statement of the lemma,

$$\begin{aligned}
\dot{D}_{\max}(\rho \|\sigma) &:= \log \inf \{ \lambda \in \mathbb{R} \mid \rho \leq_{\mathcal{C}_n^{\text{S}_{\text{eff}}}} \lambda \sigma \} \\
&\geq \log \left(1 + \frac{1}{2} \|\rho - \sigma\|_{\overline{\mathbf{M}}_n^{\text{eff}}} \right) \\
&= \log \left(1 + \dot{\Delta}(\rho, \sigma) \right) .
\end{aligned}$$

□

A.2.2. Proof of Lemma 4.11

Proof of Lemma 4.11. The properties of the computational measured Renyi divergences follow directly from [RSB25, Lemmas 2 & 5] and [Tom16, Chapter 4]. We will use $\mathbb{D}_\alpha(\mu\|\nu)$ to denote a classical Renyi divergence over probability distributions μ, ν . Recall that, when $\mu_\rho^M(i) = \text{Tr}[\rho E_i]$ and $\nu_\sigma^M(i) = \text{Tr}[\sigma E_i]$, $\check{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) = \sup_{M \in \mathbf{M}_n^{\text{eff}}} \mathbb{D}_\alpha(\mu_\rho^M\|\nu_\sigma^M)$.

(1) Follows from the positivity of the classical Renyi divergence.

(2) For every pair of probabilities distributions μ, ν , we have that for all α, β such that $0 < \alpha \leq \beta$,

$$\mathbb{D}_\alpha(\mu\|\nu) \leq \mathbb{D}_\beta(\mu\|\nu) .$$

Therefore, by taking the supremum, it directly follows that $\check{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) \leq \check{D}_\beta^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma)$.

(3) By linearity of the trace, $\mu_{\lambda\rho_1 + (1-\lambda)\rho_2}^M = \lambda\mu_{\rho_1}^M + (1-\lambda)\mu_{\rho_2}^M$. Let us first consider the case $\alpha \in (0, 1)$. Therefore,

$$\begin{aligned} & \check{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\lambda\rho_1 + (1-\lambda)\rho_2\|\lambda\sigma_1 + (1-\lambda)\sigma_2) \\ &= \inf_{M \in \mathbf{M}_n^{\text{eff}}} \mathbb{Q}_\alpha(\lambda\mu_{\rho_1}^M + (1-\lambda)\mu_{\rho_2}^M\|\lambda\nu_{\sigma_1}^M + (1-\lambda)\nu_{\sigma_2}^M) \\ &\geq \inf_{M \in \mathbf{M}_n^{\text{eff}}} \lambda\mathbb{Q}_\alpha(\mu_{\rho_1}^M\|\mu_{\rho_2}^M) + (1-\lambda)\mathbb{Q}_\alpha(\nu_{\sigma_1}^M\|\nu_{\sigma_2}^M) \\ &\geq \lambda\check{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho_1\|\rho_2) + (1-\lambda)\check{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\sigma_1\|\sigma_2) , \end{aligned}$$

where $\check{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\cdot\|\cdot)$ and $\mathbb{Q}_\alpha(\cdot\|\cdot)$ are the computational and classical counterparts of Eq. (2.5.2). The first inequality follows from joint concavity of the classical measure and the last one from the superadditivity of the infimum, proving the joint concavity of the function. Let us also set $f(t) := \frac{1}{\alpha-1} \log(t)$ (see Definition 2.15). In the case of $\alpha \in (0, 1)$, $f(t)$ is a non-increasing convex function. Therefore, it follows that $\check{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}$ is jointly convex in (ρ, σ) for $\alpha \in (0, 1)$. $\check{D}_1^{\mathbf{M}_n^{\text{eff}}}$ is jointly convex as the supremum of a family of jointly convex functionals (see [RSB25, Prop. 3, Lemma 2]). For $\alpha \in (1, \infty)$, $f(t)$ is non-decreasing and quasi-convex. Moreover, by using the fact of $\mathbb{Q}_\alpha$ being jointly convex and the supremum being superadditive, $\check{Q}_\alpha^{\mathbf{M}_n^{\text{eff}}}$ is jointly convex. Therefore, $\check{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}$ is jointly quasi-convex. Lastly, the joint quasi-convexity of $\check{D}_\infty^{\mathbf{M}_n^{\text{eff}}}$ follows by taking the supremum of a family of jointly quasi-convex functions ([RSB25, Prop. 3, Lemma 2]).

(4) By the duality of the quantum channels, $\mu_{\mathcal{E}(\rho)}^M = \mu_\rho^{\mathcal{E}^*(M)}$. Since for every admissible $\mathcal{E}$, it follows that $\mathcal{E}^*(M) \in \overline{\mathbf{M}}_n^{\text{eff}}$, then,

$$\begin{aligned} \check{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\mathcal{E}(\rho)\|\mathcal{E}(\sigma)) &= \sup_{M \in \mathbf{M}_n^{\text{eff}}} \mathbb{D}_\alpha(\mu_{\mathcal{E}(\rho)}^M\|\nu_{\mathcal{E}(\sigma)}^M) \\ &= \sup_{M \in \mathbf{M}_n^{\text{eff}}} \mathbb{D}_\alpha(\mu_\rho^{\mathcal{E}^*(M)}\|\nu_\sigma^{\mathcal{E}^*(M)}) \\ &\leq \check{D}_\alpha^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) . \end{aligned}$$

□

A.2.3. Proof of Lemma 4.22

Recall that we want to show that

$$\overset{c}{D}_{1/2}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) = -\log \overset{c}{F}(\rho, \sigma) .$$

Proof of Lemma 4.22. By definition of the computational measured Renyi divergence we have that for $\alpha = 1/2$,

$$\overset{c}{D}_{1/2}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) = -2 \log \overset{c}{Q}_{1/2}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) . \quad (\text{A.2.1})$$

Then, by Eq. (4.1.15),

$$\begin{aligned} \overset{c}{Q}_{1/2}^{\mathbf{M}_n^{\text{eff}}}(\rho\|\sigma) &= \inf_{M \in \mathbf{M}_n^{\text{eff}}} Q_{1/2}(\mathcal{M}_M(\rho)\|\mathcal{M}_M(\sigma)) \\ &= \inf_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} Q_{1/2}(\mathcal{M}_M(\rho)\|\mathcal{M}_M(\sigma)) \\ &= \inf_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} \|\mathcal{M}_M(\sigma)^{1/2} \mathcal{M}_M(\rho) \mathcal{M}_M(\sigma)^{1/2}\|_{1/2}^{1/2} \\ &= \inf_{(E_1, E_2) \in \overline{\mathbf{M}}_n^{\text{eff}}} \sum_{i=1}^2 \sqrt{\text{Tr}[E_i \rho] \text{Tr}[E_i \sigma]} \\ &= \sqrt{\overset{c}{F}(\rho, \sigma)} , \end{aligned}$$

where the second equality follows from Lemma 4.10. Together with Eq. (A.2.1), this yields the claim. $\square$

A.2.4. Proof of Lemma 4.23

Recall that we want to prove here a computational Fuchs-van de Graaf inequality

$$1 - \sqrt{\overset{c}{F}(\rho, \sigma)} \leq \overset{c}{\Delta}(\rho, \sigma) \leq \sqrt{1 - \overset{c}{F}(\rho, \sigma)} .$$

Proof of Lemma 4.23. We note that it suffices to prove that

$$\begin{aligned} 1 - \sqrt{\min_{M \in \mathbf{M}_n^{\text{eff}}} F(\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma))} &\leq \max_{M \in \mathbf{M}_n^{\text{eff}}} \Delta(\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma)) \\ &\leq \sqrt{1 - \min_{M \in \mathbf{M}_n^{\text{eff}}} F(\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma))} , \end{aligned}$$

since it directly implies our bounds. Starting with the usual information-theoretic Fuchs-van de Graaf inequality [FvdG99] applied to the states $\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma)$ for any $M \in \mathbf{M}_n^{\text{eff}}$, we have

$$1 - \sqrt{F(\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma))} \leq \Delta(\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma)) \leq \sqrt{1 - F(\mathcal{M}_M(\rho), \mathcal{M}_M(\sigma))} .$$

Now optimizing over all $M \in \mathbf{M}_n^{\text{eff}}$ yields the desired bound. $\square$

A.3. Proofs of Chapter 5

A.3.1. Proof of Theorem 5.2

Proof. Write $\rho_{AB} := |\psi_{AB}\rangle\langle\psi_{AB}|^{\otimes k}$. By Lemma 5.1, the ideal Hayashi–Schur protocol induces a CPTP map T acting only on B such that

$$\left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes T)(\rho_{AB}) \right| \Omega_{A_k A'_k} \right\rangle = \left\langle \Omega_{A_k A'_k} \left| \Lambda(\rho_{AB}) \right| \Omega_{A_k A'_k} \right\rangle. \quad (\text{A.3.1})$$

This map, which corresponds to the protocol of [MH07, LREJ25], produces $\log \dim \mathcal{V}_\lambda$ ebits conditioned on measuring the Schur label λ , with a probability distribution $\Pr(\lambda) = \text{tr}[\Pi_\lambda^A \rho_A^{\otimes k}]$. As proven in [LREJ25, Theorem 18], one has that it achieves

$$\Pr \left(\lambda : \log \dim \mathcal{V}_\lambda \geq \frac{k}{4} \min\{H_{\min}(A)_\rho, \log k\} \right) \geq \frac{2}{3}. \quad (\text{A.3.2})$$

For compactness, set

$$L := \frac{k}{4} \min\{H_{\min}(A)_\rho, \log k\}.$$

Moreover,

$$\begin{aligned} \left\langle \Omega_{A_k A'_k} \left| \Lambda(\rho_{AB}) \right| \Omega_{A_k A'_k} \right\rangle &= \mathbb{E}_\lambda \left[\frac{\dim \mathcal{V}_\lambda}{d_{A_k}} \right] \\ &\geq \frac{2}{3} \cdot \frac{2^L}{d_{A_k}}. \end{aligned}$$

where the first line is the direct computation of the overlap between the global maximally entangled state of dimension d_{A_k} and the distribution of maximally entangled states supported on a $\dim \mathcal{V}_\lambda$ -dimensional subspace. The last inequality follows by lower bounding the expected value using Eq. (A.3.2).

Now let $\tilde{T}_\delta$ be an efficient implementation of T satisfying

$$\|T - \tilde{T}_\delta\|_\diamond \leq \delta.$$

By Theorem 2.46 and the discussion after Lemma 5.1, such an implementation can be chosen with running time $O(\text{poly}(k, n, \log(1/\delta)))$. Moreover,

$$\begin{aligned} &\left| \left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes T)(\rho_{AB}) \right| \Omega_{A_k A'_k} \right\rangle - \left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes \tilde{T}_\delta)(\rho_{AB}) \right| \Omega_{A_k A'_k} \right\rangle \right| \\ &\leq \left\| (\text{Id}_{A_k} \otimes (T - \tilde{T}_\delta))(\rho_{AB}) \right\|_1 \leq \delta. \end{aligned}$$

Choose

$$\delta = \frac{2^{-n}}{d_{A_k}}.$$

Then

$$\log(1/\delta) = \log d_{A_k} + n = O(kn_A) + n = \text{poly}(n),$$

so $\tilde{T}_\delta$ is efficiently implementable. Hence

$$\left\langle \Omega_{A_k A'_k} \middle| (\text{Id}_{A_k} \otimes \tilde{T}_\delta)(\rho_{AB}) \middle| \Omega_{A_k A'_k} \right\rangle \geq \frac{2}{3} \cdot \frac{2^L}{d_{A_k}} - \frac{2^{-n}}{d_{A_k}}.$$

Finally, by the definition of $\tilde{H}_{\min}$,

$$\begin{aligned} \tilde{H}_{\min}(A^k|B^k)_{|\psi\rangle^{\otimes k}} &= -\log \left(d_{A_k} \cdot \max_T \left\langle \Omega_{A_k A'_k} \middle| (\text{Id}_{A_k} \otimes T)(\rho_{AB}) \middle| \Omega_{A_k A'_k} \right\rangle \right) \\ &\leq -\log \left(d_{A_k} \cdot \left\langle \Omega_{A_k A'_k} \middle| (\text{Id}_{A_k} \otimes \tilde{T}_\delta)(\rho_{AB}) \middle| \Omega_{A_k A'_k} \right\rangle \right) \\ &\leq -\log \left(\frac{2}{3} 2^L - 2^{-n} \right). \end{aligned}$$

Rewriting the last term gives

$$-\log \left(\frac{2}{3} 2^L - 2^{-n} \right) = -L + \log \frac{3}{2} - \log \left(1 - \frac{3 \cdot 2^{-n}}{2^{L+1}} \right).$$

Since $L \geq 0$, the ratio $\frac{3 \cdot 2^{-n}}{2^{L+1}}$ is negligible in n , and therefore so is the last term. Thus

$$\begin{aligned} \tilde{H}_{\min}(A^k|B^k)_{|\psi\rangle^{\otimes k}} &\leq -L + \log \frac{3}{2} + \text{negl}(n) \\ &= -\frac{k}{4} \min\{H_{\min}(A)_\rho, \log k\} + \log \frac{3}{2} + \text{negl}(n). \end{aligned}$$

□

A.3.2. Proof of Lemma 5.3

Proof. Since for pure states $H_{\min}(A|B)_{|\psi\rangle} = -H_{\max}(A)_\psi$, (c.f. [KRS09]), the conditional min-entropy for a pure state is given by

$$H_{\min}(A|B)_{|\psi\rangle} = -2 \log (\text{Tr} \sqrt{\rho_A}) ,$$

where $\rho_A = \text{tr}_B(|\psi_{AB}\rangle\langle\psi_{AB}|)$. Let us first prove the Haar-random case. Given $|\psi_{AB}\rangle \sim \text{Haar}(n)$, [ZPNC11, Section III.D] prove that the reduced state $\rho_A = \text{tr}_B(|\psi_{AB}\rangle\langle\psi_{AB}|)$ can be represented in the following way:

$$\rho_A = \frac{GG^\dagger}{\text{Tr}(GG^\dagger)}, \quad (\text{A.3.3})$$

where $G \in \mathcal{M}_{d_A \times d_B}(\mathbb{C})$ is an element of the Ginibre ensemble, i.e., a random matrix with independently and identically distributed complex Gaussian entries [Gin65].

Then, let us express the functional $\text{Tr} \sqrt{\rho_A}$ in terms of the rescaled empirical eigenvalue distribution of ρ_A . Writing $\{\lambda_i\}_{i=1}^{d_A}$ for the eigenvalues of ρ_A and setting

$$x_i := d_A \lambda_i(\rho_A), \quad \mu_M := \frac{1}{d_A} \sum_{i=1}^{d_A} \delta_{x_i},$$

we obtain

$$\begin{aligned} \text{Tr} \sqrt{\rho_A} &= \sum_{i=1}^{d_A} \sqrt{\lambda_i(\rho_A)} \\ &= \sqrt{d_A} \frac{1}{d_A} \sum_{i=1}^{d_A} \sqrt{x_i} \\ &= \sqrt{d_A} \int \sqrt{x} d\mu_M(x). \end{aligned}$$

Then, as proven by [Nec07, Theorem 5], the empirical spectral distribution given by μ_M converges weakly almost surely to the Marchenko-Pastur distribution μ_{MP}

$$\mu_{MP} = \max\{1 - c, 0\} \delta_0 + \frac{\sqrt{(x - a)(b - x)}}{2\pi x} \mathbf{1}_{[a, b]}(x) dx, \quad (\text{A.3.4})$$

where $a = (\sqrt{c} - 1)^2$ and $b = (\sqrt{c} + 1)^2$ and $c \in (0, \infty)$. More specifically, by taking $c = \lim_{n \rightarrow \infty} d_B/d_A = 1$, the empirical distribution given by

$$\tilde{\mu}_M := \frac{1}{d_A} \sum_{i=1}^{d_A} \delta_{y_i},$$

with $y_i = c \cdot x_i$ weakly converges to

$$\mu_{MP} = \frac{1}{2\pi} \sqrt{\frac{4 - x}{x}} \mathbf{1}_{[0, 4]}(x) dx.$$

Since weak convergence implies convergence of integrals only against bounded continuous test functions (Portmanteau theorem, e.g. [Bil99, Theorem 2.1]), it cannot be applied directly to $x \mapsto \sqrt{x}$, which is unbounded on $[0, \infty)$. We therefore first use [Nec07, Theorem 6] to control the largest eigenvalue, and hence to obtain an almost-sure bound on the support of μ_M . In particular,

$$\max_i x_i = d_A \lambda_{\max}(\rho_A) \xrightarrow{d_A \rightarrow \infty} 4,$$

since $c = 1$. Hence, for every $\varepsilon > 0$, almost surely there exists $d_0 \in \mathbb{N}$ such that for all $d_A \geq d_0$,

$$\text{supp}(\mu_M) \subset [0, 4 + \varepsilon].$$

Let g_ε be a continuous, bounded real function such that

$$g_\varepsilon(x) = \sqrt{x} \quad \text{for all } x \in [0, 4 + \varepsilon].$$

Then, for all $d_A \geq d_0$,

$$\int \sqrt{x} d\mu_M(x) = \int g_\varepsilon(x) d\mu_M(x).$$

By weak convergence of μ_M to μ_{MP} , we obtain

$$\int g_\varepsilon(x) d\mu_M(x) \xrightarrow[d_A \rightarrow \infty]{a.s.} \int g_\varepsilon(x) d\mu_{MP}(x).$$

Since $\text{supp}(\mu_{MP}) = [0, 4]$, we also have

$$\int g_\varepsilon(x) d\mu_{MP}(x) = \int \sqrt{x} d\mu_{MP}(x).$$

Therefore,

$$\int \sqrt{x} d\mu_M(x) \xrightarrow[d_A \rightarrow \infty]{} \int \sqrt{x} d\mu_{MP}(x) = \frac{1}{2\pi} \int_0^4 \sqrt{4-x} dx = \frac{8}{3\pi}.$$

Therefore, almost surely it holds that,

$$\text{Tr} \sqrt{\rho_A} = \left(\frac{8}{3\pi} + o(1) \right) \sqrt{d_A},$$

and hence,

$$\begin{aligned} H_{\min}(A|B)_{|\psi\rangle} &= -\log d_A - 2 \log \left(\frac{8}{3\pi} \right) + o(1) \\ &= -n_A + O(1), \end{aligned}$$

almost surely, since $d_A = 2^{n_A}$. Lastly, since in the informational case the conditional min-entropy is additive, one has that

$$H_{\min}(A^k|B^k)_{|\psi\rangle^{\otimes k}} = k \cdot H_{\min}(A|B)_{|\psi\rangle}.$$

□

A.3.3. Proof of Theorem 5.5

Proof. Assume $d_A = d_B$, where A_k denotes the space $A^{\otimes k}$ and B_k denotes $B^{\otimes k}$, such that $d_{A_k} = d_A^k = 2^{k \cdot n_A}$ and $d = d_A \cdot d_B = d_A^2$. Then, $T : \mathcal{B}(\mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'_k})$ acts on k copies. Given the definition of the computational conditional min-entropy

$$\begin{aligned} \tilde{H}_{\min}(A^k|B^k)_{|\psi\rangle^{\otimes k}} &= -\log \left(d_A^k \cdot \max_T \left\langle \Omega_{A_k A'_k} \middle| (\text{Id}_{A_k} \otimes T)(|\psi\rangle\langle\psi|^{\otimes k}) \middle| \Omega_{A_k A'_k} \right\rangle \right) \\ &= -\log \left(d_A^k \cdot \max_T F_T(|\psi\rangle^{\otimes k}) \right), \end{aligned}$$

where $T : \mathcal{B}(\mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'_k})$ is maximized w.r.t. the set of polynomially implementable quantum channels and $F_T(|\psi\rangle^{\otimes k}) := \langle \Omega_{A_k A'_k} | (\text{Id}_{A_k} \otimes T)(|\psi\rangle\langle\psi|^{\otimes k}) | \Omega_{A_k A'_k} \rangle$.

Note that $F_T(\cdot)$ corresponds to the following observable: given any efficient T , let us define a (non-uniform) quantum poly time (QPT) distinguisher $\mathcal{D}_T$ that, on input k copies of a pure state $|\psi\rangle$, applies $(\text{Id} \otimes T)$ and measures the two-outcome POVM $\{|\Omega\rangle\langle\Omega|, \mathbb{1} - |\Omega\rangle\langle\Omega|\}$ on AA' , outputting 1 if and only if the outcome is $|\Omega\rangle$. Then

$$\Pr[\mathcal{D}_T(|\psi\rangle^{\otimes k}) = 1] = F_T(|\psi\rangle^{\otimes k}).$$

Let us now calculate the difference in the expected value of the observable F_T for the following ensembles

$$\begin{aligned} \mathcal{E}_{\text{Haar}} &= \{|\psi_{AB}\rangle, \quad |\psi_{AB}\rangle \sim \text{Haar}(n)\}, \\ \mathcal{E}_m &= \{U_\pi |0\rangle^{n-m} |\phi_m\rangle : U_\pi \in P, |\phi_m\rangle \sim \text{Haar}(m)\}, \end{aligned}$$

where $\mathcal{E}_m$ is a Haar-subsystem state ensemble, where P is the set of unitaries implementing a random permutation on n -bit strings $\pi : \{0, 1\}^n \rightarrow \{0, 1\}^n$, $U_\pi |x\rangle = |\pi(x)\rangle$.

Let us now bound the difference between the expected values,

$$\begin{aligned} & \left| \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} F_T(|\psi\rangle^{\otimes k}) - \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_m} F_T(|\psi\rangle^{\otimes k}) \right| \\ &= \left| \text{Tr} \left[M_T \left(\mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} (|\psi\rangle\langle\psi|^{\otimes k}) - \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_m} (|\psi\rangle\langle\psi|^{\otimes k}) \right) \right] \right| \end{aligned}$$

with $M_T := (\text{Id}_{A_k} \otimes T^*)(|\Omega_{AA'}\rangle\langle\Omega_{AA'}|)$, where the Haar average of $|\psi\rangle^{\otimes k}$ can be described as

$$\mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} [|\phi\rangle\langle\phi|^{\otimes k}] = \frac{\Pi_{\text{sym}}^{(n,k)}}{\text{tr}(\Pi_{\text{sym}}^{(n,k)})} = \frac{\Pi_{\text{sym}}^{(n,k)}}{\binom{d+k-1}{k}},$$

where $\Pi_{\text{sym}}^{(n,k)}$ is the projector onto the symmetric subspace (see for example [Mel24]). On the other hand, for the Haar-subsystem ensemble $\mathcal{E}_m$, a sample can be described equivalently as follows: pick a subset $S \subseteq [d]$ of cardinality $|S| = d_S := 2^m$ uniformly at random, and then draw $|\phi\rangle$ Haar-random on the d_S -dimensional coordinate subspace $\mathcal{H}_S := \text{span}\{|x\rangle : x \in S\} \subseteq (\mathbb{C}^2)^{\otimes n}$. Then, its average over all possible permutations can be represented as

$$\mathbb{E}_{|\phi\rangle \sim \mathcal{E}_m} [|\phi\rangle\langle\phi|^{\otimes k}] = \mathbb{E}_{S: |S|=d_S} \left[\frac{\Pi_{\text{sym}}^{(S,k)}}{\text{Tr}(\Pi_{\text{sym}}^{(S,k)})} \right].$$

Moreover, let us fix a subset of basis labels $S \subseteq [d]$, with $d = d_A d_B$ with $|S| = d_S$ such that

$$(a, b) \in S \iff |a\rangle_A \otimes |b\rangle_B \in \mathcal{H}_S,$$

where $\mathcal{H}_S$ is described in the computational basis. Let us now define the set $S_B \subset [d_B]$ of B -indices appearing in S and its corresponding projector

$$S_B := \{b \in [d_B] : \exists a \in [d_A] \text{ s.t. } (a, b) \in S\}, \quad \Pi_{S_B} := \sum_{b \in S_B} |b\rangle\langle b|.$$

Then,

$$\Pi_S = \sum_{(a,b) \in S} |ab\rangle\langle ab| \leq \sum_{b \in S_B} \sum_{a \in [d_A]} |ab\rangle\langle ab| = \mathbb{1}_A \otimes \Pi_{S_B}.$$

Moreover, since $\Pi_{\text{Sym}}^{(S,k)} \leq \Pi_S^{\otimes k}$,

$$\Pi_{\text{Sym}}^{(S,k)} \leq \mathbb{1}_{A_k} \otimes \Pi_{S_B}^{\otimes k}.$$

In order to bound the expected value of F_T with respect to the Haar-random ensemble, we first use the fact that it is computationally indistinguishable from the Haar-subsystem random ensemble to estimate a bound δ . Later, we prove that this bound δ holds with overwhelming probability for computationally bounded observers via Markov's inequality.

Let us first evaluate the asymptotic behavior of the expectation values of the Haar-random ensemble introduced in Eq. (5.2.8).

$$\begin{aligned} \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} [F_T(|\psi\rangle^{\otimes k})] &= \frac{k!}{d^k} \left(1 + O\left(\frac{k^2}{d}\right) \right) \text{Tr} [M_T \Pi_{\text{sym}}^{(n,k)}] \\ &= \frac{k!}{d^k} \left(1 + O\left(\frac{k^2}{d}\right) \right) \\ &= O\left(\frac{k!}{d^k}\right). \end{aligned}$$

Similarly, the expected value of F_T for the Haar-subsystem ensemble is given by

$$\begin{aligned} \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_m} [F_T(|\psi\rangle^{\otimes k})] &= \frac{k!}{d_S^k} \left(1 + O\left(\frac{k^2}{d_S}\right) \right) \mathbb{E}_{S:|S|=d_S} \text{Tr} [M_T \Pi_{\text{sym}}^{(S,k)}] \\ &= \frac{k!}{d_S^k} \left(1 + O\left(\frac{k^2}{d_S}\right) \right) \mathbb{E}_{S:|S|=d_S} \text{Tr} [M_T (\mathbb{1}_{A_k} \otimes \Pi_{S_B}^{\otimes k})] \\ &= \frac{k!}{d_A^k d_S^k} \left(1 + O\left(\frac{k^2}{d_S}\right) \right) \mathbb{E}_{S:|S|=d_S} \text{Tr} [\Pi_{S_B}^{\otimes k}] \\ &= \frac{k!}{d_A^k} \left(1 + O\left(\frac{k^2}{d_S}\right) \right) \\ &= O\left(\frac{k!}{d_A^k}\right), \end{aligned}$$

where in the third line we use the fact that $\Pi_{\text{Sym}}^{(S,k)} \leq \mathbb{1}_{A_k} \otimes \Pi_{S_B}^{\otimes k}$. Then, the fourth line follows from the Choi constraint $\text{tr}_{A_k}[M_T] = \frac{\mathbb{1}_{B_k}}{d_A^k}$ and the fifth since $\text{Tr}[\Pi_{S_B}^{\otimes k}] = \text{rank}(\Pi_{S_B}^{\otimes k}) \leq d_S^k$ and $\text{Tr}[M_T \Pi_{\text{Sym}}^{(n,k)}] \leq \text{Tr}[M_T] = 1$.

Using the triangle inequality and the above asymptomatic inequalities, we can see that:

$$\left| \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} F_T(|\psi\rangle^{\otimes k}) - \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_m} F_T(|\psi\rangle^{\otimes k}) \right| = O\left(\frac{k!}{d_A^k}\right).$$

Therefore, it follows that

$$\mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} F_T(|\psi\rangle^{\otimes k}) = \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_m} F_T(|\psi\rangle^{\otimes k}) + O\left(\frac{k!}{d_A^k}\right),$$

where

$$\mathbb{E}_{|\psi\rangle \sim \mathcal{E}_m} F_T(|\psi\rangle^{\otimes k}) = \frac{k!}{d_A^k} \left(1 + O\left(\frac{k^2}{d_S}\right) \right).$$

Thus,

$$\begin{aligned} \mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} F_T(|\psi\rangle^{\otimes k}) &= \frac{k!}{d_A^k} \left(1 + O\left(\frac{k^2}{d_S}\right) \right) \\ &\leq \frac{2^{k \cdot m(n)}}{d_A^k}, \end{aligned}$$

where the last inequality follows by taking $m(n) = \omega(\log n)$ and $k = \text{poly}(n)$, since

$$\log(k!) \leq k(\log(k) - 1) + O(\log(k)).$$

Let us now prove that there exists a state in the Haar ensemble that satisfies this bound with high probability. To do so, fix $t = t(n) \in \mathbb{N}$. Then,

$$\left(F_T(|\psi\rangle^{\otimes k}) \right)^t = \text{tr}[M_T^{\otimes t} |\psi\rangle^{\otimes kt}].$$

One can calculate the Haar moment bound for the kt moment

$$\mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} \left[\left(F_T(|\psi\rangle^{\otimes k}) \right)^t \right] \leq \frac{(kt)!}{d^{kt}} \left(1 + O\left(\frac{(kt)^2}{d}\right) \right).$$

Hence, by Markov's inequality, for

$$\delta := \frac{2^{k m(n)}}{d_A^k},$$

one has

$$\begin{aligned}
\Pr_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} \left[F_T(|\psi\rangle^{\otimes k}) \geq \delta \right] &= \Pr_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} \left[\left(F_T(|\psi\rangle^{\otimes k}) \right)^t \geq \delta^t \right] \\
&\leq \frac{\mathbb{E}_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} \left[\left(F_T(|\psi\rangle^{\otimes k}) \right)^t \right]}{\delta^t} \\
&\leq \frac{(kt)!}{2^{kt(n_A + m(n_A))}} \left(1 + O\left(\frac{(kt)^2}{d} \right) \right).
\end{aligned}$$

Let us now fix a polynomial circuit-size bound $p(n)$ and let $\text{QPT}_{\leq p}$ denote the set of channels implementable by circuits of size at most $p(n)$. By Lemma 3.15, there exists a polynomial $r(n)$ such that

$$|\text{QPT}_{\leq p}| \leq 2^{r(n)}.$$

Therefore, by the union bound,

$$\Pr_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} \left[\exists T \in \text{QPT}_{\leq p} : F_T(|\psi\rangle^{\otimes k}) \geq \delta \right] \leq 2^{r(n)} \frac{(kt)!}{2^{kt(n_A + m(n_A))}} \left(1 + O\left(\frac{(kt)^2}{d} \right) \right).$$

Since $k \in \text{poly}(n)$, $r(n) \in \text{poly}(n)$, and $m(n_A) = \omega(\log n_A)$, there exists a polynomial $t = t(n)$

$$r(n) + \log((kt)!) = r(n) + kt(n)[\log(kt(n)) - 1] + \mathcal{O}(\log n) = o(kt(n_A + m(n_A)))$$

for all sufficiently large n . In particular, if the degree of r is g_r and the degree of k is g_k , then it suffices to take t to be a polynomial of degree strictly larger than $\max\{0, g_r - g_k - 1\}$. Hence

$$\Pr_{|\psi\rangle \sim \mathcal{E}_{\text{Haar}}} \left[\forall T \in \text{QPT}_{\leq p} : F_T(|\psi\rangle^{\otimes k}) \leq \frac{2^{k m(n_A)}}{d_A^k} \right] \geq 1 - 2^{-\Omega(n)}.$$

By the previous estimate, the computational bound holds with probability at least $1 - 2^{-\Omega(n)}$ over $|\psi\rangle \sim \mathcal{E}_{\text{Haar}}$. By Lemma 5.3, the informational bound $H_{\min}(A^k | B^k)_{|\psi\rangle^{\otimes k}} = -k(n_A - O(1))$ holds almost surely for Haar-random $|\psi\rangle$. Hence, for all sufficiently large n , these two events have nonempty intersection. Then, there exists a pure state $|\psi_{AB,n}\rangle$ in this intersection such that

$$\max_{T \in \text{QPT}_{\leq p}} F_T(|\psi_{AB,n}\rangle^{\otimes k}) \leq \frac{2^{k m(n_A)}}{d_A^k},$$

and thus

$$\begin{aligned}
\hat{H}_{\min}(A^k | B^k)_{|\psi_{AB,n}\rangle^{\otimes k}} &= -\log \left(d_A^k \max_{T \in \text{QPT}_{\leq p}} F_T(|\psi_{AB,n}\rangle^{\otimes k}) \right) \\
&\geq -\log \left(d_A^k \cdot \frac{2^{k m(n_A)}}{d_A^k} \right) \\
&= -k m(n_A).
\end{aligned}$$

□

A.3.4. Proof of Theorem 5.7

Proof. Similarly to the proof of Theorem 5.5, assume $d_A = d_B$, where A_k denotes the space $A^{\otimes k}$ and B_k denotes $B^{\otimes k}$, such that $d_{A_k} = d_A^k = 2^{k \cdot n_A}$ and $d_n = d_A \cdot d_B = d_A^2$. Then, $T : \mathcal{B}(\mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'_k})$ acts on k copies. Given the definition of the computational min-entropy,

$$\tilde{H}_{\min}(A^k|B^k)_{\rho^{\otimes k}} = -\log \left(d_A^k \cdot \max_T \left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes T)(\rho^{\otimes k}) \right| \Omega_{A_k A'_k} \right\rangle \right) \quad (\text{A.3.5})$$

$$= -\log \left(d_A^k \cdot \max_T F_T(\rho^{\otimes k}) \right), \quad (\text{A.3.6})$$

where $T : \mathcal{B}(\mathcal{H}_n^{B_k}) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'_k})$ is maximized over the set of polynomially implementable quantum channels and $F_T(\rho^{\otimes k}) := \left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes T)(\rho^{\otimes k}) \right| \Omega_{A_k A'_k} \right\rangle$.

Note that $F_T(\cdot)$ corresponds to the following observable: given any efficient T , let us define a (non-uniform) QPT distinguisher $\mathcal{D}_T$ that, on input k copies of an unknown mixed state ρ , applies $(\text{Id} \otimes T)$ and measures the two-outcome POVM $\{|\Omega\rangle\langle\Omega|, \mathbb{1} - |\Omega\rangle\langle\Omega|\}$ on AA' , outputting 1 if and only if the outcome is $|\Omega\rangle$. Then

$$\Pr[\mathcal{D}_T(\rho^{\otimes k}) = 1] = F_T(\rho^{\otimes k}).$$

Let us now bound the difference between the expected values of F_T for the Haar random and Haar subsystem ensembles introduced in Eq. (5.2.8). As stated in [BMB⁺25, Theorem D.1], the k -th moment of the GHSE ensemble can be constructed as

$$\mathbb{E}_{\mathcal{E}_{\rho \sim \eta_{n,m}}}(\rho^{\otimes k}) = \mathbb{E}_{\psi \in \text{Haar}(d_n d_m)}[\text{tr}_m(|\psi\rangle\langle\psi|)^{\otimes k}] = \frac{(d_n d_m - 1)!}{(d_n d_m + k - 1)!} \sum_{\pi \in S_k} d_m^{\text{cycles}(\pi)} \hat{\pi}_n, \quad (\text{A.3.7})$$

where S_k is the symmetric group of degree k , and $\hat{\pi}_n$ denotes the permutation unitary acting on $\mathcal{H}_n^{\otimes k}$. The function $\text{cycles}(\pi)$ denotes the number of disjoint cycles in the cycle decomposition of $\pi \in S_k$, counting fixed points as cycles of length 1. In particular,

$$\sum_{\substack{\pi \in S_k \\ \pi \neq \mathbb{1}}} d_m^{\text{cycles}(\pi)} = \frac{(d_m + k - 1)!}{(d_m - 1)!} - d_m^k. \quad (\text{A.3.8})$$

Using $M_T := (\text{Id}_{A_k} \otimes T^*)(|\Omega_{A_k A'_k}\rangle\langle\Omega_{A_k A'_k}|)$, we find that

$$\begin{aligned}
& \left| \mathbb{E}_{\rho \sim \eta_{n,m}} F_T(\rho^{\otimes k}) - F_T\left(\frac{\mathbb{1}_n^{\otimes k}}{d_n^k}\right) \right| \\
&= \left| \text{Tr} \left[M_T \left(\mathbb{E}_{\rho \sim \eta_{n,m}}(\rho^{\otimes k}) - \frac{\mathbb{1}_n^{\otimes k}}{d_n^k} \right) \right] \right| \\
&= \left| \text{Tr} \left[M_T \left(\frac{1}{d_n^k} \left(-\frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \mathbb{1}_n^{\otimes k} \right. \right. \right. \\
&\quad \left. \left. \left. + \frac{1}{(d_n d_m)^k} \left(1 - \frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \sum_{\substack{\pi \in S_k \\ \pi \neq \mathbb{1}}} d_m^{\text{cycles}(\pi)} \hat{\pi}_n \right) \right] \right| \\
&\leq \frac{1}{d_n^k} \left(\frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \text{Tr}(M_T) \\
&\quad + \frac{1}{(d_n d_m)^k} \left(1 + \frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \sum_{\substack{\pi \in S_k \\ \pi \neq \mathbb{1}}} d_m^{\text{cycles}(\pi)} |\text{Tr}(M_T \hat{\pi}_n)| \\
&\leq \frac{1}{d_n^k} \left(\frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \\
&\quad + \frac{1}{(d_n d_m)^k} \left(1 + \frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \sum_{\substack{\pi \in S_k \\ \pi \neq \mathbb{1}}} d_m^{\text{cycles}(\pi)} \\
&= \frac{1}{d_n^k} \left(\frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \\
&\quad + \frac{1}{(d_n d_m)^k} \left(1 + \frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \left(\frac{(d_m + k - 1)!}{(d_m - 1)!} - d_m^k \right).
\end{aligned}$$

The second equality follows from splitting the sum over S_k into the identity permutation and the sum over non-identity permutations, as in [BMB⁺25, Theorem D.1]. The first inequality follows from applying the triangle inequality and the second one from the fact that $|\text{Tr}(M_T \hat{\pi}_n)| \leq \text{Tr}(M_T) = 1$. The last equality follows by using Eq. (A.3.8). Finally, since

$$\left(\frac{(d_m + k - 1)!}{(d_m - 1)!} - d_m^k \right) = d_m^k \left(\frac{k(k-1)}{2d_m} + O\left(\frac{k^4}{d_m^2}\right) \right),$$

it follows that

$$\begin{aligned}
& \left| \mathbb{E}_{\rho \sim \eta_{n,m}} F_T(\rho^{\otimes k}) - F_T\left(\frac{\mathbb{1}^{\otimes k}}{d^k}\right) \right| \\
& \leq \frac{1}{d_n^k} \left(\frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \\
& \quad + \frac{1}{d_n^k} \left(1 + \frac{k(k-1)}{2d_n d_m} + O\left(\frac{k^4}{d_n^2 d_m^2}\right) \right) \left(\frac{k(k-1)}{2d_m} + O\left(\frac{k^4}{d_m^2}\right) \right) \\
& = \frac{1}{d_n^k} \left(\frac{k(k-1)}{2d_n d_m} + \frac{k(k-1)}{2d_m} \right) + O\left(\frac{k^4}{d_n^k d_m^2}\right) \\
& = O\left(\frac{k^2}{d_n^k d_m}\right).
\end{aligned}$$

Therefore we have

$$\mathbb{E}_{\rho \sim \eta_{n,m}} F_T(\rho^{\otimes k}) \leq F_T\left(\frac{\mathbb{1}^{\otimes k}}{d_n^k}\right) + O\left(\frac{k^2}{d_n^k d_m}\right).$$

On the other hand, by using the definition of the (informational) min-entropy, it follows that for any efficient CPTP map T

$$F_T(\rho^{\otimes k}) \leq \frac{2^{-H_{\min}(A^k|B^k)_{\rho^{\otimes k}}}}{d_A^k}. \quad (\text{A.3.9})$$

Let us now evaluate the conditional min-entropy for $\rho = \frac{\mathbb{1}_{AB}}{d_A d_B}$ with $d_n = d_A \cdot d_B$. Then, it directly follows that

$$H_{\min}(A|B)_{\frac{\mathbb{1}_{AB}}{d_A d_B}} = \log d_A.$$

By the additivity of the conditional min-entropy one has that

$$\begin{aligned}
\mathbb{E}_{\rho \sim \eta_{n,m}} F_T(\rho^{\otimes k}) & \leq \frac{1}{d_n^k} + O\left(\frac{k^2}{d_n^k d_m}\right) \\
& \leq \frac{1 + \varepsilon}{d_n^k},
\end{aligned}$$

for any $\varepsilon \in (0, 1]$ satisfying $\varepsilon \geq Ck^2/d_m$ for some constant $C \geq 0$.

Let us now prove that there exists a state in the GHSE ensemble that satisfies this bound with high probability. To do so, we define the t -th moment of the distribution

$$(F_T(\rho^{\otimes k}))^t = \text{Tr}\left[M_T \rho^{\otimes k}\right]^t = \text{Tr}\left(M_T^{\otimes t} \rho^{\otimes kt}\right)$$

By Markov's inequality, one has that for any moment $t \geq 1$,

$$\Pr_{\rho \sim \eta_{n,m}} [F_T(\rho^{\otimes k}) \geq \delta] = \Pr_{\rho \sim \eta_{n,m}} [(F_T(\rho^{\otimes k}))^t \geq \delta^t] \leq \frac{\mathbb{E}_{\rho \sim \eta_{n,m}} (F_T(\rho^{\otimes k}))^t}{\delta^t}.$$

Let us first calculate $\mathbb{E}_{\rho \sim \eta_{n,m}} (F_T(\rho^{\otimes k}))^t$. Taking Eq.(A.3.7), one has

$$\begin{aligned} \mathbb{E}_{\rho \sim \eta_{n,m}} (F_T(\rho^{\otimes k}))^t &= \frac{(d_n d_m - 1)!}{(d_n d_m + kt - 1)!} \sum_{\pi \in S_{kt}} d_m^{\text{cycles}(\pi)} \text{Tr}(M_T^{\otimes t} \hat{\pi}_n) \\ &\leq \frac{(d_n d_m - 1)!}{(d_n d_m + kt - 1)!} \frac{(d_m + kt - 1)!}{(d_m - 1)!} \\ &\leq \frac{1}{d_n^{kt}} \exp\left(\frac{kt(kt - 1)}{2d_m}\right). \end{aligned}$$

Then, by taking $\delta = (1 + \varepsilon)/d_n^k$, one has that for any moment $t \geq 1$,

$$\Pr_{\rho \sim \eta_{n,m}} \left[F_T(\rho^{\otimes k}) \geq \frac{(1 + \varepsilon)}{d_n^k} \right] \leq \frac{1}{(1 + \varepsilon)^t} \cdot \exp\left(\frac{kt(kt - 1)}{2d_m}\right) \leq \exp\left(\frac{k^2 t^2}{2d_m} - t \log(1 + \varepsilon)\right).$$

Fixing $\varepsilon \in (0, 1)$ and taking

$$t := \left\lfloor \frac{d_m}{k^2} \log(1 + \varepsilon) \right\rfloor \geq 1,$$

one gets

$$\Pr_{\rho \sim \eta_{n,m}} \left[F_T(\rho^{\otimes k}) \geq \frac{(1 + \varepsilon)}{d_n^k} \right] \leq \exp\left(-\frac{d_m}{2k^2} \log(1 + \varepsilon)^2\right).$$

Let us now fix a polynomial circuit-size bound $p(n)$ and let $\text{QPT}_{\leq p}$ denote the set of channels implementable by circuits of size at most $p(n)$. By Lemma 3.15, there exists a polynomial $r(n)$ such that $|\text{QPT}_{\leq p}| \leq 2^{r(n)}$. Therefore, by taking the union bound

$$\Pr_{\rho \sim \eta_{n,m}} \left[\exists T \in \text{QPT}_{\leq p} : F_T(\rho^{\otimes k}) \geq \frac{(1 + \varepsilon)}{d_n^k} \right] \leq 2^{r(n)} \exp\left(-\frac{d_m}{2k^2} \log(1 + \varepsilon)^2\right).$$

Let us now fix the variable ε such that

$$\varepsilon(n) := 2k(n) \sqrt{\frac{2(r(n) + n) \ln 2}{d_{m(n)}}}.$$

Since $m(n) = \omega(\log n)$ and $k(n), r(n) = \text{poly}(n)$, ε satisfies for a sufficiently large $n \in \mathbb{N}$ that $\varepsilon \in (0, 1]$ and $\varepsilon \geq Ck^2/d_m$. Moreover, using the standard bound $\log(1 + \varepsilon) \geq \varepsilon/2$, it follows that

$$\frac{d_m}{2k^2} \log(1 + \varepsilon)^2 \geq \frac{d_m \varepsilon^2}{8k^2} = (r(n) + n) \ln 2.$$

Hence,

$$\Pr_{\rho \sim \eta_{n,m}} \left[\forall T \in \text{QPT}_{\leq p} : F_T(\rho^{\otimes k}) \leq \frac{(1+\varepsilon)}{d_n^k} \right] \geq 1 - 2^{-n}.$$

Therefore, for $k = \text{poly}(n)$ there exists a state $\tilde{\rho} \in \eta_{n,m}$ such that for every polynomially implementable quantum channel $\text{QPT}_{\leq p} \ni T : \mathcal{B}(\mathcal{H}_n^B) \rightarrow \mathcal{B}(\mathcal{H}_n^{A'})$ with at most $p(n)$ gates,

$$F_T(\tilde{\rho}^{\otimes k}) \leq \frac{1+\varepsilon}{d_n^k}.$$

Finally, by the definition of the computational min-entropy,

$$\tilde{H}_{\min}(A^k|B^k)_{\tilde{\rho}^{\otimes k}} \geq k \cdot n_A - \log(1+\varepsilon) \geq k \cdot n_A - \text{negl}(n).$$

Let us now calculate the (informational) conditional min-entropy of a state sampled from the GHSE. Since $H_{\min}(A|B)_\rho \leq H(A|B)_\rho$ and $H(A|B)_\rho = H(AB)_\rho - H(B)_\rho$, where $H(\cdot)$ is the von-Neumann entropy (see [Tom16, Definition 5.1]), it follows that

$$H_{\min}(A|B)_\rho \leq H(M)_{|\psi\rangle} - H(B)_{|\psi\rangle},$$

where we have used the fact that $H(M)_{|\psi\rangle} = H(AB)_{|\psi\rangle}$ for the bipartite state $|\psi\rangle \in \mathbb{C}^{d_A d_B} \otimes \mathbb{C}^{d_m}$ as defined in Definition 5.6. Then, we used the fact that the von Neumann entropy is upper bounded by the rank, obtaining

$$H(M)_{|\psi\rangle} \leq \log d_m = m$$

almost surely. On the other hand, as proven in [HLW06, Lemma II.4], since $d_m = \omega(\log n)$

$$H(B)_{|\psi\rangle} \geq n_A - \text{negl}(n)$$

almost surely as well. Then, since the conditional min-entropy is additive and $k = \text{poly}(n)$,

$$H_{\min}(A^k|B^k)_{\rho^{\otimes k}} \leq -k \cdot (n_A - m) + \text{negl}(n).$$

□

A.3.5. Proof of Lemma 5.15

Proof. Fix a public matrix $A \leftarrow U_q^{m \times \ell}$. We use the standard LWE-PRG construction

$$G_{A,\ell}(s, e) = As + e \bmod q,$$

where

$$s \leftarrow U_q^\ell, \quad e \leftarrow D_{q,\sigma}^m.$$

We work in an idealized sampling model in which an exact sample $e \sim D_{q,\sigma}^m$ is assigned a randomness cost of $m \log \sigma$ bits. This is an explicit modelling assumption. Therefore, together with the $\ell \log q$ bits specifying s , the total input length is $\ell \log q + m \log \sigma$ bits.

Thus, the input size is given by

$$d_\ell = \ell \log q + m \log \sigma,$$

whereas the output of the function $G_{A,\ell}(s, e)$ has length

$$L_\ell = m \log q.$$

Let us now set the parameters of the $\text{LWE}_{\ell,m,q,\sigma}$ problem such that the corresponding PRG is length-doubling. Let δ be the constant from Assumption 5.14. Choose

$$c > \max \left\{ 1, \frac{1}{2(1-\delta)} \right\},$$

and instantiate

$$\begin{aligned} m &= \ell^\beta, \\ q &= 2^{\lceil c \log_2 m \rceil}, \\ \sigma &= 2\sqrt{m}. \end{aligned}$$

Then $q = \text{poly}(m)$ and $\sigma \geq 2\sqrt{m}$. Moreover, since $c(1-\delta) > 1/2$, for all sufficiently large m ,

$$\sigma = 2\sqrt{m} \leq q^{1-\delta}.$$

Thus these parameters lie in the range of Assumption 5.14.

Now

$$\begin{aligned} \log q &= c \log m + O(1), \\ \log \sigma &= \frac{1}{2} \log m + O(1). \end{aligned}$$

Hence

$$\begin{aligned} 2d_\ell &= 2\ell \log q + 2m \log \sigma \\ &= 2c\ell \log m + m \log m + O(m + \ell). \end{aligned}$$

On the other hand,

$$L_\ell = m \log q = cm \log m + O(m).$$

Therefore

$$L_\ell - 2d_\ell = (c-1)m \log m - 2c\ell \log m - O(m + \ell).$$

Since $m = \ell^\beta$ with $\beta > 1$, we have $\ell = o(m)$, and hence

$$2c\ell \log m + O(m + \ell) = o(m \log m).$$

Because $c > 1$, it follows that $L_\ell \geq 2d_\ell$ for all sufficiently large ℓ . Then, one can define the length-doubling generator by truncating:

$$G_{A,\ell}^{\text{dbl}}(s, e) = \text{Pref}_{2d_\ell}(As + e \bmod q).$$

It remains to prove pseudorandomness. By Assumption 5.14,

$$(A, As + e) \approx_c (A, u), \quad u \leftarrow U_q^m,$$

with distinguishing advantage at most

$$2^{-\Omega(\ell^\alpha)}$$

against quantum algorithms of size $\text{poly}(m)$. Since

$$d_\ell = \ell \log q + m \log \sigma = \text{poly}(m),$$

every QPT distinguisher of size $\text{poly}(d_\ell)$ is also of size $\text{poly}(m)$. Applying the deterministic map Pref_{2d_ℓ} to the second component cannot increase distinguishing advantage. Therefore

$$\left| \Pr_{\substack{A \leftarrow U_q^{m \times \ell} \\ r \leftarrow U_{d_\ell}}} [D_\ell(A, G_{\ell,A}^{\text{dbl}}(r)) = 1] - \Pr_{\substack{A \leftarrow U_q^{m \times \ell} \\ u \leftarrow U_q^m}} [D_\ell(A, \text{Pref}_{2d_\ell}(u)) = 1] \right| \leq 2^{-\Omega(\ell^\alpha)}.$$

Since q is a power of two, the binary encoding of $u \leftarrow U_q^m$ is exactly uniform over $\{0, 1\}^{L_\ell}$. Because $L_\ell \geq 2d_\ell$, its first $2d_\ell$ bits are distributed as U_{2d_ℓ} . This proves the claim. $\square$

A.3.6. Proof of Theorem 5.16

Proof. Let $d_A = 2^{n_A}$ and $d = d_A^2$. Let (K, G) be the scalable PRS generator constructed above from the scalable ARS construction, instantiated using the QPRF obtained from Assumption 5.14. We choose the security parameters below and write

$$|\psi_s\rangle_{AB} = G(1^n, 1^\lambda, s), \quad s \leftarrow K(1^n, 1^\lambda).$$

Similarly to Theorem 5.5, let us define, for any $T \in \text{QPT}_{\leq p}$, the following observable

$$F_T(\rho^{\otimes k}) := \left\langle \Omega_{A_k A'_k} \middle| (\text{Id}_{A_k} \otimes T)(\rho^{\otimes k}) \middle| \Omega_{A_k A'_k} \right\rangle.$$

As previously discussed, $F_T(\cdot)$ corresponds to the following observable: given any efficient T , let us define a (non-uniform) quantum poly time (QPT) distinguisher $\mathcal{D}_T$ that, on

input k copies of a pure state $|\psi\rangle$, applies $(\text{Id} \otimes T)$ and measures the two-outcome POVM $\{|\Omega\rangle\langle\Omega|, \mathbb{1} - |\Omega\rangle\langle\Omega|\}$ on AA' , outputting 1 if and only if the outcome is $|\Omega\rangle$. Then

$$\Pr[\mathcal{D}_T(|\psi\rangle^{\otimes k}) = 1] = F_T(|\psi\rangle^{\otimes k}).$$

Equivalently, if T^* denotes the Heisenberg adjoint of T , let

$$M_T := (\text{Id}_{A_k} \otimes T^*) \left(|\Omega_{A_k A'_k}\rangle\langle\Omega_{A_k A'_k}| \right),$$

so that $F_T(\omega) = \text{Tr}[M_T \omega]$ for every state ω on $A_k B_k$. Thus, for the t -th moment,

$$\left(F_T(|\psi\rangle^{\otimes k}) \right)^t = \text{tr}[M_T^{\otimes t} |\psi\rangle\langle\psi|^{\otimes kt}],$$

and, by the quantum pseudorandomness property of Definition 5.8, the expected value of the PRS state satisfies

$$\mathbb{E}_{|\psi_s\rangle \sim \mathcal{E}_{PRS}}[F_T(|\psi_s\rangle^{\otimes k})^t] \leq \mathbb{E}_{|\phi\rangle \sim \mathcal{E}_{\text{Haar}}}[F_T(\phi^{\otimes k})^t] + \varepsilon_{\text{PRS}}(n, \lambda, kt),$$

where $\varepsilon_{\text{PRS}}(n, \lambda, kt)$ denotes the distinguishing advantage of the scalable PRS against the above kt -copy QPT test.

As shown in the proof of Theorem 5.5,

$$\mathbb{E}_{|\phi\rangle \sim \mathcal{E}_{\text{Haar}}}[F_T(\phi^{\otimes k})^t] \leq \frac{(kt)!}{d^{kt}} \left(1 + O\left(\frac{(kt)^2}{d}\right) \right).$$

Set

$$\delta := \frac{2^{k m(n_A)}}{d_A^k}.$$

By Markov's inequality,

$$\begin{aligned} \Pr_{|\psi_s\rangle \sim \mathcal{E}_{PRS}}[F_T(\psi_s^{\otimes k}) \geq \delta] &\leq \frac{(kt)!}{2^{kt(n_A + m(n_A))}} \left(1 + O\left(\frac{(kt)^2}{d}\right) \right) \\ &\quad + \varepsilon_{\text{PRS}}(n, \lambda, kt) \cdot 2^{kt(n_A - m(n_A))}. \end{aligned}$$

By Lemma 3.15, there is a polynomial $r(n)$ such that $|\text{QPT}_{\leq p}| \leq 2^{r(n)}$. Therefore, by taking the union bound,

$$\begin{aligned} \Pr_{|\psi_s\rangle \sim \mathcal{E}_{PRS}} \left[\exists T \in \text{QPT}_{\leq p} : F_T(|\psi_s\rangle^{\otimes k}) \geq \delta \right] &\leq 2^{r(n)} \frac{(kt)!}{2^{kt(n_A + m(n_A))}} \left(1 + O\left(\frac{(kt)^2}{d}\right) \right) \\ &\quad + \varepsilon_{\text{PRS}}(n, \lambda, kt) \cdot 2^{r(n) + kt(n_A - m(n_A))}. \end{aligned}$$

Similarly to Theorem 5.5, since $k \in \text{poly}(n)$, $r(n) \in \text{poly}(n)$, and $m(n_A) = \omega(\log n_A)$, there exists a polynomial $t = t(n)$

$$r(n) + \log((kt)!) = r(n) + kt(n)[\log(kt(n)) - 1] + \mathcal{O}(\log n) = o(kt(n_A + m(n_A)))$$

for all sufficiently large n .

Thus, for this choice of $t = t(n)$,

$$2^{r(n)} \frac{(kt)!}{2^{kt(n_A + m(n_A))}} \left(1 + O\left(\frac{(kt)^2}{d}\right) \right) = \text{negl}(n).$$

It remains to choose the security parameters so that the second term in the union bound is negligible. Let

$$B(n) := r(n) + kt(n_A - m(n_A)) + \omega(\log n).$$

Since k, t, r, n_A are polynomially bounded and $m(n_A) \leq n_A$, we have $B(n) \in \text{poly}(n)$. It is enough to ensure that

$$\varepsilon_{\text{PRS}}(n, \lambda, kt) \leq 2^{-B(n)}.$$

By the scalable ARS construction of [BCJZ26, Theorem 4.7], together with Lemma 5.11, the scalable PRS instantiated with a quantum-secure PRF satisfies

$$\begin{aligned} \varepsilon_{\text{PRS}}(n, \lambda, kt) &\leq (1 + \sqrt{2\pi}) kt \sqrt{n} 2^{-\lambda/2} + 2\pi kt 2^{-\lambda} + 2^{-\lambda} \\ &\quad + \varepsilon_{\text{QPRF}}(n, \lambda, kt, \ell). \end{aligned} \tag{A.3.10}$$

Here λ is the scalable ARS security parameter and ℓ is the LWE security parameter used in the GGM instantiation. By the discussion following Lemma 5.15, Zhandry's quantum-security proof of GGM [Zha21, Theorem 5.5] gives

$$\varepsilon_{\text{QPRF}}(n, \lambda, kt, \ell) \leq 2^{-\Omega(\ell^\alpha)}.$$

We now choose $\lambda = \lambda(n)$ and $\ell = \ell(n)$. First choose

$$\lambda(n) \geq 2B(n) + 2\log_2(kt) + \log_2 n + O(1).$$

Then

$$(1 + \sqrt{2\pi}) kt \sqrt{n} 2^{-\lambda/2} + 2\pi kt 2^{-\lambda} + 2^{-\lambda} \leq 2^{-B(n)-1}$$

for all sufficiently large n . Since $B(n) \in \text{poly}(n)$, we have $\lambda(n) \in \text{poly}(n)$.

Next, since $B(n)$ is polynomially bounded, choose $\ell = \ell(n) \in \text{poly}(n)$ large enough so that

$$\ell(n)^\alpha \geq CB(n)$$

for a sufficiently large constant $C > 0$. Consequently,

$$\varepsilon_{\text{PRS}}(n, \lambda(n), kt) \leq 2^{-B(n)}.$$

Therefore,

$$\varepsilon_{\text{PRS}}(n, \lambda(n), kt) \cdot 2^{r(n)+kt(n_A-m(n_A))} \leq 2^{-\omega(\log n)} = \text{negl}(n).$$

Combining this with the previous union bound gives

$$\Pr_{|\psi_s\rangle \sim \mathcal{E}_{\text{PRS}}} \left[\forall T \in \text{QPT}_{\leq p} : F_T(|\psi_s\rangle^{\otimes k}) \leq \frac{2^{k m(n_A)}}{d_A^k} \right] \geq 1 - \text{negl}(n).$$

Hence, with probability $1 - \text{negl}(n)$ over the key s ,

$$\max_{T \in \text{QPT}_{\leq p}} F_T(|\psi_s\rangle^{\otimes k}) \leq \frac{2^{k m(n_A)}}{d_A^k}.$$

Using Theorem 4.36, we obtain

$$\begin{aligned} \overset{\text{c}}{H}_{\min}(A^k | B^k)_{|\psi_s\rangle^{\otimes k}} &= -\log \left(d_A^k \max_{T \in \text{QPT}_{\leq p}} F_T(|\psi_s\rangle^{\otimes k}) \right) \\ &\geq -k m(n_A). \end{aligned}$$

It remains to prove the information-theoretic bound. Let

$$\rho_{s,A} := \text{Tr}_B |\psi_s\rangle\langle\psi_s|.$$

Let $\kappa = \lceil an_A \rceil$ be the moment order, where $a > 0$ is fixed below, and let π_κ be the cyclic permutation on $A^{\otimes \kappa}$. Then

$$\text{Tr}[\pi_\kappa \rho_{s,A}^{\otimes \kappa}] = \text{Tr}(\rho_{s,A}^\kappa).$$

The Hadamard test for the controlled- π_κ estimates the real part of this quantity, which is $\text{Tr}(\rho_{s,A}^\kappa)$, and therefore has acceptance probability

$$\frac{1 + \text{Tr}(\rho_{s,A}^\kappa)}{2}.$$

This test is efficient for $\kappa = O(n_A)$, since π_κ is implemented by $\kappa - 1$ swaps [EAO⁺02]. Since the PRS indistinguishability bound applies to the acceptance probabilities of this test, and the trace moment appears with the prefactor $1/2$, we get

$$\mathbb{E}_{|\psi_s\rangle \sim \mathcal{E}_{\text{PRS}}} \text{Tr}(\rho_{s,A}^\kappa) \leq \mathbb{E}_{|\phi\rangle \sim \mathcal{E}_{\text{Haar}}} \text{Tr}(\phi_A^\kappa) + 2\varepsilon_{\text{PRS}}(n, \lambda, \kappa). \quad (\text{A.3.11})$$

Let us now bound the Haar-random term. For a Haar-random state $|\phi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$, write

$$\phi_A := \text{tr}_B[|\phi\rangle\langle\phi|_{AB}].$$

By the moment recurrence of induced random density matrices [Nec07, Sec. 3.3, Eqs. (37)–(39)], in the balanced case $d_A = d_B$, the moments

$$M_\kappa(d_A) := \mathbb{E}_{|\phi\rangle \sim \mathcal{E}_{\text{Haar}}} \text{Tr}(\phi_A^\kappa)$$

satisfy, for $\kappa \geq 3$,

$$\begin{aligned} M_\kappa(d_A) &= \frac{2(2\kappa-1)d_A}{(d_A^2 + \kappa - 1)(\kappa + 1)} M_{\kappa-1}(d_A) \\ &\quad + \frac{(\kappa-1)^2(\kappa-2)}{(d_A^2 + \kappa - 1)(d_A^2 + \kappa - 2)(\kappa + 1)} M_{\kappa-2}(d_A). \end{aligned} \quad (\text{A.3.12})$$

Moreover,

$$M_1(d_A) = 1, \quad M_2(d_A) = \frac{2d_A}{d_A^2 + 1} \leq \frac{2}{d_A}.$$

We now bound both coefficients. For $\kappa > 1$,

$$\begin{aligned} \frac{2(2\kappa-1)d_A}{(d_A^2 + \kappa - 1)(\kappa + 1)} &\leq \frac{2(2\kappa-1)d_A}{d_A^2(\kappa + 1)} \\ &= \frac{4}{d_A} - \frac{6}{d_A(\kappa + 1)} \leq \frac{4}{d_A}. \end{aligned}$$

For $\kappa > 2$,

$$\begin{aligned} \frac{(\kappa-1)^2(\kappa-2)}{(d_A^2 + \kappa - 1)(d_A^2 + \kappa - 2)(\kappa + 1)} &\leq \frac{(\kappa-1)^2(\kappa-2)}{d_A^4(\kappa + 1)} \\ &\leq \frac{\kappa^2}{d_A^4}. \end{aligned}$$

Thus

$$M_\kappa(d_A) \leq \frac{4}{d_A} M_{\kappa-1}(d_A) + \frac{\kappa^2}{d_A^4} M_{\kappa-2}(d_A).$$

Let us now prove that there exists a universal constant $C_0 \geq 8$ such that, for all moment orders $1 \leq \kappa \leq \lceil an_A \rceil = O(\log d_A)$ and all sufficiently large d_A ,

$$M_\kappa(d_A) \leq C_0^\kappa d_A^{1-\kappa}.$$

The claim holds for $\kappa = 1$, since

$$M_1(d_A) = 1 \leq C_0 d_A^0,$$

and for $\kappa = 2$, since

$$M_2(d_A) \leq \frac{2}{d_A} \leq C_0^2 d_A^{-1}.$$

Assume now that the claim holds for $\kappa - 1$ and $\kappa - 2$. Then

$$\begin{aligned} M_\kappa(d_A) &\leq \frac{4}{d_A} C_0^{\kappa-1} d_A^{2-\kappa} + \frac{\kappa^2}{d_A^4} C_0^{\kappa-2} d_A^{3-\kappa} \\ &= 4C_0^{\kappa-1} d_A^{1-\kappa} + \kappa^2 C_0^{\kappa-2} d_A^{-1-\kappa} \\ &= C_0^\kappa d_A^{1-\kappa} \left(\frac{4}{C_0} + \frac{\kappa^2}{C_0^2 d_A^2} \right). \end{aligned}$$

Since $\kappa = O(\log d_A)$, one has $\kappa^2/d_A^2 = o(1)$. Therefore, for $C_0 \geq 8$ and all sufficiently large d_A ,

$$\frac{4}{C_0} + \frac{\kappa^2}{C_0^2 d_A^2} \leq 1.$$

Hence

$$M_\kappa(d_A) \leq C_0^\kappa d_A^{1-\kappa}.$$

Therefore,

$$\mathbb{E}_{|\phi\rangle \sim \mathcal{E}_{\text{Haar}}} \text{Tr}(\phi_A^\kappa) \leq C_0^\kappa d_A^{1-\kappa}, \quad \kappa = O(\log d_A).$$

Combining the previous estimate with Eq. (A.3.11) gives

$$\mathbb{E}_{|\psi_s\rangle \sim \mathcal{E}_{\text{PRS}}} \text{Tr}(\rho_{s,A}^\kappa) \leq C_0^\kappa d_A^{1-\kappa} + 2\varepsilon_{\text{PRS}}(n, \lambda, \kappa).$$

Since

$$\|\rho_{s,A}\|_\infty^\kappa \leq \text{Tr}(\rho_{s,A}^\kappa),$$

Markov's inequality implies that, for any constant $L > C_0$,

$$\begin{aligned} \Pr_{|\psi_s\rangle \sim \mathcal{E}_{\text{PRS}}} \left[\|\rho_{s,A}\|_\infty > \frac{L}{d_A} \right] &\leq \frac{C_0^\kappa d_A^{1-\kappa} + 2\varepsilon_{\text{PRS}}(n, \lambda, \kappa)}{(L/d_A)^\kappa} \\ &= d_A \left(\frac{C_0}{L} \right)^\kappa + 2\varepsilon_{\text{PRS}}(n, \lambda, \kappa) \left(\frac{d_A}{L} \right)^\kappa. \end{aligned} \quad (\text{A.3.13})$$

Choose $L > C_0$ and $a > 0$ such that

$$a \log_2(L/C_0) > 2.$$

Since $\kappa = \lceil an_A \rceil$ and $d_A = 2^{n_A}$, the first term is exponentially small in n_A . It remains to choose the security parameters so that the second term of Eq. (A.3.13) is negligible. Since $d_A = 2^{n_A}$ and L is constant, it is enough to ensure

$$\varepsilon_{\text{PRS}}(n, \lambda, \kappa) \leq 2^{-\kappa n_A - \omega(\log n)}.$$

Indeed,

$$\begin{aligned} 2\varepsilon_{\text{PRS}}(n, \lambda, \kappa) \left(\frac{d_A}{L} \right)^\kappa &\leq 2 \cdot 2^{-\kappa n_A - \omega(\log n)} \cdot 2^{\kappa n_A} \\ &= \text{negl}(n). \end{aligned}$$

As in the computational part of the proof, by Eq. (A.3.10), we can choose $\lambda = \lambda(n) \in \text{poly}(n)$ such that also satisfies

$$\lambda(n) \geq 2\kappa n_A + 2\log_2(\kappa\sqrt{n}) + \omega(\log n).$$

Then the first three terms in Eq. (A.3.10) are bounded by $2^{-\kappa n_A - \omega(\log n) - 1}$. Next choose $\ell = \ell(n) \in \text{poly}(n)$ large enough so that it also satisfies

$$2^{-\Omega(\ell(n)^\alpha)} \leq 2^{-\kappa n_A - \omega(\log n) - 1}.$$

Therefore

$$\varepsilon_{\text{PRS}}(n, \lambda(n), \kappa) \leq 2^{-\kappa n_A - \omega(\log n)}.$$

Thus, with probability $1 - \text{negl}(n)$ over s ,

$$\|\rho_{s,A}\|_\infty \leq \frac{L}{d_A}.$$

For a pure bipartite state,

$$H_{\min}(A|B)_{\psi_s} = -H_{\max}(A)_{\psi_s} = -2\log \text{Tr} \sqrt{\rho_{s,A}}.$$

The bound $\|\rho_{s,A}\|_\infty \leq L/d_A$ implies

$$\text{Tr} \sqrt{\rho_{s,A}} \geq \sqrt{\frac{d_A}{L}},$$

while Cauchy–Schwarz gives

$$\text{Tr} \sqrt{\rho_{s,A}} \leq \sqrt{d_A}.$$

Therefore

$$-n_A \leq H_{\min}(A|B)_{\psi_s} \leq -n_A + \log L = -n_A + O(1).$$

By additivity of the conditional min-entropy for tensor powers of pure states,

$$H_{\min}(A^k|B^k)_{\psi_s^{\otimes k}} = k H_{\min}(A|B)_{\psi_s} = -k(n_A - O(1)).$$

A union bound over the informational and computational high-probability events completes the proof. □

A.3.7. Proof of Theorem 5.18

Proof. Let $d_A = 2^{n_A}$, $d_n = d_A^2$, and $d_m = 2^{m(n_A)}$. Let (K, G) be the scalable PRS generator constructed above, instantiated using the QPRF obtained from Assumption 5.14. We choose the security parameters below and define

$$|\psi_s\rangle_{ABM} = G(1^{n+m(n_A)}, 1^\lambda, s), \quad s \leftarrow K(1^{n+m(n_A)}, 1^\lambda),$$

and

$$\rho_{s,AB} = \text{Tr}_M |\psi_s\rangle\langle\psi_s|_{ABM},$$

which corresponds to the traced-PRS construction of pseudorandom density matrices of [BMB⁺25].

We first prove the computational lower bound. For $T \in \text{QPT}_{\leq p}$, define as in Theorem 5.7

$$F_T(\rho^{\otimes k}) := \left\langle \Omega_{A_k A'_k} \left| (\text{Id}_{A_k} \otimes T)(\rho^{\otimes k}) \right| \Omega_{A_k A'_k} \right\rangle.$$

For a fixed T , the test corresponding to $F_T(\rho^{\otimes k})^t$ acts on kt copies: it splits the input into t blocks of k copies, applies the F_T test to each block, and accepts iff all tests accept. Since $k, t \in \text{poly}(n)$ and T has size at most $p(n)$, this is a QPT test. Hence, by PRS security,

$$\mathbb{E}_s F_T(\rho_{s,AB}^{\otimes k})^t \leq \mathbb{E}_{\rho \sim \eta_{n,m}} F_T(\rho^{\otimes k})^t + \varepsilon_{\text{PRS}}(n + m(n_A), \lambda, kt). \quad (\text{A.3.14})$$

As shown in the proof of Theorem 5.7, using [BMB⁺25, Theorem D.1],

$$\mathbb{E}_{\rho \sim \eta_{n,m}} F_T(\rho^{\otimes k})^t \leq d_n^{-kt} \exp\left(\frac{k^2 t^2}{2d_m}\right). \quad (\text{A.3.15})$$

Set

$$\delta_n := \frac{1 + n^{-c}}{d_n^k}.$$

By Markov's inequality, (A.3.14) and (A.3.15),

$$\begin{aligned} \Pr_s \left[F_T(\rho_{s,AB}^{\otimes k}) \geq \delta_n \right] &\leq \exp\left(\frac{k^2 t^2}{2d_m} - t \log(1 + n^{-c})\right) \\ &\quad + \varepsilon_{\text{PRS}}(n + m(n_A), \lambda, kt) \delta_n^{-t}. \end{aligned} \quad (\text{A.3.16})$$

By Lemma 3.15, there exists a polynomial $r(n)$ such that $|\text{QPT}_{\leq p}| \leq 2^{r(n)}$. A union bound over $T \in \text{QPT}_{\leq p}$ gives

$$\begin{aligned} &\Pr_s \left[\exists T \in \text{QPT}_{\leq p} : F_T(\rho_{s,AB}^{\otimes k}) \geq \delta_n \right] \\ &\leq 2^{r(n)} \exp\left(\frac{k^2 t^2}{2d_m} - t \log(1 + n^{-c})\right) + 2^{r(n)} \varepsilon_{\text{PRS}}(n + m(n_A), \lambda, kt) \delta_n^{-t}. \end{aligned} \quad (\text{A.3.17})$$

Let $s(n) = \omega(\log n)$ be any polynomially bounded function, for instance $s(n) = \log^2 n$, and set

$$t(n) := \left\lceil \frac{(r(n) + 2s(n)) \log 2}{\log(1 + n^{-c})} \right\rceil. \quad (\text{A.3.18})$$

Since $\log(1 + n^{-c}) = \Theta(n^{-c})$ and $r(n), s(n) \in \text{poly}(n)$, this $t(n)$ is polynomially bounded. Moreover,

$$t \log(1 + n^{-c}) \geq (r(n) + 2s(n)) \log 2. \quad (\text{A.3.19})$$

Hence the first term in (A.3.17) is bounded by

$$2^{r(n)} \exp\left(\frac{k^2 t^2}{2d_m} - t \log(1 + n^{-c})\right) \leq \exp\left(\frac{k^2 t^2}{2d_m}\right) 2^{-2s(n)}. \quad (\text{A.3.20})$$

Since $k, t \in \text{poly}(n)$ and $d_m = 2^{m(n_A)}$, we have

$$\frac{k^2 t^2}{2d_m} = \text{negl}(n).$$

Therefore the first term is at most $\text{negl}(n)$ for all sufficiently large n . It remains to choose the security parameters so that the second term is negligible. Since

$$\delta_n^{-t} = \frac{d_n^{kt}}{(1 + n^{-c})^t} \leq d_n^{kt} = 2^{nkt},$$

it is enough to ensure

$$\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, kt) \leq 2^{-r(n) - nkt - s(n)}. \quad (\text{A.3.21})$$

Using Eq. (A.3.10), this is achieved by choosing $\lambda = \lambda(n) \in \text{poly}(n)$ such that

$$\lambda(n) \geq 2(r(n) + nkt + s(n)) + 2 \log_2(kt \sqrt{n + m(n_A)}) + O(1),$$

and then choosing $\ell = \ell(n) \in \text{poly}(n)$ large enough so that

$$2^{-\Omega(\ell(n)^\alpha)} \leq 2^{-r(n) - nkt - s(n)}.$$

This is possible because the functions k, t, r, n and $m(n_A)$ are polynomially bounded. Hence (A.3.21) holds.

Combining (A.3.17)–(A.3.21), we obtain, with probability $1 - \text{negl}(n)$ over s ,

$$\max_{T \in \text{QPT}_{\leq p}} F_T(\rho_{s,AB}^{\otimes k}) \leq \frac{1 + n^{-c}}{d_n^k}. \quad (\text{A.3.22})$$

By Theorem 4.36,

$$\begin{aligned}
\hat{H}_{\min}(A^k|B^k)_{\rho_{s,AB}^{\otimes k}} &= -\log \left(d_A^k \max_{T \in \text{QPT}_{\leq p}} F_T(\rho_{s,AB}^{\otimes k}) \right) \\
&\geq -\log \left(d_A^k \frac{1 + n^{-c}}{d_n^k} \right) \\
&= kn_A - \log(1 + n^{-c}) \\
&= kn_A - O(n^{-c}).
\end{aligned}$$

Let us now prove the information-theoretic upper bound. Let

$$\rho_{s,B} = \text{Tr}_{AM} |\psi_s\rangle\langle\psi_s|_{ABM}.$$

We will show that $\rho_{s,B}$ has almost minimal purity. Recall that the purity $\text{Tr}(\rho_{s,B}^2)$ is the expectation value of the swap operator on two copies of the B register. Similarly, $(\text{Tr}(\rho_{s,B}^2))^2$ is the expectation value of the tensor product of two such swap operators on four copies. Therefore, applying the corresponding Hadamard tests and using PRS indistinguishability from Haar-random states, we obtain

$$\left| \mathbb{E}_s \text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right| \leq 2\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 2), \quad (\text{A.3.23})$$

$$\left| \mathbb{E}_s (\text{Tr}(\rho_{s,B}^2))^2 - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} (\text{Tr}(\phi_B^2))^2 \right| \leq 2\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 4). \quad (\text{A.3.24})$$

For a Haar-random pure state on ABM , the reduced state ϕ_B is an induced random density matrix with system dimension d_A and environment dimension $d_A d_m$. Hence, by [Nec07, Eq. (39)],

$$\mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) = \frac{d_A + d_A d_m}{d_A^2 d_m + 1} \leq \frac{1 + d_m^{-1}}{d_A}. \quad (\text{A.3.25})$$

Moreover, by the exact second cumulant of the purity distribution [Gir07, Eq. (14)], applied with dimensions d_A and $d_A d_m$,

$$\begin{aligned}
\text{Var}_{\phi \sim \text{Haar}(ABM)} (\text{Tr}(\phi_B^2)) &= \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \left(\text{Tr}(\phi_B^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\
&= \frac{2(d_A^2 - 1)(d_A^2 d_m^2 - 1)}{(d_A^2 d_m + 1)^2 (d_A^2 d_m + 2)(d_A^2 d_m + 3)} \\
&\leq \frac{2}{d_A^4 d_m^2}. \quad (\text{A.3.26})
\end{aligned}$$

Equations (A.3.23) and (A.3.24) imply

$$\begin{aligned}
& \mathbb{E}_s \left(\text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\
&= \mathbb{E}_s \left(\text{Tr}(\rho_{s,B}^2) \right)^2 - 2 \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \mathbb{E}_s \text{Tr}(\rho_{s,B}^2) + \left(\mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\
&\leq \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \left(\text{Tr}(\phi_B^2) \right)^2 + 2\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 4) \\
&\quad - 2 \left(\mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 + 4 \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 2) \\
&\quad + \left(\mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\
&= \text{Var}_{\phi \sim \text{Haar}(ABM)} \left(\text{Tr}(\phi_B^2) \right) + 2\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 4) \\
&\quad + 4 \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 2). \tag{A.3.27}
\end{aligned}$$

Since $\text{Tr}(\phi_B^2) \leq 1$, we have

$$\begin{aligned}
& \mathbb{E}_s \left(\text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\
&\leq \text{Var}_{\phi \sim \text{Haar}(ABM)} \left(\text{Tr}(\phi_B^2) \right) + 2\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 4) + 4\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 2) \\
&\leq \frac{2}{d_A^4 d_m^2} + 2\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 4) + 4\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, 2). \tag{A.3.28}
\end{aligned}$$

We now choose the security parameters so that the PRS errors in (A.3.28) are of the same order as the Haar variance. By Eq. (A.3.10), applied with $Q \in \{2, 4\}$ copies, and by the QPRF estimate following Lemma 5.15, we have

$$\varepsilon_{\text{PRS}}(n + m(n_A), \lambda, Q) \leq (1 + \sqrt{2\pi})Q\sqrt{n + m(n_A)} 2^{-\lambda/2} + 2\pi Q 2^{-\lambda} + 2^{-\lambda} + 2^{-\Omega(\ell^\alpha)}.$$

For $Q \in \{2, 4\}$, the sum of the first three terms is bounded by

$$\begin{aligned}
& \sum_{Q \in \{2, 4\}} \left[(1 + \sqrt{2\pi})Q\sqrt{n + m(n_A)} 2^{-\lambda/2} + 2\pi Q 2^{-\lambda} + 2^{-\lambda} \right] \\
&\leq \left(6(1 + \sqrt{2\pi}) + 12\pi + 2 \right) \sqrt{n + m(n_A)} 2^{-\lambda/2},
\end{aligned}$$

where we used $2^{-\lambda} \leq 2^{-\lambda/2}$ and $n + m(n_A) \geq 1$. Therefore, choosing λ such that it also satisfies

$$\lambda = \lambda(n) \geq 8n_A + 4m(n_A) + 2 \log_2 \left(2 \left(6(1 + \sqrt{2\pi}) + 12\pi + 2 \right) \sqrt{n + m(n_A)} \right)$$

gives

$$\sum_{Q \in \{2, 4\}} \left[(1 + \sqrt{2\pi})Q\sqrt{n + m(n_A)} 2^{-\lambda/2} + 2\pi Q 2^{-\lambda} + 2^{-\lambda} \right] \leq \frac{1}{2d_A^4 d_m^2}.$$

Since $m(n_A) \leq n_A$, this choice satisfies $\lambda(n) \in \text{poly}(n)$.

Next choose $\ell = \ell(n) \in \text{poly}(n)$ such that it also satisfies

$$2^{-\Omega(\ell(n)^\alpha)} \leq \frac{1}{4d_A^4 d_m^2}.$$

This is possible because $\lambda(n)$, n_A , and $m(n_A)$ are polynomially bounded. Consequently,

$$\varepsilon_{\text{PRS}}(n + m(n_A), \lambda(n), 2) + \varepsilon_{\text{PRS}}(n + m(n_A), \lambda(n), 4) \leq \frac{1}{d_A^4 d_m^2}. \quad (\text{A.3.29})$$

Substituting (A.3.29) into (A.3.28), we obtain

$$\begin{aligned} & \mathbb{E}_s \left(\text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\ & \leq \frac{2}{d_A^4 d_m^2} + 4(\varepsilon_{\text{PRS}}(n + m(n_A), \lambda(n), 2) + \varepsilon_{\text{PRS}}(n + m(n_A), \lambda(n), 4)) \\ & \leq \frac{6}{d_A^4 d_m^2}. \end{aligned} \quad (\text{A.3.30})$$

By Markov's inequality applied to the squared deviation from $\delta = \frac{1}{2d_A d_m^{1/4}}$ and (A.3.30), we have that

$$\begin{aligned} & \Pr_s \left[\left(\text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \geq \frac{1}{4d_A^2 d_m^{1/2}} \right] \\ & \leq 4d_A^2 d_m^{1/2} \mathbb{E}_s \left(\text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right)^2 \\ & \leq \frac{24}{d_A^2 d_m^{3/2}} = \text{negl}(n). \end{aligned} \quad (\text{A.3.31})$$

Equivalently, with probability $1 - \text{negl}(n)$ over s ,

$$\left| \text{Tr}(\rho_{s,B}^2) - \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) \right| \leq \frac{1}{2d_A d_m^{1/4}}. \quad (\text{A.3.32})$$

Combining (A.3.32) with (A.3.25), we obtain

$$\begin{aligned} \text{Tr}(\rho_{s,B}^2) & \leq \mathbb{E}_{\phi \sim \text{Haar}(ABM)} \text{Tr}(\phi_B^2) + \frac{1}{2d_A d_m^{1/4}} \\ & \leq \frac{1 + d_m^{-1}}{d_A} + \frac{1}{2d_A d_m^{1/4}} \\ & \leq \frac{1 + d_m^{-1/4}}{d_A}, \end{aligned} \quad (\text{A.3.33})$$

for all sufficiently large n with probability $1 - \text{negl}(n)$ over s .

On the event (A.3.33), the collision entropy lower bounds the von Neumann entropy, and therefore

$$\begin{aligned} H(B)_{\rho_{s,B}} & \geq H_2(B)_{\rho_{s,B}} = -\log \text{Tr}(\rho_{s,B}^2) \\ & \geq n_A - \log(1 + d_m^{-1/4}). \end{aligned}$$

Since $\rho_{s,AB}$ is purified by M ,

$$H(AB)_{\rho_{s,AB}} = H(M)_{\psi_s} \leq \log d_m = m(n_A).$$

Consequently,

$$\begin{aligned} H_{\min}(A^k|B^k)_{\rho_{s,AB}^{\otimes k}} &\leq H(A^k|B^k)_{\rho_{s,AB}^{\otimes k}} \\ &= k H(A|B)_{\rho_{s,AB}} \\ &= k(H(AB)_{\rho_{s,AB}} - H(B)_{\rho_{s,B}}) \\ &\leq -k(n_A - m(n_A)) + k \log(1 + d_m^{-1/4}). \end{aligned}$$

Finally, since $d_m = 2^{m(n_A)}$, $m(n_A) = \omega(\log n_A)$, and $k \in \text{poly}(n)$,

$$k \log(1 + d_m^{-1/4}) = \text{negl}(n).$$

Hence

$$H_{\min}(A^k|B^k)_{\rho_{s,AB}^{\otimes k}} \leq -k(n_A - m(n_A)) + \text{negl}(n).$$

Taking the union bound with the computational event, both estimates hold simultaneously with probability $1 - \text{negl}(n)$ over the key s , completing the proof.

□

A.4. Proofs of Chapter 6

A.4.1. Proof of Theorem 6.2

Proof. The upper bound on the Stein's exponent can be proven in the following way. Let us consider for an $\varepsilon > 0$ and $m(k) \in \text{poly}(k) > 0$, a binary measurement $M := (E_{k \cdot m(k)}, \mathbb{1} - E_{k \cdot m(k)}) \in \mathbf{M}_{k \cdot m(k)}^{\text{eff}}$ that satisfies $\text{Tr}[E_{k \cdot m(k)} \rho_k^{\otimes m(k)}] \geq 1 - \varepsilon$. Recall that the infimum over all such measurements is the one that defines $\beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}})$ in Eq. (6.1.3).

Then we have

$$\begin{aligned}
& \sup_{M' \in \mathbf{M}_{k \cdot m(k)}^{\text{eff}}} D(\mathcal{M}_{M'}(\rho_k^{\otimes m(k)}) \| \mathcal{M}_{M'}(\sigma_k^{\otimes m(k)})) \\
& \geq D(\mathcal{M}_M(\rho_k^{\otimes m(k)}) \| \mathcal{M}_M(\sigma_k^{\otimes m(k)})) \\
& = \text{Tr} \left[E_{k \cdot m(k)} \rho_k^{\otimes m(k)} \right] \log \text{Tr} \left[E_{k \cdot m(k)} \rho_k^{\otimes m(k)} \right] \\
& \quad - \text{Tr} \left[E_{k \cdot m(k)} \rho_k^{\otimes m(k)} \right] \log \text{Tr} \left[E_{k \cdot m(k)} \sigma_k^{\otimes m(k)} \right] \\
& \quad + \text{Tr} \left[(\mathbb{1} - E_{k \cdot m(k)}) \rho_k^{\otimes m(k)} \right] \log \text{Tr} \left[(\mathbb{1} - E_{k \cdot m(k)}) \rho_k^{\otimes m(k)} \right] \\
& \quad - \text{Tr} \left[(\mathbb{1} - E_{k \cdot m(k)}) \rho_k^{\otimes m(k)} \right] \log \text{Tr} \left[(\mathbb{1} - E_{k \cdot m(k)}) \sigma_k^{\otimes m(k)} \right] \\
& \geq -h_2 \left(\text{Tr} \left[E_{k \cdot m(k)} \rho_k^{\otimes m(k)} \right] \right) \\
& \quad - \text{Tr} \left[E_{k \cdot m(k)} \rho_k^{\otimes m(k)} \right] \log \text{Tr} \left[E_{k \cdot m(k)} \sigma_k^{\otimes m(k)} \right] \\
& \geq -1 - (1 - \varepsilon) \log \text{Tr} \left[E_{k \cdot m(k)} \sigma_k^{\otimes m(k)} \right],
\end{aligned}$$

where $h_2(p) = -p \log(p) - (1-p) \log(1-p)$ is the binary entropy. For the second-to-last line, we use the fact that

$$\text{Tr} \left[(\mathbb{1} - E_{k \cdot m(k)}) \rho_k^{\otimes m(k)} \right] \log \text{Tr} \left[(\mathbb{1} - E_{k \cdot m(k)}) \sigma_k^{\otimes m(k)} \right] \leq 0. \quad (\text{A.4.1})$$

The last line then follows from the fact that the binary entropy is upper bounded by 1 and the constraint that $\text{Tr} \left[E_{k \cdot m(k)} \rho_k^{\otimes m(k)} \right] \geq 1 - \varepsilon$. Rearranging this inequality and subsequently taking the supremum over all such efficient measurements M yields

$$\begin{aligned}
& -\log \beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}) \\
& \leq \frac{1}{1 - \varepsilon} \left(1 + \sup_{M' \in \mathbf{M}_{k \cdot m(k)}^{\text{eff}}} D(\mathcal{M}_{M'}(\rho_k^{\otimes m(k)}) \| \mathcal{M}_{M'}(\sigma_k^{\otimes m(k)})) \right).
\end{aligned}$$

Now dividing by $m(k)$, and taking the $\limsup$ in k , gives

$$\begin{aligned}
& \limsup_{k \rightarrow \infty} -\frac{1}{m(k)} \log \beta_{m(k)}^\varepsilon(\mathbf{M}_{k \cdot m(k)}^{\text{eff}}) \\
& \leq \frac{1}{1 - \varepsilon} \limsup_{k \rightarrow \infty} \frac{1}{m(k)} \left(1 + \mathring{D}^{\mathbf{M}_{k \cdot m(k)}^{\text{eff}}}(\rho_k^{\otimes m(k)} \| \sigma_k^{\otimes m(k)}) \right) \\
& = \frac{1}{1 - \varepsilon} \limsup_{k \rightarrow \infty} \frac{1}{m(k)} \left(\mathring{D}^{\mathbf{M}_{k \cdot m(k)}^{\text{eff}}}(\rho_k^{\otimes m(k)} \| \sigma_k^{\otimes m(k)}) \right) \\
& = \frac{1}{1 - \varepsilon} \mathring{D}^{\mathbf{M}^{\text{eff}}, \infty}(\{\rho_k\}_k \| \{\sigma_k\}_k).
\end{aligned}$$

Finally taking the limit $\varepsilon \rightarrow 0$, yields the claimed result. $\square$

A.4.2. Proof of Theorem 6.7

Proof. In order to end up with the desired bound for $\mathring{D}_{\mathcal{F}_n}$, we will first swap the infimum and supremum implicitly appearing in $\mathring{D}_{\mathcal{F}_n}$. We do this by using [BaHLP14, Lemma

13], see also [SW23, Lemma 10], which, however, a priori does not directly apply to our situation, since the set of measurement maps $\mathcal{M}$ corresponding to binary POVMs $\overline{\mathbf{M}}_n^{\text{eff}}$ is not closed under flagged finite convex combinations. We say a set $\mathcal{M}$ of CPTP maps is closed under flagged finite convex combinations [SW23, BaHLP14] if for any finite probability distribution $\{p_i\}_{i=1}^m$ and any $\mathcal{M}_i \in \mathcal{M}$ it follows that $\bigoplus_{i=1}^m p_i \mathcal{M}_i \in \mathcal{M}$. So as a first step we define

$$\mathcal{M}_n^{\text{eff}} := \left\{ \bigoplus_{i=1}^m p_i \mathcal{M}_i = \sum_{i=1}^m p_i \mathcal{M}_i \otimes |i\rangle\langle i| \mid m \in \mathbb{N}, p_i \geq 0, \sum_{i=1}^m p_i = 1, \mathcal{M}_i \equiv \mathcal{M}_{M_i}, M_i \in \overline{\mathbf{M}}_n^{\text{eff}} \right\},$$

where here $\mathcal{M}_i \equiv \mathcal{M}_{M_i}$ is the measurement channel implementing the binary POVM $M_i \in \overline{\mathbf{M}}_n^{\text{eff}}$. This set of flagged measurement channels is now closed under flagged finite convex combinations by construction, and so it follows by [SW23, Lemma 10] that

$$\inf_{\sigma \in \mathcal{F}_n} \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) = \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \inf_{\sigma \in \mathcal{F}_n} D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)).$$

Importantly by expanding our set of measurement channels we do not change our objective values. It holds that

$$\sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) = \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} D(\mathcal{M}_M(\rho) \| \mathcal{M}_M(\sigma)). \quad (\text{A.4.2})$$

To prove this one easily checks that for a channel $\mathcal{M} = \bigoplus_{i=1}^m p_i \mathcal{M}_i$ it holds that

$$D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) = \sum_{i=1}^m p_i D(\mathcal{M}_i(\rho) \| \mathcal{M}_i(\sigma)),$$

which is an average and hence the claimed Eq. (A.4.2) follows. Completely analogously one shows that

$$\sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \|\mathcal{M}(\rho) - \mathcal{M}(\rho')\|_1 = \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} \|\mathcal{M}_M(\rho) - \mathcal{M}_M(\rho')\|_1 = 2\hat{\Delta}(\rho, \rho'). \quad (\text{A.4.3})$$

Now putting these observations together we get

$$\begin{aligned}
|\dot{D}_{\mathcal{F}_n}(\rho) - \dot{D}_{\mathcal{F}_n}(\rho')| &= \left| \min_{\sigma \in \mathcal{F}_n} \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} D(\mathcal{M}_M(\rho) \| \mathcal{M}_M(\sigma)) \right. \\
&\quad \left. - \min_{\sigma \in \mathcal{F}_n} \sup_{M \in \overline{\mathbf{M}}_n^{\text{eff}}} D(\mathcal{M}_M(\rho') \| \mathcal{M}_M(\sigma)) \right| \\
&= \left| \min_{\sigma \in \mathcal{F}_n} \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) - \min_{\sigma \in \mathcal{F}_n} \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} D(\mathcal{M}(\rho') \| \mathcal{M}(\sigma)) \right| \\
&= \left| \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \min_{\sigma \in \mathcal{F}_n} D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) - \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \min_{\sigma \in \mathcal{F}_n} D(\mathcal{M}(\rho') \| \mathcal{M}(\sigma)) \right| \\
&\leq \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \left| \min_{\sigma \in \mathcal{F}_n} D(\mathcal{M}(\rho) \| \mathcal{M}(\sigma)) - \min_{\sigma \in \mathcal{F}_n} D(\mathcal{M}(\rho') \| \mathcal{M}(\sigma)) \right| \\
&= \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \left| \min_{\sigma \in \mathcal{F}_n} \sum_{i=1}^m p_i D(\mathcal{M}_i(\rho) \| \mathcal{M}_i(\sigma)) \right. \\
&\quad \left. - \min_{\sigma \in \mathcal{F}_n} \sum_{i=1}^m p_i D(\mathcal{M}_i(\rho') \| \mathcal{M}_i(\sigma)) \right|, \tag{A.4.4}
\end{aligned}$$

where in the last line the supremum is over all $\mathcal{M} = \bigoplus_{i=1}^m p_i \mathcal{M}_i$, where $\mathcal{M}_i$ is a measurement channel corresponding to the binary POVM $M_i \in \overline{\mathbf{M}}_n^{\text{eff}}$. Before we now continue to bound the last term of Eq. (A.4.4) —inspired by the proof of [Win16, Lemma 7]— we introduce some notation. Fix some $\mathcal{M} = \bigoplus_{i=1}^m p_i \mathcal{M}_i \in \mathcal{M}_n^{\text{eff}}$ and define $\delta := \frac{1}{2} \|\mathcal{M}(\rho) - \mathcal{M}(\rho')\|_1$, $\delta_i := \frac{1}{2} \|\mathcal{M}_i(\rho) - \mathcal{M}_i(\rho')\|_1$, then it follows that

$$\delta = \sum_i p_i \delta_i \leq \dot{\Delta}(\rho, \rho')$$

and that $\eta_i := \frac{1}{\delta_i} (\mathcal{M}_i(\rho) - \mathcal{M}_i(\rho'))_+$ and $\eta'_i := \frac{1}{\delta_i} (\mathcal{M}_i(\rho') - \mathcal{M}_i(\rho))_+$ are both valid quantum states, where $(A)_+$ is the projection onto the positive part of A . We can now bound

$$\begin{aligned}
\mathcal{M}_i(\rho) &= \mathcal{M}_i(\rho') + (\mathcal{M}_i(\rho) - \mathcal{M}_i(\rho')) \\
&\leq \mathcal{M}_i(\rho') + \delta_i \eta_i \\
&= (1 + \delta_i) \left(\frac{1}{1 + \delta_i} \mathcal{M}_i(\rho') + \frac{\delta_i}{1 + \delta_i} \eta_i \right) \\
&=: (1 + \delta_i) \omega_i,
\end{aligned}$$

to get that

$$\omega_i = \frac{1}{1 + \delta_i} \mathcal{M}_i(\rho') + \frac{\delta_i}{1 + \delta_i} \eta_i = \frac{1}{1 + \delta_i} \mathcal{M}_i(\rho) + \frac{\delta_i}{1 + \delta_i} \eta'_i.$$

Since the relative entropy is jointly convex and $\mathcal{F}_n$ is convex, minimization over $\sigma \in \mathcal{F}_n$ preserves convexity. Applying this property to the decomposition of ω_i yields

$$\begin{aligned} \min_{\sigma \in \mathcal{F}_n} \sum_i p_i (1 + \delta_i) D(\omega_i \| \mathcal{M}_i(\sigma)) &\leq \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho') \| \mathcal{M}_i(\sigma)) \\ &\quad + \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta_i \| \mathcal{M}_i(\sigma)), \end{aligned}$$

which rearranged gives

$$\begin{aligned} \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho') \| \mathcal{M}_i(\sigma)) &\geq \min_{\sigma \in \mathcal{F}_n} \sum_i p_i (1 + \delta_i) D(\omega_i \| \mathcal{M}_i(\sigma)) \\ &\quad - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta_i \| \mathcal{M}_i(\sigma)). \end{aligned} \quad (\text{A.4.5})$$

On the other hand, for any $\sigma \in \mathcal{F}_n$ we have that,

$$\begin{aligned} D(\omega_i \| \mathcal{M}_i(\sigma)) &= -S(\omega_i) - \text{Tr}[\omega_i \log(\mathcal{M}_i(\sigma))] \\ &\geq -h\left(\frac{\delta_i}{1 + \delta_i}\right) - \frac{1}{1 + \delta_i} S(\mathcal{M}_i(\rho)) - \frac{\delta_i}{1 + \delta_i} S(\eta'_i) \\ &\quad - \frac{1}{1 + \delta_i} \text{Tr}[\mathcal{M}_i(\rho) \log(\mathcal{M}_i(\sigma))] - \frac{\delta_i}{1 + \delta_i} \text{Tr}[\eta'_i \log(\mathcal{M}_i(\sigma))] \\ &= -h\left(\frac{\delta_i}{1 + \delta_i}\right) + \frac{1}{1 + \delta_i} D(\mathcal{M}_i(\rho) \| \mathcal{M}_i(\sigma)) + \frac{\delta_i}{1 + \delta_i} D(\eta'_i \| \mathcal{M}_i(\sigma)), \end{aligned}$$

where the first inequality is from [Win16, Eq. (1)] and follows from the strong sub-additivity of the von Neumann entropy. Multiplying with $p_i(1 + \delta_i)$, then rearranging the terms, and subsequently taking the minimum yields

$$\begin{aligned} \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho) \| \mathcal{M}_i(\sigma)) &\leq \min_{\sigma \in \mathcal{F}_n} \left(\sum_i p_i (1 + \delta_i) D(\omega_i \| \mathcal{M}_i(\sigma)) \right. \\ &\quad \left. + \sum_i p_i (1 + \delta_i) h\left(\frac{\delta_i}{1 + \delta_i}\right) - \sum_i p_i \delta_i D(\eta'_i \| \mathcal{M}_i(\sigma)) \right) \\ &\leq \min_{\sigma \in \mathcal{F}_n} \sum_i p_i (1 + \delta_i) D(\omega_i \| \mathcal{M}_i(\sigma)) + \sum_i p_i (1 + \delta_i) h\left(\frac{\delta_i}{1 + \delta_i}\right) \\ &\quad - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta'_i \| \mathcal{M}_i(\sigma)). \end{aligned} \quad (\text{A.4.6})$$

Putting both inequalities together we get that,

$$\begin{aligned} \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho) \| \mathcal{M}_i(\sigma)) &- \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho') \| \mathcal{M}_i(\sigma)) \\ &\leq \sum_i p_i (1 + \delta_i) h\left(\frac{\delta_i}{1 + \delta_i}\right) + \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta_i \| \mathcal{M}_i(\sigma)) - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta'_i \| \mathcal{M}_i(\sigma)). \end{aligned}$$

Observing that swapping $\rho \leftrightarrow \rho'$ implies $\eta \leftrightarrow \eta'$ and thus effectively a minus term in the above inequality where ρ, ρ', η, η' yields the same inequality with absolute values. Then,

we get

$$\begin{aligned}
& \left| \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho) \| \mathcal{M}_i(\sigma)) - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i D(\mathcal{M}_i(\rho') \| \mathcal{M}_i(\sigma)) \right| \\
& \leq \sum_i p_i \left((1 + \delta_i) h \left(\frac{\delta_i}{1 + \delta_i} \right) \right) + \left| \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta_i \| \mathcal{M}_i(\sigma)) - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D(\eta'_i \| \mathcal{M}_i(\sigma)) \right| \\
& = \sum_i p_i \left((1 + \delta_i) h \left(\frac{\delta_i}{1 + \delta_i} \right) \right) + \left| \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\rho) - \mathcal{M}_i(\rho'))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \right. \\
& \quad \left. - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\rho') - \mathcal{M}_i(\rho))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \right| \\
& \leq \sum_i p_i \left((1 + \delta_i) h \left(\frac{\delta_i}{1 + \delta_i} \right) \right) + \kappa_{\mathcal{M}}, \tag{A.4.7}
\end{aligned}$$

where in the second equality we substituted by the definitions of η_i, η'_i and, lastly, defined

$$\begin{aligned}
\kappa_{\mathcal{M}} := & \sup_{\tau, \tau' \in \mathcal{S}(\mathcal{H}_n)} \left[\min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\tau) - \mathcal{M}_i(\tau'))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \right. \\
& \left. - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\tau') - \mathcal{M}_i(\tau))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \right]
\end{aligned}$$

where we additionally implicitly constrain τ, τ' to satisfy $\text{Tr}[(\mathcal{M}_i(\tau) - \mathcal{M}_i(\tau'))_+] = \delta_i$. This definition does not require an absolute value, as exchanging $\tau \leftrightarrow \tau'$ can introduce a global minus sign. We bound this last term in the following way:

$$\begin{aligned}
\kappa_{\mathcal{M}} := & \sup_{\tau, \tau' \in \mathcal{S}(\mathcal{H}_n)} \left[\min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\tau) - \mathcal{M}_i(\tau'))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \right. \\
& \left. - \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\tau') - \mathcal{M}_i(\tau))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \right] \\
& \leq \sup_{\tau, \tau'} \min_{\sigma \in \mathcal{F}_n} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\tau) - \mathcal{M}_i(\tau'))_+}{\delta_i} \| \mathcal{M}_i(\sigma) \right) \\
& \leq \sup_{\tau, \tau'} \sum_i p_i \delta_i D \left(\frac{(\mathcal{M}_i(\tau) - \mathcal{M}_i(\tau'))_+}{\delta_i} \| \mathcal{M}_i(\sigma^*) \right) \\
& \leq - \sum_i p_i \delta_i \log \lambda_{\min}(\mathcal{M}_i(\sigma^*)) \\
& \leq \kappa \cdot \delta + \sum_i p_i \delta_i \log \left(\frac{2}{\delta_i} \right), \tag{A.4.8}
\end{aligned}$$

where in the second inequality we upper bounded by plugging in the full rank state σ^* . The third inequality follows from the fact that $\tilde{\tau}_i = \frac{(\mathcal{M}_i(\tau) - \mathcal{M}_i(\tau'))_+}{\delta_i}$ is a valid quantum state and hence the inequality $D(\tilde{\tau}_i \| \mathcal{M}_i(\sigma^*)) \leq -\log \lambda_{\min}(\mathcal{M}_i(\sigma^*))$ holds. The fourth

inequality above stems from the following lower bound

$$\begin{aligned}
\lambda_{\min}(\mathcal{M}_i(\sigma^*)) &= \min\{\text{Tr}[E_i\sigma^*], \text{Tr}[(\mathbb{1} - E_i)\sigma^*]\} \\
&\geq \lambda_{\min}(\sigma^*) \cdot \min\{\text{Tr}[E_i], \text{Tr}[\mathbb{1} - E_i]\} \\
&\geq \lambda_{\min}(\sigma^*) \cdot \frac{\delta_i}{2},
\end{aligned} \tag{A.4.9}$$

where $M_i = (E_i, \mathbb{1} - E_i)$ and where the last inequality itself follows from the following observation. We have

$$\begin{aligned}
\delta_i &= |\text{Tr}[E_i(\rho - \rho')]| \leq \|E_i\|_{\infty} \|\rho - \rho'\|_1 \leq 2 \text{Tr}[E_i], \\
\delta_i &= |\text{Tr}[(\mathbb{1} - E_i)(\rho - \rho')]| \\
&\leq \|\mathbb{1} - E_i\|_{\infty} \|\rho - \rho'\|_1 \\
&\leq 2 \text{Tr}[\mathbb{1} - E_i],
\end{aligned}$$

which imply that $\min\{\text{Tr}[E_i], \text{Tr}[\mathbb{1} - E_i]\} \geq \frac{\delta_i}{2}$. Now setting $\kappa := -\log \lambda_{\min}(\sigma^*)$, recalling $\delta = \sum_i p_i \delta_i$ and applying the logarithm yields the desired inequality.

Now putting everything together we have that,

$$\begin{aligned}
|\bar{D}_{\mathcal{F}_n}(\rho) - \bar{D}_{\mathcal{F}_n}(\rho')| &\leq \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \sum_i p_i \underbrace{\left[(1 + \delta_i) h\left(\frac{\delta_i}{1 + \delta_i}\right) + \delta_i \left(\kappa + \log\left(\frac{2}{\delta_i}\right) \right) \right]}_{=: f(\delta_i)} \\
&\equiv \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} \sum_i p_i f(\delta_i) \\
&\leq \sup_{\mathcal{M} \in \mathcal{M}_n^{\text{eff}}} f\left(\sum_i p_i \delta_i\right) = f(\delta) \leq f(\varepsilon) \\
&= (1 + \varepsilon) h\left(\frac{\varepsilon}{1 + \varepsilon}\right) + \varepsilon \left(\kappa + \log\left(\frac{2}{\varepsilon}\right) \right),
\end{aligned} \tag{A.4.10}$$

where we have used concavity and monotonicity, since

$$f(\delta_i) := (1 + \delta_i) h\left(\frac{\delta_i}{1 + \delta_i}\right) + \delta_i \left(\kappa + \log\left(\frac{2}{\delta_i}\right) \right)$$

is a positive, continuous, concave, and monotonically increasing function from $[0, 1]$ to $[0, 3 + \kappa]$. $\square$

A.5. Proofs and Benchmark Details of Chapter 8

A.5.1. Distance bound for a malicious Alice

Since Naor's commitment is computationally hiding, and so is the extractable layer, the security against a malicious Alice is bounded by the preservation of the hiding property, as discussed in Corollary 8.12. This bound is given by the Δ -distance between the key generated by ERE-Commitment and a string of uniformly random bits of length λ_{PQS} .

Theorem A.1 (Distance bound in ERE-Commitment for a malicious receiver). *Consider ERE-Commitment (see Algorithm 3), between an honest sender and a malicious receiver with a computationally hiding and $(\eta, 2^{-\eta m})$ -relaxed statistically binding EqCommitment as base commitment. Let $K := h(r, \mathbf{x})$ be the key in $\{0, 1\}^\ell$ output by Alice, where $h(r, \mathbf{x}) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ is a universal hash function, and $\bar{T} \subset [4\lambda_{EX}]$, the set of commitments that have not been challenged. Moreover, $M \subseteq \bar{T}$ with $|M| = m$ is a randomly chosen subset of non-challenged commitments that are input to the hash function. Then, for any sampling error $\xi, \delta > 0$, a bit-flip error $\alpha \geq 0$ and a proportion $\vartheta \geq 0$ of leaked bits:*

$$\Delta(\rho_{KE}, \frac{1}{2^\ell} \mathbb{1} \otimes \rho_E) \leq \frac{1}{2} \cdot 2^{-\frac{1}{2}[(\frac{1}{2}-\xi)m-2\vartheta\lambda_{EX}-h_2(\delta+\alpha+\eta)(m-2\vartheta\lambda_{EX})-\ell-q]} + \sqrt{6} \exp(-2\lambda_{EX}\delta^2/100) + 2 \exp(-4\xi^2\lambda_{EX}), \quad (\text{A.5.1})$$

where E denotes Bob's quantum state, q the length of the syndrome needed for performing the error correction code and $\mathbb{1} \in \mathcal{B}(\mathbb{C}^{2^\ell})$ is the identity operator.

Proof. Since the sampling strategy is the same as that of the OT layer, the sampling errors will be the same. The main difference is that, for this layer, the set of unchallenged commitments $\bar{T} \subset [2\lambda_{EX}]$ is not divided into two different subsets. Therefore, the malicious Bob has less prior information:

$$\begin{aligned} H_{\min}(X|E) &\geq d_H(\boldsymbol{\theta}|_{\bar{T}}, \hat{\boldsymbol{\theta}}|_{\bar{T}}) - h_2(\delta + \alpha + \eta)|\bar{T}| \\ &\geq \left(\frac{1}{2} - \xi - \vartheta\right) 2\lambda_{EX} - h_2(\delta + \alpha + \eta) 2\lambda_{EX}(1 - \vartheta). \end{aligned} \quad (\text{A.5.2})$$

As the subset $m \in \bar{T}$ is randomly chosen from the set $\bar{T}$, the sampling error for the bases and the states will be the same ones as for the set $\bar{T}$ for a large enough m . Therefore, its min-entropy will be:

$$H_{\min}(X_m|E) \geq \left(\frac{1}{2} - \xi\right) m - 2\vartheta\lambda_{EX} - h_2(\delta + \alpha + \eta)(m - 2\vartheta\lambda_{EX}). \quad (\text{A.5.3})$$

Given an η -relaxed statistical binding EqCommitment, the probability of being relaxed statistical binding is $1 - \text{negl}(k)$ when $\eta = \omega(\log^2(k)/k)$. $\square$

A.5.2. Benchmark of QOT protocols

The numerical benchmark presented in Table 8.1 is obtained by numerically optimizing the concrete security bounds of the corresponding protocols.

While [BCKM21] does not provide explicit analytical bounds, in [BCKM21, Sec. 5.2 and Sec. 6.4] the authors refer to [BF10] for the statistical binding proof. Hence, arguing as in Lemma 8.22, one can derive the following bound for the OT layer:

$$\Delta(\rho_{K_{\bar{b}}K_bE}, \frac{1}{2^l} \mathbb{1} \otimes \rho_{K_bE}) \leq \frac{1}{2} \cdot 2^{-\frac{1}{2}[(1-\xi-h(\delta))4\lambda_{OT}-l]} + \sqrt{6} \exp(-\delta^2 8\lambda_{OT}/100) \\ + 2 \exp(-\xi^2 4\lambda_{OT}),$$

where $\xi, \delta > 0$ denote sampling parameters and $h(r, x) : \{0, 1\}^m \times \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ is a universal hash function. Similarly, following the same arguments as in Theorem A.1, one obtains the following security bound for the extractable layer:

$$\Delta(\rho_{K_{\bar{b}}K_bE}, \frac{1}{2^l} \mathbb{1} \otimes \rho_{K_bE}) \leq \frac{1}{2} \cdot 2^{-\frac{1}{2}[(\frac{1}{2}-\xi-h(\delta))\lambda_{EX}^2-1]} + \sqrt{6} \exp(-\lambda_{EX}^3 \delta^2/100) \\ + 2 \exp(-2\xi^2 \lambda_{EX}^3),$$

where the security parameters are taken following [BCKM21, Protocol 3]. Following the structure of the protocol in [BCKM21], the total number of BB84 states is given by

$$\lambda_{\text{BCKM21}} = 16 \cdot \lambda_{OT} \cdot \lambda_{eq} \cdot 4 \cdot 2 \cdot \lambda_{EX}^3 + 16 \cdot \lambda_{OT},$$

since each OT layer consists of λ_{eq} equivocal commitments, each of them compounded by four executions of the extractable commitment. Therefore, by setting $\Delta \leq \varepsilon$ with $\varepsilon = 10^{-15}$ and $\lambda_{eq} = 128$, we obtain the value $\lambda_{\text{BCKM21}} = 2.27 \times 10^{13}$ reported in Table 8.1.

We now turn to the protocol proposed in [ABKK23]. Among their constructions, we focus on the ones based on BB84 states, namely [ABKK23, Protocol 9] and [ABKK23, Protocol 10], which implement string OT with random bases. The bound for the 3-round communication protocol of [ABKK23, Protocol 9] is given in [ABKK23, Theorem B.1]. It requires a total of $\lambda_{3\text{OT ABKK22}} = 23000\lambda$ BB84 states, where λ is determined by the following security bounds:

$$\mu_{R^*} = \frac{\sqrt{5}}{2^\lambda} + \frac{4q_{RO}}{2^{18\lambda}} + \frac{148(q_{RO} + 46000\lambda + 1)^3 + 1}{2^{2\lambda}} + \frac{368000q_{RO}\lambda}{2^\lambda}, \\ \mu_{S^*} = \frac{430q_{RO}\sqrt{\lambda}}{2^\lambda},$$

where μ_{R^*} and μ_{S^*} are, respectively, the security bounds against a malicious receiver and a malicious sender, and q_{RO} is an upper bound on the total number of random-oracle queries made by the adversary.

Similarly, [ABKK23, Protocol 10], which yields a 4-round OT protocol, requires a total of $\lambda_{4\text{OT ABKK22}} = 10300\lambda$ BB84 states, where λ is determined by the bounds in [ABKK23, Theorem B.2]:

$$\mu_{R^*} = \frac{\sqrt{5}}{2^\lambda} + \frac{1}{2^{9\lambda}} + \frac{148(q_{RO} + 2n + 1)^3 + 1}{2^{2\lambda}} + \frac{16q_{RO}n}{2^\lambda}, \\ \mu_{S^*} = \frac{288q_{RO}\sqrt{\lambda}}{2^\lambda},$$

where, following [ABKK23, Appendix B.2], we take $n = 10300 \lambda$.

As discussed in [ABKK23], the random oracle (RO) can be instantiated with SHA-256. Following [HLS24, Theorem 5.1], the optimal success probability for solving the preimage search problem on an m -bit random oracle with q_{RO} quantum queries is $\Theta(q_{RO}^2/2^m)$. Taking $m = 256$ and imposing $p_{\text{succ}} \leq 2^{-128}$ yields $q_{RO} \leq 2^{64}$, corresponding to 128-bit post-quantum preimage security. This upper bound on the number of oracle queries is also consistent with the NIST recommendation of considering at most 2^{64} oracle-like queries in its post-quantum security definitions [Nat16, Sec. 4.A.2/4.A.4].

By imposing $\mu_{R^*} = \mu_{S^*} \leq \varepsilon$ with $\varepsilon = 10^{-15}$, and using $q_{RO} = 2^{64}$, we obtain the numerical values $\lambda_{3\text{OTABKK22}} = 3.22 \times 10^6$ and $\lambda_{4\text{OTABKK22}} = 1.43 \times 10^6$ BB84 states reported in Table 8.1.

For our protocol, the optimization is performed by targeting at most $\varepsilon = 10^{-15}$ and fixing $\alpha = 0.006$ and $\vartheta = 0.001$ in Equation (8.7.3) and Equation (A.5.1), with an error correction syndrome of $q = 0.1\lambda^1$. It is important to emphasize that this is the only benchmark that explicitly incorporates transmission errors, since the protocols in [BCKM21, ABKK23] are not robust to such errors. For completeness, we also include in Table 8.1 the benchmark corresponding to a noiseless implementation of our protocol.

¹In the numerical optimization, we assume that the experimental error is uniformly distributed.

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